

Filter Summary Report: CG,TIA,Automated,simple,Z1,Z4,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 Z4 Z5: $\frac{Z_1 Z_4 Z_5 g_m - Z_1 Z_4}{2Z_1 Z_4 g_m + 2Z_1 Z_5 g_m + 2Z_1 + Z_4 + 2Z_5}$

$$H(z) = \frac{Z_1 Z_4 Z_5 g_m - Z_1 Z_4}{2Z_1 Z_4 g_m + 2Z_1 Z_5 g_m + 2Z_1 + Z_4 + 2Z_5}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{s(L_4 R_1 R_5 g_m - L_4 R_1)}{2R_1 R_5 g_m + 2R_1 + 2R_5 + s^2(2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + 2C_4 L_4 R_5) + s(2L_4 R_1 g_m + L_4)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + 2\sqrt{C_4} R_5}{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}}$$

$$\text{wo: } \frac{1}{\sqrt{C_4} \sqrt{L_4}}$$

$$\text{bandwidth: } \frac{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + 2\sqrt{C_4} R_5)}$$

K-LP: 0

K-HP: 0

$$\text{K-BP: } \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$$

Qz: None

Wz: None

3.2 BP-2 $Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s(L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^2(2C_4 L_4 R_1 R_4 R_5 g_m + 2C_4 L_4 R_1 R_4 + 2C_4 L_4 R_4 R_5) + s(2L_4 R_1 R_4 g_m + 2L_4 R_1 R_5 g_m + 2L_4 R_1 + L_4 R_4 + 2L_4 R_5)}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_4} R_1 R_4 R_5 g_m + 2\sqrt{C_4} R_1 R_4 + 2\sqrt{C_4} R_4 R_5}{2\sqrt{L_4} R_1 R_4 g_m + 2\sqrt{L_4} R_1 R_5 g_m + 2\sqrt{L_4} R_1 + \sqrt{L_4} R_4 + 2\sqrt{L_4} R_5}$$

$$\text{wo: } \frac{1}{\sqrt{C_4} \sqrt{L_4}}$$

$$\text{bandwidth: } \frac{2\sqrt{L_4} R_1 R_4 g_m + 2\sqrt{L_4} R_1 R_5 g_m + 2\sqrt{L_4} R_1 + \sqrt{L_4} R_4 + 2\sqrt{L_4} R_5}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{C_4} R_1 R_4 R_5 g_m + 2\sqrt{C_4} R_1 R_4 + 2\sqrt{C_4} R_4 R_5)}$$

K-LP: 0

K-HP: 0

$$\text{K-BP: } \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$$

Qz: None

Wz: None

3.3 BP-3 $Z(s) = \left(L_1 s, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{s(L_1 R_5 g_m - L_1)}{s^2(2C_4 L_1 R_5 g_m + 2C_4 L_1) + s(2C_4 R_5 + 2L_1 g_m) + 1}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4}\sqrt{L_1} R_5 g_m \sqrt{\frac{1}{R_5 g_m + 1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1} \sqrt{\frac{1}{R_5 g_m + 1}}}{2C_4 R_5 + 2L_1 g_m}$$

$$\text{wo: } \sqrt{\frac{1}{2C_4 L_1 R_5 g_m + 2C_4 L_1}}$$

$$\text{bandwidth: } \frac{(2C_4 R_5 + 2L_1 g_m) \sqrt{\frac{1}{2C_4 L_1 R_5 g_m + 2C_4 L_1}}}{\sqrt{2}\sqrt{C_4}\sqrt{L_1} R_5 g_m \sqrt{\frac{1}{R_5 g_m + 1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1} \sqrt{\frac{1}{R_5 g_m + 1}}}$$

K-LP: 0

K-HP: 0

$$\text{K-BP: } \frac{L_1 R_5 g_m - L_1}{2C_4 R_5 + 2L_1 g_m}$$

Qz: None
Wz: None

3.4 BP-4 $Z(s) = \left(L_1 s, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{s(L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^2(2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4) + s(2C_4 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}R_5g_m\sqrt{R_4+2R_5}\sqrt{\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}\sqrt{R_4+2R_5}\sqrt{\frac{1}{R_5g_m+1}}}{2C_4R_4R_5+2L_1R_4g_m+2L_1R_5g_m+2L_1}$
wo: $\sqrt{R_4+2R_5}\sqrt{\frac{1}{2C_4L_1R_4R_5g_m+2C_4L_1R_4}}$
bandwidth: $\frac{\sqrt{R_4+2R_5}(2C_4R_4R_5+2L_1R_4g_m+2L_1R_5g_m+2L_1)\sqrt{\frac{1}{2C_4L_1R_4R_5g_m+2C_4L_1R_4}}}{\sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}R_5g_m\sqrt{R_4+2R_5}\sqrt{\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{R_4}\sqrt{R_4+2R_5}\sqrt{\frac{1}{R_5g_m+1}}}$
K-LP: 0
K-HP: 0
K-BP: $\frac{L_1R_4R_5g_m-L_1R_4}{2C_4R_4R_5+2L_1R_4g_m+2L_1R_5g_m+2L_1}$
Qz: None
Wz: None

3.5 BP-5 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{s(L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^2(C_1 L_1 R_4 + 2C_1 L_1 R_5) + s(2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_4+2\sqrt{C_1}R_5}{2\sqrt{L_1}R_4g_m+2\sqrt{L_1}R_5g_m+2\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{2\sqrt{L_1}R_4g_m+2\sqrt{L_1}R_5g_m+2\sqrt{L_1}}{\sqrt{C_1}\sqrt{L_1}(\sqrt{C_1}R_4+2\sqrt{C_1}R_5)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
Qz: None
Wz: None

3.6 BP-6 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{s(L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{R_1 R_4 + 2R_1 R_5 + s^2(C_1 L_1 R_1 R_4 + 2C_1 L_1 R_1 R_5) + s(2L_1 R_1 R_4 g_m + 2L_1 R_1 R_5 g_m + 2L_1 R_1 + L_1 R_4 + 2L_1 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1R_4+2\sqrt{C_1}R_1R_5}{2\sqrt{L_1}R_1R_4g_m+2\sqrt{L_1}R_1R_5g_m+2\sqrt{L_1}R_1+\sqrt{L_1}R_4+2\sqrt{L_1}R_5}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{2\sqrt{L_1}R_1R_4g_m+2\sqrt{L_1}R_1R_5g_m+2\sqrt{L_1}R_1+\sqrt{L_1}R_4+2\sqrt{L_1}R_5}{\sqrt{C_1}\sqrt{L_1}(\sqrt{C_1}R_1R_4+2\sqrt{C_1}R_1R_5)}$
K-LP: 0
K-HP: 0
K-BP: $\frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$
Qz: None
Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1)}{2 R_1 g_m + s^2 (2 C_4 L_4 R_1 g_m + C_4 L_4) + s (2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

Parameters:

Q: $\frac{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}}{2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + 2\sqrt{C_4} R_5}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + 2\sqrt{C_4} R_5}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{L_4} R_1 g_m + \sqrt{L_4})}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
 K-HP: $\frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

4.2 BS-2 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + C_4 L_4 R_4 + 2 C_4 L_4 R_5) + s (2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5)}$$

Parameters:

Q: $\frac{2\sqrt{L_4} R_1 R_4 g_m + 2\sqrt{L_4} R_1 R_5 g_m + 2\sqrt{L_4} R_1 + \sqrt{L_4} R_4 + 2\sqrt{L_4} R_5}{2\sqrt{C_4} R_1 R_4 R_5 g_m + 2\sqrt{C_4} R_1 R_4 + 2\sqrt{C_4} R_4 R_5}$
 wo: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$
 bandwidth: $\frac{2\sqrt{C_4} R_1 R_4 R_5 g_m + 2\sqrt{C_4} R_1 R_4 + 2\sqrt{C_4} R_4 R_5}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{L_4} R_1 R_4 g_m + 2\sqrt{L_4} R_1 R_5 g_m + 2\sqrt{L_4} R_1 + \sqrt{L_4} R_4 + 2\sqrt{L_4} R_5)}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5}$
 K-HP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_4} \sqrt{L_4}}$

4.3 BS-3 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_1 L_1 R_4 g_m + 2 C_1 L_1 R_5 g_m + 2 C_1 L_1) + s (C_1 R_4 + 2 C_1 R_5) + 2}$$

Parameters:

Q: $\frac{2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1}}{\sqrt{C_1} R_4 + 2\sqrt{C_1} R_5}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{C_1} R_4 + 2\sqrt{C_1} R_5}{\sqrt{C_1} \sqrt{L_1} (2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1})}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2}$
 K-HP: $\frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

4.4 BS-4 $Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (2 C_1 L_1 R_1 R_4 g_m + 2 C_1 L_1 R_1 R_5 g_m + 2 C_1 L_1 R_1 + C_1 L_1 R_4 + 2 C_1 L_1 R_5) + s (C_1 R_1 R_4 + 2 C_1 R_1 R_5)}$$

Parameters:

Q: $\frac{2\sqrt{L_1} R_1 R_4 g_m + 2\sqrt{L_1} R_1 R_5 g_m + 2\sqrt{L_1} R_1 + \sqrt{L_1} R_4 + 2\sqrt{L_1} R_5}{\sqrt{C_1} R_1 R_4 + 2\sqrt{C_1} R_1 R_5}$

$$\begin{aligned}
&\text{wo: } \frac{1}{\sqrt{C_1}\sqrt{L_1}} \\
&\text{bandwidth: } \frac{\sqrt{C_1}R_1R_4+2\sqrt{C_1}R_1R_5}{\sqrt{C_1}\sqrt{L_1}(2\sqrt{L_1}R_1R_4g_m+2\sqrt{L_1}R_1R_5g_m+2\sqrt{L_1}R_1+\sqrt{L_1}R_4+2\sqrt{L_1}R_5)} \\
&\text{K-LP: } \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5} \\
&\text{K-HP: } \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5} \\
&\text{K-BP: } 0 \\
&\text{Qz: None} \\
&\text{Wz: } \frac{1}{\sqrt{C_1}\sqrt{L_1}}
\end{aligned}$$

5 GE

$$5.1 \quad \text{GE-1 } Z(s) = \left(R_1, \infty, \infty, R_4, L_5s + \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{C_5L_5R_1R_4g_ms^2 - C_5R_1R_4s + R_1R_4g_m}{2R_1g_m + s^2(2C_5L_5R_1g_m + 2C_5L_5) + s(2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4) + 2}$$

Parameters:

$$\begin{aligned}
&\text{Q: } \frac{2\sqrt{L_5}R_1g_m+2\sqrt{L_5}}{2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4} \\
&\text{wo: } \frac{1}{\sqrt{C_5}\sqrt{L_5}} \\
&\text{bandwidth: } \frac{2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4}{\sqrt{C_5}\sqrt{L_5}(2\sqrt{L_5}R_1g_m+2\sqrt{L_5})} \\
&\text{K-LP: } \frac{R_1R_4g_m}{2R_1g_m+2} \\
&\text{K-HP: } \frac{R_1R_4g_m}{2R_1g_m+2} \\
&\text{K-BP: } -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\
&\text{Qz: } -\frac{\sqrt{L_5}g_m}{\sqrt{C_5}} \\
&\text{Wz: } \frac{1}{\sqrt{C_5}\sqrt{L_5}}
\end{aligned}$$

$$5.2 \quad \text{GE-2 } Z(s) = \left(R_1, \infty, \infty, R_4, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$$

$$H(s) = \frac{-C_5L_5R_1R_4s^2 + L_5R_1R_4g_ms - R_1R_4}{2R_1R_4g_m + 2R_1 + R_4 + s^2(2C_5L_5R_1R_4g_m + 2C_5L_5R_1 + C_5L_5R_4) + s(2L_5R_1g_m + 2L_5)}$$

Parameters:

$$\begin{aligned}
&\text{Q: } \frac{2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4}{2\sqrt{L_5}R_1g_m+2\sqrt{L_5}} \\
&\text{wo: } \frac{1}{\sqrt{C_5}\sqrt{L_5}} \\
&\text{bandwidth: } \frac{2\sqrt{L_5}R_1g_m+2\sqrt{L_5}}{\sqrt{C_5}\sqrt{L_5}(2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4)} \\
&\text{K-LP: } -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\
&\text{K-HP: } -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\
&\text{K-BP: } \frac{R_1R_4g_m}{2R_1g_m+2} \\
&\text{Qz: } -\frac{\sqrt{C_5}}{\sqrt{L_5}g_m} \\
&\text{Wz: } \frac{1}{\sqrt{C_5}\sqrt{L_5}}
\end{aligned}$$

$$5.3 \quad \text{GE-3 } Z(s) = \left(R_1, \infty, \infty, R_4, L_5s + R_5 + \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{C_5L_5R_1R_4g_ms^2 + R_1R_4g_m + s(C_5R_1R_4R_5g_m - C_5R_1R_4)}{2R_1g_m + s^2(2C_5L_5R_1g_m + 2C_5L_5) + s(2C_5R_1R_4g_m + 2C_5R_1R_5g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

Parameters:

$$\begin{aligned}
&\text{Q: } \frac{2\sqrt{L_5}R_1g_m+2\sqrt{L_5}}{2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1R_5g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4+2\sqrt{C_5}R_5} \\
&\text{wo: } \frac{1}{\sqrt{C_5}\sqrt{L_5}} \\
&\text{bandwidth: } \frac{2\sqrt{C_5}R_1R_4g_m+2\sqrt{C_5}R_1R_5g_m+2\sqrt{C_5}R_1+\sqrt{C_5}R_4+2\sqrt{C_5}R_5}{\sqrt{C_5}\sqrt{L_5}(2\sqrt{L_5}R_1g_m+2\sqrt{L_5})}
\end{aligned}$$

$$\begin{aligned}
\text{K-LP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\
\text{K-HP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\
\text{K-BP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{QZ: } & \frac{\sqrt{L_5 g_m}}{\sqrt{C_5 R_5 g_m} - \sqrt{C_5}} \\
\text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}}
\end{aligned}$$

$$\mathbf{5.4 \quad GE-4} \quad Z(s) = \left(R_1, \infty, \infty, R_4, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 R_5 s^2 - R_1 R_4 R_5 + s (L_5 R_1 R_4 R_5 g_m - L_5 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^2 (2C_5 L_5 R_1 R_4 R_5 g_m + 2C_5 L_5 R_1 R_5 + C_5 L_5 R_4 R_5) + s (2L_5 R_1 R_4 g_m + 2L_5 R_1 R_5 g_m + 2L_5 R_1 + L_5 R_4 + 2L_5 R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{C_5} R_1 R_4 R_5 g_m + 2\sqrt{C_5} R_1 R_5 + \sqrt{C_5} R_4 R_5}{2\sqrt{L_5} R_1 R_4 g_m + 2\sqrt{L_5} R_1 R_5 g_m + 2\sqrt{L_5} R_1 + \sqrt{L_5} R_4 + 2\sqrt{L_5} R_5} \\
\text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\
\text{bandwidth: } & \frac{2\sqrt{L_5} R_1 R_4 g_m + 2\sqrt{L_5} R_1 R_5 g_m + 2\sqrt{L_5} R_1 + \sqrt{L_5} R_4 + 2\sqrt{L_5} R_5}{\sqrt{C_5} \sqrt{L_5} (2\sqrt{C_5} R_1 R_4 R_5 g_m + 2\sqrt{C_5} R_1 R_5 + \sqrt{C_5} R_4 R_5)} \\
\text{K-LP: } & -\frac{R_1 R_4}{2R_1 R_4 g_m + 2R_1 + R_4} \\
\text{K-HP: } & -\frac{R_1 R_4}{2R_1 R_4 g_m + 2R_1 + R_4} \\
\text{K-BP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{QZ: } & -\frac{\sqrt{C_5} R_5}{\sqrt{L_5} R_5 g_m - \sqrt{L_5}} \\
\text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}}
\end{aligned}$$

$$\mathbf{5.5 \quad GE-5} \quad Z(s) = \left(R_1, \infty, \infty, R_4, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 R_4 g_m s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (2C_5 L_5 R_1 R_4 g_m + 2C_5 L_5 R_1 R_5 g_m + 2C_5 L_5 R_1 + C_5 L_5 R_4 + 2C_5 L_5 R_5) + s (2L_5 R_1 g_m + 2L_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{C_5} R_1 R_4 g_m + 2\sqrt{C_5} R_1 R_5 g_m + 2\sqrt{C_5} R_1 + \sqrt{C_5} R_4 + 2\sqrt{C_5} R_5}{2\sqrt{L_5} R_1 g_m + 2\sqrt{L_5}} \\
\text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\
\text{bandwidth: } & \frac{2\sqrt{L_5} R_1 g_m + 2\sqrt{L_5}}{\sqrt{C_5} \sqrt{L_5} (2\sqrt{C_5} R_1 R_4 g_m + 2\sqrt{C_5} R_1 R_5 g_m + 2\sqrt{C_5} R_1 + \sqrt{C_5} R_4 + 2\sqrt{C_5} R_5)} \\
\text{K-LP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{K-HP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{K-BP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\
\text{QZ: } & \frac{\sqrt{C_5} R_5 g_m - \sqrt{C_5}}{\sqrt{L_5} g_m} \\
\text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}}
\end{aligned}$$

$$\mathbf{5.6 \quad GE-6} \quad Z(s) = \left(R_1, \infty, \infty, R_4, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (2C_5 L_5 R_1 R_4 g_m + 2C_5 L_5 R_1 R_5 g_m + 2C_5 L_5 R_1 + C_5 L_5 R_4 + 2C_5 L_5 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{L_5} R_1 R_4 g_m + 2\sqrt{L_5} R_1 R_5 g_m + 2\sqrt{L_5} R_1 + \sqrt{L_5} R_4 + 2\sqrt{L_5} R_5}{2\sqrt{C_5} R_1 R_4 R_5 g_m + 2\sqrt{C_5} R_1 R_5 + \sqrt{C_5} R_4 R_5} \\
\text{wo: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}} \\
\text{bandwidth: } & \frac{2\sqrt{C_5} R_1 R_4 R_5 g_m + 2\sqrt{C_5} R_1 R_5 + \sqrt{C_5} R_4 R_5}{\sqrt{C_5} \sqrt{L_5} (2\sqrt{L_5} R_1 R_4 g_m + 2\sqrt{L_5} R_1 R_5 g_m + 2\sqrt{L_5} R_1 + \sqrt{L_5} R_4 + 2\sqrt{L_5} R_5)} \\
\text{K-LP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{K-HP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\
\text{K-BP: } & -\frac{R_1 R_4}{2R_1 R_4 g_m + 2R_1 + R_4} \\
\text{QZ: } & \frac{-\sqrt{L_5} R_5 g_m + \sqrt{L_5}}{\sqrt{C_5} R_5} \\
\text{WZ: } & \frac{1}{\sqrt{C_5} \sqrt{L_5}}
\end{aligned}$$

5.7 GE-7 $Z(s) = \left(R_1, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^2 (2 C_4 L_4 R_1 g_m + C_4 L_4) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}}{2\sqrt{C_4} R_1 R_4 g_m + 2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + \sqrt{C_4} R_4 + 2\sqrt{C_4} R_5} \\ \text{wo: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \\ \text{bandwidth: } & \frac{2\sqrt{C_4} R_1 R_4 g_m + 2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + \sqrt{C_4} R_4 + 2\sqrt{C_4} R_5}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{L_4} R_1 g_m + \sqrt{L_4})} \\ \text{K-LP: } & \frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1} \\ \text{K-HP: } & \frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1} \\ \text{K-BP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5} \\ \text{QZ: } & \frac{\sqrt{L_4}}{\sqrt{C_4} R_4} \\ \text{WZ: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \end{aligned}$$

5.8 GE-8 $Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + C_4 L_4 R_4 + 2 C_4 L_4 R_5) + s (2 L_4 R_1 g_m + L_4)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4} R_1 R_4 g_m + 2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + \sqrt{C_4} R_4 + 2\sqrt{C_4} R_5}{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}} \\ \text{wo: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \\ \text{bandwidth: } & \frac{2\sqrt{L_4} R_1 g_m + \sqrt{L_4}}{\sqrt{C_4} \sqrt{L_4} (2\sqrt{C_4} R_1 R_4 g_m + 2\sqrt{C_4} R_1 R_5 g_m + 2\sqrt{C_4} R_1 + \sqrt{C_4} R_4 + 2\sqrt{C_4} R_5)} \\ \text{K-LP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5} \\ \text{K-HP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5} \\ \text{K-BP: } & \frac{R_1 R_5 g_m - R_1}{2 R_1 g_m + 1} \\ \text{QZ: } & \frac{\sqrt{C_4} R_4}{\sqrt{L_4}} \\ \text{WZ: } & \frac{1}{\sqrt{C_4} \sqrt{L_4}} \end{aligned}$$

5.9 GE-9 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_1 L_1 R_4 g_m + 2 C_1 L_1 R_5 g_m + 2 C_1 L_1) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_1 R_5 g_m + 2 C_1 R_1 + C_1 R_4 + 2 C_1 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1}}{2\sqrt{C_1} R_1 R_4 g_m + 2\sqrt{C_1} R_1 R_5 g_m + 2\sqrt{C_1} R_1 + \sqrt{C_1} R_4 + 2\sqrt{C_1} R_5} \\ \text{wo: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \\ \text{bandwidth: } & \frac{2\sqrt{C_1} R_1 R_4 g_m + 2\sqrt{C_1} R_1 R_5 g_m + 2\sqrt{C_1} R_1 + \sqrt{C_1} R_4 + 2\sqrt{C_1} R_5}{\sqrt{C_1} \sqrt{L_1} (2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1})} \\ \text{K-LP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{K-HP: } & \frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2} \\ \text{K-BP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5} \\ \text{QZ: } & \frac{\sqrt{L_1}}{\sqrt{C_1} R_1} \\ \text{WZ: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \end{aligned}$$

5.10 GE-10 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 + C_1 L_1 R_4 + 2C_1 L_1 R_5) + s (2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

Parameters:

Q: $\frac{2\sqrt{C_1} R_1 R_4 g_m + 2\sqrt{C_1} R_1 R_5 g_m + 2\sqrt{C_1} R_1 + \sqrt{C_1} R_4 + 2\sqrt{C_1} R_5}{2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1}}$

wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

bandwidth: $\frac{2\sqrt{L_1} R_4 g_m + 2\sqrt{L_1} R_5 g_m + 2\sqrt{L_1}}{\sqrt{C_1} \sqrt{L_1} (2\sqrt{C_1} R_1 R_4 g_m + 2\sqrt{C_1} R_1 R_5 g_m + 2\sqrt{C_1} R_1 + \sqrt{C_1} R_4 + 2\sqrt{C_1} R_5)}$

K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$

K-HP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$

K-BP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$

Qz: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$

Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

6 HP

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m - 1}{2C_1 C_4 R_5 s^2 + 2g_m + s (C_1 + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_5} \sqrt{g_m}}{C_1 + 2C_4 R_5 g_m + 2C_4}$

wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_5}}$

bandwidth: $\frac{C_1 + 2C_4 R_5 g_m + 2C_4}{2C_1 C_4 R_5}$

K-LP: $\frac{R_5 g_m - 1}{2g_m}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.2 LP-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2C_1 C_4 R_4 R_5 s^2 + 2R_4 g_m + 2R_5 g_m + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

Parameters:

Q: $\frac{2\sqrt{C_1} \sqrt{C_4} \sqrt{R_4} \sqrt{R_5} \sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4}$

wo: $\frac{\sqrt{R_4 g_m + R_5 g_m + 1}}{\sqrt{C_1} \sqrt{C_4} \sqrt{R_4} \sqrt{R_5}}$

bandwidth: $\frac{C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4}{2C_1 C_4 R_4 R_5}$

K-LP: $\frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

7.3 LP-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2C_1 C_4 R_1 R_5 s^2 + 2R_1 g_m + s(C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1 g_m + 1}}{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 g_m + 1}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5}{2C_1 C_4 R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.4 LP-4 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2C_1 C_4 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5}{2C_1 C_4 R_1 R_4 R_5}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2C_4 C_5 R_1 R_5 s^2 + 2R_1 g_m + s(2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1 g_m + 1}}{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2}\sqrt{2R_1 g_m + 1}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$
 bandwidth: $\frac{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{2C_4 C_5 R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5}{2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2C_4 C_5 R_1 R_4 s^2 + 2R_1 g_m + s(2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1 g_m + 1}}{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4} \\ \text{wo: } & \frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}{2C_4 C_5 R_1 R_4} \\ \text{K-LP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_4}{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2C_4 C_5 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{2\sqrt{C_4}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{2C_4 C_5 R_1 R_4 R_5} \\ \text{K-LP: } & \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_1 R_4 R_5}{2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2(2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_4 + 2C_4 C_5 R_4 R_5) + s(2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5} \\ \text{wo: } & \sqrt{\frac{R_1g_m+1}{C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_1g_m+1}{C_4C_5R_1R_4R_5g_m+C_4C_5R_1R_4+C_4C_5R_4R_5}}(2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1R_5g_m+2C_5R_1+C_5R_4+2C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+\frac{1}{R_1R_5g_m+R_1+R_5}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_4}\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}R_5\sqrt{\frac{R_1g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}} \\ \text{K-LP: } & \frac{R_1 R_4 g_m}{2R_1 g_m + 2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4}{2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.5 X-INVALID-NUMER-5 $Z(s) = \left(L_1 s, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{s^2(2C_5 L_1 R_4 g_m + 2C_5 L_1) + s(C_5 R_4 + 2L_1 g_m) + 2}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_5}\sqrt{L_1}R_4g_m\sqrt{\frac{1}{R_4g_m+1}}+2\sqrt{C_5}\sqrt{L_1}\sqrt{\frac{1}{R_4g_m+1}}}{C_5R_4+2L_1g_m}$$

$$\text{wo: } \sqrt{\frac{1}{C_5 L_1 R_4 g_m + C_5 L_1}}$$

$$\text{bandwidth: } \frac{(C_5 R_4 + 2L_1 g_m) \sqrt{\frac{1}{C_5 L_1 R_4 g_m + C_5 L_1}}}{2\sqrt{C_5} \sqrt{L_1 R_4 g_m} \sqrt{\frac{1}{R_4 g_m + 1}} + 2\sqrt{C_5} \sqrt{L_1} \sqrt{\frac{1}{R_4 g_m + 1}}}$$

$$\text{K-LP: } 0$$

$$\text{K-HP: } -\frac{R_4}{2R_4 g_m + 2}$$

$$\text{K-BP: } \frac{L_1 R_4 g_m}{C_5 R_4 + 2L_1 g_m}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

$$\mathbf{8.6 \quad X-INVALID-NUMER-6} \quad Z(s) = \left(L_1 s, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s(L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^2(2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5) + s(C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_5} \sqrt{L_1 R_4} \sqrt{R_5 g_m} \sqrt{R_4 + 2R_5} \sqrt{\frac{1}{R_4 g_m + 1}} + \sqrt{2}\sqrt{C_5} \sqrt{L_1} \sqrt{R_5} \sqrt{R_4 + 2R_5} \sqrt{\frac{1}{R_4 g_m + 1}}}{C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1}$$

$$\text{wo: } \sqrt{R_4 + 2R_5} \sqrt{\frac{1}{2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5}}$$

$$\text{bandwidth: } \frac{\sqrt{R_4 + 2R_5} (C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1) \sqrt{\frac{1}{2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5}}}{\sqrt{2}\sqrt{C_5} \sqrt{L_1 R_4} \sqrt{R_5 g_m} \sqrt{R_4 + 2R_5} \sqrt{\frac{1}{R_4 g_m + 1}} + \sqrt{2}\sqrt{C_5} \sqrt{L_1} \sqrt{R_5} \sqrt{R_4 + 2R_5} \sqrt{\frac{1}{R_4 g_m + 1}}}$$

$$\text{K-LP: } 0$$

$$\text{K-HP: } -\frac{R_4}{2R_4 g_m + 2}$$

$$\text{K-BP: } \frac{L_1 R_4 R_5 g_m - L_1 R_4}{C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

$$\mathbf{8.7 \quad X-INVALID-NUMER-7} \quad Z(s) = \left(L_1 s, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^2(C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{s^2(2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5 g_m + 2C_5 L_1) + s(C_5 R_4 + 2C_5 R_5 + 2L_1 g_m) + 2}$$

Parameters:

$$\text{Q: } \frac{2\sqrt{C_5} \sqrt{L_1 R_4 g_m} \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}} + 2\sqrt{C_5} \sqrt{L_1} R_5 g_m \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}} + 2\sqrt{C_5} \sqrt{L_1} \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}}}{C_5 R_4 + 2C_5 R_5 + 2L_1 g_m}$$

$$\text{wo: } \sqrt{\frac{1}{C_5 L_1 R_4 g_m + C_5 L_1 R_5 g_m + C_5 L_1}}$$

$$\text{bandwidth: } \frac{(C_5 R_4 + 2C_5 R_5 + 2L_1 g_m) \sqrt{\frac{1}{C_5 L_1 R_4 g_m + C_5 L_1 R_5 g_m + C_5 L_1}}}{2\sqrt{C_5} \sqrt{L_1 R_4 g_m} \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}} + 2\sqrt{C_5} \sqrt{L_1} R_5 g_m \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}} + 2\sqrt{C_5} \sqrt{L_1} \sqrt{\frac{1}{R_4 g_m + R_5 g_m + 1}}}$$

$$\text{K-LP: } 0$$

$$\text{K-HP: } \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2}$$

$$\text{K-BP: } \frac{L_1 R_4 g_m}{C_5 R_4 + 2C_5 R_5 + 2L_1 g_m}$$

$$\text{Qz: None}$$

$$\text{Wz: None}$$

$$\mathbf{8.8 \quad X-INVALID-NUMER-8} \quad Z(s) = \left(L_1 s, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 s + L_1 g_m}{2C_4 C_5 L_1 s^2 + 2C_4 + C_5 + s(2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

Parameters:

$$\text{Q: } \frac{\sqrt{2}\sqrt{C_4} \sqrt{C_5} \sqrt{2C_4 + C_5}}{2C_4 \sqrt{L_1 g_m} + 2C_5 \sqrt{L_1 g_m}}$$

$$\text{wo: } \frac{\sqrt{2}\sqrt{2C_4 + C_5}}{2\sqrt{C_4} \sqrt{C_5} \sqrt{L_1}}$$

$$\text{bandwidth: } \frac{2C_4 \sqrt{L_1 g_m} + 2C_5 \sqrt{L_1 g_m}}{2C_4 C_5 \sqrt{L_1}}$$

$$\text{K-LP: } \frac{L_1 g_m}{2C_4 + C_5}$$

$$\text{K-HP: } 0$$

K-BP: $-\frac{C_5}{2C_4g_m+2C_5g_m}$
Qz: None
Wz: None

8.9 X-INVALID-NUMER-9 $Z(s) = \left(L_1s, \infty, \infty, \frac{1}{C_4s}, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{L_1g_m + s(C_5L_1R_5g_m - C_5L_1)}{2C_4 + C_5 + s^2(2C_4C_5L_1R_5g_m + 2C_4C_5L_1) + s(2C_4C_5R_5 + 2C_4L_1g_m + 2C_5L_1g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_5g_m\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_5g_m+1}}}{2C_4C_5R_5+2C_4L_1g_m+2C_5L_1g_m}$
wo: $\sqrt{2C_4 + C_5}\sqrt{\frac{1}{2C_4C_5L_1R_5g_m+2C_4C_5L_1}}$
bandwidth: $\frac{\sqrt{2C_4+C_5}(2C_4C_5R_5+2C_4L_1g_m+2C_5L_1g_m)\sqrt{\frac{1}{2C_4C_5L_1R_5g_m+2C_4C_5L_1}}}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_5g_m\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_5g_m+1}}}$
K-LP: $\frac{L_1g_m}{2C_4+C_5}$
K-HP: 0
K-BP: $\frac{C_5L_1R_5g_m-C_5L_1}{2C_4C_5R_5+2C_4L_1g_m+2C_5L_1g_m}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(L_1s, \infty, \infty, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{s^2(C_4L_1R_4R_5g_m - C_4L_1R_4) + s(L_1R_5g_m - L_1)}{s^2(2C_4L_1R_4g_m + 2C_4L_1R_5g_m + 2C_4L_1) + s(C_4R_4 + 2C_4R_5 + 2L_1g_m) + 1}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{L_1}R_4g_m\sqrt{\frac{1}{R_4g_m+R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1}R_5g_m\sqrt{\frac{1}{R_4g_m+R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{\frac{1}{R_4g_m+R_5g_m+1}}}{C_4R_4+2C_4R_5+2L_1g_m}$
wo: $\sqrt{\frac{1}{2C_4L_1R_4g_m+2C_4L_1R_5g_m+2C_4L_1}}$
bandwidth: $\frac{(C_4R_4+2C_4R_5+2L_1g_m)\sqrt{\frac{1}{2C_4L_1R_4g_m+2C_4L_1R_5g_m+2C_4L_1}}}{\sqrt{2}\sqrt{C_4}\sqrt{L_1}R_4g_m\sqrt{\frac{1}{R_4g_m+R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1}R_5g_m\sqrt{\frac{1}{R_4g_m+R_5g_m+1}} + \sqrt{2}\sqrt{C_4}\sqrt{L_1}\sqrt{\frac{1}{R_4g_m+R_5g_m+1}}}$
K-LP: 0
K-HP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-BP: $\frac{L_1R_5g_m-L_1}{C_4R_4+2C_4R_5+2L_1g_m}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_4s + R_4g_m}{C_1C_5R_4s^2 + 2g_m + s(2C_1 + 2C_5R_4g_m + 2C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}\sqrt{g_m}}{2C_1+2C_5R_4g_m+2C_5}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}}$
bandwidth: $\frac{2C_1+2C_5R_4g_m+2C_5}{C_1C_5R_4}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_1+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4}{C_1 C_5 R_4 R_5 s^2 + 2 R_4 g_m + 2 R_5 g_m + s (C_1 R_4 + 2 C_1 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}\sqrt{R_4 g_m + R_5 g_m + 1}}{C_1 R_4 + 2 C_1 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5}$
 wo: $\frac{\sqrt{2} R_4 g_m + 2 R_5 g_m + 2}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_4} \sqrt{R_5}}$
 bandwidth: $\frac{\sqrt{2}\sqrt{2 R_4 g_m + 2 R_5 g_m + 2}(C_1 R_4 + 2 C_1 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5)}{2 C_1 C_5 R_4 R_5 \sqrt{R_4 g_m + R_5 g_m + 1}}$
 K-LP: $\frac{R_4 R_5 g_m - R_4}{2 R_4 g_m + 2 R_5 g_m + 2}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_4 R_5}{C_1 R_4 + 2 C_1 R_5 + 2 C_5 R_4 R_5 g_m + 2 C_5 R_5}$
 Qz: None
 Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 g_m + s^2 (C_1 C_5 R_4 + 2 C_1 C_5 R_5) + s (2 C_1 + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{g_m}\sqrt{R_4 + 2 R_5}}{2 C_1 + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_4 + 2 C_1 C_5 R_5}}$
 bandwidth: $\frac{2 C_1 + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_4 + 2 R_5} \sqrt{C_1 C_5 R_4 + 2 C_1 C_5 R_5}}$
 K-LP: $\frac{R_4}{2}$
 K-HP: 0
 K-BP: $\frac{C_5 R_4 R_5 g_m - C_5 R_4}{2 C_1 + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5}$
 Qz: None
 Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{2 g_m + s^2 (2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_5) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_5}\sqrt{g_m}\sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5}}{C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_5}}$
 bandwidth: $\frac{C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m}{\sqrt{R_5} \sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5} \sqrt{2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_5}}$
 K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5}{C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m}$
 Qz: None
 Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_4 s + R_4 g_m}{2 g_m + s^2 (2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_4 C_5 R_4) + s (2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_4}\sqrt{g_m}\sqrt{2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5}}{2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_4 C_5 R_4}}$

bandwidth: $\frac{2C_1+2C_4R_4g_m+2C_5R_4g_m+2C_5}{\sqrt{R_4}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4R_4+C_1C_5R_4+2C_4C_5R_4}}$
K-LP: $\frac{R_4}{2}$
K-HP: 0
K-BP: $-\frac{C_5R_4}{2C_1+2C_4R_4g_m+2C_5R_4g_m+2C_5}$
Qz: None
Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_4R_5s + R_4R_5g_m - R_4}{2R_4g_m + 2R_5g_m + s^2(2C_1C_4R_4R_5 + C_1C_5R_4R_5 + 2C_4C_5R_4R_5) + s(C_1R_4 + 2C_1R_5 + 2C_4R_4R_5g_m + 2C_4R_4 + 2C_5R_4R_5g_m + 2C_5R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{R_4g_m+R_5g_m+1}}{C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5}$
wo: $\frac{\sqrt{2R_4g_m+2R_5g_m+2}}{\sqrt{2C_1C_4R_4R_5+C_1C_5R_4R_5+2C_4C_5R_4R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_4g_m+2R_5g_m+2}(C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5)}{2\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{R_4g_m+R_5g_m+1}\sqrt{2C_1C_4R_4R_5+C_1C_5R_4R_5+2C_4C_5R_4R_5}}$
K-LP: $\frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_4R_5}{C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4+2C_5R_4R_5g_m+2C_5R_5}$
Qz: None
Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s(C_4R_4R_5g_m - C_4R_4) - 1}{2g_m + s^2(C_1C_4R_4 + 2C_1C_4R_5) + s(C_1 + 2C_4R_4g_m + 2C_4R_5g_m + 2C_4)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_4}\sqrt{g_m}\sqrt{R_4+2R_5}}{C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$
wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1C_4R_4+2C_1C_4R_5}}$
bandwidth: $\frac{C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_4+2R_5}\sqrt{C_1C_4R_4+2C_1C_4R_5}}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_4R_4R_5g_m-C_4R_4}{C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$
Qz: None
Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, R_4, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_4s + R_1R_4g_m}{C_1C_5R_1R_4s^2 + 2R_1g_m + s(2C_1R_1 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}}{2C_1R_1+2C_5R_1R_4g_m+2C_5R_1+C_5R_4}$
wo: $\frac{\sqrt{2R_1g_m+2}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_4}}$
bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2C_1C_5R_1R_4\sqrt{R_1g_m+1}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_4}{2C_1R_1+2C_5R_1R_4g_m+2C_5R_1+C_5R_4}$
Qz: None
Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{C_1 C_5 R_1 R_4 R_5 s^2 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4} \sqrt{R_5} \sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 wo: $\frac{\sqrt{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4} \sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}{C_1 C_5 R_1 R_4 R_5}$
 K-LP: $\frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_4 R_5}{C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5}$
 Qz: None
 Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^2(C_1 C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5) + s(2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4 + 2R_5} \sqrt{R_1 g_m + 1}}{2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{C_1} C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5}$
 bandwidth: $\frac{\sqrt{2} \sqrt{2R_1 g_m + 2} (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5)}{2\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_4 + 2R_5} \sqrt{R_1 g_m + 1} \sqrt{C_1} C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5}$
 K-LP: $\frac{R_1 R_4 g_m}{2R_1 g_m + 2}$
 K-HP: 0
 K-BP: $\frac{C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4}{2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5}$
 Qz: None
 Wz: None

8.21 X-INVALID-NUMER-21 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s^2(2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5) + s(C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{2R_1 g_m + 1} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}{\sqrt{R_1} \sqrt{R_5} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5} \sqrt{2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_5}}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5}{C_1 R_1 + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.22 X-INVALID-NUMER-22 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s^2(2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4) + s(2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_1} \sqrt{R_4} \sqrt{R_1 g_m + 1} \sqrt{2C_1 C_4 + C_1 C_5 + 2C_4 C_5}}{2C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4}$
 wo: $\frac{\sqrt{2R_1 g_m + 2}}{\sqrt{2C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_4}}$

bandwidth: $\frac{\sqrt{2}\sqrt{2R_1g_m+2}(2C_1R_1+2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4)}{2\sqrt{R_1}\sqrt{R_4}\sqrt{R_1g_m+1}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4R_1R_4+C_1C_5R_1R_4+2C_4C_5R_1R_4}}$
K-LP: $\frac{R_1R_4g_m}{2R_1g_m+2}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_4}{2C_1R_1+2C_4R_1R_4g_m+2C_4R_4+2C_5R_1R_4g_m+2C_5R_1+C_5R_4}$
Qz: None
Wz: None

8.23 X-INVALID-NUMER-23 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^2(2C_1C_4R_1R_4R_5 + C_1C_5R_1R_4R_5 + 2C_4C_5R_1R_4R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_4R_5 + 2C_5R_1R_4R_5g_m + 2C_5R_1R_5 + C_5R_4R_5)}$$

Parameters:

Q: $\frac{\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{C_1R_1R_4+2C_1R_1R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}$
wo: $\frac{\sqrt{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}{\sqrt{2C_1C_4R_1R_4R_5+C_1C_5R_1R_4R_5+2C_4C_5R_1R_4R_5}}$
bandwidth: $\frac{C_1R_1R_4+2C_1R_1R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}{\sqrt{R_1}\sqrt{R_4}\sqrt{R_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4R_1R_4R_5+C_1C_5R_1R_4R_5+2C_4C_5R_1R_4R_5}}$
K-LP: $\frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$
K-HP: 0
K-BP: $-\frac{C_5R_1R_4R_5}{C_1R_1R_4+2C_1R_1R_5+2C_4R_1R_4R_5g_m+2C_4R_1R_4+2C_4R_4R_5+2C_5R_1R_4R_5g_m+2C_5R_1R_5+C_5R_4R_5}$
Qz: None
Wz: None

8.24 X-INVALID-NUMER-24 $Z(s) = \left(\frac{R_1}{C_1R_1s+1}, \infty, \infty, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_1R_5g_m - R_1 + s(C_4R_1R_4R_5g_m - C_4R_1R_4)}{2R_1g_m + s^2(C_1C_4R_1R_4 + 2C_1C_4R_1R_5) + s(C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_1R_5g_m + 2C_4R_1 + C_4R_4 + 2C_4R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4+2R_5}\sqrt{2R_1g_m+1}}{C_1R_1+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5}$
wo: $\frac{\sqrt{2R_1g_m+1}}{\sqrt{C_1C_4R_1R_4+2C_1C_4R_1R_5}}$
bandwidth: $\frac{C_1R_1+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4+2R_5}\sqrt{C_1C_4R_1R_4+2C_1C_4R_1R_5}}$
K-LP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-HP: 0
K-BP: $\frac{C_4R_1R_4R_5g_m-C_4R_1R_4}{C_1R_1+2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5}$
Qz: None
Wz: None

8.25 X-INVALID-NUMER-25 $Z(s) = \left(R_1 + \frac{1}{C_1s}, \infty, \infty, \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_5g_m + s(C_1R_1R_5g_m - C_1R_1) - 1}{2g_m + s^2(2C_1C_4R_1R_5g_m + 2C_1C_4R_1 + 2C_1C_4R_5) + s(2C_1R_1g_m + C_1 + 2C_4R_5g_m + 2C_4)}$$

Parameters:

Q: $\frac{2\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1g_m+C_1+2C_4R_5g_m+2C_4}$
wo: $\sqrt{\frac{g_m}{C_1C_4R_1R_5g_m+C_1C_4R_1+C_1C_4R_5}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1C_4R_1R_5g_m+C_1C_4R_1+C_1C_4R_5}}(2C_1R_1g_m+C_1+2C_4R_5g_m+2C_4)}{2\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{R_1R_5g_m+R_1+R_5}}}$
K-LP: $\frac{R_5g_m-1}{2g_m}$
K-HP: 0
K-BP: $\frac{C_1R_1R_5g_m-C_1R_1}{2C_1R_1g_m+C_1+2C_4R_5g_m+2C_4}$
Qz: None
Wz: None

8.26 X-INVALID-NUMER-26 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^2 (2 C_1 C_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_4 R_5) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_1 R_5 g_m + 2 C_1 R_1 + C_1 R_4 + 2 C_1 R_5 + 2 C_4 R_4 R_5 g_m + 2 C_4 R_4) + 2}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}R_5\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4} \\ \text{wo: } & \sqrt{\frac{R_4g_m+R_5g_m+1}{C_1C_4R_1R_4R_5g_m+C_1C_4R_1R_4+C_1C_4R_4R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_4g_m+R_5g_m+1}{C_1C_4R_1R_4R_5g_m+C_1C_4R_1R_4+C_1C_4R_4R_5}}(2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4)}{2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}R_5g_m\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}R_1\sqrt{R_4}\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}+2\sqrt{C_1}\sqrt{C_4}\sqrt{R_4}R_5\sqrt{\frac{R_4g_m}{R_1R_5g_m+R_1+R_5}+\frac{R_5g_m}{R_1R_5g_m+R_1+R_5}+\frac{1}{R_1R_5g_m+R_1+R_5}}} \\ \text{K-LP: } & \frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_1R_1R_4R_5g_m-C_1R_1R_4}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_4R_4R_5g_m+2C_4R_4} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.27 X-INVALID-NUMER-27 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 s + L_1 g_m}{2 C_4 + C_5 + s^2 (2 C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1) + s (2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2C_4+C_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}}{2C_4\sqrt{L_1g_m+2C_5}\sqrt{L_1g_m}} \\ \text{wo: } & \frac{\sqrt{2C_4+C_5}}{\sqrt{2C_1C_4L_1+C_1C_5L_1+2C_4C_5L_1}} \\ \text{bandwidth: } & \frac{2C_4\sqrt{L_1g_m+2C_5}\sqrt{L_1g_m}}{\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4L_1+C_1C_5L_1+2C_4C_5L_1}} \\ \text{K-LP: } & \frac{L_1g_m}{2C_4+C_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5}{2C_4g_m+2C_5g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.28 X-INVALID-NUMER-28 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 R_1 s + L_1 R_1 g_m}{2 C_4 R_1 + C_5 R_1 + s^2 (2 C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{R_1\sqrt{2C_4+C_5}\sqrt{2C_1C_4+C_1C_5+2C_4C_5}}{2C_4\sqrt{L_1}R_1g_m+2C_4\sqrt{L_1}+2C_5\sqrt{L_1}R_1g_m+C_5\sqrt{L_1}} \\ \text{wo: } & \frac{\sqrt{2C_4+C_5}}{\sqrt{2C_1C_4L_1+C_1C_5L_1+2C_4C_5L_1}} \\ \text{bandwidth: } & \frac{2C_4\sqrt{L_1}R_1g_m+2C_4\sqrt{L_1}+2C_5\sqrt{L_1}R_1g_m+C_5\sqrt{L_1}}{R_1\sqrt{2C_1C_4+C_1C_5+2C_4C_5}\sqrt{2C_1C_4L_1+C_1C_5L_1+2C_4C_5L_1}} \\ \text{K-LP: } & \frac{L_1g_m}{2C_4+C_5} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5R_1}{2C_4R_1g_m+2C_4+2C_5R_1g_m+C_5} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, R_4, R_5, \infty)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5}$$

$$\mathbf{9.2 \quad X-INVALID-ORDER-2} \quad Z(s) = \left(R_1, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s(2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.3 \quad X-INVALID-ORDER-3} \quad Z(s) = \left(R_1, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s(2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(R_1, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s(C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s(2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + s(2C_4 R_1 R_5 g_m + 2C_4 R_1 + 2C_4 R_5) + 1}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{2C_4 C_5 R_1 s^2 + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s^2(2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{2C_4 C_5 R_1 s^2 + s^3(2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 s^2 + L_5 R_1 g_m s - R_1}{2C_4 C_5 L_5 R_1 s^3 + 2C_4 R_1 s + 2R_1 g_m + s^2(2C_4 L_5 R_1 g_m + 2C_4 L_5 + 2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 + R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s^3(2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2(2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 R_5) + s(2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_5 s^2 - R_1 R_5 + s(L_5 R_1 R_5 g_m - L_5 R_1)}{2C_4 C_5 L_5 R_1 R_5 s^3 + 2R_1 R_5 g_m + R_5 + s^2(2C_4 L_5 R_1 R_5 g_m + 2C_4 L_5 R_1 + 2C_4 L_5 R_5 + 2C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s(2C_4 R_1 R_5 + 2L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_4 C_5 R_1 R_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s (2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 - C_5 R_1 R_4 s + R_1 R_4 g_m}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_4 C_5 R_1 R_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 s^2 + L_5 R_1 R_4 g_m s - R_1 R_4}{2 C_4 C_5 L_5 R_1 R_4 s^3 + 2 R_1 R_4 g_m + 2 R_1 + R_4 + s^2 (2 C_4 L_5 R_1 R_4 g_m + 2 C_4 L_5 R_4 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4) + s (2 C_4 R_1 R_4 + 2 L_5 R_1 g_m + 2 L_5)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 + R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_4 R_5 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 R_5 s^2 - R_1 R_4 R_5 + s (L_5 R_1 R_4 R_5 g_m - L_5 R_1 R_4)}{2 C_4 C_5 L_5 R_1 R_4 R_5 s^3 + 2 R_1 R_4 R_5 g_m + 2 R_1 R_5 + R_4 R_5 + s^2 (2 C_4 L_5 R_1 R_4 R_5 g_m + 2 C_4 L_5 R_1 R_4 + 2 C_4 L_5 R_4 R_5 + 2 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_5 R_1 R_5 + C_5 L_5 R_4 R_5) + s (2 C_4 R_1 R_4 R_5 + 2 L_5 R_1 R_4 g_m + 2 L_5 R_1 R_5 g_m + 2 L_5 R_1 + L_5 R_4 + 2 L_5 R_5)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 R_4 g_m s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (2 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_5 R_4 R_5) + s^2 (2 C_4 L_5 R_1 R_4 g_m + 2 C_4 L_5 R_4 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 R_5 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4 + 2 C_5 L_5 R_5) + s (2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5 + 2 L_5 R_1 R_4 R_5 g_m)}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(R_1, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (2 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_5 R_4 R_5) + s^2 (2 C_4 C_5 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 R_5 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4 + 2 C_5 L_5 R_5) + s (2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_4 + 2 C_4 R_4 R_5 + 2 C_5 R_1 R_4 R_5 g_m)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s (C_4 R_1 R_4 g_m - C_5 R_1)}{s^2 (2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^2 (2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (-C_4 C_5 R_1 R_4 + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_5 R_1)}{s^3 (2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_1 R_4 s^3 - R_1 + s^2 (C_4 L_5 R_1 R_4 g_m - C_5 L_5 R_1) + s (-C_4 R_1 R_4 + L_5 R_1 g_m)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4) + s^2 (2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 + C_4 R_4) + 1}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4 + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3 (2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_1 R_4 R_5 s^3 - R_1 R_5 + s^2 (C_4 L_5 R_1 R_4 R_5 g_m - C_4 L_5 R_1 R_4 - C_5 L_5 R_1 R_5) + s (-C_4 R_1 R_4 R_5 + L_5 R_1 R_5 g_m - L_5 R_1)}{2 R_1 R_5 g_m + R_5 + s^3 (2 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_5 + C_4 C_5 L_5 R_4 R_5) + s^2 (2 C_4 L_5 R_1 R_4 g_m + 2 C_4 L_5 R_1 R_5 g_m + 2 C_4 L_5 R_1 + C_4 L_5 R_4 + 2 C_4 L_5 R_5 + 2 C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s (2 C_4 R_1 R_4 R_5 g_m + 2 C_4 R_1 R_5 + C_4 R_4 R_5 + 2 L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_4) + s^2 (C_4 L_5 R_1 R_4 g_m + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 + L_5 R_1 g_m)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_4) + s^2 (-C_4 C_5 R_1 R_4 R_5 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1s^3 + C_4L_4R_1g_ms^2 - C_5R_1s + R_1g_m}{2C_4C_5R_1s^2 + s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1R_5s^3 - C_5R_1R_5s + R_1R_5g_m - R_1 + s^2(C_4L_4R_1R_5g_m - C_4L_4R_1)}{2R_1g_m + s^3(2C_4C_5L_4R_1R_5g_m + C_4C_5L_4R_5) + s^2(2C_4C_5R_1R_5 + 2C_4L_4R_1g_m + C_4L_4) + s(2C_4R_1R_5g_m + 2C_4R_1 + 2C_4R_5 + 2C_5R_1R_5g_m + C_5R_5) + 1}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, R_5 + \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{C_4L_4R_1g_ms^2 + R_1g_m + s^3(C_4C_5L_4R_1R_5g_m - C_4C_5L_4R_1) + s(C_5R_1R_5g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4) + s^2(2C_4C_5R_1R_5g_m + 2C_4C_5R_1 + 2C_4C_5R_5) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, L_5s + \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{C_4C_5L_4L_5R_1g_ms^4 - C_4C_5L_4R_1s^3 - C_5R_1s + R_1g_m + s^2(C_4L_4R_1g_m + C_5L_5R_1g_m)}{2C_4C_5R_1s^2 + s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4 + 2C_4C_5L_5R_1g_m + 2C_4C_5L_5) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4L_5R_1s^4 + C_4L_4L_5R_1g_ms^3 + L_5R_1g_ms - R_1 + s^2(-C_4L_4R_1 - C_5L_5R_1)}{2C_4C_5L_5R_1s^3 + 2C_4R_1s + 2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^2(2C_4L_4R_1g_m + C_4L_4 + 2C_4L_5R_1g_m + 2C_4L_5 + 2C_5L_5R_1g_m + C_5L_5) + 1}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, L_5s + R_5 + \frac{1}{C_5s}, \infty \right)$$

$$H(s) = \frac{C_4C_5L_4L_5R_1g_ms^4 + R_1g_m + s^3(C_4C_5L_4R_1R_5g_m - C_4C_5L_4R_1) + s^2(C_4L_4R_1g_m + C_5L_5R_1g_m) + s(C_5R_1R_5g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4 + 2C_4C_5L_5R_1g_m + 2C_4C_5L_5) + s^2(2C_4C_5R_1R_5g_m + 2C_4C_5R_1 + 2C_4C_5R_5) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4L_5R_1R_5s^4 - R_1R_5 + s^3(C_4L_4L_5R_1R_5g_m - C_4L_4L_5R_1) + s^2(-C_4L_4R_1R_5 - C_5L_5R_1R_5) + s(L_5R_1R_5g_m - L_5R_1)}{2R_1R_5g_m + R_5 + s^4(2C_4C_5L_4L_5R_1R_5g_m + C_4C_5L_4L_5R_5) + s^3(2C_4C_5L_5R_1R_5 + 2C_4L_4L_5R_1g_m + C_4L_4L_5) + s^2(2C_4L_4R_1R_5g_m + C_4L_4R_5 + 2C_4L_5R_1R_5g_m + 2C_4L_5R_1 + 2C_4L_5R_5 + 2C_5L_5R_1R_5g_m + C_5L_5R_5) + s(2C_4R_1R_5 + 2L_5R_1g_m + L_5)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$$

$$H(s) = \frac{C_4L_4L_5R_1g_ms^3 + L_5R_1g_ms + R_1R_5g_m - R_1 + s^4(C_4C_5L_4L_5R_1R_5g_m - C_4C_5L_4L_5R_1) + s^2(C_4L_4R_1R_5g_m - C_4L_4R_1 + C_5L_5R_1R_5g_m - C_5L_5R_1)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^3(2C_4C_5L_5R_1R_5g_m + 2C_4C_5L_5R_1 + 2C_4C_5L_5R_5) + s^2(2C_4L_4R_1g_m + C_4L_4 + 2C_4L_5R_1g_m + 2C_4L_5 + 2C_5L_5R_1g_m + C_5L_5) + s(2C_4R_1R_5g_m + 2C_4R_1 + 2C_4R_5) + 1}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1R_5s^3 - C_5R_1R_5s + R_1R_5g_m - R_1 + s^4(C_4C_5L_4L_5R_1R_5g_m - C_4C_5L_4L_5R_1) + s^2(C_4L_4R_1R_5g_m - C_4L_4R_1 + C_5L_5R_1R_5g_m - C_5L_5R_1)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^3(2C_4C_5L_4R_1R_5g_m + C_4C_5L_4R_5 + 2C_4C_5L_5R_1R_5g_m + 2C_4C_5L_5R_1 + 2C_4C_5L_5R_5) + s^2(2C_4C_5R_1R_5 + 2C_4L_4R_1g_m + C_4L_4 + 2C_5L_5R_1g_m + C_5L_5) + s(2C_4R_1R_5g_m + 2C_4R_1 + 2C_4R_5 + 2C_5R_1R_5g_m + C_5R_5)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 s^2 + L_4 R_1 g_m s}{2C_4 C_5 L_4 R_1 s^3 + 2C_5 R_1 s + 2R_1 g_m + s^2 (2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4) + 2}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_5 s^2 + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2C_4 C_5 L_4 R_1 R_5 s^3 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^2 (2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + 2C_4 L_4 R_5 + 2C_5 L_4 R_1 R_5 g_m + C_5 L_4 R_5) + s (2C_5 R_1 R_5 + 2L_4 R_1 g_m + L_4)}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 R_1 g_m s + s^2 (C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1)}{2R_1 g_m + s^3 (2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + 2C_4 C_5 L_4 R_5) + s^2 (2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4) + s (2C_5 R_1 R_5 g_m + 2C_5 R_1 + 2C_5 R_5) + 2}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 g_m s^3 - C_5 L_4 R_1 s^2 + L_4 R_1 g_m s}{2C_4 C_5 L_4 R_1 s^3 + 2C_5 R_1 s + 2R_1 g_m + s^4 (2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4 + 2C_5 L_5 R_1 g_m + 2C_5 L_5) + 2}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 s^3 + L_4 L_5 R_1 g_m s^2 - L_4 R_1 s}{2C_4 C_5 L_4 L_5 R_1 s^4 + 2R_1 + s^3 (2C_4 L_4 L_5 R_1 g_m + 2C_4 L_4 L_5 + 2C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 + 2C_5 L_5 R_1) + s (2L_4 R_1 g_m + L_4 + 2L_5 R_1 g_m + 2L_5)}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 g_m s^3 + L_4 R_1 g_m s + s^2 (C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1)}{2R_1 g_m + s^4 (2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + 2C_4 C_5 L_4 R_5) + s^2 (2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4 + 2C_5 L_5 R_1 g_m + 2C_5 L_5) + s (2C_5 R_1 R_5 g_m + 2C_5 R_1 + 2C_5 R_5) + 2}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_5 s^3 - L_4 R_1 R_5 s + s^2 (L_4 L_5 R_1 R_5 g_m - L_4 L_5 R_1)}{2C_4 C_5 L_4 L_5 R_1 R_5 s^4 + 2R_1 R_5 + s^3 (2C_4 L_4 L_5 R_1 R_5 g_m + 2C_4 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_5 + 2C_5 L_4 L_5 R_1 R_5 g_m + C_5 L_4 L_5 R_5) + s^2 (2C_4 L_4 R_1 R_5 + 2C_5 L_5 R_1 R_5 + 2L_4 L_5 R_1 g_m + L_4 L_5) + s (2L_4 R_1 R_5 g_m + L_4 R_5 + 2L_5 R_1 R_5 g_m + 2L_5 R_1 + 2L_5 R_5)}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_4 L_5 R_1 g_m s^2 + s^3 (C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_4 L_4 L_5 R_1 g_m + 2C_4 L_4 L_5 + 2C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + 2C_4 L_4 R_5 + 2C_5 L_5 R_1 R_5 g_m + 2C_5 L_5 R_1 + 2C_5 L_5 R_5) + s (2L_4 R_1 g_m + L_4 + 2L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_5 s^2 + s^3 (C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_4 C_5 L_4 R_1 R_5 + 2C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + 2C_4 L_4 R_5 + 2C_5 L_4 R_1 R_5 g_m + C_5 L_4 R_5 + 2C_5 L_5 R_1 R_5 g_m + 2C_5 L_5 R_1 + 2C_5 L_5 R_5) + s (2C_5 R_1 R_5 g_m + 2C_5 R_1 + 2C_5 R_5)}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_4C_5L_4R_1s^3 + R_1g_m + s^2(-C_4C_5R_1R_4 + C_4L_4R_1g_m) + s(C_4R_1R_4g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4) + s^2(2C_4C_5R_1R_4g_m + 2C_4C_5R_1 + C_4C_5R_4) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_4C_5L_4R_1R_5s^3 + R_1R_5g_m - R_1 + s^2(-C_4C_5R_1R_4R_5 + C_4L_4R_1R_5g_m - C_4L_4R_1) + s(C_4R_1R_4R_5g_m - C_4R_1R_4 - C_5R_1R_5)}{2R_1g_m + s^3(2C_4C_5L_4R_1R_5g_m + C_4C_5L_4R_5) + s^2(2C_4C_5R_1R_4R_5g_m + 2C_4C_5R_1R_5 + C_4C_5R_4R_5 + 2C_4L_4R_1g_m + C_4L_4) + s(2C_4R_1R_4g_m + 2C_4R_1R_5g_m + 2C_4R_1 + C_4R_4 + 2C_4R_5 + 2C_5R_1R_5g_m + C_5R_5) + 1}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{R_1g_m + s^3(C_4C_5L_4R_1R_5g_m - C_4C_5L_4R_1) + s^2(C_4C_5R_1R_4R_5g_m - C_4C_5R_1R_4 + C_4L_4R_1g_m) + s(C_4R_1R_4g_m + C_5R_1R_5g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4) + s^2(2C_4C_5R_1R_4g_m + 2C_4C_5R_1R_5g_m + 2C_4C_5R_1 + C_4C_5R_4 + 2C_4C_5R_5) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

9.51 X-INVALID-ORDER-51 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, L_5s + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_4C_5L_4L_5R_1g_ms^4 + R_1g_m + s^3(-C_4C_5L_4R_1 + C_4C_5L_5R_1R_4g_m) + s^2(-C_4C_5R_1R_4 + C_4L_4R_1g_m + C_5L_5R_1g_m) + s(C_4R_1R_4g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4 + 2C_4C_5L_5R_1g_m + 2C_4C_5L_5) + s^2(2C_4C_5R_1R_4g_m + 2C_4C_5R_1 + C_4C_5R_4) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

9.52 X-INVALID-ORDER-52 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{-C_4C_5L_4L_5R_1s^4 - R_1 + s^3(-C_4C_5L_5R_1R_4 + C_4L_4L_5R_1g_m) + s^2(-C_4L_4R_1 + C_4L_5R_1R_4g_m - C_5L_5R_1) + s(-C_4R_1R_4 + L_5R_1g_m)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^3(2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_1 + C_4C_5L_5R_4) + s^2(2C_4L_4R_1g_m + C_4L_4 + 2C_4L_5R_1g_m + 2C_4L_5 + 2C_5L_5R_1g_m + C_5L_5) + s(2C_4R_1R_4g_m + 2C_4R_1 + C_4R_4) + 1}$$

9.53 X-INVALID-ORDER-53 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, L_5s + R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_4C_5L_4L_5R_1g_ms^4 + R_1g_m + s^3(C_4C_5L_4R_1R_5g_m - C_4C_5L_4R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_4C_5R_1R_4R_5g_m - C_4C_5R_1R_4 + C_4L_4R_1g_m + C_5L_5R_1g_m) + s(C_4R_1R_4g_m + C_5R_1R_5g_m - C_5R_1)}{s^3(2C_4C_5L_4R_1g_m + C_4C_5L_4 + 2C_4C_5L_5R_1g_m + 2C_4C_5L_5) + s^2(2C_4C_5R_1R_4g_m + 2C_4C_5R_1R_5g_m + 2C_4C_5R_1 + C_4C_5R_4 + 2C_4C_5R_5) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5)}$$

9.54 X-INVALID-ORDER-54 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \infty \right)$

$$H(s) = \frac{-C_4C_5L_4L_5R_1R_5s^4 - R_1R_5 + s^3(-C_4C_5L_5R_1R_4R_5 + C_4L_4L_5R_1R_5g_m - C_4L_4L_5R_1) + s^2(-C_4L_4R_1R_5 + C_4L_5R_1R_4R_5g_m - C_4L_5R_1R_4 - C_5L_5R_1R_5) + s(-C_4R_1R_4R_5 + L_5R_1R_5g_m)}{2R_1R_5g_m + R_5 + s^4(2C_4C_5L_4L_5R_1R_5g_m + C_4C_5L_4L_5R_5) + s^3(2C_4C_5L_5R_1R_4R_5g_m + 2C_4C_5L_5R_1R_5 + C_4C_5L_5R_4R_5 + 2C_4L_4L_5R_1g_m + C_4L_4L_5) + s^2(2C_4L_4R_1R_5g_m + C_4L_4R_5 + 2C_4L_5R_1R_4g_m + 2C_4L_5R_1R_5g_m + 2C_4L_5R_1 + C_4L_5R_4 + 2C_4L_5R_5 + 2C_5L_5R_1R_5g_m) + s(2C_4R_1R_4R_5 + L_5R_1R_5g_m)}$$

9.55 X-INVALID-ORDER-55 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{R_1R_5g_m - R_1 + s^4(C_4C_5L_4L_5R_1R_5g_m - C_4C_5L_4L_5R_1) + s^3(C_4C_5L_5R_1R_4R_5g_m - C_4C_5L_5R_1R_4 + C_4L_4L_5R_1g_m) + s^2(C_4L_4R_1R_5g_m - C_4L_4R_1 + C_4L_5R_1R_4g_m + C_5L_5R_1R_5g_m - C_5L_5R_1) + s(C_4R_1R_4R_5g_m - C_4R_1R_4 + L_5R_1g_m)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^3(2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_1R_5g_m + 2C_4C_5L_5R_1 + C_4C_5L_5R_4 + 2C_4C_5L_5R_5) + s^2(2C_4L_4R_1g_m + C_4L_4 + 2C_4L_5R_1g_m + 2C_4L_5 + 2C_5L_5R_1g_m + C_5L_5) + s(2C_4R_1R_4g_m + 2C_4R_1R_5g_m + 2C_4R_1 + C_4R_4 + L_5R_1g_m)}$$

9.56 X-INVALID-ORDER-56 $Z(s) = \left(R_1, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{R_1R_5g_m - R_1 + s^4(C_4C_5L_4L_5R_1R_5g_m - C_4C_5L_4L_5R_1) + s^3(-C_4C_5L_4R_1R_5 + C_4C_5L_5R_1R_4R_5g_m - C_4C_5L_5R_1R_4) + s^2(-C_4C_5R_1R_4R_5 + C_4L_4R_1R_5g_m - C_4L_4R_1 + C_5L_5R_1R_5g_m - C_5L_5R_1) + s(C_4R_1R_4R_5g_m - C_4R_1R_4 + L_5R_1g_m)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + C_4C_5L_4L_5) + s^3(2C_4C_5L_4R_1R_5g_m + C_4C_5L_4R_5 + 2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_1R_5g_m + 2C_4C_5L_5R_1 + C_4C_5L_5R_4 + 2C_4C_5L_5R_5) + s^2(2C_4C_5R_1R_4R_5g_m + 2C_4C_5R_1R_5 + C_4C_5R_4R_5 + 2C_4L_4R_1g_m + C_4L_4 + 2C_5L_5R_1g_m + C_5L_5) + s(2C_4R_1R_4R_5g_m - C_4R_1R_4 + L_5R_1g_m)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_4 s^2 + L_4 R_1 R_4 g_m s}{2C_4 C_5 L_4 R_1 R_4 s^3 + 2R_1 R_4 g_m + 2R_4 + s^2 (2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 R_1 R_4 g_m + 2C_5 L_4 R_1 + C_5 L_4 R_4) + s (2C_5 R_1 R_4 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_4 R_5 s^2 + s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2C_4 C_5 L_4 R_1 R_4 R_5 s^3 + 2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^2 (2C_4 L_4 R_1 R_4 R_5 g_m + 2C_4 L_4 R_1 R_4 + 2C_4 L_4 R_4 R_5 + 2C_5 L_4 R_1 R_4 R_5 g_m + 2C_5 L_4 R_1 R_5 + C_5 L_4 R_4 R_5) + s (2C_5 R_1 R_4 R_5 + 2L_4 R_1 R_4 g_m + 2L_4 R_1 R_5 g_m + 2L_4 R_1 + L_4 R_4 + 2L_4 R_5)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_4 g_m s + s^2 (C_5 L_4 R_1 R_4 R_5 g_m - C_5 L_4 R_1 R_4)}{2R_1 R_4 g_m + 2R_4 + s^3 (2C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 R_1 R_4 + 2C_4 C_5 L_4 R_4 R_5) + s^2 (2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 R_1 R_4 g_m + 2C_5 L_4 R_1 R_5 g_m + 2C_5 L_4 R_1 + C_5 L_4 R_4 + 2C_5 L_4 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_4 + 2C_5 R_4 R_5 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 R_4 g_m s^3 - C_5 L_4 R_1 R_4 s^2 + L_4 R_1 R_4 g_m s}{2R_1 R_4 g_m + 2R_4 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_4 C_5 L_4 R_1 R_4 + 2C_5 L_4 L_5 R_1 g_m + 2C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 R_1 R_4 g_m + 2C_5 L_4 R_1 + C_5 L_4 R_4 + 2C_5 L_5 R_1 R_4 g_m + 2C_5 L_5 R_4) + s (2C_5 R_1 R_4 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_4 s^3 + L_4 L_5 R_1 R_4 g_m s^2 - L_4 R_1 R_4 s}{2C_4 C_5 L_4 L_5 R_1 R_4 s^4 + 2R_1 R_4 + s^3 (2C_4 L_4 L_5 R_1 R_4 g_m + 2C_4 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_1 R_4 g_m + 2C_5 L_4 L_5 R_1 + C_5 L_4 L_5 R_4) + s^2 (2C_4 L_4 R_1 R_4 + 2C_5 L_5 R_1 R_4 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (2L_4 R_1 R_4 g_m + 2L_4 R_1 + L_4 R_4 + 2L_5 R_1 R_4 g_m + 2L_5 R_4)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 R_4 g_m s^3 + L_4 R_1 R_4 g_m s + s^2 (C_5 L_4 R_1 R_4 R_5 g_m - C_5 L_4 R_1 R_4)}{2R_1 R_4 g_m + 2R_4 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 R_1 R_4 + 2C_4 C_5 L_4 R_4 R_5 + 2C_5 L_4 L_5 R_1 g_m + 2C_5 L_4 L_5) + s^2 (2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 R_1 R_4 g_m + 2C_5 L_4 R_1 R_5 g_m + 2C_5 L_4 R_1 + C_5 L_4 R_4 + 2C_5 L_4 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_4 + 2C_5 R_4 R_5 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_4 R_5 s^3 - L_4 R_1 R_4 R_5 s + s^2 (L_4 L_5 R_1 R_4 R_5 g_m - L_4 L_5 R_1 R_4)}{2C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^4 + 2R_1 R_4 R_5 + s^3 (2C_4 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 L_4 L_5 R_1 R_4 + 2C_4 L_4 L_5 R_4 R_5 + 2C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_5 L_4 L_5 R_1 R_5 + C_5 L_4 L_5 R_4 R_5) + s^2 (2C_4 L_4 R_1 R_4 R_5 + 2C_5 L_5 R_1 R_4 R_5 + 2L_4 L_5 R_1 R_4 g_m + 2L_4 L_5 R_1 R_5 g_m + 2L_4 L_5 R_1 + L_4 L_5 R_4 + 2L_4 L_5 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_4 R_5 + 2C_5 R_4 R_5 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_4 L_5 R_1 R_4 g_m s^2 + s^3 (C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_5 L_4 L_5 R_1 R_4) + s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2C_4 L_4 L_5 R_1 R_4 g_m + 2C_4 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_1 R_4 g_m + 2C_5 L_4 L_5 R_1 R_5 g_m + 2C_5 L_4 L_5 R_1 + C_5 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_5) + s^2 (2C_4 L_4 R_1 R_4 R_5 g_m + 2C_4 L_4 R_1 R_4 + 2C_4 L_4 R_4 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_4 R_5 + 2C_5 R_4 R_5 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(R_1, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_4 R_5 s^2 + s^3 (C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_5 L_4 L_5 R_1 R_4) + s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2C_4 C_5 L_4 R_1 R_4 R_5 + 2C_5 L_4 L_5 R_1 R_4 g_m + 2C_5 L_4 L_5 R_1 R_5 g_m + 2C_5 L_4 L_5 R_1 + C_5 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_5) + s^2 (2C_4 L_4 R_1 R_4 R_5 g_m + 2C_4 L_4 R_1 R_4 + 2C_4 L_4 R_4 R_5) + s (2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_4 R_5 + 2C_5 R_4 R_5 + 2L_4 R_1 g_m + 2L_4)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 s^3 + R_1 R_4 g_m + s^2 (C_4 L_4 R_1 R_4 g_m - C_5 L_4 R_1) + s (-C_5 R_1 R_4 + L_4 R_1 g_m)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4) + s (2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 R_5 s^3 + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4 - C_5 L_4 R_1 R_5) + s (-C_5 R_1 R_4 R_5 + L_4 R_1 R_5 g_m - L_4 R_1)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (2 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_4 R_4 R_5) + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + C_4 L_4 R_4 + 2 C_4 L_4 R_5 + 2 C_5 L_4 R_1 R_5 g_m + C_5 L_4 R_5) + s (2 C_5 R_1 R_4 R_5 g_m + 2 C_5 R_1 R_5 + C_5 R_4 R_5 + 2 L_4 R_1 R_5 g_m)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_4 R_1 R_4) + s^2 (C_4 L_4 R_1 R_4 g_m + C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4 + L_4 R_1 g_m)}{2 R_1 g_m + s^3 (2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4) + s (2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (-C_4 C_5 L_4 R_1 R_4 + C_5 L_4 L_5 R_1 g_m) + s^2 (C_4 L_4 R_1 R_4 g_m - C_5 L_4 R_1 + C_5 L_5 R_1 R_4 g_m) + s (-C_5 R_1 R_4 + L_4 R_1 g_m)}{2 R_1 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 s^4 - R_1 R_4 + s^3 (C_4 L_4 L_5 R_1 R_4 g_m - C_5 L_4 L_5 R_1) + s^2 (-C_4 L_4 R_1 R_4 - C_5 L_5 R_1 R_4 + L_4 L_5 R_1 g_m) + s (-L_4 R_1 + L_5 R_1 R_4 g_m)}{2 R_1 R_4 g_m + 2 R_1 + R_4 + s^4 (2 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4) + s^3 (2 C_4 L_4 L_5 R_1 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 + C_4 L_4 R_4 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4) + s (2 L_4 R_1 g_m + L_4 + 2 L_5 R_1 R_4 g_m)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_4 R_1 R_4 + C_5 L_4 L_5 R_1 g_m) + s^2 (C_4 L_4 R_1 R_4 g_m + C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1 + C_5 L_5 R_1 R_4 g_m) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4 + L_4 R_1 g_m)}{2 R_1 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4)}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^4 - R_1 R_4 R_5 + s^3 (C_4 L_4 L_5 R_1 R_4 R_5 g_m - C_4 L_4 L_5 R_1 R_4 - C_5 L_4 L_5 R_1 R_5) + s^2 (-C_4 L_4 R_1 R_4 R_5 - C_5 L_5 R_1 R_4 R_5 + L_4 L_5 R_1 R_5 g_m) + s (-L_4 R_1 R_4 R_5 + L_5 R_1 R_4 R_5 g_m)}{2 R_1 R_4 R_5 g_m + 2 R_1 R_5 + R_4 R_5 + s^4 (2 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2 C_4 L_4 L_5 R_1 R_4 g_m + 2 C_4 L_4 L_5 R_1 R_5 g_m + 2 C_4 L_4 L_5 R_1 + C_4 L_4 L_5 R_4 + 2 C_4 L_4 L_5 R_5 + 2 C_5 L_4 L_5 R_1 R_5 g_m + C_5 L_4 L_5 R_5) + s^2 (2 C_4 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_5 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_5 g_m)}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_1 R_4) + s^3 (C_4 L_4 L_5 R_1 R_4 g_m + C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4 + C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4 R_5) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4 R_5 + L_4 R_1 R_5 g_m - L_4 R_1 R_4 R_5)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^4 (2 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (2 C_4 L_4 L_5 R_1 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_5 g_m)}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(R_1, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_1 R_4) + s^3 (-C_4 C_5 L_4 R_1 R_4 R_5 + C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4 R_5 + C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4 R_5) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4 R_5 + L_4 R_1 R_5 g_m - L_4 R_1 R_4 R_5)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^4 (2 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (2 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_4 R_4 R_5 + 2 C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_5 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_5 g_m)}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1R_4s^3 + C_4L_4R_1R_4g_ms^2 - C_5R_1R_4s + R_1R_4g_m}{2R_1g_m + s^3(2C_4C_5L_4R_1R_4g_m + 2C_4C_5L_4R_1 + C_4C_5L_4R_4) + s^2(2C_4C_5R_1R_4 + 2C_4L_4R_1g_m + 2C_4L_4) + s(2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4) + 2}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{R_5}{C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1R_4R_5s^3 - C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4 + s^2(C_4L_4R_1R_4R_5g_m - C_4L_4R_1R_4)}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^3(2C_4C_5L_4R_1R_4R_5g_m + 2C_4C_5L_4R_1R_5 + C_4C_5L_4R_4R_5) + s^2(2C_4C_5R_1R_4R_5 + 2C_4L_4R_1R_4g_m + 2C_4L_4R_1R_5g_m + 2C_4L_4R_1 + C_4L_4R_4 + 2C_4L_4R_5) + s(2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_4R_5 + 2C_5R_1R_4R_5g_m + 2C_5R_1R_4 + 2C_5R_4R_5) + 2}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_4L_4R_1R_4g_ms^2 + R_1R_4g_m + s^3(C_4C_5L_4R_1R_4R_5g_m - C_4C_5L_4R_1R_4) + s(C_5R_1R_4R_5g_m - C_5R_1R_4)}{2R_1g_m + s^3(2C_4C_5L_4R_1R_4g_m + 2C_4C_5L_4R_1R_5g_m + 2C_4C_5L_4R_1 + C_4C_5L_4R_4 + 2C_4C_5L_4R_5) + s^2(2C_4C_5R_1R_4R_5g_m + 2C_4C_5R_1R_4 + 2C_4C_5R_4R_5 + 2C_4L_4R_1g_m + 2C_4L_4) + s(2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1R_5g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad L_5s + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_4C_5L_4L_5R_1R_4g_ms^4 - C_4C_5L_4R_1R_4s^3 - C_5R_1R_4s + R_1R_4g_m + s^2(C_4L_4R_1R_4g_m + C_5L_5R_1R_4g_m)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + 2C_4C_5L_4L_5) + s^3(2C_4C_5L_4R_1R_4g_m + 2C_4C_5L_4R_1 + C_4C_5L_4R_4 + 2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_4) + s^2(2C_4C_5R_1R_4 + 2C_4L_4R_1g_m + 2C_4L_4 + 2C_5L_5R_1g_m + 2C_5L_5) + s(2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{L_5s}{C_5L_5s^2+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4L_5R_1R_4s^4 + C_4L_4L_5R_1R_4g_ms^3 + L_5R_1R_4g_ms - R_1R_4 + s^2(-C_4L_4R_1R_4 - C_5L_5R_1R_4)}{2R_1R_4g_m + 2R_1 + R_4 + s^4(2C_4C_5L_4L_5R_1R_4g_m + 2C_4C_5L_4L_5R_1 + C_4C_5L_4L_5R_4) + s^3(2C_4C_5L_5R_1R_4 + 2C_4L_4L_5R_1g_m + 2C_4L_4L_5) + s^2(2C_4L_4R_1R_4g_m + 2C_4L_4R_1 + C_4L_4R_4 + 2C_4L_5R_1R_4g_m + 2C_4L_5R_4 + 2C_5L_5R_1R_4g_m + 2C_5L_5R_1 + C_5L_5R_4) + s(2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4 + 2C_5R_5) + 2}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad L_5s + R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_4C_5L_4L_5R_1R_4g_ms^4 + R_1R_4g_m + s^3(C_4C_5L_4R_1R_4R_5g_m - C_4C_5L_4R_1R_4) + s^2(C_4L_4R_1R_4g_m + C_5L_5R_1R_4g_m) + s(C_5R_1R_4R_5g_m - C_5R_1R_4)}{2R_1g_m + s^4(2C_4C_5L_4L_5R_1g_m + 2C_4C_5L_4L_5) + s^3(2C_4C_5L_4R_1R_4g_m + 2C_4C_5L_4R_1R_5g_m + 2C_4C_5L_4R_1 + C_4C_5L_4R_4 + 2C_4C_5L_4R_5 + 2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_4) + s^2(2C_4C_5R_1R_4R_5g_m + 2C_4C_5R_1R_4 + 2C_4C_5R_4R_5 + 2C_4L_4R_1g_m + 2C_4L_4 + 2C_5L_5R_1) + s(2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1R_5 + 2C_5R_4R_5) + 2}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4L_5R_1R_4R_5s^4 - R_1R_4R_5 + s^3(C_4L_4L_5R_1R_4R_5g_m - C_4L_4L_5R_1R_4) + s^2(-C_4L_4R_1R_4R_5 - C_5L_5R_1R_4R_5)}{2R_1R_4R_5g_m + 2R_1R_5 + R_4R_5 + s^4(2C_4C_5L_4L_5R_1R_4R_5g_m + 2C_4C_5L_4L_5R_1R_5 + C_4C_5L_4L_5R_4R_5) + s^3(2C_4C_5L_5R_1R_4R_5 + 2C_4L_4L_5R_1R_4g_m + 2C_4L_4L_5R_1R_5g_m + 2C_4L_4L_5R_1 + C_4L_4L_5R_4 + 2C_4L_4L_5R_5) + s^2(2C_4L_4R_1R_4R_5g_m + 2C_4L_4R_1R_5 + C_4L_4R_4R_5) + s(2C_4R_1R_4R_5g_m + 2C_4R_1R_5 + C_4R_4R_5) + 2}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \quad \infty \right)$$

$$H(s) = \frac{C_4L_4L_5R_1R_4g_ms^3 + L_5R_1R_4g_ms + R_1R_4R_5g_m - R_1R_4 + s^4(C_4C_5L_4L_5R_1R_4R_5g_m - C_4C_5L_4L_5R_1R_4) + s^3(C_4C_5L_5R_1R_4R_5g_m + 2C_4C_5L_5R_1R_4 + 2C_4C_5L_5R_4R_5 + 2C_4L_4L_5R_1g_m + 2C_4L_4L_5) + s^2(2C_4L_4R_1R_4g_m + 2C_4L_4R_1R_5 + C_4L_4R_4R_5) + s(2C_4R_1R_4g_m + 2C_4R_1R_5 + C_4R_4R_5) + 2}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(R_1, \quad \infty, \quad \infty, \quad \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \quad \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4C_5L_4R_1R_4R_5s^3 - C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4 + s^4(C_4C_5L_4L_5R_1R_4R_5g_m - C_4C_5L_4L_5R_1R_4) + s^3(C_4C_5L_4R_1R_4R_5g_m + 2C_4C_5L_4R_1R_5 + C_4C_5L_4R_4R_5 + 2C_4C_5L_5R_1R_4R_5g_m + 2C_4C_5L_5R_1R_4 + 2C_4C_5L_5R_4R_5) + s^2(2C_4L_4R_1R_4R_5g_m + 2C_4L_4R_1R_5 + C_4L_4R_4R_5) + s(2C_4R_1R_4R_5g_m + 2C_4R_1R_5 + C_4R_4R_5) + 2}$$

9.84 X-INVALID-ORDER-84 $Z(s) = (L_1s, \infty, \infty, R_4, R_5, \infty)$

$$H(s) = \frac{s(L_1R_4R_5g_m - L_1R_4)}{R_4 + 2R_5 + s(2L_1R_4g_m + 2L_1R_5g_m + 2L_1)}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(L_1s, \infty, \infty, R_4, L_5s + \frac{1}{C_5s}, \infty\right)$

$$H(s) = \frac{C_5L_1L_5R_4g_ms^3 - C_5L_1R_4s^2 + L_1R_4g_ms}{2C_5L_1L_5g_ms^3 + s^2(2C_5L_1R_4g_m + 2C_5L_1 + 2C_5L_5) + s(C_5R_4 + 2L_1g_m) + 2}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(L_1s, \infty, \infty, R_4, \frac{L_5s}{C_5L_5s^2+1}, \infty\right)$

$$H(s) = \frac{-C_5L_1L_5R_4s^3 + L_1L_5R_4g_ms^2 - L_1R_4s}{R_4 + s^3(2C_5L_1L_5R_4g_m + 2C_5L_1L_5) + s^2(C_5L_5R_4 + 2L_1L_5g_m) + s(2L_1R_4g_m + 2L_1 + 2L_5)}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(L_1s, \infty, \infty, R_4, L_5s + R_5 + \frac{1}{C_5s}, \infty\right)$

$$H(s) = \frac{C_5L_1L_5R_4g_ms^3 + L_1R_4g_ms + s^2(C_5L_1R_4R_5g_m - C_5L_1R_4)}{2C_5L_1L_5g_ms^3 + s^2(2C_5L_1R_4g_m + 2C_5L_1R_5g_m + 2C_5L_1 + 2C_5L_5) + s(C_5R_4 + 2C_5R_5 + 2L_1g_m) + 2}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(L_1s, \infty, \infty, R_4, \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \infty\right)$

$$H(s) = \frac{-C_5L_1L_5R_4R_5s^3 - L_1R_4R_5s + s^2(L_1L_5R_4R_5g_m - L_1L_5R_4)}{R_4R_5 + s^3(2C_5L_1L_5R_4R_5g_m + 2C_5L_1L_5R_5) + s^2(C_5L_5R_4R_5 + 2L_1L_5R_4g_m + 2L_1L_5R_5g_m + 2L_1L_5) + s(2L_1R_4R_5g_m + 2L_1R_5 + L_5R_4 + 2L_5R_5)}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(L_1s, \infty, \infty, R_4, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty\right)$

$$H(s) = \frac{L_1L_5R_4g_ms^2 + s^3(C_5L_1L_5R_4R_5g_m - C_5L_1L_5R_4) + s(L_1R_4R_5g_m - L_1R_4)}{R_4 + 2R_5 + s^3(2C_5L_1L_5R_4g_m + 2C_5L_1L_5R_5g_m + 2C_5L_1L_5) + s^2(C_5L_5R_4 + 2C_5L_5R_5 + 2L_1L_5g_m) + s(2L_1R_4g_m + 2L_1R_5g_m + 2L_1 + 2L_5)}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(L_1s, \infty, \infty, R_4, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty\right)$

$$H(s) = \frac{-C_5L_1R_4R_5s^2 + s^3(C_5L_1L_5R_4R_5g_m - C_5L_1L_5R_4) + s(L_1R_4R_5g_m - L_1R_4)}{R_4 + 2R_5 + s^3(2C_5L_1L_5R_4g_m + 2C_5L_1L_5R_5g_m + 2C_5L_1L_5) + s^2(2C_5L_1R_4R_5g_m + 2C_5L_1R_5 + C_5L_5R_4 + 2C_5L_5R_5) + s(C_5R_4R_5 + 2L_1R_4g_m + 2L_1R_5g_m + 2L_1)}$$

9.91 X-INVALID-ORDER-91 $Z(s) = \left(L_1s, \infty, \infty, \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty\right)$

$$H(s) = \frac{-C_5L_1R_5s^2 + s(L_1R_5g_m - L_1)}{2C_4C_5L_1R_5s^3 + s^2(2C_4L_1R_5g_m + 2C_4L_1 + 2C_5L_1R_5g_m) + s(2C_4R_5 + C_5R_5 + 2L_1g_m) + 1}$$

9.92 X-INVALID-ORDER-92 $Z(s) = \left(L_1s, \infty, \infty, \frac{1}{C_4s}, L_5s + \frac{1}{C_5s}, \infty\right)$

$$H(s) = \frac{C_5L_1L_5g_ms^2 - C_5L_1s + L_1g_m}{2C_4C_5L_1L_5g_ms^3 + 2C_4 + C_5 + s^2(2C_4C_5L_1 + 2C_4C_5L_5) + s(2C_4L_1g_m + 2C_5L_1g_m)}$$

9.93 X-INVALID-ORDER-93 $Z(s) = \left(L_1s, \infty, \infty, \frac{1}{C_4s}, \frac{L_5s}{C_5L_5s^2+1}, \infty\right)$

$$H(s) = \frac{-C_5L_1L_5s^3 + L_1L_5g_ms^2 - L_1s}{2C_4C_5L_1L_5s^4 + 2L_1g_ms + s^3(2C_4L_1L_5g_m + 2C_5L_1L_5g_m) + s^2(2C_4L_1 + 2C_4L_5 + C_5L_5) + 1}$$

$$\mathbf{9.94 \quad X-INVALID-ORDER-94} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 + L_1 g_m + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2 C_4 C_5 L_1 L_5 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_4 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_1 + 2 C_4 C_5 L_5) + s (2 C_4 C_5 R_5 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.95 \quad X-INVALID-ORDER-95} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_5 s^3 - L_1 R_5 s + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{2 C_4 C_5 L_1 L_5 R_5 s^4 + R_5 + s^3 (2 C_4 L_1 L_5 R_5 g_m + 2 C_4 L_1 L_5 + 2 C_5 L_1 L_5 R_5 g_m) + s^2 (2 C_4 L_1 R_5 + 2 C_4 L_5 R_5 + C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.96 \quad X-INVALID-ORDER-96} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 g_m s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{s^4 (2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (2 C_4 C_5 L_5 R_5 + 2 C_4 L_1 L_5 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + 2 C_4 L_5 + C_5 L_5) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.97 \quad X-INVALID-ORDER-97} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{s^4 (2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (2 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_5 R_5 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_5 g_m + C_5 L_5) + s (2 C_4 R_5 + C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.98 \quad X-INVALID-ORDER-98} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{2 C_4 C_5 L_1 R_4 s^3 + s^2 (2 C_4 L_1 R_4 g_m + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1) + s (2 C_4 R_4 + C_5 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.99 \quad X-INVALID-ORDER-99} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s (L_1 R_4 R_5 g_m - L_1 R_4)}{2 C_4 C_5 L_1 R_4 R_5 s^3 + R_4 + 2 R_5 + s^2 (2 C_4 L_1 R_4 R_5 g_m + 2 C_4 L_1 R_4 + 2 C_5 L_1 R_4 R_5 g_m + 2 C_5 L_1 R_5) + s (2 C_4 R_4 R_5 + C_5 R_4 R_5 + 2 L_1 R_4 g_m + 2 L_1 R_5 g_m + 2 L_1)}$$

$$\mathbf{9.100 \quad X-INVALID-ORDER-100} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{s^3 (2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_4) + s^2 (2 C_4 C_5 R_4 R_5 + 2 C_4 L_1 R_4 g_m + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1) + s (2 C_4 R_4 + C_5 R_4 + 2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.101 \quad X-INVALID-ORDER-101} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{2 C_4 C_5 L_1 L_5 R_4 g_m s^4 + s^3 (2 C_4 C_5 L_1 R_4 + 2 C_4 C_5 L_5 R_4 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_4 g_m + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 + 2 C_5 L_5) + s (2 C_4 R_4 + C_5 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.102 \quad X-INVALID-ORDER-102} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 s^3 + L_1 L_5 R_4 g_m s^2 - L_1 R_4 s}{2 C_4 C_5 L_1 L_5 R_4 s^4 + R_4 + s^3 (2 C_4 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5) + s^2 (2 C_4 L_1 R_4 + 2 C_4 L_5 R_4 + C_5 L_5 R_4 + 2 L_1 L_5 g_m) + s (2 L_1 R_4 g_m + 2 L_1 + 2 L_5)}$$

$$\mathbf{9.103 \quad X-INVALID-ORDER-103} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 + L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{2 C_4 C_5 L_1 L_5 R_4 g_m s^4 + s^3 (2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_4 + 2 C_4 C_5 L_5 R_4 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 C_5 R_4 R_5 + 2 C_4 L_1 R_4 g_m + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + 2 C_5 L_5) + s (2 C_4 R_4 + C_5 R_4 + 2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.104 \quad X-INVALID-ORDER-104} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 R_5 s^3 - L_1 R_4 R_5 s + s^2 (L_1 L_5 R_4 R_5 g_m - L_1 L_5 R_4)}{2 C_4 C_5 L_1 L_5 R_4 R_5 s^4 + R_4 R_5 + s^3 (2 C_4 L_1 L_5 R_4 R_5 g_m + 2 C_4 L_1 L_5 R_4 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_5) + s^2 (2 C_4 L_1 R_4 R_5 + 2 C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2 L_1 L_5 R_4 g_m + 2 L_1 L_5 R_5 g_m + 2 L_1 L_5) + s (2 L_1 R_4 R_5 g_m + 2 L_1 R_5 + L_5 R_4 + 2 L_5 R_5)}$$

$$\mathbf{9.105 \quad X-INVALID-ORDER-105} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_4 g_m s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2 R_5 + s^4 (2 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4) + s^3 (2 C_4 C_5 L_5 R_4 R_5 + 2 C_4 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5 R_5 g_m + 2 C_5 L_1 L_5) + s^2 (2 C_4 L_1 R_4 R_5 g_m + 2 C_4 L_1 R_4 + 2 C_4 L_5 R_4 + C_5 L_5 R_4 + 2 C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 C_4 R_4 R_5 + 2 L_1 R_4 g_m + 2 L_1 R_5 g_m)}$$

$$\mathbf{9.106 \quad X-INVALID-ORDER-106} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2 R_5 + s^4 (2 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4) + s^3 (2 C_4 C_5 L_1 R_4 R_5 + 2 C_4 C_5 L_5 R_4 R_5 + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5 R_5 g_m + 2 C_5 L_1 L_5) + s^2 (2 C_4 L_1 R_4 R_5 g_m + 2 C_4 L_1 R_4 + 2 C_5 L_1 R_4 R_5 g_m + 2 C_5 L_1 R_5 + C_5 L_5 R_4 + 2 C_5 L_5 R_5) + s (2 C_4 R_4 R_5 + C_5 R_4 R_5 + 2 L_1 R_5 g_m)}$$

$$\mathbf{9.107 \quad X-INVALID-ORDER-107} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 R_4 R_5 s^3 + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_5 g_m - L_1)}{s^3 (2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_5) + s^2 (C_4 C_5 R_4 R_5 + 2 C_4 L_1 R_4 g_m + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_5 g_m) + s (C_4 R_4 + 2 C_4 R_5 + C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.108 \quad X-INVALID-ORDER-108} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_4 g_m s^3 + L_1 g_m + s^2 (-C_4 C_5 L_1 R_4 + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{2 C_4 C_5 L_1 L_5 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 + 2 C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.109 \quad X-INVALID-ORDER-109} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_4 s^4 - L_1 s + s^3 (C_4 L_1 L_5 R_4 g_m - C_5 L_1 L_5) + s^2 (-C_4 L_1 R_4 + L_1 L_5 g_m)}{s^4 (2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (C_4 C_5 L_5 R_4 + 2 C_4 L_1 L_5 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_4 g_m + 2 C_4 L_1 + 2 C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.110 \quad X-INVALID-ORDER-110} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_4 g_m s^3 + L_1 g_m + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2 C_4 C_5 L_1 L_5 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_1 + 2 C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2 C_4 C_5 R_5 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.111 \quad X-INVALID-ORDER-111} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_4 R_5 s^4 - L_1 R_5 s + s^3 (C_4 L_1 L_5 R_4 R_5 g_m - C_4 L_1 L_5 R_4 - C_5 L_1 L_5 R_5) + s^2 (-C_4 L_1 R_4 R_5 + L_1 L_5 R_5 g_m - L_1 L_5)}{R_5 + s^4 (2 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_5) + s^3 (C_4 C_5 L_5 R_4 R_5 + 2 C_4 L_1 L_5 R_4 g_m + 2 C_4 L_1 L_5 R_5 g_m + 2 C_4 L_1 L_5 + 2 C_5 L_1 L_5 R_5 g_m) + s^2 (2 C_4 L_1 R_4 R_5 g_m + 2 C_4 L_1 R_5 + C_4 L_5 R_4 + 2 C_4 L_5 R_5 + C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (C_4 R_4 R_5 + 2 L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.112 \quad X-INVALID-ORDER-112} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (C_4 L_1 L_5 R_4 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 + L_1 L_5 g_m) + s (L_1 R_5 g_m - L_1)}{s^4 (2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.113 \quad X-INVALID-ORDER-113} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (-C_4 C_5 L_1 R_4 R_5 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_5 g_m - L_1)}{s^4 (2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_5 + C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_5 L_1 L_5 g_m) + s^2 (C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_5 L_1 R_5 g_m + C_5 L_5) + s (C_4 R_4 + 2C_4 R_5 + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.114 \quad X-INVALID-ORDER-114} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s (L_1 R_5 g_m - L_1)}{2C_4 L_1 L_4 g_m s^3 + s^2 (2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4) + s (2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.115 \quad X-INVALID-ORDER-115} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 s^3 + C_4 L_1 L_4 g_m s^2 - C_5 L_1 s + L_1 g_m}{2C_4 C_5 L_1 L_4 g_m s^3 + 2C_4 + C_5 + s^2 (2C_4 C_5 L_1 + C_4 C_5 L_4) + s (2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.116 \quad X-INVALID-ORDER-116} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 - C_5 L_1 R_5 s^2 + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s (L_1 R_5 g_m - L_1)}{2C_4 C_5 L_1 L_4 R_5 g_m s^4 + s^3 (2C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m) + s^2 (2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4 + 2C_5 L_1 R_5 g_m) + s (2C_4 R_5 + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.117 \quad X-INVALID-ORDER-117} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 C_5 L_1 L_4 g_m s^3 + 2C_4 + C_5 + s^2 (2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4) + s (2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.118 \quad X-INVALID-ORDER-118} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 - C_4 C_5 L_1 L_4 s^3 - C_5 L_1 s + L_1 g_m + s^2 (C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m)}{2C_4 + C_5 + s^3 (2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_4 C_5 L_1 + C_4 C_5 L_4 + 2C_4 C_5 L_5) + s (2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.119 \quad X-INVALID-ORDER-119} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 s^5 + C_4 L_1 L_4 L_5 g_m s^4 + L_1 L_5 g_m s^2 - L_1 s + s^3 (-C_4 L_1 L_4 - C_5 L_1 L_5)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + 2L_1 g_m s + s^4 (2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2C_4 L_1 L_4 g_m + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_1 + C_4 L_4 + 2C_4 L_5 + C_5 L_5) + 1}$$

$$\mathbf{9.120 \quad X-INVALID-ORDER-120} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^2 (C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 + C_5 + s^3 (2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4 + 2C_4 C_5 L_5) + s (2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.121 \quad X-INVALID-ORDER-121} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_5 s^5 - L_1 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_5 g_m - C_4 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 R_5 - C_5 L_1 L_5 R_5) + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{2 C_4 C_5 L_1 L_4 L_5 R_5 g_m s^5 + R_5 + s^4 (2 C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 g_m) + s^3 (2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_5 R_5 g_m + 2 C_4 L_1 L_5 + C_4 L_4 L_5 + 2 C_5 L_1 L_5 R_5 g_m) + s^2 (2 C_4 L_1 R_5 + C_4 L_4 R_5 + 2 C_4 L_5 R_5 + C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.122 \quad X-INVALID-ORDER-122} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 L_5 g_m s^4 + L_1 L_5 g_m s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_5 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_4 L_1 L_5 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4 + 2 C_4 L_5 + C_5 L_5) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.123 \quad X-INVALID-ORDER-123} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 - C_5 L_1 R_5 s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_5 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4 + 2 C_5 L_1 R_5 g_m + C_5 L_5) + s (2 C_4 R_5 + C_5 R_5 + 2 L_1 g_m)}$$

$$\mathbf{9.124 \quad X-INVALID-ORDER-124} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 R_5 + s^3 (2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_4 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4)}$$

$$\mathbf{9.125 \quad X-INVALID-ORDER-125} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 s^3 + L_1 L_4 g_m s^2}{2 C_4 C_5 L_1 L_4 s^4 + 2 L_1 g_m s + s^3 (2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_1 + C_5 L_4) + 2}$$

$$\mathbf{9.126 \quad X-INVALID-ORDER-126} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_5 s^3 + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 C_4 C_5 L_1 L_4 R_5 s^4 + 2 R_5 + s^3 (2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_5 g_m) + s^2 (2 C_4 L_4 R_5 + 2 C_5 L_1 R_5 + C_5 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4)}$$

$$\mathbf{9.127 \quad X-INVALID-ORDER-127} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4)}{s^4 (2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4) + s^3 (2 C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + C_5 L_4) + s (2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.128 \quad X-INVALID-ORDER-128} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 g_m s^4 - C_5 L_1 L_4 s^3 + L_1 L_4 g_m s^2}{2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + 2 L_1 g_m s + s^4 (2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_1 + C_5 L_4 + 2 C_5 L_5) + 2}$$

$$\mathbf{9.129 \quad X-INVALID-ORDER-129} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 s^3 + L_1 L_4 L_5 g_m s^2 - L_1 L_4 s}{2 C_4 C_5 L_1 L_4 L_5 s^4 + 2 L_1 + L_4 + 2 L_5 + s^3 (2 C_4 L_1 L_4 L_5 g_m + 2 C_5 L_1 L_4 L_5 g_m) + s^2 (2 C_4 L_1 L_4 + 2 C_4 L_4 L_5 + 2 C_5 L_1 L_5 + C_5 L_4 L_5) + s (2 L_1 L_4 g_m + 2 L_1 L_5 g_m)}$$

$$\mathbf{9.130 \quad X-INVALID-ORDER-130} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 g_m s^4 + L_1 L_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4)}{2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + C_5 L_4 + 2 C_5 L_5) + s (2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.131 \quad X-INVALID-ORDER-131} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_5 s^3 - L_1 L_4 R_5 s + s^2 (L_1 L_4 L_5 R_5 g_m - L_1 L_4 L_5)}{2 C_4 C_5 L_1 L_4 L_5 R_5 s^4 + 2 L_1 R_5 + L_4 R_5 + 2 L_5 R_5 + s^3 (2 C_4 L_1 L_4 L_5 R_5 g_m + 2 C_4 L_1 L_4 L_5 + 2 C_5 L_1 L_4 L_5 R_5 g_m) + s^2 (2 C_4 L_1 L_4 R_5 + 2 C_4 L_4 L_5 R_5 + 2 C_5 L_1 L_5 R_5 + C_5 L_4 L_5 R_5 + 2 L_1 L_4 L_5 g_m) + s (2 L_1 L_4 R_5 g_m + 2 L_1 L_5 R_5 g_m + 2 L_1 L_5 + L_4 L_5)}$$

$$\mathbf{9.132 \quad X-INVALID-ORDER-132} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 L_5 g_m s^3 + s^4 (C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 R_5 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5) + s^4 (2 C_4 C_5 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 g_m + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_4 L_4 L_5 + 2 C_5 L_1 L_5 R_5 g_m + 2 C_5 L_1 L_5 + C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_5 + 2 C_5 L_5 R_5 + 2 L_1 L_4 g_m + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + 2 L_1 L_5 g_m)}$$

$$\mathbf{9.133 \quad X-INVALID-ORDER-133} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_5 s^3 + s^4 (C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 R_5 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5) + s^4 (2 C_4 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_4 L_5 R_5 + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_5 R_5 g_m + 2 C_5 L_1 L_5 + C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_5 + 2 C_5 L_1 R_5 + C_5 L_4 R_5 + 2 C_5 L_5 R_5 + 2 L_1 L_4 g_m)}$$

$$\mathbf{9.134 \quad X-INVALID-ORDER-134} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4) + s (L_1 R_5 g_m - L_1)}{2 C_4 L_1 L_4 g_m s^3 + s^2 (2 C_4 L_1 R_4 g_m + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4) + s (C_4 R_4 + 2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.135 \quad X-INVALID-ORDER-135} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 s^3 + L_1 g_m + s^2 (-C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{2 C_4 C_5 L_1 L_4 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 + C_4 C_5 L_4) + s (C_4 C_5 R_4 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.136 \quad X-INVALID-ORDER-136} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 + s^3 (-C_4 C_5 L_1 R_4 R_5 + C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_5 g_m - L_1)}{2 C_4 C_5 L_1 L_4 R_5 g_m s^4 + s^3 (2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m) + s^2 (C_4 C_5 R_4 R_5 + 2 C_4 L_1 R_4 g_m + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4 + 2 C_5 L_1 R_5 g_m) + s (C_4 R_4 + 2 C_4 R_5 + C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.137 \quad X-INVALID-ORDER-137} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2 C_4 C_5 L_1 L_4 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_1 + C_4 C_5 L_4) + s (C_4 C_5 R_4 + 2 C_4 C_5 R_5 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.138 \quad X-INVALID-ORDER-138} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (-C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m) + s^2 (-C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{2 C_4 + C_5 + s^3 (2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m) + s^2 (2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 + C_4 C_5 L_4 + 2 C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.139 \quad X-INVALID-ORDER-139} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 s^5 - L_1 s + s^4 (-C_4 C_5 L_1 L_5 R_4 + C_4 L_1 L_4 L_5 g_m) + s^3 (-C_4 L_1 L_4 + C_4 L_1 L_5 R_4 g_m - C_5 L_1 L_5) + s^2 (-C_4 L_1 R_4 + L_1 L_5 g_m)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_5 R_4 + 2C_4 L_1 L_4 g_m + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_1 R_4 g_m + 2C_4 L_1 + C_4 L_4 + 2C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2L_1 g_m) + 1}$$

$$\mathbf{9.140 \quad X-INVALID-ORDER-140} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m) + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 + C_5 + s^3 (2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4 + 2C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.141 \quad X-INVALID-ORDER-141} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_5 s^5 - L_1 R_5 s + s^4 (-C_4 C_5 L_1 L_5 R_4 R_5 + C_4 L_1 L_4 L_5 R_5 g_m - C_4 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 R_5 + C_4 L_1 L_5 R_4 R_5 g_m - C_4 L_1 L_5 R_4 - C_5 L_1 L_5 R_5) + s^2 (-C_4 L_1 R_4 R_5 + L_1 L_5 R_5 g_m)}{2C_4 C_5 L_1 L_4 L_5 R_5 g_m s^5 + R_5 + s^4 (2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_5 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (C_4 C_5 L_5 R_4 R_5 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_5 R_4 g_m + 2C_4 L_1 L_5 R_5 g_m + 2C_4 L_1 L_5 + C_4 L_4 L_5 + 2C_5 L_1 L_5 R_5 g_m) + s^2 (2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_5 + C_4 L_4)}$$

$$\mathbf{9.142 \quad X-INVALID-ORDER-142} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4 + C_4 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_4 L_1 L_5 R_4 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 + L_1 L_5 g_m) + s (L_1 R_5 g_m - L_1)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_4 g_m + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4 + 2C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2C_4 R_5 + 2L_1 g_m)}$$

$$\mathbf{9.143 \quad X-INVALID-ORDER-143} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (-C_4 C_5 L_1 R_4 R_5 + C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 R_5 + L_1 L_5 R_5 g_m)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4 + 2C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2C_4 R_5 + 2L_1 g_m)}$$

$$\mathbf{9.144 \quad X-INVALID-ORDER-144} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4)}{2R_4 R_5 + s^3 (2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_4) + s^2 (2C_4 L_4 R_4 R_5 + 2L_1 L_4 R_4 g_m + 2L_1 L_4 R_5 g_m + 2L_1 L_4) + s (2L_1 R_4 R_5 g_m + 2L_1 R_4 + L_4 R_4 + 2L_4 R_5)}$$

$$\mathbf{9.145 \quad X-INVALID-ORDER-145} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_4 s^3 + L_1 L_4 R_4 g_m s^2}{2C_4 C_5 L_1 L_4 R_4 s^4 + 2R_4 + s^3 (2C_4 L_1 L_4 R_4 g_m + 2C_5 L_1 L_4 R_4 g_m + 2C_5 L_1 L_4) + s^2 (2C_4 L_4 R_4 + 2C_5 L_1 R_4 + C_5 L_4 R_4 + 2L_1 L_4 g_m) + s (2L_1 R_4 g_m + 2L_4)}$$

$$\mathbf{9.146 \quad X-INVALID-ORDER-146} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_4 R_5 s^3 + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4)}{2C_4 C_5 L_1 L_4 R_4 R_5 s^4 + 2R_4 R_5 + s^3 (2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_4 + 2C_5 L_1 L_4 R_4 R_5 g_m + 2C_5 L_1 L_4 R_5) + s^2 (2C_4 L_4 R_4 R_5 + 2C_5 L_1 R_4 R_5 + C_5 L_4 R_4 R_5 + 2L_1 L_4 R_4 g_m + 2L_1 L_4 R_5 g_m + 2L_1 L_4) + s (2L_1 R_4 R_5 g_m + 2L_1 R_4 + L_4 R_4 + 2L_4 R_5)}$$

$$\mathbf{9.147 \quad X-INVALID-ORDER-147} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_4 R_5 g_m - C_5 L_1 L_4 R_4)}{2R_4 + s^4 (2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_4) + s^3 (2C_4 C_5 L_4 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_5 L_1 L_4 R_4 g_m + 2C_5 L_1 L_4 R_5 g_m + 2C_5 L_1 L_4) + s^2 (2C_4 L_4 R_4 + 2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_4 + C_5 L_4 R_4 + 2C_5 L_4 R_5 + 2L_1 L_4 g_m) + s (2C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_4)}$$

$$\mathbf{9.148 \quad X-INVALID-ORDER-148} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_4 g_m s^4 - C_5 L_1 L_4 R_4 s^3 + L_1 L_4 R_4 g_m s^2}{2 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + 2 R_4 + s^4 (2 C_4 C_5 L_1 L_4 R_4 + 2 C_4 C_5 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 C_5 L_5 R_4 + 2 L_1 L_4 g_m) + s (2 L_1 R_4 g_m + 2 L_4)}$$

$$\mathbf{9.149 \quad X-INVALID-ORDER-149} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_4 s^3 + L_1 L_4 L_5 R_4 g_m s^2 - L_1 L_4 R_4 s}{2 C_4 C_5 L_1 L_4 L_5 R_4 s^4 + 2 L_1 R_4 + L_4 R_4 + 2 L_5 R_4 + s^3 (2 C_4 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5) + s^2 (2 C_4 L_1 L_4 R_4 + 2 C_4 L_4 L_5 R_4 + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4 + 2 L_1 L_4 L_5 g_m) + s (2 L_1 L_4 R_4 g_m + 2 L_1 L_4 + 2 L_1 L_5 R_4 g_m + 2 L_4 L_5)}$$

$$\mathbf{9.150 \quad X-INVALID-ORDER-150} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_4 g_m s^4 + L_1 L_4 R_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_4 R_5 g_m - C_5 L_1 L_4 R_4)}{2 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + 2 R_4 + s^4 (2 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_4 + 2 C_4 C_5 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_4 C_5 L_4 R_4 R_5 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4 + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_4 L_5) + s^2 (2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 R_5 g_m + 2 C_5 L_4 R_4 R_5) + s (2 L_1 R_4 R_5 + 2 L_4 R_4 R_5 + 2 L_5 R_4 R_5)}$$

$$\mathbf{9.151 \quad X-INVALID-ORDER-151} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_4 R_5 s^3 - L_1 L_4 R_4 R_5 s + s^2 (L_1 L_4 L_5 R_4 R_5 g_m - L_1 L_4 L_5 R_4)}{2 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^4 + 2 L_1 R_4 R_5 + L_4 R_4 R_5 + 2 L_5 R_4 R_5 + s^3 (2 C_4 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_1 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_4 L_5 R_5) + s^2 (2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_5 R_4 R_5 + C_5 L_4 L_5 R_4 R_5 + 2 L_1 L_4 L_5 R_4 g_m + 2 L_1 L_4 L_5 R_5 g_m + 2 L_1 L_4 L_5 R_5) + s (2 L_1 R_4 R_5 + 2 L_4 R_4 R_5 + 2 L_5 R_4 R_5)}$$

$$\mathbf{9.152 \quad X-INVALID-ORDER-152} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 L_5 R_4 g_m s^3 + s^4 (C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_5 L_1 L_4 L_5 R_4) + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4 R_5)}{2 R_4 R_5 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2 C_4 C_5 L_4 L_5 R_4 R_5 + 2 C_4 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_5 L_1 L_4 L_5) + s^3 (2 C_4 L_1 L_4 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_4 + 2 C_4 L_4 L_5 R_4 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4) + s^2 (2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4) + s (2 L_1 R_4 R_5 + 2 L_4 R_4 R_5 + 2 L_5 R_4 R_5)}$$

$$\mathbf{9.153 \quad X-INVALID-ORDER-153} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_4 R_5 s^3 + s^4 (C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_5 L_1 L_4 L_5 R_4) + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4 R_5)}{2 R_4 R_5 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2 C_4 C_5 L_1 L_4 R_4 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_5 L_1 L_4 L_5) + s^3 (2 C_4 L_1 L_4 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_4 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4) + s^2 (2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4) + s (2 L_1 R_4 R_5 + 2 L_4 R_4 R_5 + 2 L_5 R_4 R_5)}$$

$$\mathbf{9.154 \quad X-INVALID-ORDER-154} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2 R_5 + s^3 (2 C_4 L_1 L_4 R_4 g_m + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4) + s^2 (C_4 L_4 R_4 + 2 C_4 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_4 g_m + 2 L_1 R_5 g_m + 2 L_1 + L_4)}$$

$$\mathbf{9.155 \quad X-INVALID-ORDER-155} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 s^4 + L_1 R_4 g_m s + s^3 (C_4 L_1 L_4 R_4 g_m - C_5 L_1 L_4) + s^2 (-C_5 L_1 R_4 + L_1 L_4 g_m)}{s^4 (2 C_4 C_5 L_1 L_4 R_4 g_m + 2 C_4 C_5 L_1 L_4) + s^3 (C_4 C_5 L_4 R_4 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m) + s^2 (2 C_4 L_4 + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.156 \quad X-INVALID-ORDER-156} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 R_5 s^4 + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4 - C_5 L_1 L_4 R_5) + s^2 (-C_5 L_1 R_4 R_5 + L_1 L_4 R_5 g_m - L_1 L_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2 R_5 + s^4 (2 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_5) + s^3 (C_4 C_5 L_4 R_4 R_5 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_5 g_m) + s^2 (C_4 L_4 R_4 + 2 C_4 L_4 R_5 + 2 C_5 L_1 R_4 R_5 g_m + 2 C_5 L_1 R_5 + C_5 L_4 R_5 + 2 L_1 L_4 g_m) + s (C_5 R_4 R_5 + 2 L_1 R_4 g_m + 2 L_1 R_5 g_m + 2 L_1 + L_4)}$$

$$\mathbf{9.157 \quad X-INVALID-ORDER-157} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4) + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 + L_1 L_4 g_m)}{s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4) + s^3 (C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m) + s^2 (2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2C_5 R_5 + 2L_1 g_m) + 2}$$

$$\mathbf{9.158 \quad X-INVALID-ORDER-158} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (-C_4 C_5 L_1 L_4 R_4 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_4 g_m - C_5 L_1 L_4 + C_5 L_1 L_5 R_4 g_m) + s^2 (-C_5 L_1 R_4 + L_1 L_4 g_m)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_4 R_4 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4 + 2C_5 L_5) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.159 \quad X-INVALID-ORDER-159} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 s^5 - L_1 R_4 s + s^4 (C_4 L_1 L_4 L_5 R_4 g_m - C_5 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 R_4 - C_5 L_1 L_5 R_4 + L_1 L_4 L_5 g_m) + s^2 (-L_1 L_4 + L_1 L_5 R_4 g_m)}{R_4 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_4 C_5 L_4 L_5 R_4 + 2C_4 L_1 L_4 L_5 g_m + 2C_5 L_1 L_4 L_5 g_m) + s^3 (2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_4 L_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 + C_5 L_4 L_5) + s^2 (C_4 L_4 R_4 + C_5 L_5 R_4 + 2L_1 L_4 g_m + 2L_1 L_5 g_m) + s (2L_1 R_4 g_m + 2L_1 L_5 R_4 g_m)}$$

$$\mathbf{9.160 \quad X-INVALID-ORDER-160} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4 + C_5 L_1 L_5 R_4 g_m) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 + L_1 L_4 g_m)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5 g_m + 2C_5 L_1 + C_5 L_4 + 2C_5 L_5) + s (C_5 R_4 + 2C_5 R_5 + 2L_1 g_m) + 2}$$

$$\mathbf{9.161 \quad X-INVALID-ORDER-161} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^5 - L_1 R_4 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_4 R_5 g_m - C_4 L_1 L_4 L_5 R_4 - C_5 L_1 L_4 L_5 R_5) + s^3 (-C_4 L_1 L_4 R_4 R_5 - C_5 L_1 L_5 R_4 R_5 + L_1 L_4 L_5 R_5 g_m)}{R_4 R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 + 2C_4 L_1 L_4 L_5 R_4 g_m + 2C_4 L_1 L_4 L_5 R_5 g_m + 2C_4 L_1 L_4 L_5 + 2C_5 L_1 L_4 L_5 R_5 g_m) + s^3 (2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_5 + C_4 L_4 L_5 R_4 + 2C_4 L_4 L_5 R_5 + 2C_5 L_1 L_5 R_4 R_5 g_m + 2C_5 L_1 L_5 R_5 + C_5 L_4 L_5 R_5)}$$

$$\mathbf{9.162 \quad X-INVALID-ORDER-162} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (C_4 L_1 L_4 L_5 R_4 g_m + C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4 + C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4 + L_1 L_4 L_5 g_m) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 + L_1 L_4 g_m)}{R_4 + 2R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5 + 2C_4 L_1 L_4 L_5 g_m + 2C_5 L_1 L_4 L_5 g_m) + s^3 (2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_4 L_4 L_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5)}$$

$$\mathbf{9.163 \quad X-INVALID-ORDER-163} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (-C_4 C_5 L_1 L_4 R_4 R_5 + C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4 - C_5 L_1 L_4 R_5) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 + L_1 L_4 g_m)}{R_4 + 2R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5 + 2C_5 L_1 L_4 L_5 g_m) + s^3 (C_4 C_5 L_4 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_5 L_1 L_4 R_5 g_m + 2C_5 L_1 L_4 R_5)}$$

$$\mathbf{9.164 \quad X-INVALID-ORDER-164} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^3 (2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4 + C_4 L_4 R_4 + 2C_4 L_4 R_5) + s (2C_4 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

$$\mathbf{9.165 \quad X-INVALID-ORDER-165} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 s^4 + C_4 L_1 L_4 R_4 g_m s^3 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4) + s^3 (2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 L_1 L_4 g_m) + s^2 (2C_4 L_1 R_4 g_m + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1) + s (2C_4 R_4 + C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.166 \quad X-INVALID-ORDER-166} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 R_5 s^4 - C_5 L_1 R_4 R_5 s^2 + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^4 (2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_5) + s^3 (2C_4 C_5 L_1 R_4 R_5 + C_4 C_5 L_4 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4 + C_4 L_4 R_4 + 2C_4 L_4 R_5 + 2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4)}$$

$$\mathbf{9.167 \quad X-INVALID-ORDER-167} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4) + s^3 (2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m) + s^2 (2C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5 g_m + 2C_5 L_1) + s (2C_4 R_4 + C_5 R_4 + 2C_5 R_5 + 2L_1 R_4)}$$

$$\mathbf{9.168 \quad X-INVALID-ORDER-168} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 - C_4 C_5 L_1 L_4 R_4 s^4 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_5 R_4 g_m)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_5 R_4 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 L_1 R_4 g_m + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + 2C_5 L_5) + s (2C_4 R_4 + C_5 R_4 + 2L_1 R_4)}$$

$$\mathbf{9.169 \quad X-INVALID-ORDER-169} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 s^5 + C_4 L_1 L_4 L_5 R_4 g_m s^4 + L_1 L_5 R_4 g_m s^2 - L_1 R_4 s + s^3 (-C_4 L_1 L_4 R_4 - C_5 L_1 L_5 R_4)}{R_4 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_4 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_4 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_4 L_4 L_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5) + s^2 (2C_4 L_1 R_4 + C_4 L_4 R_4 + 2C_4 L_5 R_4 + C_5 L_5 R_4 + 2L_1 L_5 g_m) + s (2C_4 R_4 + C_5 R_4 + 2L_1 R_4)}$$

$$\mathbf{9.170 \quad X-INVALID-ORDER-170} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4) + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_5 R_4 g_m) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{2C_4 C_5 L_1 L_4 L_5 g_m s^5 + s^4 (2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m)}$$

$$\mathbf{9.171 \quad X-INVALID-ORDER-171} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^5 - L_1 R_4 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_4 R_5 g_m - C_4 L_1 L_4 L_5 R_4) + s^3 (-C_4 L_1 L_4 R_4 R_5 - C_5 L_1 L_5 R_4 R_5)}{R_4 R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (2C_4 C_5 L_1 L_5 R_4 R_5 + C_4 C_5 L_4 L_5 R_4 R_5 + 2C_4 L_1 L_4 L_5 R_4 g_m + 2C_4 L_1 L_4 L_5 R_5 g_m + 2C_4 L_1 L_4 L_5) + s^3 (2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_5 + 2C_4 L_1 L_5 R_4 R_5 g_m + 2C_4 L_1 L_5 R_4 + C_4 L_4 L_5 R_4 + 2C_4 L_4 L_5 R_5 + 2C_5 L_1 L_5 R_4 R_5) + s^2 (2C_4 L_1 R_4 R_5 + C_4 L_4 R_4 R_5 + 2C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2L_1 L_5 g_m) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4 R_5)}$$

$$\mathbf{9.172 \quad X-INVALID-ORDER-172} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 L_5 R_4 g_m s^4 + L_1 L_5 R_4 g_m s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4 R_5)}{R_4 + 2R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (2C_4 C_5 L_5 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_4 R_5) + s^2 (2C_4 L_1 R_4 R_5 + C_4 L_4 R_4 R_5 + 2C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2L_1 L_5 g_m) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4 R_5)}$$

$$\mathbf{9.173 \quad X-INVALID-ORDER-173} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 R_5 s^4 - C_5 L_1 R_4 R_5 s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^3 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^2 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4 R_5)}{R_4 + 2R_5 + s^5 (2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_4 C_5 L_1 R_4 R_5 + C_4 C_5 L_4 R_4 R_5 + 2C_4 C_5 L_5 R_4 R_5 + 2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_4 L_1 L_5 R_4 R_5) + s^2 (2C_4 L_1 R_4 R_5 + C_4 L_4 R_4 R_5 + 2C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2L_1 L_5 g_m) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4 R_5)}$$

$$\mathbf{9.174 \quad X-INVALID-ORDER-174} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + s (C_1 R_4 + 2C_1 R_5) + 2}$$

$$\mathbf{9.175} \quad \mathbf{X-INVALID-ORDER-175} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2C_1 C_5 L_5 s^3 + 2g_m + s^2 (C_1 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.176} \quad \mathbf{X-INVALID-ORDER-176} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 s^2 + L_5 R_4 g_m s - R_4}{C_1 C_5 L_5 R_4 s^3 + 2R_4 g_m + s^2 (2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (C_1 R_4 + 2L_5 g_m) + 2}$$

$$\mathbf{9.177} \quad \mathbf{X-INVALID-ORDER-177} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 + R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2C_1 C_5 L_5 s^3 + 2g_m + s^2 (C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_5 L_5 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.178} \quad \mathbf{X-INVALID-ORDER-178} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 R_5 s^2 - R_4 R_5 + s (L_5 R_4 R_5 g_m - L_5 R_4)}{C_1 C_5 L_5 R_4 R_5 s^3 + 2R_4 R_5 g_m + 2R_5 + s^2 (C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_5 L_5 R_4 R_5 g_m + 2C_5 L_5 R_5) + s (C_1 R_4 R_5 + 2L_5 R_4 g_m + 2L_5 R_5 g_m + 2L_5)}$$

$$\mathbf{9.179} \quad \mathbf{X-INVALID-ORDER-179} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5) + s^2 (2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_1 R_5 + 2L_5 g_m) + 2}$$

$$\mathbf{9.180} \quad \mathbf{X-INVALID-ORDER-180} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5) + s^2 (C_1 C_5 R_4 R_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

$$\mathbf{9.181} \quad \mathbf{X-INVALID-ORDER-181} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.182} \quad \mathbf{X-INVALID-ORDER-182} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{2C_1 C_4 C_5 R_5 s^3 + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.183} \quad \mathbf{X-INVALID-ORDER-183} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 g_m s^2 - C_5 s + g_m}{2C_1 C_4 C_5 L_5 s^4 + 2C_4 C_5 L_5 g_m s^3 + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.184} \quad \mathbf{X-INVALID-ORDER-184} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 s^2 + L_5 g_m s - 1}{2g_m + s^3 (2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5) + s^2 (2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4)}$$

$$\mathbf{9.185 \quad X-INVALID-ORDER-185} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 g_m s^2 + g_m + s (C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_5 s^4 + s^3 (2 C_1 C_4 C_5 R_5 + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.186 \quad X-INVALID-ORDER-186} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_5 s^2 - R_5 + s (L_5 R_5 g_m - L_5)}{2 R_5 g_m + s^3 (2 C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2 C_4 C_5 L_5 R_5) + s^2 (C_1 L_5 + 2 C_4 L_5 R_5 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2 C_4 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.187 \quad X-INVALID-ORDER-187} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_5 g_m s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_1 C_4 C_5 L_5 R_5 s^4 + 2 g_m + s^3 (2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_5 + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.188 \quad X-INVALID-ORDER-188} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_1 C_4 C_5 L_5 R_5 s^4 + 2 g_m + s^3 (C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_5 + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.189 \quad X-INVALID-ORDER-189} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 C_1 C_4 C_5 R_4 R_5 s^3 + 2 g_m + s^2 (2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4) + s (2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.190 \quad X-INVALID-ORDER-190} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2 C_1 C_4 C_5 L_5 R_4 s^4 + 2 g_m + s^3 (2 C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_4 g_m) + s^2 (2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_4 C_5 R_4 + 2 C_5 L_5 g_m) + s (2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5)}$$

$$\mathbf{9.191 \quad X-INVALID-ORDER-191} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 s^2 + L_5 R_4 g_m s - R_4}{2 R_4 g_m + s^3 (2 C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_1 L_5 + 2 C_4 L_5 R_4 g_m + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5) + s (C_1 R_4 + 2 C_4 R_4 + 2 L_5 g_m) + 2}$$

$$\mathbf{9.192 \quad X-INVALID-ORDER-192} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_4 g_m s^2 + R_4 g_m + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2 C_1 C_4 C_5 L_5 R_4 s^4 + 2 g_m + s^3 (2 C_1 C_4 C_5 R_4 R_5 + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_4 g_m) + s^2 (2 C_1 C_4 R_4 + C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_4 + 2 C_5 L_5 g_m) + s (2 C_1 + 2 C_4 R_4 g_m + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.193 \quad X-INVALID-ORDER-193} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_4 R_5 s^2 - R_4 R_5 + s (L_5 R_4 R_5 g_m - L_5 R_4)}{2 R_4 R_5 g_m + 2 R_5 + s^3 (2 C_1 C_4 L_5 R_4 R_5 + C_1 C_5 L_5 R_4 R_5 + 2 C_4 C_5 L_5 R_4 R_5) + s^2 (C_1 L_5 R_4 + 2 C_1 L_5 R_5 + 2 C_4 L_5 R_4 R_5 g_m + 2 C_4 L_5 R_4 + 2 C_5 L_5 R_4 R_5 g_m + 2 C_5 L_5 R_5) + s (C_1 R_4 R_5 + 2 C_4 R_4 R_5 + 2 L_5 R_4 g_m + 2 L_5 R_5 g_m + 2 L_5)}$$

$$\mathbf{9.194 \quad X-INVALID-ORDER-194} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2C_1 C_4 C_5 L_5 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 C_4 R_4 R_5 + 2C_1 L_5 + 2C_4 L_5 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2L_5 g_m)}$$

$$\mathbf{9.195 \quad X-INVALID-ORDER-195} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2C_1 C_4 C_5 L_5 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_4)}$$

$$\mathbf{9.196 \quad X-INVALID-ORDER-196} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 s^2 + g_m + s (C_4 R_4 g_m - C_5)}{C_1 C_4 C_5 R_4 s^3 + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.197 \quad X-INVALID-ORDER-197} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_4 R_5 s^2 + R_5 g_m + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{C_1 C_4 C_5 R_4 R_5 s^3 + 2g_m + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s (C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.198 \quad X-INVALID-ORDER-198} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.199 \quad X-INVALID-ORDER-199} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_4 g_m s^3 + g_m + s^2 (-C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m - C_5)}{2C_1 C_4 C_5 L_5 s^4 + s^3 (C_1 C_4 C_5 R_4 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.200 \quad X-INVALID-ORDER-200} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_4 s^3 + s^2 (C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{C_1 C_4 C_5 L_5 R_4 s^4 + 2g_m + s^3 (2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4 R_4 g_m + 2C_4)}$$

$$\mathbf{9.201 \quad X-INVALID-ORDER-201} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_4 g_m s^3 + g_m + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2C_1 C_4 C_5 L_5 s^4 + s^3 (C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.202 \quad X-INVALID-ORDER-202} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_4 R_5 s^3 - R_5 + s^2 (C_4 L_5 R_4 R_5 g_m - C_4 L_5 R_4 - C_5 L_5 R_5) + s (-C_4 R_4 R_5 + L_5 R_5 g_m - L_5)}{C_1 C_4 C_5 L_5 R_4 R_5 s^4 + 2R_5 g_m + s^3 (C_1 C_4 L_5 R_4 + 2C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_5) + s^2 (C_1 C_4 R_4 R_5 + C_1 L_5 + 2C_4 L_5 R_4 g_m + 2C_4 L_5 R_5 g_m + 2C_4 L_5 + 2C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_5 + 2L_5 g_m)}$$

$$\mathbf{9.203 \quad X-INVALID-ORDER-203} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 + L_5 g_m) - 1}{2g_m + s^4 (C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.204 \quad X-INVALID-ORDER-204} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (-C_4 C_5 R_4 R_5 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^4 (C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (C_1 C_4 C_5 R_4 R_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5 + 2C_5 L_5 g_m) + s (C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.205 \quad X-INVALID-ORDER-205} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_4 L_4 R_5 g_m - C_4 L_4) - 1}{C_1 C_4 L_4 s^3 + 2g_m + s^2 (2C_1 C_4 R_5 + 2C_4 L_4 g_m) + s (C_1 + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.206 \quad X-INVALID-ORDER-206} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 s^3 + C_4 L_4 g_m s^2 - C_5 s + g_m}{C_1 C_4 C_5 L_4 s^4 + 2C_4 C_5 L_4 g_m s^3 + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.207 \quad X-INVALID-ORDER-207} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^2 (C_4 L_4 R_5 g_m - C_4 L_4) - 1}{C_1 C_4 C_5 L_4 R_5 s^4 + 2g_m + s^3 (C_1 C_4 L_4 + 2C_4 C_5 L_4 R_5 g_m) + s^2 (2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_4 C_5 R_5 + 2C_4 L_4 g_m) + s (C_1 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.208 \quad X-INVALID-ORDER-208} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 g_m s^2 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s (C_5 R_5 g_m - C_5)}{C_1 C_4 C_5 L_4 s^4 + s^3 (2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.209 \quad X-INVALID-ORDER-209} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 - C_4 C_5 L_4 s^3 - C_5 s + g_m + s^2 (C_4 L_4 g_m + C_5 L_5 g_m)}{s^4 (C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5) + s^3 (2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.210 \quad X-INVALID-ORDER-210} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 s^4 + C_4 L_4 L_5 g_m s^3 + L_5 g_m s + s^2 (-C_4 L_4 - C_5 L_5) - 1}{C_1 C_4 C_5 L_4 L_5 s^5 + 2C_4 C_5 L_4 L_5 g_m s^4 + 2g_m + s^3 (C_1 C_4 L_4 + 2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5) + s^2 (2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4)}$$

$$\mathbf{9.211 \quad X-INVALID-ORDER-211} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_4 L_4 g_m + C_5 L_5 g_m) + s (C_5 R_5 g_m - C_5)}{s^4 (C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.212 \quad X-INVALID-ORDER-212} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_5 s^4 - R_5 + s^3 (C_4 L_4 L_5 R_5 g_m - C_4 L_4 L_5) + s^2 (-C_4 L_4 R_5 - C_5 L_5 R_5) + s (L_5 R_5 g_m - L_5)}{C_1 C_4 C_5 L_4 L_5 R_5 s^5 + 2 R_5 g_m + s^4 (C_1 C_4 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (C_1 C_4 L_4 R_5 + 2 C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2 C_4 C_5 L_5 R_5 + 2 C_4 L_4 L_5 g_m) + s^2 (C_1 L_5 + 2 C_4 L_4 R_5 g_m + 2 C_4 L_5 R_5 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2 C_4 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.213 \quad X-INVALID-ORDER-213} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 L_5 g_m s^3 + L_5 g_m s + R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{C_1 C_4 C_5 L_4 L_5 s^5 + 2 g_m + s^4 (2 C_1 C_4 C_5 L_5 R_5 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_5 + 2 C_4 L_4 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.214 \quad X-INVALID-ORDER-214} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{C_1 C_4 C_5 L_4 L_5 s^5 + 2 g_m + s^4 (C_1 C_4 C_5 L_4 R_5 + 2 C_1 C_4 C_5 L_5 R_5 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 + C_1 C_5 L_5 + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.215 \quad X-INVALID-ORDER-215} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_4 R_5 g_m - L_4)}{2 C_1 C_4 L_4 R_5 s^3 + 2 R_5 g_m + s^2 (C_1 L_4 + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4) + s (2 C_1 R_5 + 2 L_4 g_m) + 2}$$

$$\mathbf{9.216 \quad X-INVALID-ORDER-216} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 s^2 + L_4 g_m s}{2 g_m + s^3 (2 C_1 C_4 L_4 + C_1 C_5 L_4 + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_1 + 2 C_5)}$$

$$\mathbf{9.217 \quad X-INVALID-ORDER-217} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_5 s^2 + s (L_4 R_5 g_m - L_4)}{2 R_5 g_m + s^3 (2 C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + 2 C_4 C_5 L_4 R_5) + s^2 (C_1 L_4 + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_5 g_m) + s (2 C_1 R_5 + 2 C_5 R_5 + 2 L_4 g_m) + 2}$$

$$\mathbf{9.218 \quad X-INVALID-ORDER-218} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_4 g_m s + s^2 (C_5 L_4 R_5 g_m - C_5 L_4)}{2 C_1 C_4 C_5 L_4 R_5 s^4 + 2 g_m + s^3 (2 C_1 C_4 L_4 + C_1 C_5 L_4 + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_1 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_1 + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.219 \quad X-INVALID-ORDER-219} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 g_m s^3 - C_5 L_4 s^2 + L_4 g_m s}{2 C_1 C_4 C_5 L_4 L_5 s^5 + 2 C_4 C_5 L_4 L_5 g_m s^4 + 2 g_m + s^3 (2 C_1 C_4 L_4 + C_1 C_5 L_4 + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_4) + s^2 (2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_1 + 2 C_5)}$$

$$\mathbf{9.220 \quad X-INVALID-ORDER-220} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 s^3 + L_4 L_5 g_m s^2 - L_4 s}{s^4 (2 C_1 C_4 L_4 L_5 + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 + 2 C_1 L_5 + 2 C_4 L_4 + 2 C_5 L_5) + s (2 L_4 g_m + 2 L_5 g_m) + 2}$$

$$\mathbf{9.221 \quad X-INVALID-ORDER-221} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 g_m s^3 + L_4 g_m s + s^2 (C_5 L_4 R_5 g_m - C_5 L_4)}{2 C_1 C_4 C_5 L_4 L_5 s^5 + 2 g_m + s^4 (2 C_1 C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_4 + C_1 C_5 L_4 + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_1 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_1 + 2 C_5 R_5 g_m + 2 C_5)}$$

$$\mathbf{9.222 \quad X-INVALID-ORDER-222} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_5 s^3 - L_4 R_5 s + s^2 (L_4 L_5 R_5 g_m - L_4 L_5)}{2 R_5 + s^4 (2 C_1 C_4 L_4 L_5 R_5 + C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s^2 (C_1 L_4 R_5 + 2 C_1 L_5 R_5 + 2 C_4 L_4 R_5 + 2 C_5 L_5 R_5 + 2 L_4 L_5 g_m) + s (2 L_4 R_5 g_m + 2 L_5 R_5 g_m + 2 L_5)}$$

$$\mathbf{9.223 \quad X-INVALID-ORDER-223} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_4 L_5 g_m s^2 + s^3 (C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s (L_4 R_5 g_m - L_4)}{2 C_1 C_4 C_5 L_4 L_5 R_5 s^5 + 2 R_5 g_m + s^4 (2 C_1 C_4 L_4 L_5 + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_4 R_5 + 2 C_1 C_5 L_5 R_5 + 2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 + 2 C_1 L_5 + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_1 R_5 + 2 L_4 g_m + 2 L_5 g_m)}$$

$$\mathbf{9.224 \quad X-INVALID-ORDER-224} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_5 s^2 + s^3 (C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s (L_4 R_5 g_m - L_4)}{2 C_1 C_4 C_5 L_4 L_5 R_5 s^5 + 2 R_5 g_m + s^4 (C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + 2 C_1 C_5 L_5 R_5 + 2 C_4 C_5 L_4 R_5 + 2 C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_5 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (2 C_1 R_5 + 2 C_5 R_5 + 2 C_5)}$$

$$\mathbf{9.225 \quad X-INVALID-ORDER-225} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{C_1 C_4 L_4 s^3 + 2 g_m + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + 2 C_4 L_4 g_m) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.226 \quad X-INVALID-ORDER-226} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 s^3 + g_m + s^2 (-C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m - C_5)}{C_1 C_4 C_5 L_4 s^4 + s^3 (C_1 C_4 C_5 R_4 + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.227 \quad X-INVALID-ORDER-227} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_5 s^3 + R_5 g_m + s^2 (-C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{C_1 C_4 C_5 L_4 R_5 s^4 + 2 g_m + s^3 (C_1 C_4 C_5 R_4 R_5 + C_1 C_4 L_4 + 2 C_4 C_5 L_4 R_5 g_m) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_5 + 2 C_4 L_4 g_m) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.228 \quad X-INVALID-ORDER-228} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{C_1 C_4 C_5 L_4 s^4 + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5 + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.229 \quad X-INVALID-ORDER-229} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (-C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (-C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m - C_5)}{s^4 (C_1 C_4 C_5 L_4 + 2 C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.230 \quad X-INVALID-ORDER-230} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 s^4 + s^3 (-C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_4 L_4 + C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 + 2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4 R_4 g_m + 2C_4)}$$

$$\mathbf{9.231 \quad X-INVALID-ORDER-231} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 g_m s^4 + g_m + s^3 (C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^4 (C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.232 \quad X-INVALID-ORDER-232} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_5 s^4 - R_5 + s^3 (-C_4 C_5 L_5 R_4 R_5 + C_4 L_4 L_5 R_5 g_m - C_4 L_4 L_5) + s^2 (-C_4 L_4 R_5 + C_4 L_5 R_4 R_5 g_m - C_4 L_5 R_4 - C_5 L_5 R_5) + s (-C_4 R_4 R_5 + L_5 R_5 g_m - L_5)}{C_1 C_4 C_5 L_4 L_5 R_5 s^5 + 2R_5 g_m + s^4 (C_1 C_4 C_5 L_5 R_4 R_5 + C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (C_1 C_4 L_4 R_5 + C_1 C_4 L_5 R_4 + 2C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_5 + 2C_4 L_4 L_5 g_m) + s^2 (C_1 C_4 R_4 R_5 + C_1 L_5 + 2C_4 L_4 R_5 g_m + 2C_4 L_5 R_4 g_m + 2C_4 L_5)}$$

$$\mathbf{9.233 \quad X-INVALID-ORDER-233} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4 + C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 + L_5 g_m) - 1}{C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 + 2C_1 C_4 L_5 + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m)}$$

$$\mathbf{9.234 \quad X-INVALID-ORDER-234} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (-C_4 C_5 L_4 R_5 + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (-C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m - C_4 L_4 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 - R_5)}{C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_4 R_5 + C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 C_5 R_4 R_5 + C_1 C_4 L_4 + C_1 C_5 L_5 + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5)}$$

$$\mathbf{9.235 \quad X-INVALID-ORDER-235} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_4 R_4 R_5 g_m - L_4 R_4)}{2C_1 C_4 L_4 R_4 R_5 s^3 + 2R_4 R_5 g_m + 2R_4 + s^2 (C_1 L_4 R_4 + 2C_1 L_4 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4) + s (2C_1 R_4 R_5 + 2L_4 R_4 g_m + 2L_4 R_5 g_m + 2L_4)}$$

$$\mathbf{9.236 \quad X-INVALID-ORDER-236} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 s^2 + L_4 R_4 g_m s}{2R_4 g_m + s^3 (2C_1 C_4 L_4 R_4 + C_1 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_4) + s^2 (2C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_4) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.237 \quad X-INVALID-ORDER-237} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 R_5 s^2 + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2R_4 R_5 g_m + 2R_4 + s^3 (2C_1 C_4 L_4 R_4 R_5 + C_1 C_5 L_4 R_4 R_5 + 2C_4 C_5 L_4 R_4 R_5) + s^2 (C_1 L_4 R_4 + 2C_1 L_4 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 R_4 R_5 g_m + 2C_5 L_4 R_5) + s (2C_1 R_4 R_5 + 2C_5 R_4 R_5 + 2L_4 R_4 g_m + 2L_4 R_5 g_m + 2L_4)}$$

$$\mathbf{9.238 \quad X-INVALID-ORDER-238} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_4 g_m s + s^2 (C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4)}{2C_1 C_4 C_5 L_4 R_4 R_5 s^4 + 2R_4 g_m + s^3 (2C_1 C_4 L_4 R_4 + C_1 C_5 L_4 R_4 + 2C_1 C_5 L_4 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_4) + s^2 (2C_1 C_5 R_4 R_5 + 2C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_4 R_5 g_m + 2C_5 L_4) + s (2C_1 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.239 \quad X-INVALID-ORDER-239} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_4 g_m s^3 - C_5 L_4 R_4 s^2 + L_4 R_4 g_m s}{2C_1 C_4 C_5 L_4 L_5 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2C_1 C_4 L_4 R_4 + C_1 C_5 L_4 R_4 + 2C_1 C_5 L_5 R_4 + 2C_4 C_5 L_4 R_4 + 2C_5 L_4 L_5 g_m) + s^2 (2C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_4 + 2C_5 L_5 R_4 g_m) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.240 \quad X-INVALID-ORDER-240} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_4 s^3 + L_4 L_5 R_4 g_m s^2 - L_4 R_4 s}{2R_4 + s^4 (2C_1 C_4 L_4 L_5 R_4 + C_1 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 L_4 L_5 + 2C_4 L_4 L_5 R_4 g_m + 2C_5 L_4 L_5 R_4 g_m + 2C_5 L_4 L_5) + s^2 (C_1 L_4 R_4 + 2C_1 L_5 R_4 + 2C_4 L_4 R_4 + 2C_5 L_5 R_4 + 2L_4 L_5 g_m) + s (2L_4 R_4 g_m + 2L_4 + 2L_5 R_4 g_m)}$$

$$\mathbf{9.241 \quad X-INVALID-ORDER-241} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_4 g_m s^3 + L_4 R_4 g_m s + s^2 (C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4)}{2C_1 C_4 C_5 L_4 L_5 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2C_1 C_4 L_4 R_4 + C_1 C_5 L_4 R_4 + 2C_1 C_5 L_4 R_5 + 2C_1 C_5 L_5 R_4 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_4 + 2C_5 L_4 L_5 g_m) + s^2 (2C_1 C_5 R_4 R_5 + 2C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_5 R_4 R_5) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.242 \quad X-INVALID-ORDER-242} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_4 R_5 s^3 - L_4 R_4 R_5 s + s^2 (L_4 L_5 R_4 R_5 g_m - L_4 L_5 R_4)}{2R_4 R_5 + s^4 (2C_1 C_4 L_4 L_5 R_4 R_5 + C_1 C_5 L_4 L_5 R_4 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (C_1 L_4 L_5 R_4 + 2C_1 L_4 L_5 R_5 + 2C_4 L_4 L_5 R_4 R_5 g_m + 2C_4 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_4 R_5 g_m + 2C_5 L_4 L_5 R_5) + s^2 (C_1 L_4 R_4 R_5 + 2C_1 L_5 R_4 R_5 + 2C_4 L_4 R_4 R_5 + 2C_5 L_5 R_4 R_5 + 2L_4 L_5 R_4 g_m + 2L_4 L_5 R_5) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.243 \quad X-INVALID-ORDER-243} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_4 L_5 R_4 g_m s^2 + s^3 (C_5 L_4 L_5 R_4 R_5 g_m - C_5 L_4 L_5 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2C_1 C_4 C_5 L_4 L_5 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_4 + s^4 (2C_1 C_4 L_4 L_5 R_4 + C_1 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_4 L_5 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 C_4 L_4 R_4 R_5 + 2C_1 C_5 L_5 R_4 R_5 + 2C_1 L_4 L_5 + 2C_4 L_4 L_5 R_4 g_m + 2C_5 L_4 L_5 R_4 g_m + 2C_5 L_4 L_5 R_5 g_m + 2C_5 L_4 L_5) + s^2 (C_1 L_4 R_4 R_5 + 2C_1 L_5 R_4 R_5 + 2C_4 L_4 R_4 R_5 + 2C_5 L_5 R_4 R_5 + 2L_4 L_5 R_4 g_m + 2L_4 L_5 R_5) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.244 \quad X-INVALID-ORDER-244} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_4 R_5 s^2 + s^3 (C_5 L_4 L_5 R_4 R_5 g_m - C_5 L_4 L_5 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2C_1 C_4 C_5 L_4 L_5 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_4 + s^4 (C_1 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_4 L_5 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 C_4 L_4 R_4 R_5 + C_1 C_5 L_4 R_4 R_5 + 2C_1 C_5 L_5 R_4 R_5 + 2C_4 C_5 L_4 R_4 R_5 + 2C_5 L_4 L_5 R_4 g_m + 2C_5 L_4 L_5 R_5 g_m + 2C_5 L_4 L_5) + s^2 (C_1 L_4 R_4 + 2C_1 L_5 R_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_5 R_4) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4 g_m)}$$

$$\mathbf{9.245 \quad X-INVALID-ORDER-245} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4) + s (L_4 R_5 g_m - L_4)}{2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5) + s^2 (C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (C_1 R_4 + 2C_1 R_5 + 2L_4 g_m) + 2}$$

$$\mathbf{9.246 \quad X-INVALID-ORDER-246} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 s^3 + R_4 g_m + s^2 (C_4 L_4 R_4 g_m - C_5 L_4) + s (-C_5 R_4 + L_4 g_m)}{C_1 C_4 C_5 L_4 R_4 s^4 + 2g_m + s^3 (2C_1 C_4 L_4 + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (C_1 C_5 R_4 + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.247 \quad X-INVALID-ORDER-247} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 R_5 s^3 + R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 - C_5 L_4 R_5) + s (-C_5 R_4 R_5 + L_4 R_5 g_m - L_4)}{C_1 C_4 C_5 L_4 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s^2 (C_1 C_5 R_4 R_5 + C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4 + 2C_5 L_4 R_5 g_m) + s (C_1 R_4 + 2C_1 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_5 + 2L_4 g_m) + 2}$$

9.248 X-INVALID-ORDER-248 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_4 L_4 R_4 g_m + C_5 L_4 R_5 g_m - C_5 L_4) + s (C_5 R_4 R_5 g_m - C_5 R_4 + L_4 g_m)}{2g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 L_4 + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4) + s^2 (C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

9.249 X-INVALID-ORDER-249 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (-C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2 (C_4 L_4 R_4 g_m - C_5 L_4 + C_5 L_5 R_4 g_m) + s (-C_5 R_4 + L_4 g_m)}{2C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 + C_1 C_5 L_4 + 2C_1 C_5 L_5 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (C_1 C_5 R_4 + 2C_4 L_4 g_m + 2C_5 L_4 g_m + 2C_5 L_5 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

9.250 X-INVALID-ORDER-250 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 s^4 - R_4 + s^3 (C_4 L_4 L_5 R_4 g_m - C_5 L_4 L_5) + s^2 (-C_4 L_4 R_4 - C_5 L_5 R_4 + L_4 L_5 g_m) + s (-L_4 + L_5 R_4 g_m)}{C_1 C_4 C_5 L_4 L_5 R_4 s^5 + 2 R_4 g_m + s^4 (2 C_1 C_4 L_4 L_5 + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + C_1 C_5 L_5 R_4 + 2 C_4 L_4 L_5 g_m + 2 C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 + 2 C_1 L_5 + 2 C_4 L_4 R_4 g_m + 2 C_4 L_4 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5) + s (C_1 R_4 + 2 L_4 g_m + 2 L_5 g_m) + 1}$$

9.251 X-INVALID-ORDER-251 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2 (C_4 L_4 R_4 g_m + C_5 L_4 R_5 g_m - C_5 L_4 + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4 + L_4 g_m)}{2 C_1 C_4 C_5 L_4 L_5 s^5 + 2 g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2 C_1 C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_4 + C_1 C_5 L_4 + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (C_1 C_5 R_4 + 2 C_1 C_5 R_5 + 2 C_4 L_4 g_m + 2 C_5 L_4 g_m + 2 C_5 L_5 g_m) + s (2 C_1 + 2 C_5 R_4 g_m + 2 C_5 R_5 g_m)}$$

9.252 X-INVALID-ORDER-252 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_4 L_4 L_5 R_4 R_5 g_m - C_4 L_4 L_5 R_4 - C_5 L_4 L_5 R_5) + s^2 (-C_4 L_4 R_4 R_5 - C_5 L_5 R_4 R_5 + L_4 L_5 R_5 g_m - L_4 L_5) + C_1 C_4 C_5 L_4 L_5 R_4 R_5 s^5 + 2 R_4 R_5 g_m + 2 R_5 + s^4 (C_1 C_4 L_4 L_5 R_4 + 2 C_1 C_4 L_4 L_5 R_5 + C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_4 R_5 + C_1 C_5 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s^2 (C_1 L_4 R_4 R_5 + C_1 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s (C_1 L_4 R_4 R_5 + C_1 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s}{C_1 C_4 C_5 L_4 L_5 R_4 R_5 s^5 + 2 R_4 R_5 g_m + 2 R_5 + s^4 (C_1 C_4 L_4 L_5 R_4 + 2 C_1 C_4 L_4 L_5 R_5 + C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_4 R_5 + C_1 C_5 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s^2 (C_1 L_4 R_4 R_5 + C_1 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + s (C_1 L_4 R_4 R_5 + C_1 L_5 R_4 R_5 + C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_4 L_5 R_5 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_5 g_m) + 1}$$

9.253 X-INVALID-ORDER-253 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (C_4 L_4 L_5 R_4 g_m + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4 + L_4 L_5)}{2R_4 g_m + 2R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_4 L_5 + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 L_4 L_5 g_m + 2C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 + 2C_1 L_5}$$

9.254 X-INVALID-ORDER-254 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (-C_4 C_5 L_4 R_4 R_5 + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{2R_4 g_m + 2R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (C_1 C_4 C_5 L_4 R_4 R_5 + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s^2 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s^0 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_5)}$$

9.255 X-INVALID-ORDER-255 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5) + s^2 (2C_1 C_4 R_4 R_5 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

9.256 X-INVALID-ORDER-256 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 s^3 + C_4 L_4 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{C_1 C_4 C_5 L_4 R_4 s^4 + 2g_m + s^3 (2C_1 C_4 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_4 C_5 R_4 + 2C_4 L_4 g_m) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.257 \quad X-INVALID-ORDER-257} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{C_1 C_4 C_5 L_4 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.258 \quad X-INVALID-ORDER-258} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_4 g_m s^2 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 + 2C_4 L_4 g_m) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.259 \quad X-INVALID-ORDER-259} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 - C_4 C_5 L_4 R_4 s^3 - C_5 R_4 s + R_4 g_m + s^2 (C_4 L_4 R_4 g_m + C_5 L_5 R_4 g_m)}{2C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 + 2C_1 C_5 L_5 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 + 2C_4 L_4 g_m) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.260 \quad X-INVALID-ORDER-260} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 s^4 + C_4 L_4 L_5 R_4 g_m s^3 + L_5 R_4 g_m s - R_4 + s^2 (-C_4 L_4 R_4 - C_5 L_5 R_4)}{C_1 C_4 C_5 L_4 L_5 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_4 + 2C_4 L_4 L_5 g_m) + s^2 (2C_1 L_5 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 + 2C_4 L_5 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_4 R_4 + 2C_5 R_4)}$$

$$\mathbf{9.261 \quad X-INVALID-ORDER-261} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_4 L_4 R_4 g_m + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2C_1 C_4 C_5 L_4 L_5 s^5 + 2g_m + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_4 + 2C_1 C_5 L_5 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4 + 2C_4 L_4 g_m) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.262 \quad X-INVALID-ORDER-262} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_4 L_4 L_5 R_4 R_5 g_m - C_4 L_4 L_5 R_4) + s^2 (-C_4 L_4 R_4 R_5 - C_5 L_5 R_4 R_5) + s (L_5 R_4 R_5 g_m - L_5 R_4 R_5)}{C_1 C_4 C_5 L_4 L_5 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_5 + s^4 (C_1 C_4 L_4 L_5 R_4 + 2C_1 C_4 L_4 L_5 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_4 R_5 + 2C_1 C_4 L_5 R_4 R_5 + C_1 C_5 L_5 R_4 R_5 + 2C_4 C_5 L_5 R_4 R_5 + 2C_4 L_4 L_5 R_4 g_m + 2C_4 L_4 L_5 R_5 g_m + 2C_4 L_4 L_5) + s^2 (C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4 R_5 + 2C_4 L_4 R_5) + s (C_1 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.263 \quad X-INVALID-ORDER-263} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_4 L_5 R_4 g_m s^3 + L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_4 R_5 + 2C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_4 R_5) + s^2 (C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4 R_5 + 2C_4 L_4 R_5) + s (C_1 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.264 \quad X-INVALID-ORDER-264} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^4 (C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 C_5 L_5 R_4 R_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s^2 (C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4 R_5 + 2C_4 L_4 R_5) + s (C_1 R_4 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.265 \quad X-INVALID-ORDER-265} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s (C_1 R_1 R_4 + 2C_1 R_1 R_5)}$$

$$\mathbf{9.266 \quad X-INVALID-ORDER-266} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 - C_5 R_1 R_4 s + R_1 R_4 g_m}{2 C_1 C_5 L_5 R_1 s^3 + 2 R_1 g_m + s^2 (C_1 C_5 R_1 R_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_1 R_1 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.267 \quad X-INVALID-ORDER-267} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 s^2 + L_5 R_1 R_4 g_m s - R_1 R_4}{C_1 C_5 L_5 R_1 R_4 s^3 + 2 R_1 R_4 g_m + 2 R_1 + R_4 + s^2 (2 C_1 L_5 R_1 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4) + s (C_1 R_1 R_4 + 2 L_5 R_1 g_m + 2 L_5)}$$

$$\mathbf{9.268 \quad X-INVALID-ORDER-268} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 + R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 C_1 C_5 L_5 R_1 s^3 + 2 R_1 g_m + s^2 (C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_1 R_1 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.269 \quad X-INVALID-ORDER-269} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 R_5 s^2 - R_1 R_4 R_5 + s (L_5 R_1 R_4 R_5 g_m - L_5 R_1 R_4)}{C_1 C_5 L_5 R_1 R_4 R_5 s^3 + 2 R_1 R_4 R_5 g_m + 2 R_1 R_5 + R_4 R_5 + s^2 (C_1 L_5 R_1 R_4 + 2 C_1 L_5 R_1 R_5 + 2 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_5 R_1 R_5 + C_5 L_5 R_4 R_5) + s (C_1 R_1 R_4 R_5 + 2 L_5 R_1 R_4 g_m + 2 L_5 R_1 R_5 g_m + 2 L_5 R_1 + L_5 R_4 + 2 L_5 R_5)}$$

$$\mathbf{9.270 \quad X-INVALID-ORDER-270} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 R_4 g_m s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (C_1 C_5 L_5 R_1 R_4 + 2 C_1 C_5 L_5 R_1 R_5) + s^2 (2 C_1 L_5 R_1 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 R_5 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4 + 2 C_5 L_5 R_5) + s (C_1 R_1 R_4 + 2 C_1 R_1 R_5 + 2 L_5 R_1 g_m + 2 L_5)}$$

$$\mathbf{9.271 \quad X-INVALID-ORDER-271} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^3 (C_1 C_5 L_5 R_1 R_4 + 2 C_1 C_5 L_5 R_1 R_5) + s^2 (C_1 C_5 R_1 R_4 R_5 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 R_5 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4 + 2 C_5 L_5 R_5) + s (C_1 R_1 R_4 + 2 C_1 R_1 R_5 + 2 C_5 R_1 R_4 R_5 g_m + 2 C_5 R_1 R_5 + C_5 R_4 R_5)}$$

$$\mathbf{9.272 \quad X-INVALID-ORDER-272} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.273 \quad X-INVALID-ORDER-273} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{2 C_1 C_4 C_5 R_1 R_5 s^3 + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.274 \quad X-INVALID-ORDER-274} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{2 C_1 C_4 C_5 L_5 R_1 s^4 + s^3 (2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.275 \quad X-INVALID-ORDER-275} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 s^2 + L_5 R_1 g_m s - R_1}{2 R_1 g_m + s^3 (2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1) + s^2 (2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_4 R_1) + 1}$$

$$\mathbf{9.276 \quad X-INVALID-ORDER-276} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 + R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{2 C_1 C_4 C_5 L_5 R_1 s^4 + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.277 \quad X-INVALID-ORDER-277} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_5 s^2 - R_1 R_5 + s (L_5 R_1 R_5 g_m - L_5 R_1)}{2 R_1 R_5 g_m + R_5 + s^3 (2 C_1 C_4 L_5 R_1 R_5 + C_1 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5) + s^2 (C_1 L_5 R_1 + 2 C_4 L_5 R_1 R_5 g_m + 2 C_4 L_5 R_1 + 2 C_4 L_5 R_5 + 2 C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s (C_1 R_1 R_5 + 2 C_4 R_1 R_5 + 2 L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.278 \quad X-INVALID-ORDER-278} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2 C_1 C_4 C_5 L_5 R_1 R_5 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.279 \quad X-INVALID-ORDER-279} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2 C_1 C_4 C_5 L_5 R_1 R_5 s^4 + 2 R_1 g_m + s^3 (C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.280 \quad X-INVALID-ORDER-280} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2 R_1 g_m + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_4 R_5) + s (2 C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.281 \quad X-INVALID-ORDER-281} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 - C_5 R_1 R_4 s + R_1 R_4 g_m}{2 C_1 C_4 C_5 L_5 R_1 R_4 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_4 C_5 R_1 R_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.282 \quad X-INVALID-ORDER-282} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 s^2 + L_5 R_1 R_4 g_m s - R_1 R_4}{2 R_1 R_4 g_m + 2 R_1 + R_4 + s^3 (2 C_1 C_4 L_5 R_1 R_4 + C_1 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_5 R_1 R_4) + s^2 (2 C_1 L_5 R_1 + 2 C_4 L_5 R_1 R_4 g_m + 2 C_4 L_5 R_4 + 2 C_5 L_5 R_1 R_4 g_m + 2 C_5 L_5 R_1 + C_5 L_5 R_4) + s (C_1 R_1 R_4 + 2 C_4 R_1 R_4 + 2 L_5 R_1 g_m + 2 L_5)}$$

$$\mathbf{9.283 \quad X-INVALID-ORDER-283} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 R_4 g_m s^2 + R_1 R_4 g_m + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 C_1 C_4 C_5 L_5 R_1 R_4 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_4 R_5 + 2 C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_4 + 2 C_4 C_5 R_4 R_5 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + C_5 R_4 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.284 \quad X-INVALID-ORDER-284} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_4 R_5 s^2 - R_1 R_4 R_5 + s (L_5 R_1 R_4 R_5 g_m - L_5 R_1 R_4)}{2 R_1 R_4 R_5 g_m + 2 R_1 R_5 + R_4 R_5 + s^3 (2 C_1 C_4 L_5 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 + 2 C_4 C_5 L_5 R_1 R_4 R_5) + s^2 (C_1 L_5 R_1 R_4 + 2 C_1 L_5 R_1 R_5 + 2 C_4 L_5 R_1 R_4 R_5 g_m + 2 C_4 L_5 R_1 R_4 + 2 C_4 L_5 R_4 R_5 + 2 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_5 R_1 R_5 + C_5 L_5 R_4 R_5) + s (C_1 R_1 R_4 R_5 + 2 C_4 R_1 R_4 R_5 + 2 C_4 R_1 R_5 + 2 C_5 R_1 R_4 R_5 g_m + 2 C_5 R_1 R_5 + C_5 R_4 R_5) + 1}$$

$$\mathbf{9.285 \quad X-INVALID-ORDER-285} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 R_4 g_m s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2C_1 C_4 C_5 L_5 R_1 R_4 R_5 s^4 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (2C_1 C_4 L_5 R_1 R_4 + C_1 C_5 L_5 R_1 R_4 + 2C_1 C_5 L_5 R_1 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_5 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 + 2C_1 L_5 R_1 + 2C_4 L_5 R_1 R_4 g_m + 2C_4 L_5 R_4 + 2C_5 L_5 R_1 R_4 g_m)}$$

$$\mathbf{9.286 \quad X-INVALID-ORDER-286} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_5 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_4)}{2C_1 C_4 C_5 L_5 R_1 R_4 R_5 s^4 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (C_1 C_5 L_5 R_1 R_4 + 2C_1 C_5 L_5 R_1 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_5 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5 + 2C_5 L_5 R_1 R_4 g_m + 2C_5 L_5 R_1 R_5 g_m + 2C_5 L_5 R_1 R_4 R_5)}$$

$$\mathbf{9.287 \quad X-INVALID-ORDER-287} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 s^2 + R_1 g_m + s (C_4 R_1 R_4 g_m - C_5 R_1)}{C_1 C_4 C_5 R_1 R_4 s^3 + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.288 \quad X-INVALID-ORDER-288} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2R_1 g_m + s^2 (C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s (C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.289 \quad X-INVALID-ORDER-289} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 R_5) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.290 \quad X-INVALID-ORDER-290} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (-C_4 C_5 R_1 R_4 + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_5 R_1)}{2C_1 C_4 C_5 L_5 R_1 s^4 + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.291 \quad X-INVALID-ORDER-291} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_1 R_4 s^3 - R_1 + s^2 (C_4 L_5 R_1 R_4 g_m - C_5 L_5 R_1) + s (-C_4 R_1 R_4 + L_5 R_1 g_m)}{C_1 C_4 C_5 L_5 R_1 R_4 s^4 + 2R_1 g_m + s^3 (2C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2C_4 C_5 L_5 R_1 R_4 g_m + 2C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4) + s^2 (C_1 C_4 R_1 R_4 + 2C_4 L_5 R_1 g_m + 2C_4 L_5 + 2C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 + C_4 R_4) + 1}$$

$$\mathbf{9.292 \quad X-INVALID-ORDER-292} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_5 R_1 R_4 g_m s^3 + R_1 g_m + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4 + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{2C_1 C_4 C_5 L_5 R_1 s^4 + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5 + 2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 C_5 R_5) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.293 \quad X-INVALID-ORDER-293} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_5 R_1 R_4 R_5 s^3 - R_1 R_5 + s^2 (C_4 L_5 R_1 R_4 R_5 g_m - C_4 L_5 R_1 R_4 - C_5 L_5 R_1 R_5) + s (-C_4 R_1 R_4 R_5 + L_5 R_1 R_5 g_m - L_5 R_1)}{C_1 C_4 C_5 L_5 R_1 R_4 R_5 s^4 + 2R_1 R_5 g_m + R_5 + s^3 (C_1 C_4 L_5 R_1 R_4 + 2C_1 C_4 L_5 R_1 R_5 + C_1 C_5 L_5 R_1 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_5 R_1 R_5 + C_4 C_5 L_5 R_4 R_5) + s^2 (C_1 C_4 R_1 R_4 R_5 + C_1 L_5 R_1 + 2C_4 L_5 R_1 R_4 g_m + 2C_4 L_5 R_1 R_5 g_m + 2C_4 L_5 R_1 + C_4 L_5 R_4 + 2C_4 L_5 R_5 + 2C_5 L_5 R_1 R_4 R_5)}$$

$$\mathbf{9.294 \quad X-INVALID-ORDER-294} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_4) + s^2 (C_4 L_5 R_1 R_4 g_m + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 + L_5 R_1 g_m)}{2 R_1 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_5 R_1 R_5) + s^3 (2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + C_1 R_4 + C_1 R_5) + 1}$$

$$\mathbf{9.295 \quad X-INVALID-ORDER-295} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_4) + s^2 (-C_4 C_5 R_1 R_4 R_5 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_4 R_1 R_5 + L_5 R_1 g_m)}{2 R_1 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_5 R_1 R_5) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_5 + C_4 C_5 L_5 R_1 g_m) + s (C_1 R_1 + C_1 R_4 + C_1 R_5) + 1}$$

$$\mathbf{9.296 \quad X-INVALID-ORDER-296} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1)}{C_1 C_4 L_4 R_1 s^3 + 2 R_1 g_m + s^2 (2 C_1 C_4 R_1 R_5 + 2 C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.297 \quad X-INVALID-ORDER-297} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 s^3 + C_4 L_4 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{C_1 C_4 C_5 L_4 R_1 s^4 + s^3 (2 C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.298 \quad X-INVALID-ORDER-298} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_5 s^3 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1)}{C_1 C_4 C_5 L_4 R_1 R_5 s^4 + 2 R_1 g_m + s^3 (C_1 C_4 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_5 g_m + C_4 C_5 L_4 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_5 + 2 C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.299 \quad X-INVALID-ORDER-299} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 L_4 R_1 g_m s^2 + R_1 g_m + s^3 (C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_4 C_5 L_4 R_1 s^4 + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.300 \quad X-INVALID-ORDER-300} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 g_m s^4 - C_4 C_5 L_4 R_1 s^3 - C_5 R_1 s + R_1 g_m + s^2 (C_4 L_4 R_1 g_m + C_5 L_5 R_1 g_m)}{s^4 (C_1 C_4 C_5 L_4 R_1 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (2 C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.301 \quad X-INVALID-ORDER-301} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 s^4 + C_4 L_4 L_5 R_1 g_m s^3 + L_5 R_1 g_m s - R_1 + s^2 (-C_4 L_4 R_1 - C_5 L_5 R_1)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 + 2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1) + s^2 (2 C_4 L_4 R_1 g_m + C_4 L_4 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_4 R_1) + 1}$$

$$\mathbf{9.302 \quad X-INVALID-ORDER-302} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1) + s^2 (C_4 L_4 R_1 g_m + C_5 L_5 R_1 g_m) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^4 (C_1 C_4 C_5 L_4 R_1 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.303 \quad X-INVALID-ORDER-303} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (C_4 L_4 L_5 R_1 R_5 g_m - C_4 L_4 L_5 R_1) + s^2 (-C_4 L_4 R_1 R_5 - C_5 L_5 R_1 R_5) + s (L_5 R_1 R_5 g_m - L_5 R_1)}{C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 + 2 R_1 R_5 g_m + R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 + 2 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_1 R_5 + 2 C_1 C_4 L_5 R_1 R_5 + C_1 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 + 2 C_4 L_4 L_5 R_1 g_m + C_4 L_4 L_5) + s^2 (C_1 L_5 R_1 + 2 C_4 L_4 R_1 R_5 g_m + C_4 L_4 R_5 + 2 C_4 L_5 R_1 R_5 g_m - C_4 L_4 R_1) + s (2 C_1 R_1 R_5 + 2 C_4 R_1 g_m + L_4) + 2}$$

$$\mathbf{9.304 \quad X-INVALID-ORDER-304} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{C_4 L_4 L_5 R_1 g_m s^3 + L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^4 (C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_4 L_5 R_1) + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 + 2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + 2 C_4 L_4 R_1 g_m + C_4 L_4 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1) + s (2 C_1 R_1 R_5 + 2 C_4 R_1 g_m + L_4) + 2}$$

$$\mathbf{9.305 \quad X-INVALID-ORDER-305} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_5 s^3 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^4 (C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_4 L_5 R_1) + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 + 2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_4 R_1 R_5 g_m + C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + 2 C_4 L_4 R_1 g_m + C_4 L_4 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1) + s (2 C_1 R_1 R_5 + 2 C_4 R_1 g_m + L_4) + 2}$$

$$\mathbf{9.306 \quad X-INVALID-ORDER-306} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$$

$$H(s) = \frac{s (L_4 R_1 R_5 g_m - L_4 R_1)}{2 C_1 C_4 L_4 R_1 R_5 s^3 + 2 R_1 R_5 g_m + 2 R_1 + 2 R_5 + s^2 (C_1 L_4 R_1 + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + 2 C_4 L_4 R_5) + s (2 C_1 R_1 R_5 + 2 L_4 R_1 g_m + L_4) + 2}$$

$$\mathbf{9.307 \quad X-INVALID-ORDER-307} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 s^2 + L_4 R_1 g_m s}{2 R_1 g_m + s^3 (2 C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2 C_4 C_5 L_4 R_1) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4) + s (2 C_1 R_1 + 2 C_5 R_1) + 2}$$

$$\mathbf{9.308 \quad X-INVALID-ORDER-308} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_5 s^2 + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2 R_1 R_5 g_m + 2 R_1 + 2 R_5 + s^3 (2 C_1 C_4 L_4 R_1 R_5 + C_1 C_5 L_4 R_1 R_5 + 2 C_4 C_5 L_4 R_1 R_5) + s^2 (C_1 L_4 R_1 + 2 C_4 L_4 R_1 R_5 g_m + 2 C_4 L_4 R_1 + 2 C_4 L_4 R_5 + 2 C_5 L_4 R_1 R_5 g_m + C_5 L_4 R_5) + s (2 C_1 R_1 R_5 + 2 C_5 R_1 R_5 + 2 L_4 R_1 g_m + L_4) + 2}$$

$$\mathbf{9.309 \quad X-INVALID-ORDER-309} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 R_1 g_m s + s^2 (C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1)}{2 C_1 C_4 C_5 L_4 R_1 R_5 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_1 C_5 R_1 R_5 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4) + s (2 C_1 R_1 + 2 C_5 R_1 R_5 g_m + 2 C_5 R_1 + 2 C_5 R_5) + 2}$$

$$\mathbf{9.310 \quad X-INVALID-ORDER-310} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 g_m s^3 - C_5 L_4 R_1 s^2 + L_4 R_1 g_m s}{2 C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2 C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_4 R_1) + s^2 (2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4 + 2 C_5 L_5 R_1 g_m + 2 C_5 L_5) + s (2 C_1 R_1 + 2 C_5 R_1) + 2}$$

$$\mathbf{9.311 \quad X-INVALID-ORDER-311} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 s^3 + L_4 L_5 R_1 g_m s^2 - L_4 R_1 s}{2 R_1 + s^4 (2 C_1 C_4 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_1 + 2 C_4 C_5 L_4 L_5 R_1) + s^3 (2 C_4 L_4 L_5 R_1 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 + 2 C_1 L_5 R_1 + 2 C_4 L_4 R_1 + 2 C_5 L_5 R_1) + s (2 L_4 R_1 g_m + L_4 + 2 L_5 R_1 g_m + 2 L_5) + 2}$$

$$\mathbf{9.312 \quad X-INVALID-ORDER-312} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 g_m s^3 + L_4 R_1 g_m s + s^2 (C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1)}{2C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2R_1 g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2C_1 C_5 L_5 R_1 + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_5 R_1 R_5 + 2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4 + 2C_5 L_5 R_1 R_5) + s (L_4 R_1 R_5 g_m - L_4 R_1) + R_5}$$

$$\mathbf{9.313 \quad X-INVALID-ORDER-313} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_5 s^3 - L_4 R_1 R_5 s + s^2 (L_4 L_5 R_1 R_5 g_m - L_4 L_5 R_1)}{2R_1 R_5 + s^4 (2C_1 C_4 L_4 L_5 R_1 R_5 + C_1 C_5 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_5) + s^3 (C_1 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_1 R_5 g_m + 2C_4 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_5 + 2C_5 L_4 L_5 R_1 R_5 g_m + C_5 L_4 L_5 R_5) + s^2 (C_1 L_4 R_1 R_5 + 2C_1 L_5 R_1 R_5 + 2C_4 L_4 R_1 R_5 + 2C_5 L_5 R_1 R_5 + 2L_4 L_5 R_1 g_m + L_4 L_5) + s (L_4 R_1 R_5 g_m - L_4 R_1) + R_5}$$

$$\mathbf{9.314 \quad X-INVALID-ORDER-314} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_4 L_5 R_1 g_m s^2 + s^3 (C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_1 C_4 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 + 2C_1 C_5 L_5 R_1 R_5 + 2C_4 L_4 L_5 R_1 g_m + 2C_4 L_4 L_5 + 2C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 + 2C_4 L_4 R_1 R_5 g_m + C_4 L_4 R_1 R_5) + s (L_4 R_1 R_5 g_m - L_4 R_1) + R_5}$$

$$\mathbf{9.315 \quad X-INVALID-ORDER-315} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_5 s^2 + s^3 (C_5 L_4 L_5 R_1 R_5 g_m - C_5 L_4 L_5 R_1) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 + C_1 C_5 L_4 R_1 R_5 + 2C_1 C_5 L_5 R_1 R_5 + 2C_4 C_5 L_4 R_1 R_5 + 2C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 + 2C_4 L_4 R_1 R_5 g_m + C_4 L_4 R_1 R_5) + s (L_4 R_1 R_5 g_m - L_4 R_1) + R_5}$$

$$\mathbf{9.316 \quad X-INVALID-ORDER-316} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{C_1 C_4 L_4 R_1 s^3 + 2R_1 g_m + s^2 (C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + 2C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5) + 1}$$

$$\mathbf{9.317 \quad X-INVALID-ORDER-317} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 s^3 + R_1 g_m + s^2 (-C_4 C_5 R_1 R_4 + C_4 L_4 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_5 R_1)}{C_1 C_4 C_5 L_4 R_1 s^4 + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5) + R_5}$$

$$\mathbf{9.318 \quad X-INVALID-ORDER-318} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_5 s^3 + R_1 R_5 g_m - R_1 + s^2 (-C_4 C_5 R_1 R_4 R_5 + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{C_1 C_4 C_5 L_4 R_1 R_5 s^4 + 2R_1 g_m + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 L_4 R_1 + 2C_4 C_5 L_4 R_1 R_5 g_m + C_4 C_5 L_4 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5 + 2C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + C_4 R_1 R_4 + C_4 R_1 R_5) + R_5}$$

$$\mathbf{9.319 \quad X-INVALID-ORDER-319} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^3 (C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1) + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4 + C_4 L_4 R_1 g_m) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_4 C_5 L_4 R_1 s^4 + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 C_5 R_5) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5) + R_5}$$

$$\mathbf{9.320 \quad X-INVALID-ORDER-320} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (-C_4 C_5 L_4 R_1 + C_4 C_5 L_5 R_1 R_4 g_m) + s^2 (-C_4 C_5 R_1 R_4 + C_4 L_4 R_1 g_m + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_5 R_1)}{s^4 (C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_5 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4 + 2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5) + R_5}$$

$$\mathbf{9.321 \quad X-INVALID-ORDER-321} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 s^4 - R_1 + s^3 (-C_4 C_5 L_5 R_1 R_4 + C_4 L_4 L_5 R_1 g_m) + s^2 (-C_4 L_4 R_1 + C_4 L_5 R_1 R_4 g_m - C_5 L_5 R_1) + s (-C_4 R_1 R_4 + L_5 R_1 g_m)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 + 2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4) + s^2 (C_1 C_4 R_1 R_4 + 2 C_4 L_4 R_1 g_m + C_4 L_4 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m)}$$

$$\mathbf{9.322 \quad X-INVALID-ORDER-322} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1 + C_4 C_5 L_5 R_1 R_4 g_m) + s^2 (C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4 + C_4 L_4 R_1 g_m + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_5 R_1)}{s^4 (C_1 C_4 C_5 L_4 R_1 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2 C_1 C_4 C_5 R_1 R_5 + 2 C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + C_4 C_5 R_4 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m)}$$

$$\mathbf{9.323 \quad X-INVALID-ORDER-323} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (-C_4 C_5 L_5 R_1 R_4 R_5 + C_4 L_4 L_5 R_1 R_5 g_m - C_4 L_4 L_5 R_1) + s^2 (-C_4 L_4 R_1 R_5 + C_4 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_5) + s (C_4 R_1 R_4 R_5 g_m - C_5 R_1 R_5)}{C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 + 2 R_1 R_5 g_m + R_5 + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 R_5 + C_1 C_4 L_4 L_5 R_1 + 2 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_1 R_5 + C_1 C_4 L_5 R_1 R_4 + 2 C_1 C_4 L_5 R_1 R_5 + C_1 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_5 + C_4 C_5 L_5 R_4 R_5 + 2 C_4 L_4 L_5 R_1 R_5 g_m)}$$

$$\mathbf{9.324 \quad X-INVALID-ORDER-324} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_4 L_5 R_1) + s^3 (C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_4 + C_4 L_4 L_5 R_1 g_m) + s^2 (C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1 + C_4 L_5 R_1 R_4 g_m + C_5 L_5 R_1 R_5) + s (C_4 R_1 R_4 R_5 g_m - C_5 R_1 R_5)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 + 2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4 R_1 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 R_1 + C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5)}$$

$$\mathbf{9.325 \quad X-INVALID-ORDER-325} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_4 L_5 R_1) + s^3 (-C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_5 R_1 R_5) + s^2 (-C_4 L_4 R_1 R_5 + C_4 L_5 R_1 R_4 R_5 g_m - C_5 L_5 R_1 R_5) + s (C_4 R_1 R_4 R_5 g_m - C_5 R_1 R_5)}{C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 L_4 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_4 R_1 R_5 g_m + C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5)}$$

$$\mathbf{9.326 \quad X-INVALID-ORDER-326} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$$

$$H(s) = \frac{s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2 C_1 C_4 L_4 R_1 R_4 R_5 s^3 + 2 R_1 R_4 R_5 g_m + 2 R_1 R_4 + 2 R_4 R_5 + s^2 (C_1 L_4 R_1 R_4 + 2 C_1 L_4 R_1 R_5 + 2 C_4 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_4 R_1 R_4 + 2 C_4 L_4 R_4 R_5) + s (2 C_1 R_1 R_4 R_5 + 2 L_4 R_1 R_4 g_m + 2 L_4 R_1 R_5 g_m + 2 L_4 R_1 + L_4 R_4 + 2 L_4 R_5)}$$

$$\mathbf{9.327 \quad X-INVALID-ORDER-327} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_4 s^2 + L_4 R_1 R_4 g_m s}{2 R_1 R_4 g_m + 2 R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_4 + C_1 C_5 L_4 R_1 R_4 + 2 C_4 C_5 L_4 R_1 R_4) + s^2 (2 C_1 L_4 R_1 + 2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4 + 2 C_5 L_4 R_1 R_4 g_m + 2 C_5 L_4 R_1 + C_5 L_4 R_4) + s (2 C_1 R_1 R_4 + 2 C_5 R_1 R_4 + 2 L_4 R_1 g_m + 2 L_4)}$$

$$\mathbf{9.328 \quad X-INVALID-ORDER-328} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 R_1 R_4 R_5 s^2 + s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2 R_1 R_4 R_5 g_m + 2 R_1 R_4 + 2 R_4 R_5 + s^3 (2 C_1 C_4 L_4 R_1 R_4 R_5 + C_1 C_5 L_4 R_1 R_4 R_5 + 2 C_4 C_5 L_4 R_1 R_4 R_5) + s^2 (C_1 L_4 R_1 R_4 + 2 C_1 L_4 R_1 R_5 + 2 C_4 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_4 R_1 R_4 + 2 C_4 L_4 R_4 R_5 + 2 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_5 L_4 R_1 R_5 + C_5 L_4 R_4 R_5) + s (2 C_1 R_1 R_4 R_5 + 2 C_5 R_1 R_4 R_5 + 2 L_4 R_1 R_4 g_m + 2 L_4 R_1 R_5 g_m + 2 L_4 R_1 + L_4 R_4 + 2 L_4 R_5)}$$

$$\mathbf{9.329 \quad X-INVALID-ORDER-329} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_4 R_1 R_4 g_m s + s^2 (C_5 L_4 R_1 R_4 R_5 g_m - C_5 L_4 R_1 R_4)}{2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + 2 R_1 R_4 g_m + 2 R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_4 + C_1 C_5 L_4 R_1 R_4 + 2 C_1 C_5 L_4 R_1 R_5 + 2 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_1 R_4 + 2 C_4 C_5 L_4 R_4 R_5) + s^2 (2 C_1 C_5 R_1 R_4 R_5 + 2 C_1 L_4 R_1 + 2 C_4 L_4 R_1 R_4 g_m + 2 C_4 L_4 R_4 + 2 C_5 L_4 R_1 R_4 g_m + 2 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 R_4 R_5 g_m)}$$

$$\mathbf{9.330 \quad X-INVALID-ORDER-330} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 R_4 g_m s^3 - C_5 L_4 R_1 R_4 s^2 + L_4 R_1 R_4 g_m s}{2C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 + 2R_1 R_4 g_m + 2R_4 + s^4 (2C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_4 + C_1 C_5 L_4 R_1 R_4 + 2C_1 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_4 R_1 R_4 + 2C_5 L_4 L_5 R_1 g_m + 2C_5 L_4 L_5) + s^2 (2C_1 L_4 R_1 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_4 + 2C_5 L_4 L_5 R_1 R_4 g_m + 2C_5 L_4 L_5 R_4) + s (L_4 R_1 R_4 g_m + L_4 R_1 R_4) + R_4}$$

$$\mathbf{9.331 \quad X-INVALID-ORDER-331} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_4 s^3 + L_4 L_5 R_1 R_4 g_m s^2 - L_4 R_1 R_4 s}{2R_1 R_4 + s^4 (2C_1 C_4 L_4 L_5 R_1 R_4 + C_1 C_5 L_4 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_1 R_4) + s^3 (2C_1 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_1 R_4 g_m + 2C_4 L_4 L_5 R_4 + 2C_5 L_4 L_5 R_1 R_4 g_m + 2C_5 L_4 L_5 R_1 + C_5 L_4 L_5 R_4) + s^2 (C_1 L_4 R_1 R_4 + 2C_1 L_5 R_1 R_4 + 2C_4 L_4 R_1 R_4 + 2C_5 L_5 R_1 R_4 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (L_4 R_1 R_4 g_m + L_4 R_1 R_4) + R_4}$$

$$\mathbf{9.332 \quad X-INVALID-ORDER-332} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_4 L_5 R_1 R_4 g_m s^3 + L_4 R_1 R_4 g_m s}{2C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 + 2R_1 R_4 g_m + 2R_4 + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 R_5 + 2C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_4 + C_1 C_5 L_4 R_1 R_4 + 2C_1 C_5 L_4 R_1 R_5 + 2C_1 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 R_1 R_4 + 2C_4 C_5 L_4 R_4 R_5) + s^2 (C_1 L_4 R_1 R_4 + 2C_1 L_5 R_1 R_4 + 2C_4 L_4 R_1 R_4 + 2C_5 L_5 R_1 R_4 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (L_4 R_1 R_4 g_m + L_4 R_1 R_4) + R_4}$$

$$\mathbf{9.333 \quad X-INVALID-ORDER-333} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_4 L_5 R_1 R_4 R_5 s^3 - L_4 R_1 R_4 R_5 s + s^2 (L_4 L_5 R_1 R_4 R_5 g_m - L_4 R_1 R_4 R_5)}{2R_1 R_4 R_5 + s^4 (2C_1 C_4 L_4 L_5 R_1 R_4 R_5 + C_1 C_5 L_4 L_5 R_1 R_4 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5) + s^3 (C_1 L_4 L_5 R_1 R_4 + 2C_1 L_4 L_5 R_1 R_5 + 2C_4 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 L_4 L_5 R_1 R_4 + 2C_4 L_4 L_5 R_4 R_5 + 2C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_5 L_4 L_5 R_1 R_5 + C_5 L_4 L_5 R_4 R_5) + s^2 (C_1 L_4 R_1 R_4 R_5 + 2C_1 L_5 R_1 R_4 R_5 + 2C_4 L_4 R_1 R_4 R_5 + 2C_5 L_5 R_1 R_4 R_5 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (L_4 R_1 R_4 R_5 g_m + L_4 R_1 R_4 R_5) + R_4}$$

$$\mathbf{9.334 \quad X-INVALID-ORDER-334} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{2C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (2C_1 C_4 L_4 L_5 R_1 R_4 + C_1 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_5 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_4 R_5 + 2C_1 C_5 L_5 R_1 R_4 R_5 + 2C_1 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 L_4 L_5 R_1 R_4 + 2C_4 L_4 L_5 R_4 R_5) + s^2 (C_1 L_4 R_1 R_4 R_5 + 2C_1 L_5 R_1 R_4 R_5 + 2C_4 L_4 R_1 R_4 R_5 + 2C_5 L_5 R_1 R_4 R_5 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (L_4 R_1 R_4 R_5 g_m + L_4 R_1 R_4 R_5) + R_4}$$

$$\mathbf{9.335 \quad X-INVALID-ORDER-335} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_5 (C_5 L_5 s^2 + 1)}{2C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (C_1 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_5 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_4 R_5 + C_1 C_5 L_4 R_1 R_4 R_5 + 2C_1 C_5 L_5 R_1 R_4 R_5 + 2C_4 C_5 L_4 R_1 R_4 R_5 + 2C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_5 L_4 L_5 R_1 R_5 + C_5 L_4 L_5 R_4 R_5) + s^2 (C_1 L_4 R_1 R_4 R_5 + 2C_1 L_5 R_1 R_4 R_5 + 2C_4 L_4 R_1 R_4 R_5 + 2C_5 L_5 R_1 R_4 R_5 + 2L_4 L_5 R_1 g_m + 2L_4 L_5) + s (L_4 R_1 R_4 R_5 g_m + L_4 R_1 R_4 R_5) + R_4}$$

$$\mathbf{9.336 \quad X-INVALID-ORDER-336} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5) + s^2 (C_1 L_4 R_1 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + C_4 L_4 R_4 + 2C_4 L_4 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2L_4 R_1 g_m + L_4) + R_4}$$

$$\mathbf{9.337 \quad X-INVALID-ORDER-337} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 s^3 + R_1 R_4 g_m + s^2 (C_4 L_4 R_1 R_4 g_m - C_5 L_4 R_1) + s (-C_5 R_1 R_4 + L_4 R_1 g_m)}{C_1 C_4 C_5 L_4 R_1 R_4 s^4 + 2R_1 g_m + s^3 (2C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4) + s^2 (C_1 C_5 R_1 R_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4) + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2R_1}$$

$$\mathbf{9.338 \quad X-INVALID-ORDER-338} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 R_5 s^3 + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4 - C_5 L_4 R_1 R_5) + s (-C_5 R_1 R_4 R_5 + L_4 R_1 R_5 g_m - L_4 R_1 R_4)}{C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5 + C_1 C_5 L_4 R_1 R_5 + 2C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_4 R_4 R_5) + s^2 (C_1 C_5 R_1 R_4 R_5 + C_1 L_4 R_1 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + C_4 L_4 R_4 + 2C_4 L_4 R_5) + s (C_1 R_1 R_4 R_5 + L_4 R_1 R_5 g_m + L_4 R_1 R_4) + R_4}$$

9.339 X-INVALID-ORDER-339 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_4 R_1 R_4) + s^2 (C_4 L_4 R_1 R_4 g_m + C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4 + L_4 R_1 g_m)}{2 R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_4 R_1 R_5) + s^3 (2 C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_4 R_5) + s^2 (C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4 + 2 C_5 L_4 R_1 g_m + C_5 L_4) + s (2 C_1 R_1}$$

9.340 X-INVALID-ORDER-340 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (-C_4 C_5 L_4 R_1 R_4 + C_5 L_4 L_5 R_1 g_m) + s^2 (C_4 L_4 R_1 R_4 g_m - C_5 L_4 R_1 + C_5 L_5 R_1 R_4 g_m) + s (-C_5 R_1 R_4 + L_4 R_1 g_m)}{2C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2C_1 C_5 L_5 R_1 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4) + s^2 (C_1 C_5 R_1 R_4 + 2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4 + 2C_5 L_5 R_1)}$$

9.341 X-INVALID-ORDER-341 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 s^4 - R_1 R_4 + s^3 (C_4 L_4 L_5 R_1 R_4 g_m - C_5 L_4 L_5 R_1) + s^2 (-C_4 L_4 R_1 R_4 - C_5 L_5 R_1 R_4 + L_4 L_5 R_1 g_m) + s (-L_4 R_1 + L_5 R_1)}{C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 + 2 R_1 R_4 g_m + 2 R_1 + R_4 + s^4 (2 C_1 C_4 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_1 + 2 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4) + s^3 (C_1 C_4 L_4 R_1 R_4 + C_1 C_5 L_5 R_1 R_4 + 2 C_4 L_4 L_5 R_1 g_m + 2 C_4 L_4 L_5 + 2 C_5 L_4 L_5 R_1 g_m + C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 + 2 C_1 L_5 R_1)}$$

9.342 X-INVALID-ORDER-342 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_4 R_1 R_4 + C_5 L_4 L_5 R_1 g_m) + s^2 (C_4 L_4 R_1 R_4 g_m + C_5 L_4 R_1 R_5 g_m - C_5 L_4 R_1 + C_5 L_5 R_1)}{2C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 + C_1 C_5 L_4 R_1 + 2C_1 C_5 L_5 R_1 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5) + s^2 (C_1 C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5) + s (C_1 C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5 + C_4 C_5 L_4 R_1 R_4 g_m + C_4 C_5 L_4 R_1 R_5 g_m + C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + C_4 C_5 L_4 R_5) + C_4 C_5 L_4 L_5 R_1 R_4 g_m + R_1 R_4 g_m}.$$

9.343 X-INVALID-ORDER-343 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^4 - R_1 R_4 R_5 + s^3 (C_4 L_4 L_5 R_1 R_4 R_5 g_m + C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 + 2C_1 C_4 L_4 L_5 R_1 R_5 + C_1 C_5 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_4 R_5)) + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 + C_1 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_1 R_4 g_m)}{C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 + 2C_1 C_4 L_4 L_5 R_1 R_5 + C_1 C_5 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_4 R_5)) + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 + C_1 L_4 L_5 R_1 + 2C_4 L_4 L_5 R_1 R_4 g_m)}$$

9.344 X-INVALID-ORDER-344 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_1 R_4) + s^3 (C_4 L_4 L_5 R_1 R_4 g_m}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5) + s^4 (2C_1 C_4 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1$$

9.345 X-INVALID-ORDER-345 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_4 C_5 L_4 L_5 R_1 \cdot}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5) + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 + C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5) + s^2 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5) + s (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5) + C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5}.$$

9.346 X-INVALID-ORDER-346 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + C_4 L_4 R_4 + 2C_4 L_4 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5)}$$

9.347 X-INVALID-ORDER-347 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 s^3 + C_4 L_4 R_1 R_4 g_m s^2 - C_5 R_1 R_4 s + R_1 R_4 g_m}{C_1 C_4 C_5 L_4 R_1 R_4 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4) + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_4 C_5 R_1 R_4 + 2 C_4 L_4 R_1 g_m + 2 C_4 L_4) + s (2 C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_4 + 2 C_5 R_1 R_4 g_m + 2 C_5 R_1 + C_5 R_4) + 2}$$

9.348 X-INVALID-ORDER-348 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 R_1 R_4 R_5 s^3 - C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_4 L_4 R_1 R_4 R_5 g_m - C_4 L_4 R_1 R_4)}{C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5 + 2C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_4 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 + C_1 C_5 R_1 R_4 R_5 + 2C_4 C_5 R_1 R_4 R_5 + 2C_4 L_4 R_1 R_4 g_m + 2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 R_4)}.$$

9.349 X-INVALID-ORDER-349 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 L_4 R_1 R_4 g_m s^2 + R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_4 R_1 R_4) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2 R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_4 R_1 R_5) + s^3 (2 C_1 C_4 C_5 R_1 R_4 R_5 + 2 C_1 C_4 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 + 2 C_1 C_5 R_1 R_5 + 2 C_4 C_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 R_1 R_4 +$$

9.350 X-INVALID-ORDER-350 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 - C_4 C_5 L_4 R_1 R_4 s^3 - C_5 R_1 R_4 s + R_1 R_4 g_m + s^2 (C_4 L_4 R_1 R_4 g_m + C_5 L_5 R_1 R_4 g_m)}{2 C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_4 R_1 + 2 C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_1 C_4 R_1 R_4 + C_1 C_5 R_1 R_4 +$$

$$\text{9.351} \quad \text{X-INVALID-ORDER-351} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 s^4 + C_4 L_4 L_5 R_1 R_4 g_m s^3 + L_5 R_1 R_4 g_m s - R_1 R_4 + s^2 (-C_4 L_4 R_1 R_4 - C_5 L_5 R_1 R_4)}{C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 + 2 R_1 R_4 g_m + 2 R_1 + R_4 + s^4 (2 C_1 C_4 L_4 L_5 R_1 + 2 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4) + s^3 (C_1 C_4 L_4 R_1 R_4 + 2 C_1 C_4 L_5 R_1 R_4 + C_1 C_5 L_5 R_1 R_4 + 2 C_4 C_5 L_5 R_1 R_4 + 2 C_4 L_4 L_5 R_1 g_m + 2 C_4 L_4 L_5) + s^2 (2 C_1 L_5 R_1 + 2 C_4 L_4 R_1 R_4 g_m}$$

9.352 X-INVALID-ORDER-352 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_4 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_4 C_5 L_4 R_1 R_4 R_5 g_m}{2C_1 C_4 C_5 L_4 L_5 R_1 s^5 + 2R_1 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_4 R_1 + 2C_1 C_5 L_5 R_1 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 + 2C_4 C_5$$

9.353 X-INVALID-ORDER-353 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^4 - R_1 R_4}{C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 + 2C_1 C_4 L_4 L_5 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 + 2C_1 C_4 L_5 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 + 2C_4 L_4 L_5 R_1 R_4 R_5)}$$

9.354 X-INVALID-ORDER-354 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 + 2C_1 C_4 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C$$

9.355 X-INVALID-ORDER-355 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_4C_5L_4L_5R_1R_5) + s^4(C_1C_4C_5L_4R_1R_4R_5 + 2C_1C_4C_5L_5R_1R_4R_5 + 2C_4C_5L_4L_5R_1R_4g_m + 2C_4C_5L_4L_5R_1R_5g_m + 2C_4C_5L_4L_5R_1 + C_4C_5L_4L_5R_4 + 2C_4C_5L_4L_5R_5) + s^3(C_1C_4L_4R_1R_4 + 2$$

9.356 X-INVALID-ORDER-356 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s(C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s(2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5) + 2}$$

$$\mathbf{9.357 \quad X-INVALID-ORDER-357} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (-C_1 C_5 R_1 R_4 + C_5 L_5 R_4 g_m) + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^3 (2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5) + s^2 (2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.358 \quad X-INVALID-ORDER-358} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_4 s^3 - R_4 + s^2 (C_1 L_5 R_1 R_4 g_m - C_5 L_5 R_4) + s (-C_1 R_1 R_4 + L_5 R_4 g_m)}{2R_4 g_m + s^3 (2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4) + s^2 (2C_1 L_5 R_1 g_m + 2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 + C_1 R_4 + 2L_5 g_m) + 2}$$

$$\mathbf{9.359 \quad X-INVALID-ORDER-359} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4 + C_5 L_5 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5) + s^2 (2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.360 \quad X-INVALID-ORDER-360} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_4 R_5 s^3 - R_4 R_5 + s^2 (C_1 L_5 R_1 R_4 R_5 g_m - C_1 L_5 R_1 R_4 - C_5 L_5 R_4 R_5) + s (-C_1 R_1 R_4 R_5 + L_5 R_4 R_5 g_m - L_5 R_4)}{2R_4 R_5 g_m + 2R_5 + s^3 (2C_1 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_5 L_5 R_1 R_5 + C_1 C_5 L_5 R_4 R_5) + s^2 (2C_1 L_5 R_1 R_4 g_m + 2C_1 L_5 R_1 R_5 g_m + 2C_1 L_5 R_1 + C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_5 L_5 R_4 R_5 g_m + 2C_5 L_5 R_5) + s (2C_1 R_1 R_4 R_5 g_m + 2C_1 R_1 R_5 + C_1 R_4 R_5 + 2L_5 R_4 g_m + 2L_5 R_5 g_m + 2L_5)}$$

$$\mathbf{9.361 \quad X-INVALID-ORDER-361} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4) + s^2 (C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 + L_5 R_4 g_m)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 R_5 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5) + s^2 (2C_1 L_5 R_1 g_m + 2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2L_5 g_m) + 2}$$

$$\mathbf{9.362 \quad X-INVALID-ORDER-362} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4) + s^2 (-C_1 C_5 R_1 R_4 R_5 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 R_5 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5) + s^2 (2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2L_5 g_m) + 2}$$

$$\mathbf{9.363 \quad X-INVALID-ORDER-363} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s (C_1 R_1 g_m - C_5)}{2C_1 C_4 C_5 R_1 s^3 + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.364 \quad X-INVALID-ORDER-364} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2C_1 C_4 C_5 R_1 R_5 s^3 + 2g_m + s^2 (2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2C_4 C_5 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.365 \quad X-INVALID-ORDER-365} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.366 \quad X-INVALID-ORDER-366} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 g_m s^3 + g_m + s^2 (-C_1 C_5 R_1 + C_5 L_5 g_m) + s (C_1 R_1 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.367 \quad X-INVALID-ORDER-367} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 s^3 + s^2 (C_1 L_5 R_1 g_m - C_5 L_5) + s (-C_1 R_1 + L_5 g_m) - 1}{2C_1 C_4 C_5 L_5 R_1 s^4 + 2g_m + s^3 (2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4)}$$

$$\mathbf{9.368 \quad X-INVALID-ORDER-368} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 g_m s^3 + g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.369 \quad X-INVALID-ORDER-369} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_5 s^3 - R_5 + s^2 (C_1 L_5 R_1 R_5 g_m - C_1 L_5 R_1 - C_5 L_5 R_5) + s (-C_1 R_1 R_5 + L_5 R_5 g_m - L_5)}{2C_1 C_4 C_5 L_5 R_1 R_5 s^4 + 2R_5 g_m + s^3 (2C_1 C_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_5 R_1 + 2C_1 C_4 L_5 R_5 + 2C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_5) + s^2 (2C_1 C_4 R_1 R_5 + 2C_1 L_5 R_1 g_m + C_1 L_5 + 2C_4 L_5 R_5 g_m + 2C_4 L_5 + 2C_5 L_5 R_5 g_m) + s (2C_1 R_1 R_5 g_m + C_1 R_5 + 2C_4 R_5 + 2L_5 g_m)}$$

$$\mathbf{9.370 \quad X-INVALID-ORDER-370} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1) + s^2 (C_1 L_5 R_1 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + L_5 g_m) - 1}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m)}$$

$$\mathbf{9.371 \quad X-INVALID-ORDER-371} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1) + s^2 (-C_1 C_5 R_1 R_5 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2C_4 C_5 R_5 + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m)}$$

$$\mathbf{9.372 \quad X-INVALID-ORDER-372} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2C_1 C_4 C_5 R_1 R_4 s^3 + 2g_m + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_4 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.373 \quad X-INVALID-ORDER-373} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2C_1 C_4 C_5 R_1 R_4 R_5 s^3 + 2R_4 g_m + 2R_5 g_m + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5 + 2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + 2C_4 C_5 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m + 2C_5 R_4 R_5)}$$

$$\mathbf{9.374 \quad X-INVALID-ORDER-374} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_5 g_m)}$$

$$\mathbf{9.375 \quad X-INVALID-ORDER-375} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (-C_1 C_5 R_1 R_4 + C_5 L_5 R_4 g_m) + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_4 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_4 R_4 g_m + 2C_5 R_4)}$$

$$\mathbf{9.376 \quad X-INVALID-ORDER-376} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_4 s^3 - R_4 + s^2 (C_1 L_5 R_1 R_4 g_m - C_5 L_5 R_4) + s (-C_1 R_1 R_4 + L_5 R_4 g_m)}{2C_1 C_4 C_5 L_5 R_1 R_4 s^4 + 2R_4 g_m + s^3 (2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_4 + 2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 C_4 R_1 R_4 + 2C_1 L_5 R_1 g_m + 2C_1 L_5 + 2C_4 L_5 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 + C_1 R_4 + 2C_4 R_4)}$$

$$\mathbf{9.377 \quad X-INVALID-ORDER-377} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 R_4 g_m s^3 + R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4 + C_5 L_5 R_4 g_m) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_4 R_4 R_5)} + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 + C_1 R_4 + 2C_4 R_4 R_5)}$$

$$\mathbf{9.378 \quad X-INVALID-ORDER-378} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_4 R_5 s^3 - R_4 R_5 + s^2 (C_1 L_5 R_1 R_4 R_5 g_m - C_1 L_5 R_1 R_4 - C_5 L_5 R_4 R_5) + s (-C_1 R_1 R_4 R_5 + L_5 R_4 R_5 g_m)}{2C_1 C_4 C_5 L_5 R_1 R_4 R_5 s^4 + 2R_4 R_5 g_m + 2R_5 + s^3 (2C_1 C_4 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 L_5 R_1 R_4 + 2C_1 C_4 L_5 R_4 R_5 + 2C_1 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_5 L_5 R_1 R_5 + C_1 C_5 L_5 R_4 R_5 + 2C_4 C_5 L_5 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 + 2C_1 L_5 R_1 R_4 g_m + 2C_1 L_5 R_1 R_5 g_m + 2C_1 L_5 R_1 + C_1 L_5 R_4 R_5)}$$

$$\mathbf{9.379 \quad X-INVALID-ORDER-379} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4) + s^2 (C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_4 + 2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 R_5 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_4 R_5)}$$

$$\mathbf{9.380 \quad X-INVALID-ORDER-380} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4) + s^2 (-C_1 C_5 R_1 R_4 R_5 + C_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_5 L_5 R_1 R_4 g_m + 2C_1 C_5 L_5 R_1 R_5 g_m + 2C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_4 R_5)}$$

$$\mathbf{9.381 \quad X-INVALID-ORDER-381} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 s^3 + g_m + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.382 \quad X-INVALID-ORDER-382} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 R_1 R_4 R_5 s^3 + R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 - C_1 C_5 R_1 R_5 - C_4 C_5 R_4 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2g_m + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 R_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4 + 2C_5 R_5)}$$

$$\mathbf{9.383 \quad X-INVALID-ORDER-383} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.384 \quad X-INVALID-ORDER-384} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_5 R_1 R_4 g_m s^4 + g_m + s^3 (-C_1 C_4 C_5 R_1 R_4 + C_1 C_5 L_5 R_1 g_m + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.385 \quad X-INVALID-ORDER-385} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_5 R_1 R_4 s^4 + s^3 (C_1 C_4 L_5 R_1 R_4 g_m - C_1 C_5 L_5 R_1 - C_4 C_5 L_5 R_4) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m + C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_1 R_1 - C_4 R_4 + L_5 g_m) - 1}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4) + s^3 (2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_4 g_m)}$$

$$\mathbf{9.386 \quad X-INVALID-ORDER-386} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_5 R_1 R_4 g_m s^4 + g_m + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4 + C_1 C_5 L_5 R_1 g_m + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.387 \quad X-INVALID-ORDER-387} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_5 R_1 R_4 R_5 s^4 - R_5 + s^3 (C_1 C_4 L_5 R_1 R_4 R_5 g_m - C_1 C_4 L_5 R_1 R_4 - C_1 C_5 L_5 R_1 R_5 - C_4 C_5 L_5 R_4 R_5) + s^2 (-C_1 C_4 R_1 R_4 R_5 + C_1 L_5 R_1 R_5 g_m - C_1 L_5 R_1 + C_4 L_5 R_4 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}{2R_5 g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 + C_1 C_4 C_5 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_5 R_1 + C_1 C_4 L_5 R_4 + 2C_1 C_4 L_5 R_5 + 2C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_4 R_5 g_m + 2C_4 C_5 L_5 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 R_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}$$

$$\mathbf{9.388 \quad X-INVALID-ORDER-388} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_5 R_1 R_4) + s^3 (C_1 C_4 L_5 R_1 R_4 g_m + C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m + C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}$$

$$\mathbf{9.389 \quad X-INVALID-ORDER-389} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_5 R_1 R_4) + s^3 (-C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m + C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}{2g_m + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 R_4 R_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m + C_4 L_5 R_4 g_m + C_5 L_5 R_5 g_m) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4 R_5)}$$

$$\mathbf{9.390 \quad X-INVALID-ORDER-390} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1) + s^2 (C_4 L_4 R_5 g_m - C_4 L_4) + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2g_m + s^3 (2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.391 \quad X-INVALID-ORDER-391} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 s^4 + g_m + s^3 (C_1 C_4 L_4 R_1 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_4 L_4 g_m) + s (C_1 R_1 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4) + s^3 (2C_1 C_4 C_5 R_1 + 2C_4 C_5 L_4 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.392 \quad X-INVALID-ORDER-392} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_5 s^4 + R_5 g_m + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1 - C_4 C_5 L_4 R_5) + s^2 (-C_1 C_5 R_1 R_5 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_4 C_5 L_4 R_5 g_m) + s^2 (2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2C_4 C_5 R_5 + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_5 g_m + 2C_4 + 2C_5 g_m)}$$

$$\mathbf{9.393 \quad X-INVALID-ORDER-393} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_4 R_1) + s^3 (C_1 C_4 L_4 R_1 g_m + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4) + s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.394 \quad X-INVALID-ORDER-394} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 g_m s^5 + g_m + s^4 (-C_1 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_5 L_5 R_1 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.395 \quad X-INVALID-ORDER-395} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 s^5 + s^4 (C_1 C_4 L_4 L_5 R_1 g_m - C_4 C_5 L_4 L_5) + s^3 (-C_1 C_4 L_4 R_1 - C_1 C_5 L_5 R_1 + C_4 L_4 L_5 g_m) + s^2 (C_1 L_5 R_1 g_m - C_4 L_4 - C_5 L_5) + s (-C_1 R_1 + L_5 g_m) - 1}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + 2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + 2C_5 g_m)}$$

$$\mathbf{9.396 \quad X-INVALID-ORDER-396} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 g_m s^5 + g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_4 R_1 + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_5 L_5 R_1 g_m + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.397 \quad X-INVALID-ORDER-397} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 - R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 R_5 g_m - C_1 C_4 L_4 L_5 R_1 - C_4 C_5 L_4 L_5 R_5) + s^3 (-C_1 C_4 L_4 R_1 R_5 - C_1 C_5 L_5 R_1 R_5 + C_4 L_4 L_5 R_5 g_m - C_4 L_4 L_5) + s^2 (C_1 L_5 R_1 g_m + C_4 L_4 R_5)}{2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 + 2C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + C_1 C_4 L_4 R_5 + 2C_1 C_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_5 R_1 + 2C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_5)}{2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 + 2C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + C_1 C_4 L_4 R_5 + 2C_1 C_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_5 R_1 + 2C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_5 + 2C_4 C_5 L_5 R_5)}$$

$$\mathbf{9.398 \quad X-INVALID-ORDER-398} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (C_1 C_4 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1 + C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 + C_4 L_4 L_5 g_m) + s^2 (C_1 L_5 R_1 g_m + C_4 L_4 R_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + C_1 C_4 L_4 + 2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + 2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m)}$$

$$\mathbf{9.399 \quad X-INVALID-ORDER-399} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (-C_1 C_4 C_5 L_4 R_1 R_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1 + C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 + C_4 L_4 L_5 g_m) + s^2 (C_1 L_5 R_1 g_m + C_4 L_4 R_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_4 R_1 + 2C_4 L_4 g_m + 2C_4 L_5 g_m + 2C_5 L_5 g_m)}$$

$$\mathbf{9.400 \quad X-INVALID-ORDER-400} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 R_1 R_5 g_m - C_1 L_4 R_1) + s (L_4 R_5 g_m - L_4)}{2R_5 g_m + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5) + s^2 (2C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (2C_1 R_1 R_5 g_m + 2C_1 R_1 + 2C_1 R_5 + 2L_4 g_m) + 2}$$

$$\mathbf{9.401 \quad X-INVALID-ORDER-401} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 R_1 s^3 + L_4 g_m s + s^2 (C_1 L_4 R_1 g_m - C_5 L_4)}{2C_1 C_4 C_5 L_4 R_1 s^4 + 2g_m + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2C_4 C_5 L_4) + s^2 (2C_1 C_5 R_1 + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5)}$$

$$\mathbf{9.402 \quad X-INVALID-ORDER-402} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 R_1 R_5 s^3 + s^2 (C_1 L_4 R_1 R_5 g_m - C_1 L_4 R_1 - C_5 L_4 R_5) + s (L_4 R_5 g_m - L_4)}{2C_1 C_4 C_5 L_4 R_1 R_5 s^4 + 2R_5 g_m + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_4 R_1 R_5 g_m + C_1 C_5 L_4 R_5 + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_5 R_1 R_5 + 2C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_5 g_m + 2C_4 L_4 + 2C_5 L_4 R_5 g_m) + s (2C_1 R_1 R_5 g_m + 2C_1 R_1 + 2C_1 R_5 + 2C_5 R_5)}$$

$$\mathbf{9.403 \quad X-INVALID-ORDER-403} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_4 g_m s + s^3 (C_1 C_5 L_4 R_1 R_5 g_m - C_1 C_5 L_4 R_1) + s^2 (C_1 L_4 R_1 g_m + C_5 L_4 R_5 g_m - C_5 L_4)}{2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + 2C_1 C_5 R_5 + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_5)}$$

$$\mathbf{9.404 \quad X-INVALID-ORDER-404} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_4 L_5 R_1 g_m s^4 + L_4 g_m s + s^3 (-C_1 C_5 L_4 R_1 + C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 R_1 g_m - C_5 L_4)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5 + 2C_4 C_5 L_4) + s^2 (2C_1 C_5 R_1 + 2C_4 L_4 g_m + 2C_5 L_4 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_5)}$$

$$\mathbf{9.405 \quad X-INVALID-ORDER-405} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 L_5 R_1 s^4 - L_4 s + s^3 (C_1 L_4 L_5 R_1 g_m - C_5 L_4 L_5) + s^2 (-C_1 L_4 R_1 + L_4 L_5 g_m)}{2C_1 C_4 C_5 L_4 L_5 R_1 s^5 + s^4 (2C_1 C_4 L_4 L_5 R_1 g_m + 2C_1 C_4 L_4 L_5 + 2C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 + 2C_1 C_5 L_5 R_1 + 2C_4 L_4 L_5 g_m + 2C_5 L_4 L_5 g_m) + s^2 (2C_1 L_4 R_1 g_m + C_1 L_4 + 2C_1 L_5 R_1 g_m + 2C_1 L_5 + 2C_4 L_4 + 2C_5 L_5) + s (2C_1 R_1 + 2L_4 g_m)}$$

$$\mathbf{9.406 \quad X-INVALID-ORDER-406} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_4 L_5 R_1 g_m s^4 + L_4 g_m s + s^3 (C_1 C_5 L_4 R_1 R_5 g_m - C_1 C_5 L_4 R_1 + C_5 L_4 L_5 g_m) + s^2 (C_1 L_4 R_1 g_m + C_5 L_4 R_5 g_m - C_5 L_4)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_4 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5 + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_5 R_1 + 2C_4 L_4 g_m + 2C_5 L_4 g_m + 2C_5 L_5 g_m) + s (2C_1 R_1 + 2L_4 g_m)}$$

$$\mathbf{9.407 \quad X-INVALID-ORDER-407} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 L_5 R_1 R_5 s^4 - L_4 R_5 s + s^3 (C_1 L_4 L_5 R_1 R_5 g_m - C_1 L_4 L_5 R_1 - C_5 L_4 L_5 R_5) + s^2 (-C_1 L_4 R_1 R_5 + L_4 L_5 R_5 g_m)}{2C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 + 2R_5 + s^4 (2C_1 C_4 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_4 L_5 R_1 + 2C_1 C_4 L_4 L_5 R_5 + 2C_1 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_5 L_4 L_5 R_5 + 2C_4 C_5 L_4 L_5 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 + 2C_1 C_5 L_5 R_1 R_5 + 2C_1 L_4 L_5 R_1 g_m + C_1 L_4 L_5 + 2C_4 L_4 L_5 R_5 g_m + 2C_4 L_4 L_5 + 2C_5 L_4 L_5 R_5 g_m) + s^2 (2C_1 C_5 R_1 R_5 + 2C_4 L_4 R_5 g_m + 2C_5 L_4 R_5 g_m + 2C_5 L_5 R_5) + s (2C_1 R_1 R_5 + 2L_4 R_5 g_m)}$$

$$\mathbf{9.408 \quad X-INVALID-ORDER-408} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_1 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_5 L_4 L_5 R_1) + s^3 (C_1 L_4 L_5 R_1 g_m + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 g_m + C_5 L_4 R_5 g_m - C_5 L_4)}{2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_4 L_5 R_1 g_m + 2C_1 C_4 L_4 L_5 + 2C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5) + s^2 (2C_1 C_5 R_1 R_5 + 2C_4 L_4 R_5 g_m + 2C_5 L_4 R_5 g_m + 2C_5 L_5 R_5) + s (2C_1 R_1 R_5 + 2L_4 R_5 g_m)}$$

$$\mathbf{9.409 \quad X-INVALID-ORDER-409} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_1 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_5 L_4 L_5 R_1) + s^3 (-C_1 C_5 L_4 R_1 R_5 + C_5 L_4 L_5 R_5 g_m - C_5 L_4 L_5) + s^2 (C_1 L_4 R_1 g_m + C_5 L_4 R_5 g_m - C_5 L_4)}{2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_4 R_1 R_5 g_m + C_1 C_5 L_4 R_5 + 2C_4 C_5 L_4 R_5 g_m) + s^2 (2C_1 C_5 R_1 R_5 + 2C_4 L_4 R_5 g_m + 2C_5 L_4 R_5 g_m + 2C_5 L_5 R_5) + s (2C_1 R_1 R_5 + 2L_4 R_5 g_m)}$$

$$\mathbf{9.410 \quad X-INVALID-ORDER-410} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1) + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^3 (2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

$$\mathbf{9.411 \quad X-INVALID-ORDER-411} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 s^4 + g_m + s^3 (-C_1 C_4 C_5 R_1 R_4 + C_1 C_4 L_4 R_1 g_m - C_4 C_5 L_4) + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_4 C_5 L_4 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.412 \quad X-INVALID-ORDER-412} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_5 s^4 + R_5 g_m + s^3 (-C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1 - C_4 C_5 L_4 R_5) + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4 - C_1 C_5 R_1 R_5 - C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_1 R_1 R_5 g_m + C_4 R_4 R_5 g_m - C_5)}{2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_4 C_5 L_4 R_5 g_m) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5 + 2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5) + s (C_1 R_1 R_5 g_m + C_4 R_4 R_5 g_m - C_5)}$$

$$\mathbf{9.413 \quad X-INVALID-ORDER-413} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_4 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4 + C_1 C_4 L_4 R_1 g_m + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.414 \quad X-INVALID-ORDER-414} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 g_m s^5 + g_m + s^4 (-C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_5 R_1 R_4 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (-C_1 C_4 C_5 R_1 R_4 + C_1 C_4 L_4 R_1 g_m + C_1 C_5 L_5 R_1 g_m - C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (2C_4 g_m + 2C_5 g_m)}$$

$$\mathbf{9.415 \quad X-INVALID-ORDER-415} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 s^5 + s^4 (-C_1 C_4 C_5 L_5 R_1 R_4 + C_1 C_4 L_4 L_5 R_1 g_m - C_4 C_5 L_4 L_5) + s^3 (-C_1 C_4 L_4 R_1 + C_1 C_4 L_5 R_1 R_4 g_m - C_1 C_5 L_5 R_1 - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m - C_4 L_4 + C_4 L_5 R_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}$$

$$\mathbf{9.416 \quad X-INVALID-ORDER-416} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 g_m s^5 + g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_5 R_1 R_4 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 R_1 R_4 + C_1 C_4 L_4 R_1 g_m + C_1 C_5 L_5 R_1 g_m + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m + C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^4 (2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 g_m + 2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + C_1 C_4 C_5 R_4 + 2C_1 C_4 C_5 R_5 + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_1 R_1 g_m + C_4 R_4 g_m + C_5 R_5 g_m - C_5)}$$

$$\mathbf{9.417 \quad X-INVALID-ORDER-417} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 R_5 s^5 - R_5 + s^4 (-C_1 C_4 C_5 L_5 R_1 R_4 R_5 + C_1 C_4 L_4 L_5 R_1 R_5 g_m - C_1 C_4 L_4 L_5 R_1 - C_4 C_5 L_4 L_5 R_5) + s^3 (-C_1 C_4 L_4 R_1 R_5 + C_1 C_4 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m - C_4 L_4 + C_4 L_5 R_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 + C_1 C_4 C_5 L_5 R_4 R_5 + 2C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + C_1 C_4 L_4 R_5 + 2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m - C_4 L_4 + C_4 L_5 R_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}$$

$$\mathbf{9.418 \quad X-INVALID-ORDER-418} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_5 R_1 R_4 + C_1 C_4 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_4 L_4 R_1 + C_1 C_4 L_5 R_1 R_4 g_m + C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 C_4 R_1 R_4 g_m - C_1 C_5 R_1 - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s^2 (C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5 + 2C_4 C_5 R_4 g_m + 2C_4 C_5 R_5 g_m + 2C_4 C_5) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}$$

$$\mathbf{9.419 \quad X-INVALID-ORDER-419} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (-C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_5 R_1 R_4 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (-C_1 C_4 C_5 R_1 R_4 R_5 + C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_5 L_5 R_1 R_4 g_m + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m - C_4 L_4 + C_4 L_5 R_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_5 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_5 + C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_5 R_1 R_4 g_m + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (-C_1 C_4 R_1 R_4 + C_1 L_5 R_1 g_m - C_4 L_4 + C_4 L_5 R_4 g_m) + s (C_1 R_1 g_m + C_4 R_4 g_m - C_5)}$$

$$\mathbf{9.420 \quad X-INVALID-ORDER-420} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (C_1 L_4 R_1 R_4 R_5 g_m - C_1 L_4 R_1 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2 R_4 R_5 g_m + 2 R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 L_4 R_1 R_4 + 2 C_1 C_4 L_4 R_4 R_5) + s^2 (2 C_1 L_4 R_1 R_4 g_m + 2 C_1 L_4 R_1 R_5 g_m + 2 C_1 L_4 R_1 + C_1 L_4 R_4 + 2 C_1 L_4 R_5 + 2 C_4 L_4 R_4 R_5 g_m + 2 C_4 L_4 R_4) + s (2 C_1 R_1 R_4 R_5 g_m + 2 C_1 R_1 R_4 + 2 C_1 R_4 R_5 + 2 L_4 R_4 g_m + 2 L_4 R_5 g_m + 2 L_4 R_4)}$$

$$\mathbf{9.421 \quad X-INVALID-ORDER-421} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 R_1 R_4 s^3 + L_4 R_4 g_m s + s^2 (C_1 L_4 R_1 R_4 g_m - C_5 L_4 R_4)}{2 C_1 C_4 C_5 L_4 R_1 R_4 s^4 + 2 R_4 g_m + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_5 L_4 R_1 + C_1 C_5 L_4 R_4 + 2 C_4 C_5 L_4 R_4) + s^2 (2 C_1 C_5 R_1 R_4 + 2 C_1 L_4 R_1 g_m + 2 C_1 L_4 + 2 C_4 L_4 R_4 g_m + 2 C_5 L_4 R_4 g_m + 2 C_5 L_4) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_4 + 2 C_5 R_4 + 2 L_4 g_m)}$$

$$\mathbf{9.422 \quad X-INVALID-ORDER-422} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 R_1 R_4 R_5 s^3 + s^2 (C_1 L_4 R_1 R_4 R_5 g_m - C_1 L_4 R_1 R_4 - C_5 L_4 R_4 R_5) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + 2 R_4 R_5 g_m + 2 R_4 + s^3 (2 C_1 C_4 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 L_4 R_1 R_4 + 2 C_1 C_4 L_4 R_4 R_5 + 2 C_1 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_5 L_4 R_1 R_5 + C_1 C_5 L_4 R_4 R_5 + 2 C_4 C_5 L_4 R_4 R_5) + s^2 (2 C_1 C_5 R_1 R_4 R_5 + 2 C_1 L_4 R_1 R_4 g_m + 2 C_1 L_4 R_1 R_5 g_m + 2 C_1 L_4 R_1 + C_1 L_4 R_4)}$$

$$\mathbf{9.423 \quad X-INVALID-ORDER-423} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_4 R_4 g_m s + s^3 (C_1 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_5 L_4 R_1 R_4) + s^2 (C_1 L_4 R_1 R_4 g_m + C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4 R_5)}{2 R_4 g_m + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_4 R_4 R_5) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_5 L_4 R_1 R_5 g_m + 2 C_1 C_5 L_4 R_1 + C_1 C_5 L_4 R_4 + 2 C_1 C_5 L_4 R_5 + 2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_4) + s^2 (2 C_1 C_5 R_1 R_4 R_5 g_m + 2 C_1 C_5 R_1 R_4 R_5)}$$

$$\mathbf{9.424 \quad X-INVALID-ORDER-424} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_4 L_5 R_1 R_4 g_m s^4 + L_4 R_4 g_m s + s^3 (-C_1 C_5 L_4 R_1 R_4 + C_5 L_4 L_5 R_4 g_m) + s^2 (C_1 L_4 R_1 R_4 g_m - C_5 L_4 L_5 R_4)}{2 R_4 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_5 L_4 L_5 R_1 g_m + 2 C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_5 L_4 R_1 + C_1 C_5 L_4 R_4 + 2 C_1 C_5 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_5 R_4)}$$

$$\mathbf{9.425 \quad X-INVALID-ORDER-425} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 L_5 R_1 R_4 s^4 - L_4 R_4 s + s^3 (C_1 L_4 L_5 R_1 R_4 g_m - C_5 L_4 L_5 R_4) + s^2 (-C_1 L_4 R_1 R_4 + L_4 L_5 R_4 g_m)}{2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 + 2 R_4 + s^4 (2 C_1 C_4 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 L_4 L_5 R_4 + 2 C_1 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_4 + 2 C_4 C_5 L_4 L_5 R_4) + s^3 (2 C_1 C_4 L_4 R_1 R_4 + 2 C_1 C_5 L_5 R_1 R_4 + 2 C_1 L_4 L_5 R_1 g_m + 2 C_1 L_4 L_5 + 2 C_4 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_4 g_m + 2 C_5 L_4 L_5 R_4)}$$

$$\mathbf{9.426 \quad X-INVALID-ORDER-426} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_4 L_5 R_1 R_4 g_m s^4 + L_4 R_4}{2 R_4 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_4 R_4 R_5 + 2 C_1 C_5 L_4 L_5 R_1 g_m + 2 C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_5 L_4 R_1 R_5 + C_1 C_5 L_4 L_5 R_4 + 2 C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4)}$$

$$\mathbf{9.427 \quad X-INVALID-ORDER-427} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_4 L_5 R_1 R_4}{2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 + 2 R_4 R_5 + s^4 (2 C_1 C_4 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 L_4 L_5 R_1 R_4 + 2 C_1 C_4 L_4 L_5 R_4 R_5 + 2 C_1 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_1 C_5 L_4 L_5 R_1 R_5 + C_1 C_5 L_4 L_5 R_4 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (2 C_1 C_4 L_4 R_1 R_4 R_5 + 2 C_1 C_5 L_5 R_1 R_4 R_5 + 2 C_1 L_4 L_5 R_1 R_4 g_m + 2 C_1 L_4 L_5 R_1 R_5 + 2 C_4 L_4 L_5 R_4 R_5 g_m + 2 C_5 L_4 L_5 R_4 R_5)}$$

$$\mathbf{9.428 \quad X-INVALID-ORDER-428} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{2 R_4 R_5 g_m + 2 R_4 + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_4 R_5) + s^4 (2 C_1 C_4 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 L_4 L_5 R_4 + 2 C_1 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_5 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_4 + 2 C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4)}$$

$$\mathbf{9.429 \quad X-INVALID-ORDER-429} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{2 R_4 R_5 g_m + 2 R_4 + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_4 R_5) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 + 2 C_1 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_5 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_4 + 2 C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4)}$$

$$\mathbf{9.430 \quad X-INVALID-ORDER-430} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_4 R_1 R_4) + s^2 (C_1 L_4 R_1 R_5 g_m - C_1 L_4 R_1 + C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 + L_4 R_5 g_m - L_4)}{2 R_4 g_m + 2 R_5 g_m + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_1 R_5 g_m + 2 C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2 C_1 C_4 L_4 R_5) + s^2 (2 C_1 L_4 R_1 g_m + C_1 L_4 + 2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4) + s (2 C_1 R_1 R_4 g_m + 2 C_1 R_1 R_5 g_m + 2 C_1 R_1 + C_1 R_4 + 2 C_1 R_5 + 2 L_4 g_m) + 2}$$

$$\mathbf{9.431 \quad X-INVALID-ORDER-431} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_4 s^4 + R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_4 g_m - C_1 C_5 L_4 R_1 - C_4 C_5 L_4 R_4) + s^2 (-C_1 C_5 R_1 R_4 + C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m - C_5 L_4) + s (C_1 R_1 R_4 g_m - C_5 R_4 + L_4 g_m)}{2 g_m + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4) + s^3 (2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 + C_1 C_5 R_4 + 2 C_4 L_4 g_m + 2 C_5 L_4 g_m) + s (2 C_1 R_1 g_m + 2 C_1 + 2 C_5 R_4 g_m)}$$

$$\mathbf{9.432 \quad X-INVALID-ORDER-432} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + R_4 R_5 g_m - R_4 + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_4 R_1 R_4 - C_1 C_5 L_4 R_1 R_5 - C_4 C_5 L_4 R_4 R_5) + s^2 (-C_1 C_5 R_1 R_4 R_5 + C_1 L_4 R_1 R_5 g_m)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_1 R_5 g_m + 2 C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2 C_1 C_4 L_4 R_5 + 2 C_1 C_5 L_4 R_1 R_5 g_m + C_1 C_5 L_4 R_5 + 2 C_4 C_5 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_4 R_5) + s^2 (2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 R_5 g_m + 2 C_1 C_5 R_1 R_4 R_5 + C_1 L_4 R_1 R_5 g_m)}$$

$$\mathbf{9.433 \quad X-INVALID-ORDER-433} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 R_1 R_4) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_1 C_5 L_4 R_1 R_5 g_m - C_1 C_5 L_4 R_1 + C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4 + C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m + C_5 L_4)}{2 g_m + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2 C_1 C_4 C_5 L_4 R_5) + s^3 (2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 R_5 g_m + 2 C_1 C_5 R_1 R_4 R_5 + C_1 L_4 R_1 g_m)}$$

$$\mathbf{9.434 \quad X-INVALID-ORDER-434} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (-C_1 C_4 C_5 L_4 R_1 R_4 + C_1 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m - C_1 C_5 L_4 R_1 + C_1 C_5 L_5 R_1 R_4 g_m - C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2 (-C_1 C_5 R_1 R_4 + C_1 L_4 R_1 g_m + C_4 L_4 R_4 g_m + C_5 L_4)}{2 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2 C_1 C_5 L_5 R_1 g_m + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4) + s^2 (2 C_1 C_5 R_1 R_4 g_m + 2 C_1 C_5 R_1 R_5 g_m + 2 C_1 C_5 R_1 R_4 R_5 + C_1 L_4 R_1 g_m)}$$

$$\mathbf{9.435 \quad X-INVALID-ORDER-435} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 - R_4 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 g_m - C_1 C_5 L_4 L_5 R_1 - C_4 C_5 L_4 L_5 R_4) + s^3 (-C_1 C_4 L_4 R_1 R_4 - C_1 C_5 L_5 R_1 R_4 + C_1 L_4 L_5 R_1 g_m + C_4 L_4 L_5 R_4 g_m)}{2 R_4 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4) + s^4 (2 C_1 C_4 L_4 L_5 R_1 g_m + 2 C_1 C_4 L_4 L_5 + 2 C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_5 R_1 R_4 + C_1 L_4 L_5 R_1 g_m)}$$

$$\mathbf{9.436 \quad X-INVALID-ORDER-436} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 R_1 R_4 + C_1 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_1 C_5 L_4 R_1 R_5 g_m - C_1 C_5 L_4 R_1 + C_1 C_5 L_5 R_1 R_4 g_m + C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4)}{2 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2 C_1 C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_4 R_1 g_m + 2 C_1 C_4 L_4 + 2 C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2 C_1 C_5 L_5 R_1 g_m + 2 C_1 C_5 L_5 + 2 C_4 C_5 L_4 R_4 g_m + 2 C_4 C_5 L_4 R_5 + C_1 L_4 L_5 R_1 g_m)}$$

$$\mathbf{9.437 \quad X-INVALID-ORDER-437} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 s^5 - R_4 R_5 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 R_5 g_m - C_1 C_4 L_4 L_5 R_1 R_4 - C_1 C_5 L_4 L_5 R_1 R_5 - C_4 C_5 L_4 L_5 R_4 R_5) + s^3 (-C_1 C_5 R_1 R_4 R_5 + C_1 L_4 L_5 R_1 R_5 g_m + C_4 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4 R_5)}{2 R_4 R_5 g_m + 2 R_5 + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_4 L_5 R_4 R_5) + s^4 (2 C_1 C_4 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_4 L_4 L_5 R_1 + C_1 C_4 L_4 L_5 R_4 + 2 C_1 C_4 L_4 L_5 R_5 + 2 C_1 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_5 L_4 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_4 R_5 + C_1 L_4 L_5 R_1 R_5 g_m)}$$

$$\mathbf{9.438 \quad X-INVALID-ORDER-438} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1 R_4) + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 g_m + C_1 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2 C_1 C_4 L_4 L_5 R_1 g_m + 2 C_1 C_4 L_4 L_5 + 2 C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5)}$$

$$\mathbf{9.439 \quad X-INVALID-ORDER-439} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1 R_4) + s^4 (-C_1 C_4 C_5 L_4 R_1 R_4 R_5 + C_1 L_4 L_5 R_1 R_5 g_m + C_4 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_4 L_5 R_4 R_5)}{2 R_4 g_m + 2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2 C_1 C_5 L_4 L_5 R_1 g_m + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_5 + C_1 L_4 L_5 R_1 R_5 g_m)}$$

9.440 X-INVALID-ORDER-440 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_4 R_1 R_4) + s^2 (C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4) + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5)}$$

9.441 X-INVALID-ORDER-441 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_4 s^4 + R_4 g_m + s^3 (C_1 C_4 L_4 R_1 R_4 g_m - C_4 C_5 L_4 R_4) + s^2 (-C_1 C_5 R_1 R_4 + C_4 L_4 R_4 g_m) + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_4 + 2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_4 C_5 R_4 + 2C_4 L_4 g_m) + s (2C_1 R_1 g_m)}$$

9.442 X-INVALID-ORDER-442 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 R_1 R_4 R_5 s^4 + R_4 R_5 g_m - R_4 + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_4 R_1 R_4 - C_4 C_5 L_4 R_4 R_5) + s^2}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_4 C_5 L_4 R_4 R_5 g_m + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 R_5)}$$

9.443 X-INVALID-ORDER-443 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 R_1 R_4) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_1 C_5 R_1 R_4 R_5)}{2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 R_4 R_5)}$$

9.444 X-INVALID-ORDER-444 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (-C_1 C_4 C_5 L_4 R_1 R_4 + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m + C_1 C_5 L_5 R_1 R_4 g_m - C_4 C_5 L_4 R_4)}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_5 R_1 g_m + 2C_1 C_5 L_5 + 2C_4 C_5 L_4}$$

9.445 X-INVALID-ORDER-445 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R_1 R_4 s^5 - R_4 + s^4 (C_1 C_4 L_4 L_5 R_1 R_4 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (-C_1 C_4 L_4 R_1 R_4 - C_1 C_5 L_5 R_1 R_4) + s^2 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4) + s (2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_4 L_5 R_1 g_m + 2C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^0 (2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4)}{2R_4 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_4 L_5 R_1 g_m + 2C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5 R_4)}$$

9.446 X-INVALID-ORDER-446 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m s^5 + R_4 g_m + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 R_1 R_4 + C_4 C_5 L}{2g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4 + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 R_1 R_4 + 2C_1$$

9.447 X-INVALID-ORDER-447 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_4 L_5 R}{2R_4 R_5 g_m + 2R_5 + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_4 L_5 R_4 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 + 2C_1 C_4 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_4 L_5 R_1 + C_1 C_4 L_4 L_5 R_4 + 2C_1 C_4 L_4 L_5 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_4 L_5 R_5)}$$

9.448 X-INVALID-ORDER-448 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^5 (C_1}{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 R_5 + 2C_1 C_4 L_4 L_5 R_1 g_m + 2C_1 C_4 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2$$

$$\mathbf{9.449 \quad X-INVALID-ORDER-449} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 R_5)}$$

$$\mathbf{9.450 \quad X-INVALID-ORDER-450} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 s^3 + C_1 L_1 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2g_m + s^3 (2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1) + s^2 (C_1 C_5 R_4 + 2C_1 L_1 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.451 \quad X-INVALID-ORDER-451} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_5 L_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_5) + s^2 (C_1 C_5 R_4 R_5 + 2C_1 L_1 R_4 g_m + 2C_1 L_1 R_5 g_m + 2C_1 L_1) + s (C_1 R_4 + 2C_1 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

$$\mathbf{9.452 \quad X-INVALID-ORDER-452} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + R_4 g_m + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^3 (2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1) + s^2 (C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_1 L_1 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.453 \quad X-INVALID-ORDER-453} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_4 g_m s^4 - C_1 C_5 L_1 R_4 s^3 - C_5 R_4 s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_5 L_5 R_4 g_m)}{2C_1 C_5 L_1 L_5 g_m s^4 + 2g_m + s^3 (2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + 2C_1 C_5 L_5) + s^2 (C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_5 L_5 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.454 \quad X-INVALID-ORDER-454} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_4 s^4 + C_1 L_1 L_5 R_4 g_m s^3 + L_5 R_4 g_m s - R_4 + s^2 (-C_1 L_1 R_4 - C_5 L_5 R_4)}{2R_4 g_m + s^4 (2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5) + s^3 (C_1 C_5 L_5 R_4 + 2C_1 L_1 L_5 g_m) + s^2 (2C_1 L_1 R_4 g_m + 2C_1 L_1 + 2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (C_1 R_4 + 2L_5 g_m) + 2}$$

$$\mathbf{9.455 \quad X-INVALID-ORDER-455} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4) + s^2 (C_1 L_1 R_4 g_m + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2C_1 C_5 L_1 L_5 g_m s^4 + 2g_m + s^3 (2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1 + 2C_1 C_5 L_5) + s^2 (C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_1 L_1 g_m + 2C_5 L_5 g_m) + s (2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

$$\mathbf{9.456 \quad X-INVALID-ORDER-456} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_1 L_1 L_5 R_4 R_5 g_m - C_1 L_1 L_5 R_4) + s^2 (-C_1 L_1 R_4 R_5 - C_5 L_5 R_4 R_5) + s (L_5 R_4 R_5 g_m - L_5 R_4)}{2R_4 R_5 g_m + 2R_5 + s^4 (2C_1 C_5 L_1 L_5 R_4 R_5 g_m + 2C_1 C_5 L_1 L_5 R_5) + s^3 (C_1 C_5 L_5 R_4 R_5 + 2C_1 L_1 L_5 R_4 g_m + 2C_1 L_1 L_5 R_5 g_m + 2C_1 L_1 L_5) + s^2 (2C_1 L_1 R_4 R_5 g_m + 2C_1 L_1 R_5 + C_1 L_5 R_4 + 2C_1 L_5 R_5 + 2C_5 L_5 R_4 R_5 g_m + 2C_5 L_5 R_5) + s (C_1 R_4 R_5 + 2L_5 R_4 g_m + 2L_5 R_5 g_m + 2L_5)}$$

$$\mathbf{9.457 \quad X-INVALID-ORDER-457} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_5 R_4 g_m s^3 + L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5 R_5 g_m + 2C_1 C_5 L_1 L_5) + s^3 (C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_1 L_1 L_5 g_m) + s^2 (2C_1 L_1 R_4 g_m + 2C_1 L_1 R_5 g_m + 2C_1 L_1 + 2C_1 L_5 + 2C_5 L_5 R_4 g_m + 2C_5 L_5 R_5 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_1 R_5 + 2L_5 g_m) + 2}$$

$$\mathbf{9.458 \quad X-INVALID-ORDER-458} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_5 + C_1 C_5 L_5 R_4 + 2 C_1 C_5 L_5 R_5) + s^2 (C_1 C_5 R_4 R_5 + 2 C_1 L_1 R_4 g_m + 2 C_1 L_1 R_5 g_m + 2 C_1 L_1 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_5 g_m + 2 C_5 L_5) + s (C_1 R_4 + 2 C_1 R_5)}$$

$$\mathbf{9.459 \quad X-INVALID-ORDER-459} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) - 1}{2 g_m + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1) + s^2 (2 C_1 C_4 R_5 + 2 C_1 L_1 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.460 \quad X-INVALID-ORDER-460} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 s^3 + C_1 L_1 g_m s^2 - C_5 s + g_m}{2 C_1 C_4 C_5 L_1 s^4 + s^3 (2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.461 \quad X-INVALID-ORDER-461} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) - 1}{2 C_1 C_4 C_5 L_1 R_5 s^4 + 2 g_m + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_5 g_m) + s^2 (2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_1 L_1 g_m + 2 C_4 C_5 R_5) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.462 \quad X-INVALID-ORDER-462} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 g_m s^2 + g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s (C_5 R_5 g_m - C_5)}{s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1) + s^3 (2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.463 \quad X-INVALID-ORDER-463} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 - C_1 C_5 L_1 s^3 - C_5 s + g_m + s^2 (C_1 L_1 g_m + C_5 L_5 g_m)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.464 \quad X-INVALID-ORDER-464} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 s^4 + C_1 L_1 L_5 g_m s^3 + L_5 g_m s + s^2 (-C_1 L_1 - C_5 L_5) - 1}{2 C_1 C_4 C_5 L_1 L_5 s^5 + 2 g_m + s^4 (2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5) + s^2 (2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4)}$$

$$\mathbf{9.465 \quad X-INVALID-ORDER-465} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 + g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s^2 (C_1 L_1 g_m + C_5 L_5 g_m) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.466 \quad X-INVALID-ORDER-466} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_5 s^4 - R_5 + s^3 (C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5) + s^2 (-C_1 L_1 R_5 - C_5 L_5 R_5) + s (L_5 R_5 g_m - L_5)}{2 C_1 C_4 C_5 L_1 L_5 R_5 s^5 + 2 R_5 g_m + s^4 (2 C_1 C_4 L_1 L_5 R_5 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_5 g_m) + s^3 (2 C_1 C_4 L_1 R_5 + 2 C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2 C_1 L_1 L_5 g_m + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 L_1 R_5 g_m + C_1 L_5 + 2 C_4 L_5 R_5 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2 C_4 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.467 \quad X-INVALID-ORDER-467} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_5 g_m s^3 + L_5 g_m s + R_5 g_m + s^4 (C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_5 + 2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4)}$$

$$\mathbf{9.468 \quad X-INVALID-ORDER-468} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^4 (C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_5 + 2C_1 C_4 C_5 L_5 R_5 + 2C_1 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_5 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_5 g_m + C_1 C_5 L_5 + 2C_4 C_5 L_5 R_5 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_5 + C_1 C_5 R_5 + 2C_1 L_1 g_m + 2C_4 C_5 R_5 + 2C_5 L_5 g_m) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.469 \quad X-INVALID-ORDER-469} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4)}{2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4) + s^2 (2C_1 C_4 R_4 R_5 + 2C_1 L_1 R_4 g_m + 2C_1 L_1 R_5 g_m + 2C_1 L_1) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4) + 2}$$

$$\mathbf{9.470 \quad X-INVALID-ORDER-470} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 s^3 + C_1 L_1 R_4 g_m s^2 - C_5 R_4 s + R_4 g_m}{2C_1 C_4 C_5 L_1 R_4 s^4 + 2g_m + s^3 (2C_1 C_4 L_1 R_4 g_m + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_4 C_5 R_4) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5)}$$

$$\mathbf{9.471 \quad X-INVALID-ORDER-471} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4)}{2C_1 C_4 C_5 L_1 R_4 R_5 s^4 + 2R_4 g_m + 2R_5 g_m + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + 2C_1 C_5 L_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_5) + s^2 (2C_1 C_4 R_4 R_5 + C_1 C_5 R_4 R_5 + 2C_1 L_1 R_4 g_m + 2C_1 L_1 R_5 g_m + 2C_1 L_1 + 2C_4 C_5 R_4 R_5) + s (C_1 R_4 + 2C_1 R_5 + 2C_4 R_4 R_5 g_m + 2C_4 R_4 + 2C_5 R_4 R_5 g_m) + 2}$$

$$\mathbf{9.472 \quad X-INVALID-ORDER-472} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_4 g_m s^2 + R_4 g_m + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4) + s^3 (2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 C_5 R_5 + 2C_1 L_1 g_m + 2C_4 C_5 R_4 R_5 g_m + 2C_4 C_5 R_4) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 R_4 g_m + 2C_5 R_4) + 2}$$

$$\mathbf{9.473 \quad X-INVALID-ORDER-473} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_4 g_m s^4 - C_1 C_5 L_1 R_4 s^3 - C_5 R_4 s + R_4 g_m + s^2 (C_1 L_1 R_4 g_m + C_5 L_5 R_4 g_m)}{2C_1 C_4 C_5 L_1 L_5 R_4 g_m s^5 + 2g_m + s^4 (2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_4 g_m + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + 2C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_4 C_5 R_4 + 2C_5 L_5 g_m) + s (2C_1 + 2C_4 R_4 g_m + 2C_5 L_5 g_m) + 2}$$

$$\mathbf{9.474 \quad X-INVALID-ORDER-474} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_4 s^4 + C_1 L_1 L_5 R_4 g_m s^3 + L_5 R_4 g_m s - R_4 + s^2 (-C_1 L_1 R_4 - C_5 L_5 R_4)}{2C_1 C_4 C_5 L_1 L_5 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_4 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_4 + 2C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2C_1 L_1 L_5 g_m + 2C_4 C_5 L_5 R_4) + s^2 (2C_1 L_1 R_4 g_m + 2C_1 L_1 + 2C_1 L_5 + 2C_4 L_5 R_4 g_m + 2C_5 L_5 R_4 g_m + 2C_5 L_5) + s (C_1 R_4 + 2C_4 R_4 g_m + 2C_5 L_5 g_m) + 2}$$

$$\mathbf{9.475 \quad X-INVALID-ORDER-475} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_4 g_m s^4 + R_4 g_m + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4) + s^2 (C_1 L_1 R_4 g_m + C_5 L_5 R_4 g_m) + s (C_5 R_4 R_5 g_m - C_5 R_4)}{2C_1 C_4 C_5 L_1 L_5 R_4 g_m s^5 + 2g_m + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1 + 2C_1 C_5 L_5 + 2C_4 C_5 L_5 R_4 g_m) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_4 C_5 R_4 R_5) + s (C_1 R_4 + 2C_4 R_4 g_m + 2C_5 L_5 g_m) + 2}$$

$$\mathbf{9.476 \quad X-INVALID-ORDER-476} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_4 R_5 s^4 - R_4 R_5 + s^3 (C_1 L_1 L_5 R_4 R_5 g_m - C_1 L_1 L_5 R_4) + s^2 (-C_1 L_1 R_4 R_5 - C_5 L_5 R_4 R_5) + s (L_5 R_4 R_5 g_m - L_5 R_4 R_5)}{2C_1 C_4 C_5 L_1 L_5 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_5 + s^4 (2C_1 C_4 L_1 L_5 R_4 R_5 g_m + 2C_1 C_4 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_4 R_5 g_m + 2C_1 C_5 L_1 L_5 R_5) + s^3 (2C_1 C_4 L_1 R_4 R_5 + 2C_1 C_4 L_5 R_4 R_5 + C_1 C_5 L_5 R_4 R_5 + 2C_1 L_1 L_5 R_4 g_m + 2C_1 L_1 L_5 R_5 g_m + 2C_1 L_1 L_5 + 2C_4 C_5 L_5 R_4 R_5) + s^2 (2C_1 L_1 R_4 R_5 + 2C_1 L_1 R_5 + 2C_4 C_5 L_5 R_4 R_5) + s (C_1 R_4 + 2C_4 R_4 g_m + 2C_5 L_5 g_m) + 2}$$

$$\mathbf{9.477 \quad X-INVALID-ORDER-477} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_5 R_4 g_m s^3 + L_5 R_4 g_m s + R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_4 R_5 + C_5 L_5 R_4 R_5) + s (L_5 R_4 R_5 g_m - L_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2C_1 C_4 C_5 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5 R_5 g_m + 2C_1 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + 2C_1 C_4 L_5 R_4 + C_1 C_5 L_5 R_4 + 2C_1 C_5 L_5 R_5 + 2C_1 L_1 L_5) + s^2 (2C_1 L_1 R_4 R_5 + 2C_1 L_1 R_5 + 2C_4 C_5 L_5 R_4 R_5) + s (C_1 R_4 + 2C_4 R_4 g_m + 2C_5 L_5 g_m) + 2}$$

$$\mathbf{9.478 \quad X-INVALID-ORDER-478} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_4 R_5 s^3 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4 R_5) - C_1 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4}{2 R_4 g_m + 2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 + 2 C_1 C_4 C_5 L_5 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_4 L_1 R_4 R_5 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_5 + C_1 C_5 L_5 R_4 + 2 C_1 C_5 L_5 R_5) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + 2 C_1 L_1 g_m) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.479 \quad X-INVALID-ORDER-479} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 g_m + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + 2 C_1 L_1 g_m) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.480 \quad X-INVALID-ORDER-480} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_4 s^4 + g_m + s^3 (C_1 C_4 L_1 R_4 g_m - C_1 C_5 L_1) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m - C_5)}{s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.481 \quad X-INVALID-ORDER-481} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_4 R_5 s^4 + R_5 g_m + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4 - C_1 C_5 L_1 R_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 - C_4 C_5 R_4 R_5) + s (C_4 R_4 R_5 g_m - C_4 R_4 - C_5 R_5) - 1}{2 g_m + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_5) + s^3 (C_1 C_4 C_5 R_4 R_5 + 2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_5 g_m) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_1 L_1 g_m + 2 C_4 C_5 R_4 R_5 g_m + 2 C_4 C_5 R_5) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5)}$$

$$\mathbf{9.482 \quad X-INVALID-ORDER-482} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_4) + s^3 (C_1 C_4 L_1 R_4 g_m + C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s^2 (C_1 L_1 g_m + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.483 \quad X-INVALID-ORDER-483} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_5 R_4 g_m s^5 + g_m + s^4 (-C_1 C_4 C_5 L_1 R_4 + C_1 C_5 L_1 L_5 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m - C_1 C_5 L_1 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.484 \quad X-INVALID-ORDER-484} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_5 R_4 s^5 + s^4 (C_1 C_4 L_1 L_5 R_4 g_m - C_1 C_5 L_1 L_5) + s^3 (-C_1 C_4 L_1 R_4 + C_1 L_1 L_5 g_m - C_4 C_5 L_5 R_4) + s^2 (-C_1 L_1 + C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5) + s^2 (C_1 C_4 R_4 + 2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_4)}$$

$$\mathbf{9.485 \quad X-INVALID-ORDER-485} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_5 R_4 g_m s^5 + g_m + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_4 + C_1 C_5 L_1 L_5 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m + C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 L_1 g_m + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.486 \quad X-INVALID-ORDER-486} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_5 R_4 R_5 s^5 - R_5 + s^4 (C_1 C_4 L_1 L_5 R_4 R_5 g_m - C_1 C_4 L_1 L_5 R_4 - C_1 C_5 L_1 L_5 R_5) + s^3 (-C_1 C_4 L_1 R_4 R_5 + C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5 - C_4 C_5 L_5 R_4 R_5) + s^2 (-C_1 L_1 + C_4 L_5 R_4 g_m - C_5 L_5) + s (-C_4 R_4 + L_5 g_m) - 1}{2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (C_1 C_4 C_5 L_5 R_4 R_5 + 2 C_1 C_4 L_1 L_5 R_4 g_m + 2 C_1 C_4 L_1 L_5 R_5 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_5 g_m) + s^3 (2 C_1 C_4 L_1 R_4 R_5 g_m + 2 C_1 C_4 L_1 R_5 + C_1 C_4 L_5 R_4 + 2 C_1 C_4 L_5 R_5 + C_1 C_5 L_5 R_5 + 2 C_1 L_1 L_5 g_m + 2 C_4 C_5 L_5 R_4 R_5) + s^2 (C_1 C_4 R_4 + 2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_4)}$$

$$\mathbf{9.487 \quad X-INVALID-ORDER-487} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (C_1 C_4 L_1 L_5 R_4 g_m + C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4 + C_1 L_1 L_5 g_m + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 C_5 L_5 R_4 R_5) + s (-C_4 R_4 + L_5 g_m) - 1}{2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_4 g_m + 2 C_4 C_5 L_5 R_5 g_m + 2 C_4 C_5 L_5 R_4 R_5) + s^2 (C_1 C_4 R_4 + 2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_4 R_4)}$$

$$\mathbf{9.488 \quad X-INVALID-ORDER-488} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (-C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4 - C_1 C_5 L_1 R_5 + C_4 C_5 L_5 R_4 R_5 g_m - C_4 C_5 L_5 R_4) + s^2 (C_1 C_4 L_1 R_4 R_5 - C_1 C_4 L_1 + C_4 C_5 L_5 R_5 g_m - C_4 C_5 L_5) + s (C_1 C_4 L_1 R_4 R_5 - C_1 C_4 L_1 + C_4 C_5 L_5 R_5 g_m - C_4 C_5 L_5)}{2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (C_1 C_4 C_5 R_4 R_5 + 2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_5 g_m + C_1 C_5 L_1 L_5) + s^2 (C_1 C_4 L_1 R_4 R_5 - C_1 C_4 L_1 + C_4 C_5 L_5 R_5 g_m - C_4 C_5 L_5) + s (C_1 C_4 L_1 R_4 R_5 - C_1 C_4 L_1 + C_4 C_5 L_5 R_5 g_m - C_4 C_5 L_5)}$$

$$\mathbf{9.489 \quad X-INVALID-ORDER-489} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 L_4 R_5 g_m - C_4 L_4) - 1}{2 C_1 C_4 L_1 L_4 g_m s^4 + 2 g_m + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4) + s^2 (2 C_1 C_4 R_5 + 2 C_1 L_1 g_m + 2 C_4 L_4 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.490 \quad X-INVALID-ORDER-490} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 s^5 + C_1 C_4 L_1 L_4 g_m s^4 - C_5 s + g_m + s^3 (-C_1 C_5 L_1 - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m + C_4 L_4 g_m)}{2 C_1 C_4 C_5 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4) + s^3 (2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.491 \quad X-INVALID-ORDER-491} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_5 s^5 - C_5 R_5 s + R_5 g_m + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4) + s^3 (-C_1 C_5 L_1 R_5 - C_4 C_5 L_4 R_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 L_4 R_5 g_m - C_4 L_4) - 1}{2 C_1 C_4 C_5 L_1 L_4 R_5 g_m s^5 + 2 g_m + s^4 (2 C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_5 + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4 + 2 C_1 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_4 R_5 g_m) + s^2 (2 C_1 C_4 R_5 + C_1 C_5 R_5 + 2 C_1 L_1 g_m + 2 C_4 C_5 R_5 + 2 C_4 L_4 g_m) + s (C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 g_m)}$$

$$\mathbf{9.492 \quad X-INVALID-ORDER-492} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 g_m s^4 + g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4) + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m + C_4 L_4 g_m) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4) + s^3 (2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.493 \quad X-INVALID-ORDER-493} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 - C_1 C_4 C_5 L_1 L_4 s^5 - C_5 s + g_m + s^4 (C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (-C_1 C_5 L_1 - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4 + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.494 \quad X-INVALID-ORDER-494} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 s^6 + C_1 C_4 L_1 L_4 L_5 g_m s^5 + L_5 g_m s + s^4 (-C_1 C_4 L_1 L_4 - C_1 C_5 L_1 L_5 - C_4 C_5 L_4 L_5) + s^3 (C_1 L_1 L_5 g_m + C_4 L_4 L_5 g_m) + s^2 (-C_1 L_1 - C_4 L_4 - C_5 L_5) - 1}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 + C_1 C_4 C_5 L_4 L_5) + s^4 (2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_1 + C_1 C_4 L_4 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5) + s^2 (2 C_1 L_1 g_m + 2 C_4 L_4 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.495 \quad X-INVALID-ORDER-495} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4) + s^4 (C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_5 R_5 g_m - C_5)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4 + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.496 \quad X-INVALID-ORDER-496} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 - R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_5 g_m - C_1 C_4 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_5 - C_1 C_5 L_1 L_5 R_5 - C_4 C_5 L_4 L_5 R_5) + s^3 (C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5 + C_4 L_4 L_5) + s^2 (C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5 + C_4 L_4 L_5) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m s^6 + 2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_5 + 2 C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_5 g_m + 2 C_1 C_4 L_1 L_5 R_5 g_m + 2 C_1 C_4 L_1 L_5 + C_1 C_4 L_4 L_5 + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (2 C_1 C_4 L_1 R_5 + C_1 C_4 L_4 R_5 + 2 C_1 C_4 L_5 R_5 + 2 C_4 C_5 L_5 R_5) + s^2 (2 C_1 C_4 L_1 R_5 + C_1 C_4 L_4 R_5 + 2 C_1 C_4 L_5 R_5) + s (C_5 R_5 g_m - C_5)}$$

$$\mathbf{9.497 \quad X-INVALID-ORDER-497} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 L_5 g_m s^5 + L_5 g_m s + R_5 g_m + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_1 L_1 L_5 g_m + C_4 L_4 L_5 g_m) + s^2 (C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5 + C_4 L_4 L_5) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 + C_1 C_4 C_5 L_4 L_5) + s^4 (2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 L_1 R_5 + C_1 C_4 L_4 R_5 + 2 C_1 C_4 L_5 R_5) + s (C_5 R_5 g_m - C_5)}$$

$$\mathbf{9.508 \quad X-INVALID-ORDER-508} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_5 s^4 - C_5 L_4 R_5 s^2 + s^5 (C_1 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_5 L_1 L_4 L_5) + s^3 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 R_5 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_4 L_5 R_5 + 2 C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_5 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_5 g_m + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_5 L_1 L_5 + C_1 C_5 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}$$

$$\mathbf{9.509 \quad X-INVALID-ORDER-509} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_4 L_4 R_5 g_m - C_4 L_4) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 C_1 C_4 L_1 L_4 g_m s^4 + 2 g_m + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + 2 C_1 L_1 g_m + 2 C_4 L_4 g_m) + s (C_1 + 2 C_4 R_4 g_m + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.510 \quad X-INVALID-ORDER-510} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 s^5 + g_m + s^4 (-C_1 C_4 C_5 L_1 R_4 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m - C_1 C_5 L_1 - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.511 \quad X-INVALID-ORDER-511} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_5 s^5 + R_5 g_m + s^4 (-C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_4 L_1 R_4 - C_1 C_5 L_1 R_5 - C_4 C_5 L_4 R_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 - C_4 C_5 R_4 R_5 + C_4 L_4 R_5 g_m) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2 C_1 C_4 C_5 L_1 L_4 R_5 g_m s^5 + 2 g_m + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_5 + 2 C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_4 C_5 R_4 R_5 + 2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4 + 2 C_1 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_4 R_5 g_m) + s^2 (C_1 C_4 R_4 + 2 C_1 C_4 R_5 + C_1 C_5 R_5) + s (C_4 R_4 R_5 g_m - C_4 R_4) - 1}$$

$$\mathbf{9.512 \quad X-INVALID-ORDER-512} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4) + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_4 + C_1 C_4 L_1 L_4 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m + C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (C_1 L_1 g_m + C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4 + C_4 L_4 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.513 \quad X-INVALID-ORDER-513} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 + C_1 C_4 C_5 L_1 L_5 R_4 g_m) + s^4 (-C_1 C_4 C_5 L_1 R_4 + C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m - C_1 C_5 L_1 - C_4 C_5 L_4 + C_4 C_5 L_5 R_4 g_m) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4 + 2 C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.514 \quad X-INVALID-ORDER-514} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (-C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (-C_1 C_4 L_1 L_4 + C_1 C_4 L_1 L_5 R_4 g_m - C_1 C_5 L_1 L_5 - C_4 C_5 L_4 L_5) + s^3 (-C_1 C_4 L_1 R_4 + C_1 L_1 L_5 g_m - C_4 C_5 L_5 R_4 + C_4 L_4 L_5 g_m) + s^2 (-C_1 L_1 - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 + C_1 C_4 C_5 L_4 L_5) + s^4 (C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 + C_1 C_4 L_4 + 2 C_1 C_4 L_5 + C_1 C_5 L_5 + 2 C_4 C_5 L_5 R_4) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.515 \quad X-INVALID-ORDER-515} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 + C_1 C_4 C_5 L_1 L_5 R_4 g_m) + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_4 + C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_1 R_4 g_m + C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4 + C_4 L_4 g_m) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_4 + 2 C_1 C_4 C_5 L_5) + s^3 (C_1 C_4 C_5 R_4 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_4 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 + C_1 C_5 + 2 C_4 C_5 R_4 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.516 \quad X-INVALID-ORDER-516} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 - R_5 + s^5 (-C_1 C_4 C_5 L_1 L_5 R_4 R_5 + C_1 C_4 L_1 L_4 L_5 R_5 g_m - C_1 C_4 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_5 + C_1 C_4 L_1 L_5 R_4 R_5 g_m - C_1 C_4 L_1 L_5 R_4) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m s^6 + 2 R_5 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_5 + 2 C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 C_5 L_5 R_4 R_5 + 2 C_1 C_4 L_1 L_4 R_5 g_m + 2 C_1 C_4 L_1 L_5 R_4 g_m + 2 C_1 C_4 L_1 L_5 R_5 g_m + 2 C_1 C_4 L_1 L_5 + C_1 C_4 L_4 L_5 + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_4 L_5 R_5 g_m) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}$$

$$\mathbf{9.517 \quad X-INVALID-ORDER-517} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4 + C_1 C_4 L_1 L_5 R_4 g_m + C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_1 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 + C_1 C_4 C_5 L_4 L_5) + s^4 (C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_4 g_m + 2 C_1 C_4 L_1 R_5 g_m - C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4 L_5) + s^2 (C_1 L_1 g_m - C_4 C_5 R_4 + C_4 L_4 g_m + C_5 L_5 g_m) + s (C_4 R_4 g_m + C_5 R_5 g_m - C_5)}$$

9.518 X-INVALID-ORDER-518 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (-C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (-C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4 + C_1 C_4 L_1 L_5 R_4) + s^3 (-C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_1 L_5 R_4) + s^2 (-C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_1 L_5 R_4) + s (-C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_1 L_5 R_4) + C_1 C_4 C_5 L_1 L_4 R_5}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 + C_1 C_4 C_5 L_4 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_5 + C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_5 L_1 L_4) + s^3 (2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s^2 (2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s (2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_4) + C_1 C_4 C_5 L_1 L_4 R_5}$$

9.519 X-INVALID-ORDER-519 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_4 R_5 g_m - C_1 L_1 L_4 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4)}{2R_4 R_5 g_m + 2R_4 + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_4) + s^3 (2C_1 C_4 L_4 R_4 R_5 + 2C_1 L_1 L_4 R_4 g_m + 2C_1 L_1 L_4 R_5 g_m + 2C_1 L_1 L_4) + s^2 (2C_1 L_1 R_4 R_5 g_m + 2C_1 L_1 R_4 + C_1 L_4 R_4 + 2C_1 L_4 R_5 + 2C_4 L_4 R_4 R_5 g_m + 2C_4 L_4 R_4) + s (2C_1 R_4 R_5 + 2L_4 R_4 g_m + 2L_4 R_5 g_m + 2L_4)}$$

9.520 X-INVALID-ORDER-520 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_4 s^4 + C_1 L_1 L_4 R_4 g_m s^3 - C_5 L_4 R_4 s^2 + L_4 R_4 g_m s}{2C_1 C_4 C_5 L_1 L_4 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_4 R_4 + 2C_1 C_5 L_1 R_4 + C_1 C_5 L_4 R_4 + 2C_1 L_1 L_4 g_m + 2C_4 C_5 L_4 R_4) + s^2 (2C_1 L_1 R_4 g_m + 2C_1 L_4 + 2C_4 L_4 R_4 g_m + 2C_5 L_4 R_4 g_m + 2C_5 L_4) + s (2C_1 R_4 + 2C_5 R_4 + 2L_4)}$$

9.521 X-INVALID-ORDER-521 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_4 R_5 s^4 - C_5 L_4 R_4 R_5 s^2 + s^3 (C_1 L_1 L_4 R_4 R_5 g_m - C_1 L_1 L_4 R_4) + s (L_4 R_4 R_5 g_m - L_4 R_4}{2C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_4 + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_4 R_4 R_5 + 2C_1 C_5 L_1 R_4 R_5 + C_1 C_5 L_4 R_4 R_5 + 2C_1 L_1 L_4 R_4 g_m + 2C_1 L_1 L_4 R_5 g_m + 2C_1 L_1 L_4 + 2C_4 C_5 L_4 R_4 R_5) + s^2 (2C_1 L_1$$

9.522 X-INVALID-ORDER-522 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_4 g_m s^3 + L_4 R_4 g_m s + s^4 (C_1 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_5 L_1 L_4 R_4) + s^2 (C_5 L_4 R_4 R_5 g_m - C_5 L_4 R_4 R_5)}{2 R_4 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (2 C_1 C_4 C_5 L_4 R_4 R_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_5 L_1 L_4 R_4 g_m + 2 C_1 C_5 L_1 L_4 R_5 g_m + 2 C_1 C_5 L_1 L_4) + s^3 (2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_4 + C_1 C_5 L_4 R_4 + 2 C_1 C_5 L_4 R_5 + 2 C_1 L_1 L_4 g_m + 2 C_4$$

9.523 X-INVALID-ORDER-523 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_4 g_m s^5 - C_1 C_5 L_1 L_4 R_4 s^4 - C_5 L_4 R_4 s^2 + L_4 R_4 g_m s + s^3 (C_1 L_1 L_4 R_4 g_m + C_5 L_4 L_1 R_4)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + 2R_4 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2C_1 C_4 L_4 R_4 + 2C_1 C_5 L_1 R_4 + C_1 C_5 L_4 R_4)}$$

9.524 X-INVALID-ORDER-524 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_4 s^5 + C_1 L_1 L_4 L_5 R_4 g_m s^4 + L_4 L_5 R_4 g_m s^2 - L_4 R_4 s + s^3 (-C_1 L_1 L_4 R_4 - C_5 L_4 L_5 R_4)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + 2R_4 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_4 g_m + 2C_1 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_4 + C_1 C_5 L_4 L_5 R_4 + 2C_1 L_1 L_4 L_5 g_m + 2C_4 C_5 L_4 L_5 R_4) + s^3 (2C_1 L_1 L_4 R_4 g_m + 2C_1 L_1 L_4 + 2C_1 L_1 L_5 R_4 g_m + 2C_1 L_1 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_4)}$$

9.525 X-INVALID-ORDER-525 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_4 R_4 g_m}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + 2R_4 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_5 g_m + 2C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_4 L_5}.$$

9.526 X-INVALID-ORDER-526 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_4}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + 2R_4 R_5 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_1 C_5 L_1 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 + 2C_1 C_4 L_4 L_5 R_4 R_5 + 2C_1 C_5 L_1 L_5 R_4 R_5 + C_1 C_5 L_4 L_5 R_4 R_5 + 2C_1 L_1 L_4 L_5 R_4 g_m + 2C_1 L_1 L_4 L_5 R_5 g_m + 2C_1 L_1 L_4 L_5 R_5)}.$$

9.527 X-INVALID-ORDER-527 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_4R_5q_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5q_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_4L_1L_4L_5R_4q_m + 2C_1C_5L_1L_4L_5R_4q_m + 2C_1C_5L_1L_4L_5R_5q_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_4R_5q_m + 2C_1C_4L_1L_4R_4 + 2C_1C_4L_4L_5R_4 + 2C_1C_5L_1L_4L_5)}{2R_4R_5q_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5q_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_4L_1L_4L_5R_4q_m + 2C_1C_5L_1L_4L_5R_4q_m + 2C_1C_5L_1L_4L_5R_5q_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_4R_5q_m + 2C_1C_4L_1L_4R_4 + 2C_1C_4L_4L_5R_4 + 2C_1C_5L_1L_4L_5)}$$

9.528 X-INVALID-ORDER-528 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_4R_5g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_4R_5g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_4R_5g_m + 2C_1C_5L_1L_4R_4R_5)}{1 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_4R_5g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_4R_5g_m + 2C_1C_5L_1L_4R_4R_5) + s^3(2C_1C_4L_1L_4R_4R_5 + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_4R_5 + 2C_1C_5L_1L_4R_4) + s^2(2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_4) + s(C_1C_4L_1L_4 + C_1C_5L_1L_4) + 1}$$

9.529 X-INVALID-ORDER-529 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 L_1 L_4 R_5 g_m - C_1 L_1 L_4) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4) + s (L_4 R_5 g_m - L_4)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5 g_m + 2 C_1 C_4 L_1 L_4) + s^3 (C_1 C_4 L_4 R_4 + 2 C_1 C_4 L_4 R_5 + 2 C_1 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_4 g_m + 2 C_1 L_1 R_5 g_m + 2 C_1 L_1 + C_1 L_4 + 2 C_4 L_4 R_4 g_m + 2 C_4 L_4 R_5 g_m + 2 C_4 L_4) + s (C_1 R_4 + 2 C_1 R_5 + 2 L_4 g_m) + 2}$$

9.530 X-INVALID-ORDER-530 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 s^5 + R_4 g_m + s^4 (C_1 C_4 L_1 L_4 R_4 g_m - C_1 C_5 L_1 L_4) + s^3 (-C_1 C_5 L_1 R_4 + C_1 L_1 L_4 g_m - C_4 C_5 L_4 R_4) + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m - C_5 L_4) + s (-C_5 R_4 + L_4 g_m)}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 + 2C_5 R_4)}$$

9.531 X-INVALID-ORDER-531 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + R_4 R_5 g_m - R_4 + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_4 - C_1 C_5 L_1 L_4 R_5) + s^3 (-C_1 C_5 L_1 R_4 R_5 + C_1 L_1 L_4 R_5 g_m - C_1 L_1 L_4 - C_4 C_5)}{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_5 g_m) + s^3 (C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_5 + C_1 C_5 L_4 R_5 + 2C_1 L_1 L_4 g_r)}$$

9.532 X-INVALID-ORDER-532 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (C_1 C_4 L_1 L_4 R_4 g_m + C_1 C_5 L_1 L_4 R_5 g_m - C_1 C_5 L_1 L_4) + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4 + C_1 L_1 L_4 g_m + C_4 C_5 L_4 R_4 R_5 g_m - C_4 C_5 L_4 R_4) + s^2 (C_1 L_1 R_4 g_m + C_1 L_1 L_4 R_4 g_m - C_1 L_1 L_4 R_4) + s (C_1 L_1 L_4 R_4 g_m - C_1 L_1 L_4 R_4) + C_1 L_1 L_4 R_4}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1 + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_4) + s^2 (2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1 C_5 L_1 + C_1 C_5 L_4 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4 R_5 g_m + 2C_4 C_5 L_4 R_4) + s (C_1 L_1 L_4 R_4 g_m - C_1 L_1 L_4 R_4) + C_1 L_1 L_4 R_4}$$

9.533 X-INVALID-ORDER-533 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + R_4 g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_4 g_m - C_1 C_5 L_1 L_4 + C_1 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (-C_1 C_5 L_1 R_4 + C_1 L_1 L_4 g_m - C_4 C_5 L_4 R_4 + C_5 L_4 L_5 g_m) + s^2}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_4 L_5) + s^4 (C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + C_1 C_5 L_4 + 2C_1 C_5 L_5 + 2C_4 C_5 L_5}$$

9.534 X-INVALID-ORDER-534 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 - R_4 + s^5 (C_1 C_4 L_1 L_4 L_5 R_4 g_m - C_1 C_5 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_4 - C_1 C_5 L_1 L_5 R_4 + C_1 L_1 L_4 L_5 g_m - C_4 C_5 L_4 L_5 R_4) + s^3 (-C_1 L_1 L_4}{2R_4 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 g_m + 2C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_4 L_5 + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_1 L_5 + C_1 C_5 L_4 L_5 + 2C_4 C_5 L_4 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) +}$$

9.535 **X-INVALID-ORDER-535** $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_4 g_m + C_1 C_5 L_1 L_4 R_5 g_m - C_1 C_5 L_1 L_4 + C_1 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4 + C_1 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^2 (C_1 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s (C_4 C_5 L_4 L_5 g_m) + R_4 g_m}{2 C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_4 L_5) + s^4 (C_1 C_4 C_5 L_4 R_4 + 2 C_1 C_4 C_5 L_4 R_5 + 2 C_1 C_4 L_1 L_4 g_m + 2 C_1 C_5 L_1 L_4 g_m + 2 C_1 C_5 L_1 L_5 g_m + 2 C_4 C_5 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_4 + 2 C_1 C_5 L_1 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_4 g_m) + s^2 (2 C_1 C_4 L_4 + 2 C_1 C_5 L_1 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5 R_4 g_m) + s (C_4 C_5 L_4 L_5 g_m) + R_4 g_m}.$$

9.536 X-INVALID-ORDER-536 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 - R_4 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_4 R_5 g_m - C_1 C_4 L_1 L_4 L_5 R_4 - 2R_4 R_5 g_m + 2R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 L_1 L_4 L_5 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_4 L_4 L_5 R_4 + 2C_1 C_4 L_4 L_5 R_5))}{2R_4 R_5 g_m + 2R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 L_1 L_4 L_5 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_4 L_4 L_5 R_4 + 2C_1 C_4 L_4 L_5 R_5)}$$

9.537 X-INVALID-ORDER-537 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_4) + s^5 (C_1 C_4 L_1 L_4 L_5 R_4 g_m + C_1 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4)}{2R_4 g_m + 2R_5 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_5 + 2C_1 C_4 L_1 L_4 L_5 g_m + 2C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5)}$$

9.538 X-INVALID-ORDER-538 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_4) + s^5 (-C_1 C_4 C_5 L_1 L_4 R_4 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 + C_1 C_4 L_1 L_4 R_4 g_m + C_1 C_4 L_1 L_4 R_5)}{2 R_4 g_m + 2 R_5 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_5 + 2 C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 C_5 L_4 R_4 R_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5)}$$

9.539 X-INVALID-ORDER-539 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_4) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_4 L_4 R_4 R_5 g_m - C_4 L_4 R_4)}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5) + s^2 (2C_1 C_4 R_4 R_5 + 2C_1 L_1 R_4 g_m + 2C_1 L_1 R_5 g_m + 2C_1 L_1 + 2C_4 L_4 R_4 g_m + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (C_1 R_4 + 2C_1 L_1)}$$

9.540 X-INVALID-ORDER-540 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 s^5 + C_1 C_4 L_1 L_4 R_4 g_m s^4 - C_5 R_4 s + R_4 g_m + s^3 (-C_1 C_5 L_1 R_4 - C_4 C_5 L_4 R_4) + s^2 (C_1 L_1 R_4 g_m + C_4 L_4 R_4 g_m)}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + 2C_4 C_5 L_4 R_4 g_m + 2C_4 C_5 L_4) + s^2 (2C_1 C_4 R_4 + C_1 C_5 R_4 + 2C_1 L_1 g_m + 2C_4 C_5 R_4 + 2C_4 L_4 g_m)}$$

9.541 X-INVALID-ORDER-541 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 - C_5 R_4 R_5 s + R_4 R_5 g_m - R_4 + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_4) + s^3 (-C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_4 R_5)}$$

9.542 X-INVALID-ORDER-542 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 R_4 g_m s^4 + R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_4) + s^3 (C_1 C_5 L_1 R_4 R_5 g_m - C_1 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 R_5 g_m}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_4 C_5 R_4 R_5 + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 R_5 g_m + 2C_1$$

9.543 X-INVALID-ORDER-543 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 - C_1 C_4 C_5 L_1 L_4 R_4 s^5 - C_5 R_4 s + R_4 g_m + s^4 (C_1 C_4 L_1 L_4 R_4 g_m + C_1 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_4 g_m) + s^3 (2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + 2C_1 C_4 L_5))}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 + 2C_1 C_4 L_5)}$$

9.544 X-INVALID-ORDER-544 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + C_1 C_4 L_1 L_4 L_5 R_4 g_m s^5 + L_5 R_4 g_m s - R_4 + s^4 (-C_1 C_4 L_1 L_4 R_4 - C_1 C_5 L_1 L_5 R_4 - C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m)}{2 R_4 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_4 g_m + 2 C_1 C_4 L_4 L_5 + 2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_4 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5)}$$

9.545 X-INVALID-ORDER-545 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 L_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 L_5)}$$

9.546 X-INVALID-ORDER-546 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5}{2R_4 R_5 g_m + 2R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_4 R_5 + C_1 C_4 C_5 L_4 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 L_1 L_4 L_5 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_4 R_5 g_m + 2C_1 C_4 L_1 L_5 R_5 g_m + 2C_1 C_4 L_1 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_1 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_5 R_5) + s^2 (2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_4 + 2C_1 C_4 L_1 L_5 R_5) + s (2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 + 2C_1 C_4 L_4 L_5) + 2C_1 C_4 L_1 L_4 L_5}.$$

$$\mathbf{9.547 \quad X-INVALID-ORDER-547} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 L_5 R_4 g_m s^5}{2 R_4 g_m + 2 R_5 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_5 + 2 C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_5 R_4 R_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_4 L_1 L_4 R_5 g_m) + s^3 (2 C_1 C_4 C_5 L_5 R_4 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_4 L_1 L_4 R_5 g_m) + s^2 (2 C_1 C_4 C_5 L_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_4 L_1 L_4 R_5 g_m) + s (2 C_1 C_4 C_5 L_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_4 L_1 L_4 R_5 g_m) + 2}$$

$$\mathbf{9.548 \quad X-INVALID-ORDER-548} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 L_1 L_4 L_5 R_4 g_m s^5}{2 R_4 g_m + 2 R_5 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_4 + 2 C_1 C_4 C_5 L_4 L_5 R_5) + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 g_m) + s^3 (2 C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 g_m) + s^2 (2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 g_m) + s (2 C_1 C_4 C_5 L_1 + C_1 C_4 C_5 L_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 g_m) + 2}$$

$$\mathbf{9.549 \quad X-INVALID-ORDER-549} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{C_1 C_5 L_1 R_4 s^3 + s^2 (2 C_1 L_1 + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1) + s (C_5 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.550 \quad X-INVALID-ORDER-550} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s (L_1 R_4 R_5 g_m - L_1 R_4)}{C_1 C_5 L_1 R_4 R_5 s^3 + R_4 + 2 R_5 + s^2 (C_1 L_1 R_4 + 2 C_1 L_1 R_5 + 2 C_5 L_1 R_4 R_5 g_m + 2 C_5 L_1 R_5) + s (C_5 R_4 R_5 + 2 L_1 R_4 g_m + 2 L_1 R_5 g_m + 2 L_1)}$$

$$\mathbf{9.551 \quad X-INVALID-ORDER-551} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{s^3 (C_1 C_5 L_1 R_4 + 2 C_1 C_5 L_1 R_5) + s^2 (2 C_1 L_1 + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1) + s (C_5 R_4 + 2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.552 \quad X-INVALID-ORDER-552} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{2 C_1 C_5 L_1 L_5 s^4 + s^3 (C_1 C_5 L_1 R_4 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_1 L_1 + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 + 2 C_5 L_5) + s (C_5 R_4 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.553 \quad X-INVALID-ORDER-553} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 s^3 + L_1 L_5 R_4 g_m s^2 - L_1 R_4 s}{C_1 C_5 L_1 L_5 R_4 s^4 + R_4 + s^3 (2 C_1 L_1 L_5 + 2 C_5 L_1 L_5 R_4 g_m + 2 C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 + C_5 L_5 R_4 + 2 L_1 L_5 g_m) + s (2 L_1 R_4 g_m + 2 L_1 + 2 L_5)}$$

$$\mathbf{9.554 \quad X-INVALID-ORDER-554} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 + L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{2 C_1 C_5 L_1 L_5 s^4 + s^3 (C_1 C_5 L_1 R_4 + 2 C_1 C_5 L_1 R_5 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_1 L_1 + 2 C_5 L_1 R_4 g_m + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + 2 C_5 L_5) + s (C_5 R_4 + 2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.555 \quad X-INVALID-ORDER-555} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 R_5 s^3 - L_1 R_4 R_5 s + s^2 (L_1 L_5 R_4 R_5 g_m - L_1 L_5 R_4)}{C_1 C_5 L_1 L_5 R_4 R_5 s^4 + R_4 R_5 + s^3 (C_1 L_1 L_5 R_4 + 2 C_1 L_1 L_5 R_5 + 2 C_5 L_1 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_4 R_5 + C_5 L_5 R_4 R_5 + 2 L_1 L_5 R_4 g_m + 2 L_1 L_5 R_5 g_m + 2 L_1 L_5) + s (2 L_1 R_4 R_5 g_m + 2 L_1 R_5 + L_5 R_4 + 2 L_5 R_5)}$$

$$\mathbf{9.556 \quad X-INVALID-ORDER-556} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_4 g_m s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^4 (C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^3 (2C_1 L_1 L_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + C_5 L_5 R_4 + 2C_5 L_5 R_5 + 2L_1 L_5 g_m) + s (2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1 + 2L_5)}$$

$$\mathbf{9.557 \quad X-INVALID-ORDER-557} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^4 (C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^3 (C_1 C_5 L_1 R_4 R_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + 2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5 + C_5 L_5 R_4 + 2C_5 L_5 R_5) + s (C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

$$\mathbf{9.558 \quad X-INVALID-ORDER-558} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_1 R_5 g_m - L_1)}{2C_1 C_4 L_1 R_5 s^3 + s^2 (C_1 L_1 + 2C_4 L_1 R_5 g_m + 2C_4 L_1) + s (2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.559 \quad X-INVALID-ORDER-559} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s (L_1 R_5 g_m - L_1)}{s^3 (2C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_5) + s^2 (C_1 L_1 + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_5 L_1 R_5 g_m) + s (2C_4 R_5 + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.560 \quad X-INVALID-ORDER-560} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 g_m + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2C_1 C_4 C_5 L_1 R_5 s^3 + 2C_4 + C_5 + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1) + s (2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.561 \quad X-INVALID-ORDER-561} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 - C_5 L_1 s + L_1 g_m}{2C_1 C_4 C_5 L_1 L_5 s^4 + 2C_4 C_5 L_1 L_5 g_m s^3 + 2C_4 + C_5 + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 + 2C_4 C_5 L_5) + s (2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.562 \quad X-INVALID-ORDER-562} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 s^3 + L_1 L_5 g_m s^2 - L_1 s}{2L_1 g_m s + s^4 (2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5) + s^3 (2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 + 2C_4 L_5 + C_5 L_5) + 1}$$

$$\mathbf{9.563 \quad X-INVALID-ORDER-563} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 + L_1 g_m + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2C_1 C_4 C_5 L_1 L_5 s^4 + 2C_4 + C_5 + s^3 (2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_5) + s (2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.564 \quad X-INVALID-ORDER-564} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_5 s^3 - L_1 R_5 s + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{R_5 + s^4 (2C_1 C_4 L_1 L_5 R_5 + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_5) + s^3 (C_1 L_1 L_5 + 2C_4 L_1 L_5 R_5 g_m + 2C_4 L_1 L_5 + 2C_5 L_1 L_5 R_5 g_m) + s^2 (C_1 L_1 R_5 + 2C_4 L_1 R_5 + 2C_4 L_5 R_5 + C_5 L_5 R_5 + 2L_1 L_5 g_m) + s (2L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.565 \quad X-INVALID-ORDER-565} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 g_m s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2C_1 C_4 C_5 L_1 L_5 R_5 s^5 + s^4 (2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_5 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_4 L_5 + C_5 L_5) + s (2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.566 \quad X-INVALID-ORDER-566} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2C_1 C_4 C_5 L_1 L_5 R_5 s^5 + s^4 (C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_5 + 2C_4 C_5 L_5 R_5 + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_5 L_1 R_5 g_m + C_5 L_5) + s (2C_4 R_5 + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.567 \quad X-INVALID-ORDER-567} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_1 R_4 R_5 g_m - L_1 R_4)}{2C_1 C_4 L_1 R_4 R_5 s^3 + R_4 + 2R_5 + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + 2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4) + s (2C_4 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

$$\mathbf{9.568 \quad X-INVALID-ORDER-568} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_4) + s^2 (2C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_5 L_1 R_4 g_m + 2C_5 L_1) + s (2C_4 R_4 + C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.569 \quad X-INVALID-ORDER-569} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^3 (2C_1 C_4 L_1 R_4 R_5 + C_1 C_5 L_1 R_4 R_5 + 2C_4 C_5 L_1 R_4 R_5) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + 2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4 + 2C_5 L_1 R_4 R_5 g_m + 2C_5 L_1 R_5) + s (2C_4 R_4 R_5 + C_5 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

$$\mathbf{9.570 \quad X-INVALID-ORDER-570} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{2C_1 C_4 C_5 L_1 R_4 R_5 s^4 + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_4) + s^2 (2C_1 L_1 + 2C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5 g_m + 2C_5 L_1) + s (2C_4 R_4 + C_5 R_4 + 2C_5 R_5 + 2L_1 g_m) + 2}$$

$$\mathbf{9.571 \quad X-INVALID-ORDER-571} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{2C_1 C_4 C_5 L_1 L_5 R_4 s^5 + s^4 (2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_4 + 2C_4 C_5 L_5 R_4 + 2C_5 L_1 L_5 g_m) + s^2 (2C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + 2C_5 L_5) + s (2C_4 R_4 + C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.572 \quad X-INVALID-ORDER-572} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 s^3 + L_1 L_5 R_4 g_m s^2 - L_1 R_4 s}{R_4 + s^4 (2C_1 C_4 L_1 L_5 R_4 + C_1 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_5 R_4) + s^3 (2C_1 L_1 L_5 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 + 2C_4 L_1 R_4 + 2C_4 L_5 R_4 + C_5 L_5 R_4 + 2L_1 L_5 g_m) + s (2L_1 R_4 g_m + 2L_1 + 2L_5)}$$

$$\mathbf{9.573 \quad X-INVALID-ORDER-573} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_4 g_m s^3 + L_1 R_4 g_m s + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{2C_1 C_4 C_5 L_1 L_5 R_4 s^5 + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_4 + 2C_4 C_5 L_5 R_4 + 2C_5 L_1 L_5 g_m) + s^2 (2C_1 L_1 + 2C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_5 L_1 R_4 g_m + 2C_5 L_1 R_5) + s (2C_4 R_4 + C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.574 \quad X-INVALID-ORDER-574} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_4 R_5 s^3 - L_1 R_4 R_5 s + s^2 (L_1 L_5 R_4 R_5 g_m - L_1 L_5 R_4)}{R_4 R_5 + s^4 (2C_1 C_4 L_1 L_5 R_4 R_5 + C_1 C_5 L_1 L_5 R_4 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5) + s^3 (C_1 L_1 L_5 R_4 + 2C_1 L_1 L_5 R_5 + 2C_4 L_1 L_5 R_4 R_5 g_m + 2C_4 L_1 L_5 R_4 + 2C_5 L_1 L_5 R_4 R_5 g_m + 2C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_4 R_5 + 2C_4 L_1 R_4 R_5 + 2C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2L_1 L_5 R_4 g_m + 2L_1 L_5 R_5) + s (2L_1 R_4 R_5 + 2L_1 R_5) + 1}$$

$$\mathbf{9.575 \quad X-INVALID-ORDER-575} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_4 g_m s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{2C_1 C_4 C_5 L_1 L_5 R_4 R_5 s^5 + R_4 + 2R_5 + s^4 (2C_1 C_4 L_1 L_5 R_4 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 + 2C_1 L_1 L_5 + 2C_4 C_5 L_5 R_4 R_5 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 R_5 + 2C_4 L_1 R_4 R_5 + 2C_4 L_5 R_4 R_5 + C_5 L_5 R_4 R_5 + 2L_1 L_5 R_4 g_m + 2L_1 L_5 R_5) + s (2L_1 R_4 R_5 + 2L_1 R_5) + 1}$$

$$\mathbf{9.576 \quad X-INVALID-ORDER-576} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_4 R_5 s^2 + s^3 (C_5 L_1 L_5 R_4 R_5 g_m - C_5 L_1 L_5 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{2C_1 C_4 C_5 L_1 L_5 R_4 R_5 s^5 + R_4 + 2R_5 + s^4 (C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 + C_1 C_5 L_1 R_4 R_5 + 2C_4 C_5 L_1 R_4 R_5 + 2C_4 C_5 L_5 R_4 R_5 + 2C_5 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5)}$$

$$\mathbf{9.577 \quad X-INVALID-ORDER-577} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4) + s (L_1 R_5 g_m - L_1)}{s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1) + s (C_4 R_4 + 2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.578 \quad X-INVALID-ORDER-578} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 R_4 s^2 + L_1 g_m + s (C_4 L_1 R_4 g_m - C_5 L_1)}{C_1 C_4 C_5 L_1 R_4 s^3 + 2C_4 + C_5 + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1) + s (C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.579 \quad X-INVALID-ORDER-579} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 R_4 R_5 s^3 + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 C_5 L_1 R_4 R_5 s^4 + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_5) + s^2 (C_1 L_1 + C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + 2C_5 L_1 R_5 g_m) + s (C_4 R_4 + 2C_4 R_5 + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.580 \quad X-INVALID-ORDER-580} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 g_m + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 + C_5 + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.581 \quad X-INVALID-ORDER-581} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_4 g_m s^3 + L_1 g_m + s^2 (-C_4 C_5 L_1 R_4 + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{2C_1 C_4 C_5 L_1 L_5 s^4 + 2C_4 + C_5 + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.582 \quad X-INVALID-ORDER-582} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_4 s^4 - L_1 s + s^3 (C_4 L_1 L_5 R_4 g_m - C_5 L_1 L_5) + s^2 (-C_4 L_1 R_4 + L_1 L_5 g_m)}{C_1 C_4 C_5 L_1 L_5 R_4 s^5 + s^4 (2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5) + s^3 (C_1 C_4 L_1 R_4 + C_4 C_5 L_5 R_4 + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 + 2C_4 L_5 + C_5 L_5) + s (C_4 R_4 + 2L_1 g_m) + 1}$$

$$\mathbf{9.583 \quad X-INVALID-ORDER-583} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_4 g_m s^3 + L_1 g_m + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2C_1 C_4 C_5 L_1 L_5 s^4 + 2C_4 + C_5 + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.584 \quad X-INVALID-ORDER-584} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_4 R_5 s^4 - L_1 R_5 s + s^3 (C_4 L_1 L_5 R_4 R_5 g_m - C_4 L_1 L_5 R_4 - C_5 L_1 L_5 R_5) + s^2 (-C_4 L_1 R_4 R_5 + L_1 L_5 R_5 g_m - L_1 L_5)}{C_1 C_4 C_5 L_1 L_5 R_4 R_5 s^5 + R_5 + s^4 (C_1 C_4 L_1 L_5 R_4 + 2C_1 C_4 L_1 L_5 R_5 + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_5) + s^3 (C_1 C_4 L_1 R_4 R_5 + C_1 L_1 L_5 + C_4 C_5 L_5 R_4 R_5 + 2C_4 L_1 L_5 R_4 g_m + 2C_4 L_1 L_5 R_5 g_m + 2C_4 L_1 L_5 + 2C_5 L_1 L_5 R_5 g_m) + s^2 (C_1 L_1 R_5 + 2C_4 L_1 R_4)}$$

$$\mathbf{9.585 \quad X-INVALID-ORDER-585} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (C_4 L_1 L_5 R_4 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 + L_1 L_5 g_m) + s (L_1 R_5 g_m - L_1)}{s^5 (C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m)}$$

$$\mathbf{9.586 \quad X-INVALID-ORDER-586} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (-C_4 C_5 L_1 R_4 R_5 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_4 R_5 g_m - L_1 R_4 - L_5 R_5) + R_5}{s^5 (C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_5 R_4 + 2 C_4 C_5 L_5 R_5 + 2 C_5 L_1 L_5 g_m) + s^2 (C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1 + C_4 C_5 L_4) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.587 \quad X-INVALID-ORDER-587} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 L_1 L_4 s^4 + s^3 (2 C_1 C_4 L_1 R_5 + 2 C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.588 \quad X-INVALID-ORDER-588} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 s^3 + C_4 L_1 L_4 g_m s^2 - C_5 L_1 s + L_1 g_m}{C_1 C_4 C_5 L_1 L_4 s^4 + 2 C_4 C_5 L_1 L_4 g_m s^3 + 2 C_4 + C_5 + s^2 (2 C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1 + C_4 C_5 L_4) + s (2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.589 \quad X-INVALID-ORDER-589} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 - C_5 L_1 R_5 s^2 + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 C_5 L_1 L_4 R_5 s^5 + s^4 (C_1 C_4 L_1 L_4 + 2 C_4 C_5 L_1 L_4 R_5 g_m) + s^3 (2 C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + C_4 L_4 + 2 C_5 L_1 R_5 g_m) + s (2 C_4 R_5 + C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.590 \quad X-INVALID-ORDER-590} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 g_m s^2 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s (C_5 L_1 R_5 g_m - C_5 L_1)}{C_1 C_4 C_5 L_1 L_4 s^4 + 2 C_4 + C_5 + s^3 (2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m) + s^2 (2 C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_1 + C_4 C_5 L_4) + s (2 C_4 C_5 R_5 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.591 \quad X-INVALID-ORDER-591} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 - C_4 C_5 L_1 L_4 s^3 - C_5 L_1 s + L_1 g_m + s^2 (C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m)}{2 C_4 + C_5 + s^4 (C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5) + s^3 (2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m) + s^2 (2 C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1 + C_4 C_5 L_4 + 2 C_4 C_5 L_5) + s (2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.592 \quad X-INVALID-ORDER-592} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 s^5 + C_4 L_1 L_4 L_5 g_m s^4 + L_1 L_5 g_m s^2 - L_1 s + s^3 (-C_4 L_1 L_4 - C_5 L_1 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + 2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + 2 L_1 g_m s + s^4 (C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_1 L_4 g_m + 2 C_4 L_1 L_5 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 + C_4 L_4 + 2 C_4 L_5 + C_5 L_5) + 1}$$

$$\mathbf{9.593 \quad X-INVALID-ORDER-593} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^2 (C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2 C_4 + C_5 + s^4 (C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5) + s^3 (2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m) + s^2 (2 C_1 C_4 L_1 + C_1 C_5 L_1 + 2 C_4 C_5 L_1 R_5 g_m + 2 C_4 C_5 L_1 + C_4 C_5 L_4 + 2 C_4 C_5 L_5) + s (2 C_4 C_5 R_5 + 2 C_4 L_1 g_m + 2 C_5 L_1 g_m)}$$

$$\mathbf{9.594 \quad X-INVALID-ORDER-594} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_5 s^5 - L_1 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_5 g_m - C_4 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 R_5 - C_5 L_1 L_5 R_5) + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 + R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 + 2 C_4 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_4 L_1 L_5 R_5 + C_1 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_5 + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_5 R_5 g_m + 2 C_4 L_1 L_5 + C_4 L_4 L_5 + 2 C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 + 2 C_4 L_1 + C_4 L_4 + 2 C_4 L_5 + C_5 L_5) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.595 \quad X-INVALID-ORDER-595} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 L_5 g_m s^4 + L_1 L_5 g_m s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_1 R_5 + 2 C_4 C_5 L_5 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_4 L_1 L_5 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 L_5 g_m + 2 C_4 L_1 + C_4 L_4 + 2 C_5 L_1 L_5 R_5) + s (2 C_4 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.596 \quad X-INVALID-ORDER-596} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 - C_5 L_1 R_5 s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 + L_4 R_5) + 2}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2 C_4 C_5 L_5 R_5 + 2 C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^2 (2 C_1 L_1 R_5 + 2 C_4 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4) + 2}$$

$$\mathbf{9.597 \quad X-INVALID-ORDER-597} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 C_1 C_4 L_1 L_4 R_5 s^4 + 2 R_5 + s^3 (C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4) + s^2 (2 C_1 L_1 R_5 + 2 C_4 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4) + 2}$$

$$\mathbf{9.598 \quad X-INVALID-ORDER-598} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 s^3 + L_1 L_4 g_m s^2}{2 L_1 g_m s + s^4 (2 C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_4) + s^3 (2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_4 L_4 + 2 C_5 L_1 + C_5 L_4) + 2}$$

$$\mathbf{9.599 \quad X-INVALID-ORDER-599} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_5 s^3 + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 R_5 + s^4 (2 C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_4 R_5) + s^3 (C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_5 g_m) + s^2 (2 C_1 L_1 R_5 + 2 C_4 L_4 R_5 + 2 C_5 L_1 R_5 + C_5 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4) + 2}$$

$$\mathbf{9.600 \quad X-INVALID-ORDER-600} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4)}{2 C_1 C_4 C_5 L_1 L_4 R_5 s^5 + s^4 (2 C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4) + s^3 (2 C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m) + s^2 (2 C_1 L_1 + 2 C_4 L_4 + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + C_5 L_4) + s (2 C_5 R_5 + 2 L_1 g_m) + 2}$$

$$\mathbf{9.601 \quad X-INVALID-ORDER-601} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 g_m s^4 - C_5 L_1 L_4 s^3 + L_1 L_4 g_m s^2}{2 C_1 C_4 C_5 L_1 L_4 L_5 s^6 + 2 C_4 C_5 L_1 L_4 L_5 g_m s^5 + 2 L_1 g_m s + s^4 (2 C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_1 L_1 + 2 C_4 L_4 + 2 C_5 L_1 + C_5 L_4 + 2 C_5 L_5) + 2}$$

$$\mathbf{9.602 \quad X-INVALID-ORDER-602} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 s^3 + L_1 L_4 L_5 g_m s^2 - L_1 L_4 s}{2 L_1 + L_4 + 2 L_5 + s^4 (2 C_1 C_4 L_1 L_4 L_5 + C_1 C_5 L_1 L_4 L_5 + 2 C_4 C_5 L_1 L_4 L_5) + s^3 (2 C_4 L_1 L_4 L_5 g_m + 2 C_5 L_1 L_4 L_5 g_m) + s^2 (C_1 L_1 L_4 + 2 C_1 L_1 L_5 + 2 C_4 L_1 L_4 + 2 C_4 L_4 L_5 + 2 C_5 L_1 L_5 + C_5 L_4 L_5) + s (2 L_1 L_4 g_m + 2 L_1 L_5 g_m) + 2}$$

$$\mathbf{9.603 \quad X-INVALID-ORDER-603} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 g_m s^4 + L_1 L_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4)}{2 C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_4 R_5 + 2 C_4 L_1 L_4 g_m + 2 C_5 L_1 L_4 g_m + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_1 L_1 + 2 C_4 L_4 + 2 C_5 L_1 R_5 g_m + 2 C_5 L_1 + C_5 L_4) + s (2 L_1 R_5 g_m + 2 L_1 + L_4) + 2}$$

$$\mathbf{9.604 \quad X-INVALID-ORDER-604} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_5 s^3 - L_1 L_4 R_5 s + s^2 (L_1 L_4 L_5 R_5 g_m - L_1 L_4 L_5)}{2 L_1 R_5 + L_4 R_5 + 2 L_5 R_5 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_5 + C_1 C_5 L_1 L_4 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_5) + s^3 (C_1 L_1 L_4 L_5 + 2 C_4 L_1 L_4 L_5 R_5 g_m + 2 C_4 L_1 L_4 L_5 + 2 C_5 L_1 L_4 L_5 R_5 g_m) + s^2 (C_1 L_1 L_4 R_5 + 2 C_1 L_1 L_5 R_5 + 2 C_4 L_1 L_4 R_5 + 2 C_4 L_4 L_5 R_5 + 2 C_5 L_1 L_5 R_5 + C_5 L_4 L_5 R_5 + 2 L_1 L_4 L_5) + s (2 L_1 R_5 g_m + 2 L_1 L_5 g_m) + 2}$$

$$\mathbf{9.605 \quad X-INVALID-ORDER-605} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 L_5 g_m s^3 + s^4 (C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 + 2 R_5 + s^5 (2 C_1 C_4 L_1 L_4 L_5 + C_1 C_5 L_1 L_4 L_5 + 2 C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5) + s^4 (2 C_1 C_4 L_1 L_4 R_5 + 2 C_1 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 g_m + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_4 + 2 C_1 L_1 L_5 + 2 C_4 L_1 L_4 R_5 g_m + 2 C_4 L_1 L_4 + 2 C_4 L_4 L_5 R_5 + 2 C_5 L_1 L_5 R_5 + C_5 L_4 L_5 R_5 + 2 L_1 L_4 L_5) + s^2 (2 C_1 L_1 R_5 + 2 C_4 L_4 R_5 + 2 L_1 L_4 g_m) + s (2 L_1 R_5 g_m + 2 L_1 + L_4) + 2}$$

$$\mathbf{9.606 \quad X-INVALID-ORDER-606} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_5 s^3 + s^4 (C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 + 2R_5 + s^5 (C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + 2C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_4 L_5 R_5 + 2C_5 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_4 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_5 L_1 L_4 R_5 g_m + C_5 L_1 L_4 L_5)} + 1$$

$$\mathbf{9.607 \quad X-INVALID-ORDER-607} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 L_1 L_4 s^4 + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + 2C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4) + s (C_4 R_4 + 2C_4 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.608 \quad X-INVALID-ORDER-608} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 s^3 + L_1 g_m + s^2 (-C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{C_1 C_4 C_5 L_1 L_4 s^4 + 2C_4 + C_5 + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 L_4 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4) + s (C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.609 \quad X-INVALID-ORDER-609} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_5 s^4 + s^3 (-C_4 C_5 L_1 R_4 R_5 + C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 - C_5 L_1 R_5) + s (L_1 R_5 g_m - L_1)}{C_1 C_4 C_5 L_1 L_4 R_5 s^5 + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_5 g_m) + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 + C_4 C_5 R_4 R_5 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1 + C_4 L_4 + C_5 L_1 L_4)} + 1$$

$$\mathbf{9.610 \quad X-INVALID-ORDER-610} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{C_1 C_4 C_5 L_1 L_4 s^4 + 2C_4 + C_5 + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_4 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.611 \quad X-INVALID-ORDER-611} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (-C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m) + s^2 (-C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m - C_5 L_1)}{2C_4 + C_5 + s^4 (C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5) + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4 + 2C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.612 \quad X-INVALID-ORDER-612} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 s^5 - L_1 s + s^4 (-C_4 C_5 L_1 L_5 R_4 + C_4 L_1 L_4 L_5 g_m) + s^3 (-C_4 L_1 L_4 + C_4 L_1 L_5 R_4 g_m - C_5 L_1 L_5) + s^2 (-C_4 L_1 R_4 + L_1 L_5 g_m)}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_1 R_4 + C_4 C_5 L_5 R_4 + 2C_4 L_1 L_4 g_m + 2C_4 L_1 L_5 g_m + 2C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + C_5 L_1 L_4 L_5)} + 1$$

$$\mathbf{9.613 \quad X-INVALID-ORDER-613} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 g_m s^4 + L_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m) + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 + C_5 + s^4 (C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5) + s^3 (C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^2 (2C_1 C_4 L_1 + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + C_4 C_5 L_4 + 2C_4 C_5 L_5) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

$$\mathbf{9.614 \quad X-INVALID-ORDER-614} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_5 s^5 - L_1 R_5 s + s^4 (-C_4 C_5 L_1 L_5 R_4 R_5 + C_4 L_1 L_4 L_5 R_5 g_m - C_4 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 + C_4 L_1 L_5 R_4 g_m - C_5 L_1 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 + R_5 + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 R_5 + C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_5 + C_1 C_4 L_1 L_5 R_4 + 2C_1 C_4 L_1 L_5 R_5 + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_5 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (C_1 C_4 L_1 R_4 R_5 + C_4 C_5 L_5 R_4 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5 R_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + C_5 L_1 L_4 L_5)} + 1$$

$$\mathbf{9.615 \quad X-INVALID-ORDER-615} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4 + C_4 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4 + C_4 L_1 L_5 R_4 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + C_4 C_5 L_5 R_4 + 2C_4 C_5 L_5 R_5 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_5 R_5 g_m + 2C_5 L_1 L_5 R_5 g_m) + s^2 (C_1 L_1 + 2C_4 L_1 R_4 g_m + C_5 L_1 L_4 L_5)} + 1$$

$$\mathbf{9.616 \quad X-INVALID-ORDER-616} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^2 (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4)}{C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^3 (C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s^2 (C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + s (C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + (-C_4 C_5 L_1 L_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4)}$$

$$\mathbf{9.617 \quad X-INVALID-ORDER-617} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4)}{2 C_1 C_4 L_1 L_4 R_4 R_5 s^4 + 2 R_4 R_5 + s^3 (C_1 L_1 L_4 R_4 + 2 C_1 L_1 L_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_4) + s^2 (2 C_1 L_1 R_4 R_5 + 2 C_4 L_4 R_4 R_5 + 2 L_1 L_4 R_4 g_m + 2 L_1 L_4 R_5 g_m + 2 L_1 L_4) + s (2 L_1 R_4 R_5 g_m + 2 L_1 R_4 + L_4 R_4 + 2 L_4 R_5)}$$

$$\mathbf{9.618 \quad X-INVALID-ORDER-618} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_4 s^3 + L_1 L_4 R_4 g_m s^2}{2 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 R_4 + 2 C_4 C_5 L_1 L_4 R_4) + s^3 (2 C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4) + s^2 (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + s (2 L_1 R_4 g_m + 2 L_4)}$$

$$\mathbf{9.619 \quad X-INVALID-ORDER-619} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_4 R_5 s^3 + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4)}{2 R_4 R_5 + s^4 (2 C_1 C_4 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_4 R_4 R_5 + 2 C_4 C_5 L_1 L_4 R_4 R_5) + s^3 (C_1 L_1 L_4 R_4 + 2 C_1 L_1 L_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_4 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5) + s^2 (2 C_1 L_1 R_4 R_5 + 2 C_4 L_4 R_4 R_5 + 2 C_5 L_1 R_4 R_5 + C_5 L_4 R_4 R_5 + 2 L_1 L_4 R_4 g_m + 2 L_1 L_4 R_5 g_m + 2 L_1 L_4 R_5) + s (2 C_1 L_1 R_4 R_5 + 2 C_4 L_4 R_4 R_5 + 2 C_5 L_1 R_4 R_5 + C_5 L_4 R_4 R_5 + 2 L_1 L_4 R_4 g_m + 2 L_1 L_4 R_5 g_m + 2 L_1 L_4 R_5) + (-C_5 L_1 L_4 R_4 R_5 s^3 + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4))}$$

$$\mathbf{9.620 \quad X-INVALID-ORDER-620} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_4 g_m s^2 + s^3 (C_5 L_1 L_4 R_4 R_5 g_m - C_5 L_1 L_4 R_4)}{2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + 2 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 R_4 + 2 C_1 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_4) + s^3 (2 C_1 C_5 L_1 R_4 R_5 + 2 C_1 L_1 L_4 + 2 C_4 C_5 L_4 R_4 R_5 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4) + s^2 (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + s (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + (-C_5 L_1 L_4 R_4 R_5 s^3 + s^2 (L_1 L_4 R_4 R_5 g_m - C_5 L_1 L_4 R_4))}$$

$$\mathbf{9.621 \quad X-INVALID-ORDER-621} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_4 g_m s^4 - C_5 L_1 L_4 R_4 s^3 + L_1 L_4 R_4 g_m s^2}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + 2 R_4 + s^5 (2 C_1 C_5 L_1 L_4 L_5 + 2 C_4 C_5 L_1 L_4 L_5 R_4 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 R_4 + 2 C_1 C_5 L_1 L_5 R_4 + 2 C_4 C_5 L_1 L_4 R_4 + 2 C_4 C_5 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_1 L_1 L_4 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 + 2 C_5 L_1 L_5 R_4 g_m) + s^2 (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + s (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + (-C_5 L_1 L_4 L_5 R_4 s^4 + C_5 L_1 L_4 R_4 s^3 - L_1 L_4 R_4 g_m s^2)}$$

$$\mathbf{9.622 \quad X-INVALID-ORDER-622} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_4 s^3 + L_1 L_4 L_5 R_4 g_m s^2 - L_1 L_4 R_4 s}{2 L_1 R_4 + L_4 R_4 + 2 L_5 R_4 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_4 + C_1 C_5 L_1 L_4 L_5 R_4 + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^3 (2 C_1 L_1 L_4 L_5 + 2 C_4 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5) + s^2 (C_1 L_1 L_4 R_4 + 2 C_1 L_1 L_5 R_4 + 2 C_4 L_1 L_4 R_4 + 2 C_4 L_4 L_5 R_4 + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4 + 2 L_1 L_4) + s (C_1 L_1 L_4 R_4 + 2 C_1 L_1 L_5 R_4 + 2 C_4 L_1 L_4 R_4 + 2 C_4 L_4 L_5 R_4 + 2 C_5 L_1 L_5 R_4 + C_5 L_4 L_5 R_4 + 2 L_1 L_4) + (-C_5 L_1 L_4 L_5 R_4 s^3 + L_1 L_4 L_5 R_4 g_m s^2 - L_1 L_4 R_4 s)}$$

$$\mathbf{9.623 \quad X-INVALID-ORDER-623} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_4 g_m s^4 + L_1 L_4 R_4 g_m s^3 - C_5 L_1 L_4 L_5 R_4 s^2 + L_1 L_4 R_4 s}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + 2 R_4 + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 + 2 C_1 C_5 L_1 L_4 L_5 + 2 C_4 C_5 L_1 L_4 L_5 R_4 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_4 + C_1 C_5 L_1 L_4 R_4 + 2 C_1 C_5 L_1 L_4 R_5 + 2 C_1 C_5 L_1 L_5 R_4 + 2 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_4 + 2 C_4 C_5 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_1 C_5 L_1 R_4 R_5 + 2 C_1 L_1 L_4 + 2 C_4 C_5 L_4 R_4 R_5 + 2 C_4 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_4 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4) + s^2 (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + s (2 C_1 L_1 R_4 + 2 C_4 L_4 R_4 + 2 C_5 L_1 R_4 + C_5 L_4 R_4 + 2 L_1 L_4 g_m) + (-C_5 L_1 L_4 L_5 R_4 s^4 + L_1 L_4 R_4 g_m s^3 - C_5 L_1 L_4 L_5 R_4 s^2 + L_1 L_4 R_4 s)}$$

$$\mathbf{9.624 \quad X-INVALID-ORDER-624} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_4 R_5 s^3 - L_1 L_4 R_4 R_5 s + s^2 (L_1 L_4 L_5 R_4 R_5 g_m - L_1 L_4 R_4 R_5)}{2 L_1 R_4 R_5 + L_4 R_4 R_5 + 2 L_5 R_4 R_5 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_4 R_5 + C_1 C_5 L_1 L_4 L_5 R_4 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_4 R_5) + s^3 (C_1 L_1 L_4 L_5 R_4 + 2 C_1 L_1 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_1 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_4 L_5 R_5) + s^2 (C_1 L_1 L_4 R_4 R_5 + 2 C_1 L_1 L_5 R_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4 R_5) + s (C_1 L_1 L_4 R_4 R_5 + 2 C_1 L_1 L_5 R_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4 R_5) + (-C_5 L_1 L_4 L_5 R_4 R_5 s^3 - L_1 L_4 R_4 R_5 s + s^2 (L_1 L_4 L_5 R_4 R_5 g_m - L_1 L_4 R_4 R_5))}$$

$$\mathbf{9.625 \quad X-INVALID-ORDER-625} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + 2 R_4 R_5 + s^5 (2 C_1 C_4 L_1 L_4 L_5 R_4 + C_1 C_5 L_1 L_4 L_5 R_4 + 2 C_1 C_5 L_1 L_4 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_4 R_5 + 2 C_1 L_1 L_4 L_5 + 2 C_4 C_5 L_4 L_5 R_4 R_5 + 2 C_4 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_5) + s^3 (C_1 L_1 L_4 L_5 R_4 + 2 C_1 L_1 L_4 L_5 R_5 + 2 C_4 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 L_1 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_5 L_1 L_4 L_5 R_5) + s^2 (C_1 L_1 L_4 R_4 R_5 + 2 C_1 L_1 L_5 R_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4 R_5) + s (C_1 L_1 L_4 R_4 R_5 + 2 C_1 L_1 L_5 R_4 R_5 + 2 C_4 L_1 L_4 R_4 R_5 + 2 C_4 L_4 L_5 R_4 R_5 + 2 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_5 g_m + 2 C_5 L_1 L_4 R_5) + (-C_5 L_5 R_5 s^2 + L_5 s + R_5)}$$

$$\mathbf{9.626 \quad X-INVALID-ORDER-626} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + 2R_4 R_5 + s^5 (C_1 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_4 R_4 R_5 + 2C_1 C_5 L_1 L_5 R_4 R_5 + 2C_4 C_5 L_1 L_4 R_4 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 + 2C_5 L_1 L_4 L_5 R_4 g_m + 2C_5 L_1 L_4 L_5 R_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + 2R_4 R_5 + s^5 (C_1 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2C_1 C_4 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_4 R_4 R_5 + 2C_1 C_5 L_1 L_5 R_4 R_5 + 2C_4 C_5 L_1 L_4 R_4 R_5 + 2C_4 C_5 L_4 L_5 R_4 R_5 + 2C_5 L_1 L_4 L_5 R_4 g_m + 2C_5 L_1 L_4 L_5 R_5)}$$

$$\mathbf{9.627 \quad X-INVALID-ORDER-627} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5) + s^3 (C_1 L_1 L_4 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + C_4 L_4 R_4 + 2C_4 L_4 R_5 + 2L_1 L_4 g_m) + s (2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1 + L_4)}$$

$$\mathbf{9.628 \quad X-INVALID-ORDER-628} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 s^4 + L_1 R_4 g_m s + s^3 (C_4 L_1 L_4 R_4 g_m - C_5 L_1 L_4) + s^2 (-C_5 L_1 R_4 + L_1 L_4 g_m)}{C_1 C_4 C_5 L_1 L_4 R_4 s^5 + s^4 (2C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4) + s^3 (C_1 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.629 \quad X-INVALID-ORDER-629} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 R_5 s^4 + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4 - C_5 L_1 L_4 R_5) + s^2 (-C_5 L_1 R_4 R_5 + L_1 L_4 R_5 g_m - L_1 L_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + R_4 + 2R_5 + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_5) + s^3 (C_1 C_5 L_1 R_4 R_5 + C_1 L_1 L_4 + C_4 C_5 L_4 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 + 2C_5 L_1 L_4 R_5 g_m) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + C_4 L_4 R_4 + 2C_4 L_4 R_5 + 2L_1 L_4 g_m) + s (L_1 R_4 R_5 g_m - L_1 R_4)}$$

$$\mathbf{9.630 \quad X-INVALID-ORDER-630} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4) + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 + L_1 L_4 g_m)}{s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4) + s^3 (C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.631 \quad X-INVALID-ORDER-631} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (-C_4 C_5 L_1 L_4 R_4 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_4 g_m - C_5 L_1 L_4 + C_5 L_1 L_5 R_4 g_m) + s^2 (-C_5 L_1 R_4 + L_1 L_4 g_m)}{2C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m + 2C_5 L_1 L_5 g_m) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.632 \quad X-INVALID-ORDER-632} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 s^5 - L_1 R_4 s + s^4 (C_4 L_1 L_4 L_5 R_4 g_m - C_5 L_1 L_4 L_5) + s^3 (-C_4 L_1 L_4 R_4 - C_5 L_1 L_5 R_4 + L_1 L_4 L_5 g_m) + s^2 (-L_1 L_4 + L_1 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + R_4 + s^5 (2C_1 C_4 L_1 L_4 L_5 + C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 + C_1 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_4 + 2C_4 L_1 L_4 L_5 g_m + 2C_5 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_4 + 2C_1 L_1 L_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_4 L_5) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.633 \quad X-INVALID-ORDER-633} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_4 R_5 g_m - C_5 L_1 L_4 + C_5 L_1 L_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_4 L_5) + s^3 (C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.634 \quad X-INVALID-ORDER-634} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^5 - L_1 R_4 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_4 g_m - C_5 L_1 L_4 L_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + R_4 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 R_5 + C_1 C_5 L_1 L_4 L_5 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_5 R_4 R_5 + C_1 L_1 L_4 L_5 + C_4 C_5 L_4 L_5 R_4 R_5 + 2C_4 L_1 L_4 L_5 R_4 g_m + 2C_4 L_1 L_4 L_5 R_5 g_m + 2C_4 L_1 L_4 L_5) + s^3 (C_1 L_1 L_4 + 2C_1 L_1 L_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_4 L_5) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

$$\mathbf{9.635 \quad X-INVALID-ORDER-635} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (C_4 L_1 L_4 L_5 R_4 g_m + C_5 L_1 L_4 L_5 R_5 g_m - C_5 L_1 L_4 L_5)}{R_4 + 2R_5 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 L_1 L_4 L_5 + C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_4 + 2C_4 C_5 L_4 L_5) + s^3 (C_1 L_1 L_4 + 2C_1 L_1 L_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_4 L_5) + s^2 (2C_1 L_1 + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1 + C_5 L_4) + s (C_5 R_4 + 2L_1 g_m) + 2}$$

9.636 X-INVALID-ORDER-636 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4 + 2R_5 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^3 (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^2 (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5)}{s^5 (C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_4 + 2R_5 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 + C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^3 (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^2 (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s (C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + C_1 C_4 L_1 L_4 R_4 + C_1 C_4 L_1 L_4 R_5 + C_1 C_5 L_1 L_4 R_5 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5)}$$

9.637 X-INVALID-ORDER-637 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{R_4 + 2R_5 + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_1 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 + 2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4 + C_4 L_4 R_4 + 2C_4 L_4 R_5) + s (2C_4 R_4 R_5 + 2L_1 R_4 g_m + 2L_1 R_5 g_m + 2L_1)}$$

9.638 X-INVALID-ORDER-638 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 s^4 + C_4 L_1 L_4 R_4 g_m s^3 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s}{C_1 C_4 C_5 L_1 L_4 R_4 s^5 + s^4 (2C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 L_1 L_4 g_m) + s^2 (2C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_4 + 2C_5 L_1 R_4 g_m + 2C_5 L_1) + s (2C_4 R_4 + C_5 R_4 + 2L_1 g_m) + 2}$$

$$\text{9.639 X-INVALID-ORDER-639 } Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_4 R_5 s^4 - C_5 L_1 R_4 R_5 s^2 + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + R_4 + 2R_5 + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_1 R_4 R_5 + C_1 C_5 L_1 R_4 R_5 + 2C_4 C_5 L_1 R_4 R_5 + C_4 C_5 L_4 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (C_1 L_1 R_4 + 2C_1 L_1 R_5 +$$

9.640 X-INVALID-ORDER-640 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_4 R_4 g_m s^3 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_4) + s^2 (C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4)}{s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 R_5 g_m + 2C_4 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_4 R_5 + 2C_4 L_1 L_4 g_m}$$

9.641 X-INVALID-ORDER-641 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 - C_4 C_5 L_1 L_4 R_4 s^4 - C_5 L_1 R_4 s^2 + L_1 R_4 g_m s + s^3 (C_4 L_1 L_4 R_4 g_m + C_5 L_1 L_5 R_4 g_m)}{2C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 R_4 + C_1 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_4 + 2C_4 C_5 L_5 R_4)}$$

9.642 X-INVALID-ORDER-642 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 s^5 + C_4 L_1 L_4 L_5 R_4 g_m s^4 + L_1 L_5 R_4 g_m s^2 - L_1 R_4 s + s^3 (-C_4 L_1 L_4 R_4 - C_5 L_1 L_5 R_4)}{C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + R_4 + s^5 (2C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_5 R_4 + C_1 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_4 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (2C_1 L_1 L_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_4 L_1 L_5 R_4) + s^2 (C_1 L_1 L_4 R_4 + C_1 L_1 L_5 R_4 + C_4 L_1 L_4 R_4 g_m + C_4 L_1 L_5 R_4 g_m) + s (C_1 L_1 R_4 + C_4 L_1 R_4 g_m) + R_4}$$

9.643 X-INVALID-ORDER-643 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_4 R_5 g_m}{2 C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 + 2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_4 R_5 + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 R_4 g_m + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4$$

9.644 X-INVALID-ORDER-644 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^5 - L_1 R_4 F}{C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5 s^6 + R_4 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_4 + 2 C_1 C_4 L_1 L_4 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 + 2 C_1 C_4 L_1 L_5 R_4 R_5 + C_1 C_5 L_1 L_5 R_4 R_5 + 2 C_4 C_5 L_1 L_5 R_4 R_5 + C_4 C_5 L_4 L_5 R_4 R_5 + 2 C_4 L_1 L_4 L_5 R_4 g_m + 2 C_4 L_1 L_4 L_5 R_5)}$$

$$\mathbf{9.654 \quad X-INVALID-ORDER-654} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4) + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4 + C_1 L_1 L_5 R_4 g_m) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4) + s (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_5 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_5 R_1 R_5 g_m + 2 C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2 C_1 C_5 L_5 R_5 + 2 C_1 L_1 L_5 g_m) + s^2 (2 C_1 L_1 R_4 g_m + 2 C_1 L_1 R_5 g_m + 2 C_1 L_1 + 2 C_1 L_5 R_1 g_m + 2 C_1 L_5 + 2 C_5 L_5 R_4 g_m + 2 C_5 L_5 R_4) + s (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}$$

$$\mathbf{9.655 \quad X-INVALID-ORDER-655} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_1 C_5 L_1 L_5 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_4) + s^3 (-C_1 C_5 L_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_5 R_1 R_4) + s^2 (-C_1 C_5 R_1 R_4 R_5 + C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4) + s (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}{2 R_4 g_m + 2 R_5 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_5 g_m + 2 C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_5 L_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_5 + 2 C_1 C_5 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_5 R_1 R_5 g_m + 2 C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_4 + 2 C_1 C_5 L_5 R_5) + s^2 (2 C_1 C_5 R_1 R_4 R_5 g_m + 2 C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5 + 2 C_1 L_1 R_4 R_5 g_m + 2 C_1 L_1 R_5 + C_1 L_5 R_1 R_4 g_m + C_5 L_5 R_4 R_5 g_m - C_5 L_5 R_4)}$$

$$\mathbf{9.656 \quad X-INVALID-ORDER-656} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2 g_m + s^3 (2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1) + s^2 (2 C_1 C_4 R_1 R_5 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_5 + 2 C_1 L_1 g_m) + s (2 C_1 R_1 g_m + C_1 + 2 C_4 R_5 g_m + 2 C_4)}$$

$$\mathbf{9.657 \quad X-INVALID-ORDER-657} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 s^3 + g_m + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m) + s (C_1 R_1 g_m - C_5)}{2 C_1 C_4 C_5 L_1 s^4 + s^3 (2 C_1 C_4 C_5 R_1 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.658 \quad X-INVALID-ORDER-658} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 s^3 + R_5 g_m + s^2 (-C_1 C_5 R_1 R_5 + C_1 L_1 R_5 g_m - C_1 L_1) + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2 C_1 C_4 C_5 L_1 R_5 s^4 + 2 g_m + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_1 C_4 L_1 R_5 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_5 g_m) + s^2 (2 C_1 C_4 R_1 R_5 g_m + 2 C_1 C_4 R_1 + 2 C_1 C_4 R_5 + 2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2 C_1 L_1 g_m + 2 C_4 C_5 R_5) + s (2 C_1 R_1 g_m + C_1 + 2 C_4 R_5 g_m + 2 C_4 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.659 \quad X-INVALID-ORDER-659} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1) + s^3 (2 C_1 C_4 C_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 R_1 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.660 \quad X-INVALID-ORDER-660} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 + g_m + s^3 (-C_1 C_5 L_1 + C_1 C_5 L_5 R_1 g_m) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 C_5 R_1 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.661 \quad X-INVALID-ORDER-661} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 s^4 + s^3 (-C_1 C_5 L_5 R_1 + C_1 L_1 L_5 g_m) + s^2 (-C_1 L_1 + C_1 L_5 R_1 g_m - C_5 L_5) + s (-C_1 R_1 + L_5 g_m) - 1}{2 C_1 C_4 C_5 L_1 L_5 s^5 + 2 g_m + s^4 (2 C_1 C_4 C_5 L_5 R_1 + 2 C_1 C_4 L_1 L_5 g_m + 2 C_1 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 + 2 C_1 C_4 L_5 R_1 g_m + 2 C_1 C_4 L_5 + 2 C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5 + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + 2 C_1 L_1 g_m + 2 C_4 L_5 g_m + 2 C_5 L_5 g_m) + s (2 C_1 R_1 g_m + C_1 + 2 C_4)}$$

$$\mathbf{9.662 \quad X-INVALID-ORDER-662} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 + g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_1 C_5 L_5 R_1 g_m) + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_1 L_1 g_m + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{2 C_1 C_4 C_5 L_1 L_5 g_m s^5 + s^4 (2 C_1 C_4 C_5 L_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 + 2 C_1 C_4 C_5 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_5) + s^3 (2 C_1 C_4 C_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 R_1 + 2 C_1 C_4 C_5 R_5 + 2 C_1 C_4 L_1 g_m + 2 C_1 C_5 L_1 g_m + 2 C_4 C_5 L_5 g_m) + s^2 (2 C_1 C_4 R_1 g_m + 2 C_1 C_4 + 2 C_1 C_5 R_1 g_m + C_1 C_5 + 2 C_4 C_5 R_5 g_m + 2 C_4 C_5) + s (2 C_4 g_m + 2 C_5 g_m)}$$

$$\mathbf{9.690 \quad X-INVALID-ORDER-690} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 + C_1 C_4 C_5 L_4 L_5 R_1 g_m) + s^4 (-C_1 C_4 C_5 L_4 R_1 + C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 R_1 g_m - C_1 C_5 L_1 + C_1 C_5 L_5 R_1 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{s^5 (2C_1 C_4 C_5 L_1 L_4 g_m + 2C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2C_1 C_4 C_5 L_1 + 2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 + 2C_1 C_4 L_1 g_m + 2C_1 C_5 L_1 g_m + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.691 \quad X-INVALID-ORDER-691} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 s^6 + s^5 (-C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (-C_1 C_4 L_1 L_4 + C_1 C_4 L_4 L_5 R_1 g_m - C_1 C_5 L_1 L_5 - C_4 C_5 L_4 L_5) + s^3 (-C_1 C_4 L_4 R_1 - C_1 C_5 L_5 R_1 + C_1 L_1 L_5 g_m + C_4 L_4 L_5 g_m) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_4 L_1 L_5 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 + 2C_1 C_4 L_4 R_1 g_m + C_1 C_4 L_4 + 2C_1 C_4 L_5 R_1 g_m + 2C_1 C_4 L_5 + 2C_1 C_5 L_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.692 \quad X-INVALID-ORDER-692} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 + C_1 C_4 C_5 L_4 L_5 R_1 g_m) + s^4 (C_1 C_4 C_5 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_4 R_1 + C_1 C_4 L_1 L_4 g_m + C_1 C_5 L_1 L_5 g_m + C_4 C_5 L_4 L_5 g_m) + s^3 (C_1 C_4 L_4 R_1 g_m + C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1 + C_1 C_5 L_5 R_1 g_m + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{s^5 (2C_1 C_4 C_5 L_1 L_4 g_m + 2C_1 C_4 C_5 L_1 L_5 g_m) + s^4 (2C_1 C_4 C_5 L_1 R_5 g_m + 2C_1 C_4 C_5 L_1 + 2C_1 C_4 C_5 L_4 R_1 g_m + C_1 C_4 C_5 L_4 + 2C_1 C_4 C_5 L_5 R_1 g_m + 2C_1 C_4 C_5 L_5) + s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_1 C_4 L_1 g_m + 2C_1 C_5 L_1 g_m + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.693 \quad X-INVALID-ORDER-693} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_5 s^6 - R_5 + s^5 (-C_1 C_4 C_5 L_4 L_5 R_1 R_5 + C_1 C_4 L_1 L_4 L_5 R_5 g_m - C_1 C_4 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_5 + C_1 C_4 L_4 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m s^6 + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_5 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_4 L_5 R_5 + 2C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_5 R_5 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_4 L_4 L_5 + 2C_1 C_5 L_1 L_5 R_5 g_m + 2C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^3 (2C_1 C_4 C_5 R_1 R_5 g_m + 2C_1 C_4 C_5 R_1 + 2C_1 C_4 C_5 R_5 + 2C_1 C_4 L_1 g_m + 2C_1 C_5 L_1 g_m + 2C_4 C_5 L_4 g_m + 2C_4 C_5 L_5 g_m) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.694 \quad X-INVALID-ORDER-694} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_4 L_1 L_4 + C_1 C_4 L_4 L_5 R_1 g_m + C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_4 L_1 L_5 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.695 \quad X-INVALID-ORDER-695} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (-C_1 C_4 C_5 L_1 L_4 R_5 + C_1 C_4 C_5 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (-C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 L_1 L_4 R_5 g_m - C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5 R_5 g_m - C_4 C_5 L_4) + s^3 (C_1 C_4 L_4 R_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (-C_1 C_5 R_1 + C_1 L_1 g_m + C_4 L_4 g_m + C_5 L_5 g_m) + s (-C_1 C_4 R_1 g_m + C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_5 + 2C_1 C_4 C_5 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_5 R_1 + 2C_1 C_4 C_5 L_5 R_5 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_4 L_1 L_5 g_m + 2C_1 C_5 L_1 L_5 g_m + 2C_4 C_5 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 g_m + 2C_1 C_4 + 2C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.696 \quad X-INVALID-ORDER-696} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_5 g_m - C_1 L_1 L_4) + s^2 (C_1 L_4 R_1 R_5 g_m - C_1 L_4 R_1) + s (L_4 R_5 g_m - L_4)}{2R_5 g_m + s^4 (2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5 + 2C_1 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_5 g_m + 2C_1 L_1 + 2C_1 L_4 R_1 g_m + C_1 L_4 + 2C_4 L_4 R_5 g_m + 2C_4 L_4) + s (2C_1 R_1 R_5 g_m + 2C_1 R_1 + 2C_1 R_5 + 2L_4 g_m) + 2}$$

$$\mathbf{9.697 \quad X-INVALID-ORDER-697} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 s^4 + L_4 g_m s + s^3 (-C_1 C_5 L_4 R_1 + C_1 L_1 L_4 g_m) + s^2 (C_1 L_4 R_1 g_m - C_5 L_4)}{2C_1 C_4 C_5 L_1 L_4 s^5 + 2g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 L_1 L_4 g_m + 2C_1 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_1 + 2C_1 C_5 L_4 R_1 g_m + C_1 C_5 L_4 + 2C_4 C_5 L_4) + s^2 (2C_1 C_5 R_1 + 2C_1 L_1 g_m + 2C_4 L_4 g_m + 2C_5 L_4 g_m) + s (2C_1 R_1 g_m + 2C_1 + 2C_5)}$$

$$\mathbf{9.698 \quad X-INVALID-ORDER-698} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_5 s^4 + s^3 (-C_1 C_5 L_4 R_1 R_5 + C_1 L_1 L_4 R_5 g_m - C_1 L_1 L_4) + s^2 (C_1 L_4 R_1 R_5 g_m - C_1 L_4 R_1 - C_5 L_4 R_5) + s (L_4 R_5 g_m - L_4)}{2C_1 C_4 C_5 L_1 L_4 R_5 s^5 + 2R_5 g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_5 g_m) + s^3 (2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + 2C_1 C_4 L_4 R_5 + 2C_1 C_5 L_1 R_5 + 2C_1 C_5 L_4 R_1 R_5 g_m + C_1 C_5 L_4 R_5 + 2C_1 L_1 L_4 g_m + 2C_4 C_5 L_4 R_5) + s^2 (2C_1 C_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_1 + C_4 C_5 L_4 R_5 g_m - C_4 C_5 L_4) + s (2C_1 R_1 R_5 g_m + 2C_1 R_1 + 2C_1 R_5 + 2L_4 g_m) + 2}$$

9.717 X-INVALID-ORDER-717 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_4 s^4 + L_4 R_4 g_m s + s^3 (-C_1 C_5 L_4 R_1 R_4 + C_1 L_1 L_4 R_4 g_m) + s^2 (C_1 L_4 R_1 R_4 g_m - C_5 L_4 R_4)}{2C_1 C_4 C_5 L_1 L_4 R_4 s^5 + 2R_4 g_m + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_4 + 2C_1 C_5 L_1 R_4 + 2C_1 C_5 L_4 R_1 R_4 g_m + 2C_1 C_5 L_4 R_1 + C_1 C_5 L_4 R_4 + 2C_1 L_1 L_4 g_m + 2C_4 C_5 L_4 R_4) + s^2 (2C_1 C_5 R_1 R_4}$$

9.718 X-INVALID-ORDER-718 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_4 R_5 s^4 + s^3}{2C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + 2R_4 R_5 g_m + 2R_4 + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_4 R_5 + 2C_1 C_5 L_1 R_4 R_5 + 2C_1 C_5 L_4 R_1 R_4 R_5 g_m + 2C_1 C_5 L_4 R_1 R_4 R_5)}.$$

9.719 X-INVALID-ORDER-719 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_4 R_4 g_m s + s^4 (C_1 C_5 L_1 L_4 R_4)}{2 R_4 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_4 R_1 R_4 + 2 C_1 C_4 C_5 L_4 R_4 R_5 + 2 C_1 C_4 L_1 L_4 R_4 g_m + 2 C_1 C_5 L_1 L_4 R_4 g_m + 2 C_1 C_5 L_1 L_4 R_5 g_m + 2 C_1 C_5 L_1 L_4) + s^3 (2 C_1 C_4 L_4 R_1 R_4 g_m + 2 C_1 C_4 L_4 R_4 + 2 C_1 C_5 L_1 R_4 R_5 g_m)}$$

9.720 X-INVALID-ORDER-720 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_4 g_m s^5 + L_4 R_4 g_n}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + 2R_4 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 R_4 g_m + 2C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_4 g_m + 2C_1 C_5 L_4 L_5 R_1 g_m + 2C_1 C_5 L_4 L_5 -}$$

9.721 X-INVALID-ORDER-721 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_4 s^5 - L_4 R_4}{2C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 + 2R_4 + s^5 (2C_1 C_4 C_5 L_4 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 L_5 R_4 g_m + 2C_1 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_4 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_5 L_4 L_5 R_1 + C_1 C_5 L_4 L_5 R_4 + 2C_1 C_5 L_4 L_5 R_4 g_m)}$$

9.722 X-INVALID-ORDER-722 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_4g_ms^6 + 2R_4g_ms^5 + s^5(2C_1C_4C_5L_1L_4R_4R_5g_m + 2C_1C_4C_5L_1L_4R_4 + 2C_1C_4C_5L_4L_5R_1R_4g_m + 2C_1C_4C_5L_4L_5R_4 + 2C_1C_5L_1L_4L_5g_m) + s^4(2C_1C_4C_5L_4R_1R_4R_5g_m + 2C_1C_4C_5L_4R_1R_4 + 2C_1C_4C_5L_4R_4R_5 + 2C_1C_4L_1L_4R_4g_m + 2C_1C_5L_1L_4R_4g_r)}{1}$$

9.723 X-INVALID-ORDER-723 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_4R_5s^6 + 2R_4R_5 + s^5(2C_1C_4C_5L_4L_5R_1R_4R_5 + 2C_1C_4L_1L_4L_5R_4R_5g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_4R_5g_m + 2C_1C_5L_1L_4L_5R_5) + s^4(2C_1C_4L_1L_4R_4R_5 + 2C_1C_4L_4L_5R_1R_4R_5g_m + 2C_1C_4L_4L_5R_1R_4 + 2C_1C_4L_4L_5R_4R_5 + 2C_1C_5L_1L_5}$$

9.724 X-INVALID-ORDER-724 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_4R_5g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_4L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_4R_5)}$$

9.725 X-INVALID-ORDER-725 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_4R_5g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4C_5L_4R_1R_4)}{2R_4R_5g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4C_5L_4R_1R_4)}$$

9.736 X-INVALID-ORDER-736 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_4 R_5 g_m - R_4 + s^4 (C_1 C_4 L_1 L_4 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_4) + s^3 (C_1 C_4 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_4 R_1 R_4) + s^2 (C_1 L_1 R_4 R_5 g_m - C_1 L_1 R_4 + C_4 L_4 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^4 (2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 L_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_4 + 2C_1 C_4 L_4 R_1 R_4 g_m + 2C_1 C_4 L_4 R_1 R_5 g_m + 2C_1 C_4 L_4 R_1 + C_1 C_4 L_4 R_4 + 2C_1 C_4 L_4 R_5) + s^2 (2C_1 C_4 R_1 R_4 R_5 g_m + 2C_1 C_4 R_1 R_4 + 2C_1 C_4 R_4 R_5 + 2C_1}$$

9.737 X-INVALID-ORDER-737 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 s^5 + R_4 g_m + s^4 (-C_1 C_4 C_5 L_4 R_1 R_4 + C_1 C_4 L_1 L_4 R_4 g_m) + s^3 (C_1 C_4 L_4 R_1 R_4 g_m - C_1 C_5 L_1 R_4 - C_4 C_5 L_4 R_4) + s^2 (-C_1 C_5 R_4 g_m + C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + 2C_4 C_5 L_4 R_4 g_m)}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_4 g_m + 2C_1 C_4 L_4 R_1 g_m + 2C_1 C_4 L_4 + 2C_1 C_5 L_1 R_4 g_m + 2C_1 C_5 L_1 + 2C_4 C_5 L_4 R_4 g_m)}$$

9.738 X-INVALID-ORDER-738 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_4 R_5 s^5 + R_4 R_5 g_m - R_4 + 2R_4 g_m + 2R_5 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 + 2C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_4 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 R_5 g_m + 2C_1 C_4 L_1 L_4) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 R_4 R_5 g_m + 2$$

9.739 X-INVALID-ORDER-739 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (C_1 C_4 C_5 L_4 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_4 R_1 R_4 R_5)}{2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_4 R_5 + 2C_1 C_4 L_1 L_4 g_m) + s^3 (2C_1 C_4 C_5 R_1 R_4 R_5 g_m +$$

9.740 X-INVALID-ORDER-740 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + R_4 g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 R_4 + C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m) + s^4 (-2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m))}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_4 R_4 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_5 R_4 + 2C_1 C_4 C_5 L_5 R_1 R_4 g_m)}$$

9.741 X-INVALID-ORDER-741 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_4 s^6 - R_4 + s^5 (-C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 g_m)) + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_4 g_m + 2C_1 C_4 L_4 L_5)}{2R_4 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 g_m)} + s^4 (2C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_4 g_m + 2C_1 C_4 L_4 L_5)}$$

9.742 X-INVALID-ORDER-742 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_4 g_m s^6 + R_4 g_m + s}{2C_1 C_4 C_5 L_1 L_4 L_5 g_m s^6 + 2g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 g_m + 2C_1 C_4 C_5 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_4 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_4 R_1 R_5 g_m + 2C_1 C_4$$

9.743 X-INVALID-ORDER-743 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2R_4R_5g_m + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_4L_5R_1R_5 + C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_4L_1L_4L_5R_4g_m + 2C_1C_4L_1L_4L_5R_5g_m + 2C_1C_4L_1L_4L_5) + s^4(2C_1C_4C_5L_5R_1R_4R_5)}{2R_4R_5g_m + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_4L_5R_1R_5 + C_1C_4C_5L_4L_5R_4R_5 + 2C_1C_4L_1L_4L_5R_4g_m + 2C_1C_4L_1L_4L_5R_5g_m + 2C_1C_4L_1L_4L_5) + s^4(2C_1C_4C_5L_5R_1R_4R_5)}$$

9.744 X-INVALID-ORDER-744 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

[illegible]

9.745 X-INVALID-ORDER-745 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \frac{R_4(C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_4g_m + 2R_5g_m + s^6(2C_1C_4C_5L_1L_4L_5R_4g_m + 2C_1C_4C_5L_1L_4L_5R_5g_m + 2C_1C_4C_5L_1L_4L_5) + s^5(2C_1C_4C_5L_1L_4R_4R_5g_m + 2C_1C_4C_5L_1L_4R_5 + 2C_1C_4C_5L_1L_5R_4R_5g_m + 2C_1C_4C_5L_1L_5R_4 + 2C_1C_4C_5L_4L_5R_1R_4g_m + 2C_1C_4C_5L_4L_5R_1R_5g_m + 2C_1C_4C_5L_4L_5R_1 + 2C_1C_4C_5L_4L_5R_4 + 2C_1C_4C_5L_4L_5R_5 + 2C_1C_4C_5L_4L_5R_1R_4R_5)}{(s^2 + s(R_4 + R_5) + R_4R_5)(s^2 + s(R_1 + R_2 + R_3) + R_1R_2 + R_1R_3 + R_2R_3)(s^2 + s(R_4 + R_5) + R_4R_5)}$$

9.746 X-INVALID-ORDER-746 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 s^2 + L_1 R_1 R_4 g_m s}{C_1 C_5 L_1 R_1 R_4 s^3 + 2R_1 + s^2 (2C_1 L_1 R_1 + 2C_5 L_1 R_1 R_4 g_m + 2C_5 L_1 R_1 + C_5 L_1 R_4) + s (C_5 R_1 R_4 + 2L_1 R_1 g_m + 2L_1)}$$

9.747 X-INVALID-ORDER-747 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 R_5 s^2 + s(L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{C_1 C_5 L_1 R_1 R_4 R_5 s^3 + R_1 R_4 + 2R_1 R_5 + s^2(C_1 L_1 R_1 R_4 + 2C_1 L_1 R_1 R_5 + 2C_5 L_1 R_1 R_4 R_5 g_m + 2C_5 L_1 R_1 R_5 + C_5 L_1 R_4 R_5) + s(C_5 R_1 R_4 R_5 + 2L_1 R_1 R_4 g_m + 2L_1 R_1 R_5 g_m + 2L_1 R_1 + L_1 R_4 + 2L_1 R_5)}$$

9.748 X-INVALID-ORDER-748 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 R_1 R_4 g_m s + s^2 (C_5 L_1 R_1 R_4 R_5 g_m - C_5 L_1 R_1 R_4)}{2R_1 + s^3 (C_1 C_5 L_1 R_1 R_4 + 2C_1 C_5 L_1 R_1 R_5) + s^2 (2C_1 L_1 R_1 + 2C_5 L_1 R_1 R_4 g_m + 2C_5 L_1 R_1 R_5 g_m + 2C_5 L_1 R_1 + C_5 L_1 R_4 + 2C_5 L_1 R_5) + s (C_5 R_1 R_4 + 2C_5 R_1 R_5 + 2L_1 R_1 g_m + 2L_1)}$$

9.749 X-INVALID-ORDER-749 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_1 L_5 R_1 R_4 g_m s^3 - C_5 L_1 R_1 R_4 s^2 + L_1 R_1 R_4 g_m s}{2C_1 C_5 L_1 L_5 R_1 s^4 + 2R_1 + s^3 (C_1 C_5 L_1 R_1 R_4 + 2C_5 L_1 L_5 R_1 g_m + 2C_5 L_1 L_5) + s^2 (2C_1 L_1 R_1 + 2C_5 L_1 R_1 R_4 g_m + 2C_5 L_1 R_1 + C_5 L_1 R_4 + 2C_5 L_5 R_1) + s (C_5 R_1 R_4 + 2L_1 R_1 g_m + 2L_1)}$$

9.750 X-INVALID-ORDER-750 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_4 s^3 + L_1 L_5 R_1 R_4 g_m s^2 - L_1 R_1 R_4 s}{C_1 C_5 L_1 L_5 R_1 R_4 s^4 + R_1 R_4 + s^3 (2C_1 L_1 L_5 R_1 + 2C_5 L_1 L_5 R_1 R_4 g_m + 2C_5 L_1 L_5 R_1 + C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_1 R_4 + C_5 L_5 R_1 R_4 + 2L_1 L_5 R_1 g_m + 2L_1 L_5) + s (2L_1 R_1 R_4 g_m + 2L_1 R_1 + L_1 R_4 + 2L_5 R_1)}$$

9.751 X-INVALID-ORDER-751 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_1 L_5 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s + s^2 (C_5 L_1 R_1 R_4 R_5 g_m - C_5 L_1 R_1 R_4)}{2C_1 C_5 L_1 L_5 R_1 s^4 + 2R_1 + s^3 (C_1 C_5 L_1 R_1 R_4 + 2C_1 C_5 L_1 R_1 R_5 + 2C_5 L_1 L_5 R_1 g_m + 2C_5 L_1 L_5) + s^2 (2C_1 L_1 R_1 + 2C_5 L_1 R_1 R_4 g_m + 2C_5 L_1 R_1 R_5 g_m + 2C_5 L_1 R_1 + C_5 L_1 R_4 + 2C_5 L_1 R_5 + 2C_5 L_5 R_1) + s (C_5 R_1 R_4 + 2C_5 R_1 R_5 + 2L_1 R_1 g_m + 2L_1)}$$

9.752 X-INVALID-ORDER-752 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_4 R_5 s^3 - L_1 R_1 R_4 R_5 s + s^2 (L_1 L_5 R_1 R_4 R_5 g_m - L_1 L_5 R_1 R_4)}{C_1 C_5 L_1 L_5 R_1 R_4 R_5 s^4 + R_1 R_4 R_5 + s^3 (C_1 L_1 L_5 R_1 R_4 + 2C_1 L_1 L_5 R_1 R_5 + 2C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_5 L_1 L_5 R_1 R_5 + C_5 L_1 L_5 R_4 R_5) + s^2 (C_1 L_1 R_1 R_4 R_5 + C_5 L_5 R_1 R_4 R_5 + 2L_1 L_5 R_1 R_4 g_m + 2L_1 L_5 R_1 R_5 g_m + 2L_1 L_5 R_1 + L_1 L_5 R_4 + 2L_1 L_5 R_5) + s (2L_1 R_1 R_4 R_5 g_m + 2L_1 R_1 R_4 R_5)}$$

9.753 X-INVALID-ORDER-753 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, R_4, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{L_1 L_5 R_1 R_4 g_m s^2 + s^3 (C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_5 L_1 L_5 R_1 R_4) + s (L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_5 L_1 L_5 R_1 R_5) + s^3 (2C_1 L_1 L_5 R_1 + 2C_5 L_1 L_5 R_1 R_4 g_m + 2C_5 L_1 L_5 R_1 R_5 g_m + 2C_5 L_1 L_5 R_1 + C_5 L_1 L_5 R_4 + 2C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_1 R_4 + 2C_1 L_1 R_1 R_5 + C_5 L_5 R_1 R_4 + 2C_5 L_5 R_1 R_5 + 2L_1 L_5 R_1 g_m + 2L_1 L_5) + s (2L_1 R_1 R_4 + 2L_1 R_1 R_5)}$$

$$\mathbf{9.754 \quad X-INVALID-ORDER-754} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 R_5 s^2 + s^3 (C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_5 L_1 L_5 R_1 R_4) + s (L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_5 L_1 L_5 R_1 R_5) + s^3 (C_1 C_5 L_1 R_1 R_4 R_5 + 2C_5 L_1 L_5 R_1 R_4 g_m + 2C_5 L_1 L_5 R_1 R_5 g_m + 2C_5 L_1 L_5 R_1 + C_5 L_1 L_5 R_4 + 2C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_1 R_4 + 2C_1 L_1 R_1 R_5 + 2C_5 L_1 R_1 R_4 R_5 g_m + 2C_5 L_1 R_1 R_5 + C_5 L_1 R_4 R_5 + C_5 L_5 R_1 R_4)}$$

$$\mathbf{9.755 \quad X-INVALID-ORDER-755} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_1 R_1 R_5 g_m - L_1 R_1)}{2C_1 C_4 L_1 R_1 R_5 s^3 + R_1 + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 + 2C_4 L_1 R_5) + s (2C_4 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.756 \quad X-INVALID-ORDER-756} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_5 s^2 + s (L_1 R_1 R_5 g_m - L_1 R_1)}{R_1 + s^3 (2C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_5) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 + 2C_4 L_1 R_5 + 2C_5 L_1 R_1 R_5 g_m + C_5 L_1 R_5) + s (2C_4 R_1 R_5 + C_5 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.757 \quad X-INVALID-ORDER-757} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 g_m + s (C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{2C_1 C_4 C_5 L_1 R_1 R_5 s^3 + 2C_4 R_1 + C_5 R_1 + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_5) + s (2C_4 C_5 R_1 R_5 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.758 \quad X-INVALID-ORDER-758} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 g_m s^2 - C_5 L_1 R_1 s + L_1 R_1 g_m}{2C_1 C_4 C_5 L_1 L_5 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 + 2C_4 C_5 L_5 R_1) + s (2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.759 \quad X-INVALID-ORDER-759} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 s^3 + L_1 L_5 R_1 g_m s^2 - L_1 R_1 s}{R_1 + s^4 (2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1) + s^3 (2C_4 L_1 L_5 R_1 g_m + 2C_4 L_1 L_5 + 2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 + 2C_4 L_5 R_1 + C_5 L_5 R_1) + s (2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.760 \quad X-INVALID-ORDER-760} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 g_m s^2 + L_1 R_1 g_m + s (C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{2C_1 C_4 C_5 L_1 L_5 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_5 + 2C_4 C_5 L_5 R_1) + s (2C_4 C_5 R_1 R_5 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.761 \quad X-INVALID-ORDER-761} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_5 s^3 - L_1 R_1 R_5 s + s^2 (L_1 L_5 R_1 R_5 g_m - L_1 L_5 R_1)}{R_1 R_5 + s^4 (2C_1 C_4 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 R_5) + s^3 (C_1 L_1 L_5 R_1 + 2C_4 L_1 L_5 R_1 R_5 g_m + 2C_4 L_1 L_5 R_1 + 2C_4 L_1 L_5 R_5 + 2C_5 L_1 L_5 R_1 R_5 g_m + C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_1 R_5 + 2C_4 L_1 R_1 R_5 + 2C_4 L_5 R_1 R_5 + C_5 L_5 R_1 R_5 + 2L_1 L_5 R_1 g_m + L_1 L_5) + s (L_1 R_1 R_5 g_m + L_1 R_1)}$$

$$\mathbf{9.762 \quad X-INVALID-ORDER-762} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_1 g_m s^2 + s^3 (C_5 L_1 L_5 R_1 R_5 g_m - C_5 L_1 L_5 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{2C_1 C_4 C_5 L_1 L_5 R_1 R_5 s^5 + R_1 + s^4 (2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_5 + 2C_4 C_5 L_5 R_1 R_5 + 2C_4 L_1 L_5 R_1 g_m + 2C_4 L_1 L_5 + 2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 R_5 + 2C_4 L_5 R_1 R_5 + C_5 L_5 R_1 R_5 + 2L_1 L_5 R_1 g_m + L_1 L_5) + s (L_1 R_1 R_5 g_m + L_1 R_1)}$$

$$\mathbf{9.763 \quad X-INVALID-ORDER-763} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_1 R_5 g_m - C_5 L_1 L_5 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 s^5 + R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 + 2 C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_1 + 2 C_4 L_1 R_5) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.764 \quad X-INVALID-ORDER-764} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s (L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{2 C_1 C_4 L_1 R_1 R_4 R_5 s^3 + R_1 R_4 + 2 R_1 R_5 + s^2 (C_1 L_1 R_1 R_4 + 2 C_1 L_1 R_1 R_5 + 2 C_4 L_1 R_1 R_4 R_5 g_m + 2 C_4 L_1 R_1 R_4 + 2 C_4 L_1 R_4 R_5) + s (2 C_4 R_1 R_4 R_5 + 2 L_1 R_1 R_4 g_m + 2 L_1 R_1 R_5 g_m + 2 L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.765 \quad X-INVALID-ORDER-765} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 s^2 + L_1 R_1 R_4 g_m s}{2 R_1 + s^3 (2 C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2 C_4 C_5 L_1 R_1 R_4) + s^2 (2 C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4 + 2 C_5 L_1 R_1 R_4 g_m + 2 C_5 L_1 R_1 + C_5 L_1 R_4) + s (2 C_4 R_1 R_4 + C_5 R_1 R_4 + 2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.766 \quad X-INVALID-ORDER-766} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 R_5 s^2 + s (L_1 R_1 R_4 R_5 g_m - L_1 R_1 R_4)}{R_1 R_4 + 2 R_1 R_5 + s^3 (2 C_1 C_4 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 R_1 R_4 R_5 + 2 C_4 C_5 L_1 R_1 R_4 R_5) + s^2 (C_1 L_1 R_1 R_4 + 2 C_1 L_1 R_1 R_5 + 2 C_4 L_1 R_1 R_4 R_5 g_m + 2 C_4 L_1 R_1 R_4 + 2 C_4 L_1 R_4 R_5 + 2 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_5 L_1 R_1 R_5 + C_5 L_1 R_4 R_5) + s (2 C_4 R_1 R_4 R_5 + C_5 R_1 R_4 R_5 + 2 L_1 R_1 R_4 g_m + 2 L_1 R_1 R_5 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.767 \quad X-INVALID-ORDER-767} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 R_4 g_m s + s^2 (C_5 L_1 R_1 R_4 R_5 g_m - C_5 L_1 R_1 R_4)}{2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 s^4 + 2 R_1 + s^3 (2 C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2 C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_1 R_4 + 2 C_4 C_5 L_1 R_4 R_5) + s^2 (2 C_1 L_1 R_1 + 2 C_4 C_5 R_1 R_4 R_5 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4 + 2 C_5 L_1 R_1 R_4 g_m + 2 C_5 L_1 R_1 R_5 g_m + 2 C_5 L_1 R_1 + C_5 L_1 R_4) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.768 \quad X-INVALID-ORDER-768} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 R_4 g_m s^3 - C_5 L_1 R_1 R_4 s^2 + L_1 R_1 R_4 g_m s}{2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 + 2 R_1 + s^4 (2 C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_4) + s^3 (2 C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2 C_4 C_5 L_1 R_1 R_4 + 2 C_4 C_5 L_5 R_1 R_4 + 2 C_5 L_1 L_5 R_1 g_m + 2 C_5 L_1 L_5) + s^2 (2 C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_4 g_m + 2 C_4 L_1 R_4 + 2 C_5 L_1 R_1 R_4 g_m + 2 C_5 L_1 R_1 R_5 g_m + 2 C_5 L_1 R_1 + C_5 L_1 R_4) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.769 \quad X-INVALID-ORDER-769} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_4 s^3 + L_1 L_5 R_1 R_4 g_m s^2 - L_1 R_1 R_4 s}{R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_5 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 R_4 + 2 C_4 C_5 L_1 L_5 R_1 R_4) + s^3 (2 C_1 L_1 L_5 R_1 + 2 C_4 L_1 L_5 R_1 R_4 g_m + 2 C_4 L_1 L_5 R_4 + 2 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_5 L_1 L_5 R_1 + C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_1 R_4 + 2 C_4 L_1 R_1 R_4 + 2 C_4 L_5 R_1 R_4 + C_5 L_5 R_1 R_4 + 2 L_1 L_5 R_1 g_m + 2 L_1 L_5) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.770 \quad X-INVALID-ORDER-770} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g_m s + s^2 (C_5 L_1 R_1 R_4 R_5 g_m - C_5 L_1 R_1 R_4 R_5)}{2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 + 2 R_1 + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_4) + s^3 (2 C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2 C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_1 R_4 + 2 C_4 C_5 L_5 R_1 R_4 + 2 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_5 L_1 L_5 R_1 + C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_1 R_4 + 2 C_4 L_1 R_1 R_4 + 2 C_4 L_5 R_1 R_4 + C_5 L_5 R_1 R_4 + 2 L_1 L_5 R_1 g_m + 2 L_1 L_5) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.771 \quad X-INVALID-ORDER-771} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_4 R_5 s^3 - L_1 R_1 R_4 R_5 s + s^2 (L_1 L_5 R_1 R_4 R_5 g_m - L_1 L_5 R_1 R_4)}{R_1 R_4 R_5 + s^4 (2 C_1 C_4 L_1 L_5 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 R_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_4 R_5) + s^3 (C_1 L_1 L_5 R_1 R_4 + 2 C_1 L_1 L_5 R_1 R_5 + 2 C_4 L_1 L_5 R_1 R_4 R_5 g_m + 2 C_4 L_1 L_5 R_1 R_4 + 2 C_4 L_1 L_5 R_4 R_5 + 2 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2 C_5 L_1 L_5 R_1 R_5 + C_5 L_1 L_5 R_4 R_5) + s^2 (C_1 L_1 R_1 R_4 R_5 + 2 C_4 L_1 R_1 R_4 R_5 + 2 C_4 L_5 R_1 R_4 R_5 + C_5 L_5 R_1 R_4 R_5 + 2 L_1 L_5 R_1 g_m + 2 L_1 L_5) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.772 \quad X-INVALID-ORDER-772} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_1 R_4}{2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 + R_1 R_4 + 2 R_1 R_5 + s^4 (2 C_1 C_4 L_1 L_5 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_4 C_5 L_1 L_5 R_4 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_4 R_5 + 2 C_1 L_1 L_5 R_1 + 2 C_4 C_5 L_5 R_1 R_4 R_5 + 2 C_4 L_1 L_5 R_1 R_4 g_m + 2 C_4 L_1 L_5 R_1 + C_5 L_1 L_5 R_4) + s^2 (C_1 L_1 R_1 R_4 R_5 + 2 C_4 L_1 R_1 R_4 R_5 + 2 C_4 L_5 R_1 R_4 R_5 + C_5 L_5 R_1 R_4 R_5 + 2 L_1 L_5 R_1 g_m + 2 L_1 L_5) + s (L_1 R_1 + L_1 R_4 + 2 L_1 R_5)}$$

$$\mathbf{9.773 \quad X-INVALID-ORDER-773} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_4 R_5 s^2 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 + R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 R_4 + 2C_4 C_5 L_1 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 + 2C_5 L_1 L_5 R_1 R_4 g_m + 2C_5 L_1 L_5 R_1 R_4 R_5)}{2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 + R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 R_4 + 2C_4 C_5 L_1 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_5 R_1 R_4 R_5 + 2C_5 L_1 L_5 R_1 R_4 g_m + 2C_5 L_1 L_5 R_1 R_4 R_5)}$$

$$\mathbf{9.774 \quad X-INVALID-ORDER-774} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (C_4 L_1 R_1 R_4 R_5 g_m - C_4 L_1 R_1 R_4) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 + C_4 L_1 R_4 + 2C_4 L_1 R_5) + s (C_4 R_1 R_4 + 2C_4 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.775 \quad X-INVALID-ORDER-775} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 R_1 R_4 s^2 + L_1 R_1 g_m + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 R_1 R_4 s^3 + 2C_4 R_1 + C_5 R_1 + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4) + s (C_4 C_5 R_1 R_4 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.776 \quad X-INVALID-ORDER-776} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 R_1 R_4 R_5 s^3 + s^2 (C_4 L_1 R_1 R_4 R_5 g_m - C_4 L_1 R_1 R_4 - C_5 L_1 R_1 R_5) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_4 C_5 L_1 R_1 R_4 R_5 s^4 + R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_1 R_5 + C_4 C_5 L_1 R_4 R_5) + s^2 (C_1 L_1 R_1 + C_4 C_5 R_1 R_4 R_5 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 + C_4 L_1 R_4 + 2C_4 L_1 R_5 + 2C_5 L_1 R_1 R_5 g_m + 2C_5 L_1 R_1 R_4 R_5)}$$

$$\mathbf{9.777 \quad X-INVALID-ORDER-777} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 g_m + s^2 (C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_4 C_5 L_1 R_1 R_4) + s (C_4 L_1 R_1 R_4 g_m + C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{2C_4 R_1 + C_5 R_1 + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 C_5 L_1 R_1 R_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_5) + s (C_4 C_5 R_1 R_4 + 2C_4 C_5 R_1 R_5 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.778 \quad X-INVALID-ORDER-778} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_1 R_4 g_m s^3 + L_1 R_1 g_m + s^2 (-C_4 C_5 L_1 R_1 R_4 + C_5 L_1 L_5 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{2C_1 C_4 C_5 L_1 L_5 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + 2C_4 C_5 L_5 R_1) + s (C_4 C_5 R_1 R_4 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.779 \quad X-INVALID-ORDER-779} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_1 R_4 s^4 - L_1 R_1 s + s^3 (C_4 L_1 L_5 R_1 R_4 g_m - C_5 L_1 L_5 R_1) + s^2 (-C_4 L_1 R_1 R_4 + L_1 L_5 R_1 g_m)}{C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 + R_1 + s^4 (2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_4 C_5 L_5 R_1 R_4 + 2C_4 L_1 L_5 R_1 g_m + 2C_4 L_1 L_5 + 2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_1)}$$

$$\mathbf{9.780 \quad X-INVALID-ORDER-780} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_5 R_1 R_4 g_m s^3 + L_1 R_1 g_m + s^2 (C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_4 C_5 L_1 R_1 R_4 + C_5 L_1 L_5 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m + C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{2C_1 C_4 C_5 L_1 L_5 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_5 + 2C_4 C_5 L_5 R_1) + s (C_4 C_5 R_1 R_4 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.781 \quad X-INVALID-ORDER-781} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^4 - L_1 R_1 R_5 s + s^3 (C_4 L_1 L_5 R_1 R_4 R_5 g_m - C_4 L_1 L_5 R_1)}{C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 + R_1 R_5 + s^4 (C_1 C_4 L_1 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 R_5 + C_4 C_5 L_1 L_5 R_4 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 + C_1 L_1 L_5 R_1 + C_4 C_5 L_5 R_1 R_4 R_5 + 2C_4 L_1 L_5 R_1 R_4 g_m + 2C_4 L_1 L_5 R_1 R_5 g_m + 2C_4 L_1 L_5 R_1 R_4 R_5)}$$

$$\mathbf{9.782 \quad X-INVALID-ORDER-782} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_1 R_4) + s^3 (C_4 L_1 L_5 R_1 R_4 g_m + C_5 L_1 L_5 R_1 R_5 g_m - C_5 L_1 L_5 R_1) + s^2 (C_4 L_1 L_5 R_1 R_4 R_5 g_m + C_4 L_1 L_5 R_1 R_4 R_5)}{R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5) + s^4 (2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_5 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5 + C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_5 R_1 R_5)}$$

$$\mathbf{9.783 \quad X-INVALID-ORDER-783} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^4 (C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_1 R_4) + s^3 (-C_4 C_5 L_1 R_1 R_4 R_5) + s^2 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 + 2 C_4 C_5 L_1 L_5 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_4 C_5 L_1 R_1 R_5 + C_4 C_5 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_1 + 2 C_4 L_1 R_5 + C_4 L_4 R_1) + s (2 C_4 R_1 R_5 + 2 L_1 R_1 g_m + L_1)}{R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 + 2 C_4 C_5 L_1 L_5 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 + 2 C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_4 C_5 L_1 R_1 R_5 + C_4 C_5 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_1 + 2 C_4 L_1 R_5 + C_4 L_4 R_1) + s (2 C_4 R_1 R_5 + 2 L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.784 \quad X-INVALID-ORDER-784} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_1 R_5 g_m - C_4 L_1 L_4 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_4 L_1 L_4 R_1 s^4 + R_1 + s^3 (2 C_1 C_4 L_1 R_1 R_5 + 2 C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_1 + 2 C_4 L_1 R_5 + C_4 L_4 R_1) + s (2 C_4 R_1 R_5 + 2 L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.785 \quad X-INVALID-ORDER-785} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 s^3 + C_4 L_1 L_4 R_1 g_m s^2 - C_5 L_1 R_1 s + L_1 R_1 g_m}{C_1 C_4 C_5 L_1 L_4 R_1 s^4 + 2 C_4 R_1 + C_5 R_1 + s^3 (2 C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4) + s^2 (2 C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_1 + C_4 C_5 L_4 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.786 \quad X-INVALID-ORDER-786} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_5 s^4 - C_5 L_1 R_1 R_5 s^2 + s^3 (C_4 L_1 L_4 R_1 R_5 g_m - C_4 L_1 L_4 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_4 C_5 L_1 L_4 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 R_1 R_5 + C_4 C_5 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_1 + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1) + s (2 C_4 R_1 R_5 + 2 L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.787 \quad X-INVALID-ORDER-787} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 R_1 g_m s^2 + L_1 R_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_4 C_5 L_1 L_4 R_1) + s (C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 R_1 s^4 + 2 C_4 R_1 + C_5 R_1 + s^3 (2 C_1 C_4 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4) + s^2 (2 C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_4 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_1) + s (2 C_4 C_5 R_1 R_5 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.788 \quad X-INVALID-ORDER-788} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 g_m s^4 - C_4 C_5 L_1 L_4 R_1 s^3 - C_5 L_1 R_1 s + L_1 R_1 g_m + s^2 (C_4 L_1 L_4 R_1 g_m + C_5 L_1 L_5 R_1 g_m)}{2 C_4 R_1 + C_5 R_1 + s^4 (C_1 C_4 C_5 L_1 L_4 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_1) + s^3 (2 C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_4 C_5 L_1 L_5) + s^2 (2 C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_1 + C_4 C_5 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.789 \quad X-INVALID-ORDER-789} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 s^5 + C_4 L_1 L_4 L_5 R_1 g_m s^4 + L_1 L_5 R_1 g_m s^2 - L_1 R_1 s + s^3 (-C_4 L_1 L_4 R_1 - C_5 L_1 L_5 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_4 L_5 R_1) + s^3 (2 C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4 + 2 C_4 L_1 L_5 R_1 g_m + 2 C_4 L_1 L_5 + 2 C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 + 2 C_4 L_1 R_1 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.790 \quad X-INVALID-ORDER-790} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 g_m s^4 + L_1 R_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_4 C_5 L_1 L_4 R_1) + s^2 (C_4 L_1 L_4 R_1 g_m + C_5 L_1 L_5 R_1 g_m) + s (C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{2 C_4 R_1 + C_5 R_1 + s^4 (C_1 C_4 C_5 L_1 L_4 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_1) + s^3 (2 C_1 C_4 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_4 C_5 L_1 L_5) + s^2 (2 C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_4 C_5 L_1 R_1 + 2 C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.791 \quad X-INVALID-ORDER-791} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^5 - L_1 R_1 R_5 s + s^4 (C_4 L_1 L_4 L_5 R_1 R_5 g_m - C_4 L_1 L_4 L_5 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 + R_1 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 + 2 C_1 C_4 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_1 R_5 + 2 C_4 L_1 L_4 L_5 R_1 g_m + C_4 L_1 L_4 L_5) + s^3 (C_1 L_1 L_5 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s^2 (C_1 L_1 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.792 \quad X-INVALID-ORDER-792} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_4 L_1 L_4 L_5 R_1 g_m s^4 + L_1 L_5 R_1 g_m s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_1) + s^3 (C_4 L_1 L_4 R_1 R_5 g_m - C_4 L_1 L_4 L_5 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + 2 C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_1) + s^3 (2 C_1 C_4 L_1 R_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_5 + C_4 C_5 L_4 L_5 R_1) + s^2 (C_1 L_1 L_5 R_1 + 2 C_4 L_1 R_1 R_5 g_m + 2 C_4 L_1 R_5 + C_4 L_4 R_1 + 2 C_4 C_5 L_5 R_1) + s (2 C_4 L_1 R_1 g_m + 2 C_4 L_1 + 2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.793 \quad X-INVALID-ORDER-793} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + \frac{1}{C_4 s}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_5 s^4 - C_5 L_1 R_1 R_5 s^2 + s^5 (C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_5 L_1 L_4 L_5 R_1 R_5) + C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_4 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_1 R_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_4 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_5 R_5 + C_4 C_5 L_4 L_5 R_1 R_5)}$$

$$\mathbf{9.794 \quad X-INVALID-ORDER-794} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^2 (L_1 L_4 R_1 R_5 g_m - L_1 L_4 R_1)}{2 C_1 C_4 L_1 L_4 R_1 R_5 s^4 + 2 R_1 R_5 + s^3 (C_1 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_1 R_5 g_m + 2 C_4 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_5) + s^2 (2 C_1 L_1 R_1 R_5 + 2 C_4 L_4 R_1 R_5 + 2 L_1 L_4 R_1 g_m + L_1 L_4) + s (2 L_1 R_1 R_5 g_m + 2 L_1 R_1 + 2 L_1 R_5 + L_4 R_1)}$$

$$\mathbf{9.795 \quad X-INVALID-ORDER-795} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_1 s^3 + L_1 L_4 R_1 g_m s^2}{2 R_1 + s^4 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_1 L_4 R_1) + s^3 (2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_1 g_m + C_5 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + 2 C_4 L_4 R_1 + 2 C_5 L_1 R_1 + C_5 L_4 R_1) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.796 \quad X-INVALID-ORDER-796} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_1 R_5 s^3 + s^2 (L_1 L_4 R_1 R_5 g_m - L_1 L_4 R_1)}{2 R_1 R_5 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_5) + s^3 (C_1 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_1 R_5 g_m + 2 C_4 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_5 + 2 C_5 L_1 L_4 R_1 R_5 g_m + C_5 L_1 L_4 R_5) + s^2 (2 C_1 L_1 R_1 R_5 + 2 C_4 L_4 R_1 R_5 + 2 C_5 L_1 R_1 R_5 + C_5 L_4 R_1 R_5 + 2 L_1 L_4 R_1 g_m + L_1 L_4) + s (2 L_1 R_1 R_5 g_m + 2 L_1 R_1 + 2 L_1 R_5 + L_4 R_1)}$$

$$\mathbf{9.797 \quad X-INVALID-ORDER-797} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 R_1 g_m s^2 + s^3 (C_5 L_1 L_4 R_1 R_5 g_m - C_5 L_1 L_4 R_1)}{2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + 2 R_1 + s^4 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_1 L_4 R_5) + s^3 (2 C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_1 g_m + C_5 L_1 L_4) + s^2 (2 C_1 L_1 R_1 + 2 C_4 L_4 R_1 + 2 C_5 L_1 R_1 + C_5 L_4 R_1) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.798 \quad X-INVALID-ORDER-798} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_1 g_m s^4 - C_5 L_1 L_4 R_1 s^3 + L_1 L_4 R_1 g_m s^2}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2 R_1 + s^5 (2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_1 L_4 L_5) + s^4 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2 C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_4 L_5 R_1) + s^3 (2 C_4 L_1 L_4 R_1 g_m + 2 C_4 L_1 L_4 + 2 C_5 L_1 L_4 R_1 g_m + C_5 L_1 L_4 + 2 C_5 L_1 L_5 R_1 g_m + 2 C_5 L_1 L_5) + s^2 (2 C_1 L_1 R_1 + 2 C_4 L_4 R_1 + 2 C_5 L_1 R_1 + C_5 L_4 R_1) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.799 \quad X-INVALID-ORDER-799} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_1 s^3 + L_1 L_4 L_5 R_1 g_m s^2 - L_1 L_4 R_1 s}{2 L_1 R_1 + L_4 R_1 + 2 L_5 R_1 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_1 + C_1 C_5 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_1) + s^3 (2 C_4 L_1 L_4 L_5 R_1 g_m + 2 C_4 L_1 L_4 L_5 + 2 C_5 L_1 L_4 L_5 R_1 g_m + C_5 L_1 L_4 L_5) + s^2 (C_1 L_1 L_4 R_1 + 2 C_1 L_1 L_5 R_1 + 2 C_4 L_1 L_4 R_1 + 2 C_4 L_4 L_5 R_1 + 2 C_5 L_1 L_5 R_1 + C_5 L_4 L_5 R_1) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.800 \quad X-INVALID-ORDER-800} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_1 g_m s^4 + L_1 L_4 R_1 g_m s^2 + s^3 (C_5 L_1 L_4 R_1 R_5 g_m - C_5 L_1 L_4 R_1 R_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2 R_1 + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_1 L_4 L_5) + s^4 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2 C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_4 L_5 R_1) + s^3 (2 C_1 C_5 L_1 R_1 R_5 + 2 C_4 C_5 L_4 R_1 R_5) + s^2 (2 C_1 L_1 R_1 R_5 + 2 C_4 L_4 R_1 R_5 + 2 C_5 L_1 R_1 R_5 + C_5 L_4 R_1 R_5) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.801 \quad X-INVALID-ORDER-801} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_1 R_5 s^3 - L_1 L_4 R_1 R_5 s + s^2 (L_1 L_4 L_5 R_1 R_5 g_m - L_1 L_4 L_5 R_1)}{2 L_1 R_1 R_5 + L_4 R_1 R_5 + 2 L_5 R_1 R_5 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_1 R_5 + C_1 C_5 L_1 L_4 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_5) + s^3 (C_1 L_1 L_4 L_5 R_1 + 2 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2 C_4 L_1 L_4 L_5 R_1 + 2 C_4 L_1 L_4 L_5 R_5 + 2 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_5 L_1 L_4 L_5 R_5) + s^2 (C_1 L_1 L_4 R_1 R_5 + 2 C_1 L_1 L_5 R_1 R_5 + 2 C_4 L_4 R_1 R_5 + 2 C_5 L_1 R_1 R_5 + C_5 L_4 R_1 R_5) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.802 \quad X-INVALID-ORDER-802} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_4 L_5 R_1}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2 R_1 R_5 + s^5 (2 C_1 C_4 L_1 L_4 L_5 R_1 + C_1 C_5 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_5 + 2 C_1 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_4 L_5 R_1 R_5 + 2 C_4 L_1 L_4 L_5 R_1 g_m + 2 C_4 L_1 L_4 L_5 + 2 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_5 L_1 L_4 L_5 R_5) + s^3 (C_1 L_1 L_4 L_5 R_1 + 2 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2 C_4 L_1 L_4 L_5 R_1 + 2 C_4 L_1 L_4 L_5 R_5 + 2 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_5 L_1 L_4 L_5 R_5) + s^2 (C_1 L_1 L_4 R_1 R_5 + 2 C_1 L_1 L_5 R_1 R_5 + 2 C_4 L_4 R_1 R_5 + 2 C_5 L_1 R_1 R_5 + C_5 L_4 R_1 R_5) + s (2 L_1 R_1 g_m + 2 L_1)}$$

$$\mathbf{9.803 \quad X-INVALID-ORDER-803} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad \frac{L_4 s}{C_4 L_4 s^2 + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_4 R_1 R_5 s^3 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 + 2R_1 R_5 + s^5 (C_1 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_5 + 2C_5 L_1 L_4 L_5 R_1 g_m + C_5 L_1 L_4 L_5 R_1)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 + 2R_1 R_5 + s^5 (C_1 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 R_5 + 2C_4 C_5 L_4 L_5 R_1 R_5 + 2C_5 L_1 L_4 L_5 R_1 g_m + C_5 L_1 L_4 L_5 R_1)}$$

$$\mathbf{9.804 \quad X-INVALID-ORDER-804} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{s^3 (C_4 L_1 L_4 R_1 R_5 g_m - C_4 L_1 L_4 R_1) + s^2 (C_4 L_1 R_1 R_4 R_5 g_m - C_4 L_1 R_1 R_4) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_4 L_1 L_4 R_1 s^4 + R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5 + 2C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_1 R_5 g_m + 2C_4 L_1 R_1 + C_4 L_1 R_4 + 2C_4 L_1 R_5 + C_4 L_4 R_1) + s (C_4 R_1 R_4 + 2C_4 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.805 \quad X-INVALID-ORDER-805} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 s^3 + L_1 R_1 g_m + s^2 (-C_4 C_5 L_1 R_1 R_4 + C_4 L_1 L_4 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_1) + s (C_4 C_5 R_1 R_4 + 2C_4 L_1 R_1 g_m + 2C_4 L_1 + 2C_5 L_1 R_1 g_m + C_5 L_1)}$$

$$\mathbf{9.806 \quad X-INVALID-ORDER-806} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_5 s^4 + s^3 (-C_4 C_5 L_1 R_1 R_4 R_5 + C_4 L_1 L_4 R_1 R_5 g_m - C_4 L_1 L_4 R_1) + s^2 (C_4 L_1 R_1 R_4 R_5 g_m - C_4 L_1 R_1 R_4) + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + R_1 + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_4 C_5 L_1 L_4 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5 + C_1 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 R_1 R_5 + C_4 C_5 L_1 R_4 R_5 + C_4 C_5 L_4 R_1 R_5 + 2C_4 L_1 L_4 R_1 g_m + C_4 L_1 L_4 R_1)}$$

$$\mathbf{9.807 \quad X-INVALID-ORDER-807} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{L_1 R_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_4 C_5 L_1 L_4 R_1) + s^2 (C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_4 C_5 L_1 R_1 R_4 + C_4 L_1 L_4 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m + C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 R_1 s^4 + 2C_4 R_1 + C_5 R_1 + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_5 + C_4 C_5 L_4 R_1) + s (C_4 C_5 R_1 R_4)}$$

$$\mathbf{9.808 \quad X-INVALID-ORDER-808} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 g_m s^4 + L_1 R_1 g_m + s^3 (-C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_5 R_1 R_4 g_m) + s^2 (-C_4 C_5 L_1 R_1 R_4 + C_4 L_1 L_4 R_1 g_m + C_5 L_1 L_5 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{2C_4 R_1 + C_5 R_1 + s^4 (C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_5 R_1) + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_1 + 2C_4 C_5 L_5 R_1) + s (C_4 C_5 R_1 R_4)}$$

$$\mathbf{9.809 \quad X-INVALID-ORDER-809} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 s^5 - L_1 R_1 s + s^4 (-C_4 C_5 L_1 L_5 R_1 R_4 + C_4 L_1 L_4 L_5 R_1 g_m) + s^3 (-C_4 L_1 L_4 R_1 + C_4 L_1 L_5 R_1 R_4 g_m) + s^2 (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 + C_4 C_5 L_4 L_5 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}$$

$$\mathbf{9.810 \quad X-INVALID-ORDER-810} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 g_m s^4 + L_1 R_1 g_m + s^3 (C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_5 R_1 R_4 g_m) + s^2 (C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_4 C_5 L_1 R_1 R_4 + C_4 L_1 L_4 R_1 g_m + C_5 L_1 L_5 R_1 g_m) + s (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{2C_4 R_1 + C_5 R_1 + s^4 (C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_5 R_1) + s^3 (C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_5 R_1 g_m + 2C_4 C_5 L_1 L_5) + s^2 (2C_1 C_4 L_1 R_1 + C_1 C_5 L_1 R_1 + 2C_4 C_5 L_1 R_1 R_4 g_m + 2C_4 C_5 L_1 R_1 R_5 g_m + 2C_4 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 + 2C_4 C_5 L_4 R_1 + 2C_4 C_5 L_5 R_1) + s (C_4 C_5 R_1 R_4)}$$

$$\mathbf{9.811 \quad X-INVALID-ORDER-811} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 s^5 - L_1 R_1 s + s^4 (-C_4 C_5 L_1 L_5 R_1 R_4 + C_4 L_1 L_4 L_5 R_1 g_m) + s^3 (-C_4 L_1 L_4 R_1 + C_4 L_1 L_5 R_1 R_4 g_m) + s^2 (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 R_5 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 + C_1 C_4 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_4 L_1 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 R_5 + C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}$$

$$\mathbf{9.812 \quad X-INVALID-ORDER-812} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \quad \infty, \quad \infty, \quad L_4 s + R_4 + \frac{1}{C_4 s}, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_4 C_5 L_1 L_4 L_5 R_1) + s^4 (C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_1 R_4 + C_4 L_1 L_4 L_5 R_1 g_m) + s^3 (C_4 L_1 R_1 R_4 R_5 g_m - C_4 L_1 R_1 R_4) + s^2 (C_4 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_4 L_5 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 + C_4 C_5 L_5 R_1 R_4 + 2C_4 C_5 L_1 R_1 R_4 g_m - C_5 L_1 R_1)}$$

9.813 X-INVALID-ORDER-813 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{s^5 (C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5)) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 + 2 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_4 C_5 L_1 L_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_1 R_4 R_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5)}$$

9.814 X-INVALID-ORDER-814 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s^2 (L_1 L_4 R_1 R_4 R_5 g_m - L_1 L_4 R_1 R_4)}{2 C_1 C_4 L_1 L_4 R_1 R_4 R_5 s^4 + 2 R_1 R_4 R_5 + s^3 (C_1 L_1 L_4 R_1 R_4 + 2 C_1 L_1 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_1 R_4 + 2 C_4 L_1 L_4 R_4 R_5) + s^2 (2 C_1 L_1 R_1 R_4 R_5 + 2 C_4 L_4 R_1 R_4 R_5 + 2 L_1 L_4 R_1 R_4 g_m + 2 L_1 L_4 R_1 R_5 g_m + 2 L_1 L_4 R_1 + L_1 L_4 R_4 + 2 L_1 L_4 R_5) + s (2 L_1 R_1 R_4 R_5 g_m + 2 L_1 R_1 R_4 R_5)}$$

9.815 X-INVALID-ORDER-815 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 L_4 R_1 R_4 s^3 + L_1 L_4 R_1 R_4 g_m s^2}{2 R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_5 L_1 L_4 R_1 R_4 + 2 C_4 C_5 L_1 L_4 R_1 R_4) + s^3 (2 C_1 L_1 L_4 R_1 + 2 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_4 + 2 C_5 L_1 L_4 R_1 R_4 g_m + 2 C_5 L_1 L_4 R_1 + C_5 L_1 L_4 R_4) + s^2 (2 C_1 L_1 R_1 R_4 + 2 C_4 L_4 R_1 R_4 + 2 C_5 L_1 R_1 R_4 + C_5 L_4 R_1 R_4 + 2 L_1 L_4 R_1 g_m + 2 L_1 L_4)}$$

9.816 X-INVALID-ORDER-816 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 L_4 R_1 R_4 R_5 s^3 + s^2 (L_1 L_4 R_1 R_4 R_5 g_m - L_1 L_4 R_1 R_4)}{2 R_1 R_4 R_5 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_4 R_5 + C_1 C_5 L_1 L_4 R_1 R_4 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5) + s^3 (C_1 L_1 L_4 R_1 R_4 + 2 C_1 L_1 L_4 R_1 R_5 + 2 C_4 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_1 R_4 + 2 C_4 L_1 L_4 R_4 R_5 + 2 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_5 L_1 L_4 R_1 R_5 + C_5 L_1 L_4 R_4 R_5) + s^2 (2 C_1 L_1 R_1 R_4 R_5 + 2 C_4 L_4 R_1 R_4 R_5 + 2 L_1 L_4 R_1 R_4 g_m + 2 L_1 L_4 R_1 R_5 g_m + 2 L_1 L_4 R_1 + L_1 L_4 R_4 + 2 L_1 L_4 R_5)}$$

9.817 X-INVALID-ORDER-817 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 L_4 R_1 R_4 g_m s^2 + s^3 (C_1 L_1 L_4 R_1 R_4 + C_1 L_1 L_4 R_1 R_5 + C_4 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 L_1 L_4 R_1 R_4 + 2 C_4 L_1 L_4 R_4 R_5)}{2 C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^5 + 2 R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_5 L_1 L_4 R_1 R_4 + 2 C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 R_4 + 2 C_4 C_5 L_1 L_4 R_4 R_5) + s^3 (2 C_1 C_5 L_1 R_1 R_4 R_5 + 2 C_1 L_1 L_4 R_1 + 2 C_4 C_5 L_4 R_1 R_4 R_5 + 2 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_4)}$$

9.818 X-INVALID-ORDER-818 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_1 R_4 g_m s^4}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^6 + 2 R_1 R_4 + s^5 (2 C_1 C_5 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_5 L_1 L_4 R_1 R_4 + 2 C_1 C_5 L_1 L_5 R_1 R_4 + 2 C_4 C_5 L_1 L_4 R_1 R_4 + 2 C_4 C_5 L_4 L_5 R_1 R_4 + 2 C_5 L_1 L_4 L_5 R_1 g_m + 2 C_5 L_1 L_4 L_5) + s^3 (2 C_1 L_1 L_4 R_1 R_4 R_5 + 2 C_4 L_4 R_1 R_4 R_5 + 2 L_1 L_4 R_1 R_4 g_m + 2 L_1 L_4 R_1 R_5 g_m + 2 L_1 L_4 R_1 + L_1 L_4 R_4 + 2 L_1 L_4 R_5)}$$

9.819 X-INVALID-ORDER-819 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 L_4 L_5 R_1 R_4 s^3 + L_1 L_4 L_5 R_1 R_4 g_m s^2 - L_1 L_4 R_1 R_4 s}{2 L_1 R_1 R_4 + L_4 R_1 R_4 + 2 L_5 R_1 R_4 + s^4 (2 C_1 C_4 L_1 L_4 L_5 R_1 R_4 + C_1 C_5 L_1 L_4 L_5 R_1 R_4 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_4) + s^3 (2 C_1 L_1 L_4 L_5 R_1 + 2 C_4 L_1 L_4 L_5 R_1 R_4 g_m + 2 C_4 L_1 L_4 L_5 R_4 + 2 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2 C_5 L_1 L_4 L_5 R_1 + C_5 L_1 L_4 L_5 R_4) + s^2 (C_1 L_1 L_4 R_1 R_4 + 2 C_1 L_1 L_5 R_1 R_4 + 2 C_4 L_4 R_1 R_4 R_5 + 2 L_1 L_4 R_1 R_4 g_m + 2 L_1 L_4 R_1 R_5 g_m + 2 L_1 L_4 R_1 + L_1 L_4 R_4 + 2 L_1 L_4 R_5)}$$

9.820 X-INVALID-ORDER-820 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_5 L_1 L_4 L_5 R_1 R_4 g_m s^4}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^6 + 2 R_1 R_4 + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 + 2 C_1 C_5 L_1 L_4 L_5 R_1 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_5 L_1 L_4 R_1 R_4 + 2 C_1 C_5 L_1 L_5 R_1 R_4 + 2 C_4 C_5 L_1 L_4 R_1 R_4 + 2 C_4 C_5 L_4 L_5 R_1 R_4 + 2 C_5 L_1 L_4 L_5 R_1 g$$

9.823 X-INVALID-ORDER-823 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1R_4R_5s^6 + 2R_1R_4R_5 + s^5(C_1C_5L_1L_4L_5R_1R_4 + 2C_1C_5L_1L_4L_5R_1R_5 + 2C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_4C_5L_1L_4L_5R_1R_4 + 2C_4C_5L_1L_4L_5R_4R_5) + s^4(2C_1C_4L_1L_4R_1R_4R_5 + C_1C_5L_1L_4R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_4R_5 + 2C_4C_5L_1L_4R_1R_4R_5 +$$

9.824 X-INVALID-ORDER-824 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{s^3(C_4L_1L_4R_1R_4R_5g_m - C_4L_1L_4R_1R_4) + s^2(L_1L_4R_1R_5g_m - L_1L_4R_1) + s(L_1R_1R_4R_5g_m - L_1R_1R_4)}{R_1R_4 + 2R_1R_5 + s^4(C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_1L_4R_1R_5) + s^3(C_1L_1L_4R_1 + 2C_4L_1L_4R_1R_4g_m + 2C_4L_1L_4R_1R_5g_m + 2C_4L_1L_4R_1 + C_4L_1L_4R_4 + 2C_4L_1L_4R_5) + s^2(C_1L_1R_1R_4 + 2C_1L_1R_1R_5 + C_4L_4R_1R_4 + 2C_4L_4R_1R_5 + 2L_1L_4R_1g_m + L_1L_4) + s(2L_1R_1R_4}$$

9.825 X-INVALID-ORDER-825 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_4 s^4 + L_1 R_1 R_4 g_m s + s^3 (C_4 L_1 L_4 R_1 R_4 g_m - C_5 L_1 L_4 R_1) + s^2 (-C_5 L_1 R_1 R_4 + L_1 L_4 R_1 g_m)}{C_1 C_4 C_5 L_1 L_4 R_1 R_4 s^5 + 2R_1 + s^4 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4) + s^3 (C_1 C_5 L_1 R_1 R_4 + C_4 C_5 L_4 R_1 R_4 + 2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4 + 2C_5 L_1 L_4 R_1 g_m + C_5 L_1 L_4) + s^2 (2C_1 L_1 R_1 + 2C_4 L_4 R_1 + 2C_5 L_1 R_1 R_4)}$$

9.826 X-INVALID-ORDER-826 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^4 + s^3 (C_4 L_1 L_4 R_1 R_4 R_5 g_m - C_4 L_1 L_4 R_1 R_4 R_5)}{C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^5 + R_1 R_4 + 2 R_1 R_5 + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5) + s^3 (C_1 C_5 L_1 R_1 R_4 R_5 + C_1 L_1 L_4 R_1 + C_4 C_5 L_4 R_1 R_4 R_5 + 2 C_4 L_1 L_4 R_1 R_4 g_m + 2 C_4 L_1 L_4 R_1 R_4 R_5) + s^2 (C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5) + s (C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5) + C_1 C_4 L_1 L_4 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_5 L_1 L_4 R_1 R_5 + 2 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5}$$

9.827 X-INVALID-ORDER-827 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 R_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m - C_4 C_5 L_1 L_4 R_1 R_4) + s^3 (C_4 L_1 L_4 R_1 R_4 g_m + C_5 L_1 L_4 R_1 R_5 g_m - C_4 C_5 L_1 L_4 R_1 R_4 R_5)}{2R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_4 R_5) + s^3 (C_1 C_5 L_1 R_1 R_4 + 2C_1 C_5 L_1 R_1 R_5 + C_4 C_5 L_4 R_1 R_4 + 2C_4 C_5 L_4 R_1 R_5)}$$

9.828 X-INVALID-ORDER-828 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^5 + L_1 R_1 R_4 g_m s + s^4 (-C_4 C_5 L_1 L_4 R_1 R_4 + C_5 L_1 L_4 L_5 R_1 g_m) + s^3 (C_4 L_1 L_4 R_1 R_4 g_m - C_5 L_1 L_4 L_5 R_1 g_m)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_4 L_5 R_1) + s^3 (C_1 C_5 L_1 R_1 R_4 + C_4 C_5 L_4 R_1 R_4)}$$

9.829 X-INVALID-ORDER-829 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^5 - L_1 R_1 R_4 s + s^4 (C_4 L_1 L_4 L_5 R_1 R_4 + C_5 L_1 L_4 L_5 R_1 R_4)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^6 + R_1 R_4 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_1 + C_1 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_4 q_m + 2C_4 C_5 L_1 L_4 L_5 R_1 + C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 R_4 + C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 L_1 L_4 L_5 R_1 q_m + 2C_4 L_1 L_4 L_5 + 2C_5 L_1 L_4 L_5 R_1 q_m + C_5 L_1 L_4 L_5 R_4)}$$

9.830 X-INVALID-ORDER-830 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^5 + L_1 R_1 R_4 g_m s + s^4 (C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m - 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5 + 2C_4 C_5 L_1 L_4 L_5 R_1 q_m + 2C_4 C_5 L_1 L_4 L_5)) + s^4 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 q_m + 2C_4 C_5 L_1 L_4 R_1 R_5 q_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_4 R_5)}{}$$

9.831 X-INVALID-ORDER-831 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 B_1 B_4 B_5 s^6 + B_1 B_4 B_5 + s^5 (C_1 C_4 L_1 L_4 L_5 B_1 B_4 + 2 C_1 C_4 L_1 L_4 L_5 B_1 B_5 + C_1 C_5 L_1 L_4 L_5 B_1 B_5 + 2 C_4 C_5 L_1 L_4 L_5 B_1 B_4 g_{\text{m}} + 2 C_4 C_5 L_1 L_4 L_5 B_1 B_5 + C_4 C_5 L_1 L_4 L_5 B_4 B_5)}{C_1 C_4 C_5 L_1 L_4 L_5 B_1 B_4 B_5 + B_1 B_4 B_5 + s^5 (C_1 C_4 L_1 L_4 L_5 B_1 B_4 + 2 C_1 C_4 L_1 L_4 L_5 B_1 B_5 + C_1 C_5 L_1 L_4 L_5 B_1 B_5 + 2 C_4 C_5 L_1 L_4 L_5 B_1 B_4 g_{\text{m}} + 2 C_4 C_5 L_1 L_4 L_5 B_1 B_5 + C_4 C_5 L_1 L_4 L_5 B_4 B_5)} + s^4 (C_1 C_4 L_1 L_4 B_1 B_4 B_5 + C_1 C_5 L_1 L_5 B_1 B_4 B_5 + C_1 L_1 L_4 L_5 B_1 + C_4 C_5 L_4 L_5 B_1 B_4 B_5 + C_4 C_5 L_4 L_5 B_4 B_5)$$

9.832 X-INVALID-ORDER-832 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_4 + 2R_1 R_5 + s^6 (C_1 C_4 C_5 L_3 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_3 L_4 L_5 R_1 R_5) + s^5 (2C_1 C_4 L_3 L_4 L_5 R_1 + C_1 C_5 L_3 L_4 L_5 R_1 + 2C_4 C_5 L_3 L_4 L_5 R_1 R_4 q_{\infty} + 2C_4 C_5 L_3 L_4 L_5 R_1 R_5 q_{\infty} + 2C_4 C_5 L_3 L_4 L_5 R_1 + C_4 C_5 L_3 L_4 L_5 R_4 + 2C_4 C_5 L_3 L_4 L_5 R_5) + s^4 (C_1 C_4 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_5) + s^3 (C_1 C_4 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_5) + s^2 (C_1 C_4 L_3 L_4 R_1 + C_1 C_4 L_3 L_4 R_4) + s (C_1 C_4 L_3 L_4 R_4 + C_1 C_4 L_3 L_4 R_5) + C_1 C_4 L_3 L_4 R_1}{s^6 (C_1 C_4 C_5 L_3 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_3 L_4 L_5 R_1 R_5) + s^5 (2C_1 C_4 L_3 L_4 L_5 R_1 + C_1 C_5 L_3 L_4 L_5 R_1 + 2C_4 C_5 L_3 L_4 L_5 R_1 R_4 q_{\infty} + 2C_4 C_5 L_3 L_4 L_5 R_1 R_5 q_{\infty} + 2C_4 C_5 L_3 L_4 L_5 R_1 + C_4 C_5 L_3 L_4 L_5 R_4 + 2C_4 C_5 L_3 L_4 L_5 R_5) + s^4 (C_1 C_4 L_3 L_4 R_1 R_4 + 2C_1 C_4 L_3 L_4 R_1 R_5) + s^3 (C_1 C_4 L_3 L_4 R_4 + 2C_1 C_4 L_3 L_4 R_5) + s^2 (C_1 C_4 L_3 L_4 R_1 + C_1 C_4 L_3 L_4 R_4) + s (C_1 C_4 L_3 L_4 R_4 + C_1 C_4 L_3 L_4 R_5) + C_1 C_4 L_3 L_4 R_1}.$$

9.833 X-INVALID-ORDER-833 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{1}{R_1 R_4 + 2R_1 R_5 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5) + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 + C_1 C_5 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_4 C_5 L_1 L_4 L_5 R_1 + C_4 C_5 L_1 L_4 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4$$

9.834 X-INVALID-ORDER-834 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{s^3(C_4L_1L_4R_1R_4R_5g_m - C_4L_1L_4R_1R_4) + s(L_1R_1R_4R_5g_m - L_1R_1R_4)}{R_1R_4 + 2R_1R_5 + s^4(C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_1L_4R_1R_5) + s^3(2C_1C_4L_1R_1R_4R_5 + 2C_4L_1L_4R_1R_4g_m + 2C_4L_1L_4R_1R_5g_m + 2C_4L_1L_4R_1 + C_4L_1L_4R_4 + 2C_4L_1L_4R_5) + s^2(C_1L_1R_1R_4 + 2C_1L_1R_1R_5 + 2C_4L_1R_1R_4R_5g_m + 2C_4L_1R_1R_4 + 2C_4L_1R_4R_5 + C_4L_4R_1)}$$

9.835 X-INVALID-ORDER-835 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_4 s^4 + C_4 L_1 L_4 R_1 R_4 g_m s^3 - C_5 L_1 R_1 R_4 s^2 + L_1 R_1 R_4 g_m s}{C_1 C_4 C_5 L_1 L_4 R_1 R_4 s^5 + 2R_1 + s^4(2C_1 C_4 L_1 L_4 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4) + s^3(2C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2C_4 C_5 L_1 R_1 R_4 + C_4 C_5 L_4 R_1 R_4 + 2C_4 L_1 L_4 R_1 g_m + 2C_4 L_1 L_4) + s^2(2C_1 L_1 R_1 + 2C_4 L_1 R_1 R_4 g_m + 2C_4 L_1 R_4 + 2C_4 L_1 L_4)}$$

9.836 X-INVALID-ORDER-836 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^4 - C_5 L_1 R_1 R_4 R_5 s^3 + C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^5 + R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 + C_4 C_5 L_4 R_1 R_4 R_5 + 2C_4 L_1 L_4 R_1 R_4 g_m + 2C_4 L_1 L_4 R_1 R_4 R_5)}{C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 s^5 + R_1 R_4 + 2R_1 R_5 + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 R_5 + 2C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 R_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 R_1 R_4 R_5 + 2C_4 C_5 L_1 R_1 R_4 R_5 + C_4 C_5 L_4 R_1 R_4 R_5 + 2C_4 L_1 L_4 R_1 R_4 g_m + 2C_4 L_1 L_4 R_1 R_4 R_5)}$$

9.837 X-INVALID-ORDER-837 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 L_1 L_4 R_1 R_4 g_m s^3 + L_1 R_1 R_4 g}{2R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 + C_1 C_5 L_1 R_1 R_4 + 2C_1 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_4 + 2C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_5 + 2C_4 C_5 L_1 R_1 R_5)}$$

9.838 X-INVALID-ORDER-838 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^5 - C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^4 + C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^3 - C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^2 + C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s - C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2R_1 + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_4 + 2C_4 L_1 L_4 L_5 R_1 R_4 g_m) + s^3 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_4 + 2C_4 L_1 L_4 L_5 R_1 R_4 g_m) + s^2 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_4 + 2C_4 L_1 L_4 L_5 R_1 R_4 g_m) + s (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_5 R_4 + 2C_4 L_1 L_4 L_5 R_1 R_4 g_m) - C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m}$$

9.839 X-INVALID-ORDER-839 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^5 + C_4 L_1 L_4}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 s^6 + R_1 R_4 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5 R_1 + C_4 C_5 L_1 L_4 L_5 R_4) + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 R_4 + 2C_4 C_5 L_1 L_5 R_1 R_4 + C_4 C_5 L_4 L_5 R_1 R_4 + 2C_4 L_1 L_4 L_5 R_1 g_m + 2C_4 L_1 L_4 L_5) +}$$

9.840 X-INVALID-ORDER-840 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1s^6 + 2R_1 + s^5(C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_4C_5L_1L_4L_5R_1g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5 + 2C_1C_4L_1L_4R_1 + 2C_1C_5L_1L_5R_1 + 2C_4C_5L_1L_4R_1R_4g_m + 2C_4C_5L_1L_4R_1R_5g_m + 2C_4C_5L_1L_4R_1R_4R_5)}{2C_1C_4C_5L_1L_4L_5R_1s^6 + 2R_1 + s^5(C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_4C_5L_1L_4L_5R_1g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5 + 2C_1C_4L_1L_4R_1 + 2C_1C_5L_1L_5R_1 + 2C_4C_5L_1L_4R_1R_4g_m + 2C_4C_5L_1L_4R_1R_5g_m + 2C_4C_5L_1L_4R_1R_4R_5)}$$

9.841 X-INVALID-ORDER-841 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 s^6 + R_1 R_4 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 L_1 L_4 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_4 C_5 L_1 L_4 L_5 R_4 R_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 R_5 + 2 C_1 C_4 L_1 L_5 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 R_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5)}{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 s^6 + R_1 R_4 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 L_1 L_4 L_5 R_1 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_4 C_5 L_1 L_4 L_5 R_4 R_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 R_5 + 2 C_1 C_4 L_1 L_5 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 R_4 R_5 + 2 C_4 C_5 L_1 L_5 R_1 R_4 R_5 + C_4 C_5 L_1 L_5 R_4 R_5)}$$

9.860 X-INVALID-ORDER-860 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (C_1 L_1 L_5 R_1 R_5 g_m - C_1 L_1 L_5 R_1 - C_5 L_1 L_5 R_5) + s^2 (-C_1 L_1 R_1 R_5 - C_5 L_5 R_1 R_5 + L_1 L_5 R_5)}{2C_1 C_4 C_5 L_1 L_5 R_1 R_5 s^5 + 2R_1 R_5 g_m + R_5 + s^4 (2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_5 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_5 + 2C_1 L_1 L_5 R_1 g_m + C_1 L_1 L_5 + 2C_4 C_5 L_5 R_1 R_5 + 2C_4 L_1 L_5 R_5 g_m + 2C_4 L_1 L_5 + 2C_4 C_5 L_5 R_1 R_5 + 2C_4 L_1 L_5 R_5)}$$

9.861 X-INVALID-ORDER-861 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^3 (C_1 L_1 L_5 R_1 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 R_5) + s (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 g_m - 2 C_4 C_5 L_5 R_1 R_5) + s^2 (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 g_m - 2 C_4 C_5 L_5 R_1 R_5) + s (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5 + 2 C_4 C_5 L_5 R_1 R_5 g_m - 2 C_4 C_5 L_5 R_1 R_5) + 2 C_1 C_4 L_1 R_1 R_5 g_m - 2 C_1 C_4 L_1 R_1 - 2 C_1 C_4 L_1 R_5 - 2 C_4 C_5 L_5 R_1 R_5 g_m + 2 C_4 C_5 L_5 R_1 R_5}$$

9.862 X-INVALID-ORDER-862 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^3 (-C_1 C_5 L_1 R_1 R_5 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + 2C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_5 g_m + 2C_4 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + 2C_1 C_4 L_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 +$$

9.863 X-INVALID-ORDER-863 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4) + s (L_1 R_4 R_5 g_m - L_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_4 R_5) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 + C_1 L_1 R_4 + 2C_1 L_1 R_5 + 2C_4 L_1 R_4 R_5 g_m + 2C_4 L_1 R_4) + s (2C_4 R_1 R_4 R_5 g_m + 2C_4 R_1 R_4 + 2C_4 R_4 R_5 + 2L_1}$$

9.864 X-INVALID-ORDER-864 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_4 s^3 + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m - C_5 L_1 R_4) + s (-C_5 R_1 R_4 + L_1 R_4 g_m)}{2C_1 C_4 C_5 L_1 R_1 R_4 s^4 + 2R_1 g_m + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_4 + 2C_1 C_5 L_1 R_1 R_4 g_m + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2C_4 C_5 L_1 R_4) + s^2 (2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_4 C_5 R_1 R_4 + 2C_4 L_1 R_4 g_m + 2C_5 L_1 R_4 g_m + 2C_5 L_1) + s (2C_4 R_1 R_4 g_m + 2C_4 R_4 + 2C_5 R_1 R_4 g_m +$$

9.865 X-INVALID-ORDER-865 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_4 R_5 s^3 + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4 - C_5 L_1 R_4)}{2C_1 C_4 C_5 L_1 R_1 R_4 R_5 s^4 + 2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_4 R_5 + 2C_1 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_1 R_5 + C_1 C_5 L_1 R_4 R_5 + 2C_4 C_5 L_1 R_4 R_5) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 + C_1)}$$

9.866 X-INVALID-ORDER-866 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s^3 (C_1 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_5 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 R_4 g_m + C_5 L_1 R_4 R_5 g_m - C_5 L_1 R_4 R_5)}{2 R_1 g_m + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_4 + 2 C_1 C_4 C_5 L_1 R_4 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2 C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_4) + s^2 (2 C_1 L_1 R_1 g_m + 2 C_1 L_1 R_1 R_4 + 2 C_1 L_1 R_4 R_5 g_m + 2 C_1 L_1 R_4 R_5) + s (C_1 R_1 g_m + C_1 R_1 R_4 + C_1 R_4 R_5 g_m + C_1 R_4 R_5) + C_5 L_1 R_4 R_5 g_m + C_5 L_1 R_4 R_5}$$

9.867 X-INVALID-ORDER-867 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (-C_1 C_5 L_1 R_1 R_4 + C_5 L_1 L_5 R_4 g_m) + s^2 (C_1 L_1 R_1 R_4 g_m - C_5 L_1 R_4 g_m)}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2 C_4 C_5 L_1 R_4 + 2 C_4 C_5 L_5 R_1 R_4 g_m)}$$

9.868 X-INVALID-ORDER-868 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_4 s^4 - R_1 R_4 + s^3 (C_1 L_1 L_5 R_1 R_4 g_m - C_5 L_1 L_5 R_4) + s^2 (-C_1 L_1 R_1 R_4 - C_5 L_5 R_1 R_4) + s (C_1 L_1 L_5 R_1 R_4 g_m - C_5 L_1 L_5 R_4) + C_1 C_5 L_1 L_5 R_1 R_4}{2C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 + 2R_1 R_4 g_m + 2R_1 + R_4 + s^4 (2C_1 C_4 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_5 R_4) + s^3 (2C_1 C_4 L_1 R_1 R_4 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5 + 2C_4 C_5 L_5 R_1 R_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_1 R_4) + s^2 (2C_1 C_4 L_1 R_1 R_4 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5 + 2C_4 C_5 L_5 R_1 R_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_1 R_4) + s (2C_1 C_4 L_1 R_1 R_4 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5 + 2C_4 C_5 L_5 R_1 R_4 + 2C_4 L_1 L_5 R_4 g_m + 2C_5 L_1 L_5 R_1 R_4) + C_1 C_5 L_1 L_5 R_1 R_4}$$

9.869 X-INVALID-ORDER-869 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + 2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_4 + 2 C_1 C_4 C_5 L_1 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_5 L_1 R_1 R_5)}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_4 + 2 C_1 C_4 C_5 L_1 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_5 L_1 R_1 R_5)}$$

9.870 X-INVALID-ORDER-870 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_5R_1R_4R_5s^5 + 2R_1R_4R_5g_m + 2R_1R_5 + R_4R_5 + s^4(2C_1C_4L_1L_5R_1R_4R_5g_m + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5 + 2C_1C_5L_1L_5R_1R_4R_5g_m + 2C_1C_5L_1L_5R_1R_5 + C_1C_5L_1L_5R_4R_5 + 2C_4C_5L_1L_5R_4R_5) + s^3(2C_1C_4L_1R_1R_4R_5 + 2C_1L_1L_5R_1R_4g_m +$$

9.871 X-INVALID-ORDER-871 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_4 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_5)}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_4 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_5)}$$

9.872 X-INVALID-ORDER-872 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_5 + 2C_4C_5L_1L_5R_5)}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_5 + 2C_4C_5L_1L_5R_5)}$$

9.873 X-INVALID-ORDER-873 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 + L_1 R_5 g_m - L_1)}{2R_1 g_m + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5) + s^2 (2C_1 L_1 R_1 g_m + C_1 L_1 + 2C_4 L_1 R_4 g_m + 2C_4 L_1 R_5 g_m + 2C_4 L_1) + s (2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2L_1 g_m) + 1}$$

9.874 X-INVALID-ORDER-874 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_1 R_4 s^4 + R_1 g_m + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_1 R_4) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4 + C_4 L_1 R_4 g_m - C_5 L_1) + s (C_4 R_1 R_4 g_m - C_5 R_1 + L_1 g_m)}{s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1) + s^2 (2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.875 X-INVALID-ORDER-875 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_1 R_4 R_5 s^4 + R_1 R_5 g_m - R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4 - C_1 C_5 L_1 R_1 R_5 - C_4 C_5 L_1 R_4 R_5) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 - C_4 C_5 R_1 R_4)}{2 R_1 g_m + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_5 + 2 C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_1 R_4 R_5 g_m + 2 C_4 C_5 L_1 R_5) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 R_1 + C_4 C_5 R_1 R_4)}.$$

9.876 X-INVALID-ORDER-876 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4) + s^2 (C_1 L_1 R_1 g_m + C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4 + C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1 R_1)}{s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1) + s^2 (2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1)}$$

9.877 X-INVALID-ORDER-877 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m s^5 + R_1 g_m + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_1 R_4 + C_4 C_5 L_5 R_1 R_4 g_m + C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4 + C_4 L_1 R_4 g_m)}{s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2 (2C_4 C_5 R_1 R_4)}$$

9.878 X-INVALID-ORDER-878 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 - R_1 + s^4 (C_1 C_4 L_1 L_5 R_1 R_4 g_m - C_1 C_5 L_1 L_5 R_1 - C_4 C_5 L_1 L_5 R_4) + s^3 (-C_1 C_4 L_1 R_1 R_4 + C_1 L_1 L_5 R_1 g_m - C_4 C_5 L_5 R_1 R_4 + C_4 L_1 L_5 R_4 g_m - C_1 C_4 L_1 L_5 R_1 R_4) + s^2 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4) + s (2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5) + s^0 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_4 C_5 L_5 R_1 R_4 g_m + 2C_4 C_5 L_5 R_1 R_4 + C_4 C_5 L_5 R_1 R_4)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 R_4 g_m + 2C_4 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_4 C_5 L_5 R_1 R_4 g_m + 2C_4 C_5 L_5 R_1 R_4 + C_4 C_5 L_5 R_1 R_4)}$$

9.879 X-INVALID-ORDER-879 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m s^5 + R_1 g_m + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 C_5 L_5 R_1 R_4 g_m + C_5 L_1 R_5 g_m)}{s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m +$$

9.880 X-INVALID-ORDER-880 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 - R_1 R_5 + s^4 (C_1 C_4 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 L_1 L_5 R_1 R_4 - C_1 C_5 L_1 L_5 R_1 R_4 R_5 g_m + C_1 C_4 L_1 L_5 R_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + C_1 C_4 C_5 L_1 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_4 R_5) + s^4 (2C_1 C_4 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 R_4 + 2C_1 C_4 L_1 L_5 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4 R_5)}{2R_1 R_5 g_m + R_5 + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_4 R_5) + s^4 (2C_1 C_4 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 R_4 + 2C_1 C_4 L_1 L_5 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5 + 2C_4 C_5 L_1 L_5 R_4 R_5 g_m + 2C_4 C_5 L_1 L_5 R_4 R_5)}$$

9.881 X-INVALID-ORDER-881 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4) + s^4 (C_1 C_4 L_1 L_5 R_1 R_4 g_m + C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^3 (C_1 C_4 L_1 L_5 R_1 R_4 g_m + C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s^2 (C_1 C_4 L_1 L_5 R_1 R_4 g_m + C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + s (C_1 C_4 L_1 L_5 R_1 R_4 g_m + C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_1 L_5 R_4 R_5 g_m - C_4 C_5 L_1 L_5 R_4) + C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^3 (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s^2 (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + s (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_5 g_m + 2 C_4 C_5 L_1 L_5) + C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4}$$

9.882 X-INVALID-ORDER-882 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4) + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_5 L_1 L_5 R_1 R_4 R_5)}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_5 R_4 g_m + 2 C_4 C_5 L_1 L_5 R_4 R_5)}$$

9.883 X-INVALID-ORDER-883 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1) + s^3 (C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (L_1 R_5 g_m - L_1)}{2 R_1 g_m + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5 + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + 2 C_4 L_4 R_1 g_m + C_4 L_4) + s (2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5 + 2 L_1 g_m) + 1}$$

9.884 X-INVALID-ORDER-884 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 s^5 + R_1 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 g_m - C_4 C_5 L_1 L_4) + s^3 (-C_1 C_5 L_1 R_1 - C_4 C_5 L_4 R_1 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_4 L_4 R_1 g_m - C_5 L_1) + s (-C_5 R_1 + L_1 g_m)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_1 + 2C_4 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_4 C_5 R_1 + 2C_4 L_1 g_m + 2C_5 L_1 g_m) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m +$$

9.885 X-INVALID-ORDER-885 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + R_1 R_5 g_m - R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1 - C_4 C_5 L_1 L_4 R_5) + s^3 (-C_1 C_5 L_1 R_1 R_5 - C_4 C_5 L_4 R_1 R_5 + C_4 L_1 L_4 R_5 g_m - C_4 L_1 R_5 g_m) + s^2 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5) + s (2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + 2C_1 C_4 L_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_5 + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 R_5)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + 2C_1 C_4 L_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 + 2C_4 C_5 L_1 R_5 + 2C_4 C_5 L_4 R_1 R_5 g_m + 2C_4 C_5 L_4 R_1 R_5)}$$

9.886 X-INVALID-ORDER-886 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1) + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m + C_4 L_4 R_1 g_m)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_4 C_5 R_1 R_5 g_m + 2C_4 C_5 R_1 + 2C_4 C_5 L_4) + s (2C_4 C_5 L_4 R_1 g_m + 2C_4 C_5 L_4) + s^0 (2C_4 C_5 L_4)}$$

9.887 X-INVALID-ORDER-887 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (-C_1 C_5 L_1 R_1 - C_4 C_5 L_4 R_1 + C_4 L_1 L_4 g_m + C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 R_1 g_m}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 + 2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4 + 2C_4 C_5 L_5 R_1 g_m +$$

9.888 X-INVALID-ORDER-888 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 - R_1 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_1 - C_1 C_5 L_1 L_5 R_1 - C_4 C_5 L_4 L_5 R_1 + C_4 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_5 R_1 g_m - C_1 C_4 L_1 L_4 L_5 g_m + C_1 C_5 L_1 L_5 g_m - C_4 C_5 L_4 L_5 g_m) + s^2 (C_1 C_4 L_1 L_4 L_5 g_m - C_4 C_5 L_4 L_5 g_m) + s (C_1 C_4 L_1 L_4 L_5 g_m - C_4 C_5 L_4 L_5 g_m) + C_1 C_4 L_1 L_4 L_5 g_m}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5) + s^2 (2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5) + s (2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5) + 2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + C_4 C_5 L_4 L_5}$$

9.889 X-INVALID-ORDER-889 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1 + C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_1 R_5 g_m - C_4 C_5 L_1 R_1 + C_4 C_5 L_4 L_5 R_1 g_m) + s^2 (C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s (2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^0 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 g_m)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + 2C_1 C_4 C_5 L_1 R_5 + 2C_4 C_5 L_1 L_4 g_m + 2C_4 C_5 L_1 L_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 g_m) + s^2 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 g_m) + s (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 g_m) + s^0 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_4 L_5 R_1 g_m)}$$

9.890 X-INVALID-ORDER-890 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 - R_1 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 L_1 L_4 L_5 R_1 - C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_5 R_5)}{2R_1 R_5 g_m + R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_5 R_5)}$$

9.891 X-INVALID-ORDER-891 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 g_m + C_5 C_4 L_1 L_5 R_5 g_m - C_5 C_4 L_1 L_5) + s^3 (C_1 C_4 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^2 (C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 g_m + C_5 C_4 L_1 L_5 R_5 g_m - C_5 C_4 L_1 L_5) + s (C_1 C_4 L_1 L_4 R_1 + C_1 C_5 L_1 L_5 R_1 g_m + C_5 C_4 L_1 L_5 R_5 g_m - C_5 C_4 L_1 L_5) + C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^3 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 L_5) + s^2 (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 L_5) + s (2 C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 + C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 L_5) + C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1}$$

9.892 X-INVALID-ORDER-892 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^5 (-C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_4 C_5 L_1 L_4 L_5)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1$$

9.893 X-INVALID-ORDER-893 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_1 R_5 g_m - C_1 L_1 L_4 R_1) + s^2 (L_1 L_4 R_5 g_m - L_1 L_4) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5) + s^3 (2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 + 2C_1 L_1 R_5 + 2C_4 L_4 R_1 R_5 g_m + 2C_4 L_4 R_1 + 2C_4 L_4 R_5 + 2L_1 L_4 g_m) + s (2L_1 R_5)}$$

9.894 X-INVALID-ORDER-894 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 s^4 + L_4 R_1 g_m s + s^3 (C_1 L_1 L_4 R_1 g_m - C_5 L_1 L_4) + s^2 (-C_5 L_4 R_1 + L_1 L_4 g_m)}{2C_1 C_4 C_5 L_1 L_4 R_1 s^5 + 2R_1 g_m + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_4 C_5 L_1 L_4) + s^3 (2C_1 C_5 L_1 R_1 + 2C_4 C_5 L_4 R_1 + 2C_4 L_1 L_4 g_m + 2C_5 L_1 L_4 g_m) + s^2 (2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_1 + 2C_5 L_4 R_1 g_m + C_5 L_4)}.$$

9.895 X-INVALID-ORDER-895 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 R_5 s^4 + s^3 (C_1 L_1 L_4 R_1 R_5 g_m - C_1 L_1 L_4 R_1 - C_5 L_1 L_4 R_5) + s^2 (-C_5 L_4 R_1 R_5 + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_5 L_1 R_1 R_5 + 2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_4 C_5 L_4 R_1 R_5 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 R_5))}{2C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_5 L_1 R_1 R_5 + 2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_4 C_5 L_4 R_1 R_5 + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4 R_5)}$$

9.896 X-INVALID-ORDER-896 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_4 R_1 g_m s + s^4 (C_1 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_5 L_1 L_4 R_1) + s^3 (C_1 L_1 L_4 R_1 g_m + C_5 L_1 L_4 R_5 g_m - C_5 L_1}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_1 + 2 C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_4 C_5 L_1 L_4 R_5 g_m + 2 C_4 C_5 L_1 L_4) + s^3 (2 C_1 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_5 L_1 R_1 + 2 C_1 C_5 L_1 R_5 + 2 C_4 C_5 L_4 R_1 R_5 g_m -$$

9.897 X-INVALID-ORDER-897 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_1 g_m s^5 + L_4 R_1 g_m s + s^4 (-C_1 C_5 L_1 L_4 R_1 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_4 R_1 g_m - C_5 L_1 L_4 L_5 g_m)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^3 (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s^2 (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + s (2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 + 2 C_4 C_5 L_1 L_4 + 2 C_4 C_5 L_4 L_5 R_1 g_m + 2 C_4 C_5 L_4 L_5) + C_1 C_5 L_1 L_4 L_5 R_1 g_m s + L_4 R_1 g_m s + s^4 (-C_1 C_5 L_1 L_4 R_1 + C_5 L_1 L_4 L_5 g_m) + s^3 (C_1 L_1 L_4 R_1 g_m - C_5 L_1 L_4 L_5 g_m)}$$

9.898 X-INVALID-ORDER-898 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_1 s^5 - L_4 R_1 s + s^4 (C_1 L_1 L_4 L_5 R_1 g_m - C_5 L_1 L_4 L_5) + s^3 (-C_1 L_1 L_4 R_1 - C_5 L_4 L_5)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2R_1 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_5 L_1 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 L_1 L_4 L_5 g_m + 2C_5 L_1 L_4 L_5 g_m) + s^3 (2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_1 L_1 L_4)}$$

9.899 X-INVALID-ORDER-899 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_1 g_m s^5 + L_4 R_1 g_m s + 2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5 g_m)}{s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + s^2 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + s (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + 2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_4 R_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5 g_m) + s^3 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + s^2 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + s (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 L_1 L_4 L_5) + 2R_1 g_m}$$

9.900 X-INVALID-ORDER-900 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_2 C_3 C_4 C_5 L_1 L_2 L_3 L_4 L_5 R_1 R_2 R_3 R_4 R_5 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2 R_1 R_5) + s^5 (2 C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_4 L_1 L_4 L_5 R_1 + 2 C_1 C_4 L_1 L_4 L_5 R_5 + 2 C_1 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_4 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_5)}{2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 + 2 R_1 R_5 + s^5 (2 C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2 C_1 C_4 L_1 L_4 L_5 R_1 + 2 C_1 C_4 L_1 L_4 L_5 R_5 + 2 C_1 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_4 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 R_5) + s^4 (2 C_1 C_4 L_1 L_4 R_1 R_5 + 2 C_1 C_5 L_1 L_5 R_1 R_5 + 2 C_1 L_1 L_4 L_5 R_1 g_m + C_1 L_1 L_4 L_5 + 2 C_4 C_5 L_4 L_5)}$$

9.901 X-INVALID-ORDER-901 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4L_1L_4L_5R_1g_m + 2C_1C_4L_1L_4L_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1R_5 + 2C_1C_4L_1L_4R_5g_m + 2C_1C_4L_1L_4R_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5R_5 + 2C_4C_5L_1L_4L_5)}{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4L_1L_4L_5R_1g_m + 2C_1C_4L_1L_4L_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1R_5 + 2C_1C_4L_1L_4R_5g_m + 2C_1C_4L_1L_4R_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5R_5 + 2C_4C_5L_1L_4L_5)}$$

9.902 X-INVALID-ORDER-902 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5g_m + 2C_1C_4L_1L_4R_5 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5R_5 + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5)}{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5g_m + 2C_1C_4L_1L_4R_5 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5R_5 + C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_5g_m + 2C_4C_5L_1L_4L_5)}$$

9.903 X-INVALID-ORDER-903 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4 + C_4 L_1 L_4 R_5 g_m - C_4 L_1 L_4) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_4 L_1 R_4 R_5 g_m - C_4 L_1 R_4 + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4) + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1}{2 R_1 g_m + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_5 + 2 C_4 L_1 L_4 g_m) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 L_1 R_4 g_m + 2 C_4 L_1 R_5 g_m + 2 C_4 L_1 + 2 C_4 L_4 R_1 g_m + C_4 L_4) + s (2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_4) + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1}$$

9.904 X-INVALID-ORDER-904 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 s^5 + R_1 g_m + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m - C_4 C_5 L_1 L_4) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_1 R_4 - C_4 C_5 L_4 R_1 + C_4 L_1 L_4 g_m) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4 + C_4 L_1 R_4 g_m + C_4 L_4 l)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_4 C_5 L_1 L_4 g_m) + s^3 (2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_4 C_5 R_1 R_4 g_r}$$

9.905 X-INVALID-ORDER-905 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + R_1 R_5 g_m - R_1 + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1 - C_4 C_5 L_1 L_4 R_5) + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m + C_1 C_4 L_1 R_1 R_4 R_5 - C_1 C_4 L_1 R_1 R_4 - C_4 C_5 L_1 R_1 R_5) + s^2 (C_1 C_4 L_1 R_1 R_4 R_5 - C_1 C_4 L_1 R_1 R_4 - C_4 C_5 L_1 R_1 R_5) + s (C_1 C_4 L_1 R_1 R_4 R_5 - C_1 C_4 L_1 R_1 R_4 - C_4 C_5 L_1 R_1 R_5) + C_1 C_4 L_1 R_1 R_4 R_5}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_5 g_m) + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5) + s^2 (2C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_1 R_5 + C_1 C_4 L_1 R_4 + C_1 C_4 L_1 R_5) + s (C_1 C_4 L_1 R_1 R_4 + C_1 C_4 L_1 R_1 R_5 + C_1 C_4 L_1 R_4 + C_1 C_4 L_1 R_5) + C_1 C_4 L_1 R_1 R_4 R_5}$$

9.906 X-INVALID-ORDER-906 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4 + C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4 R_1) + s^2 (C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m) + s (2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 R_5 g_m) + 2 C_4 C_5 L_1 L_4}{s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m) + s^3 (2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_4 C_5 L_1 R_4 g_m + 2 C_4 C_5 L_1 R_5 g_m) + s^2 (2 C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m) + 2 C_4 C_5 L_1 L_4}$$

9.907 X-INVALID-ORDER-907 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m + C_4 C_5 L_4 L_5 R_1 g_m) + s^2 (2 C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m) + s^3 (2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_4 C_5 L_1 R_4 g_m -$$

9.908 X-INVALID-ORDER-908 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 - R_1 + s^5 (-C_1 C_4 C_5 L_1 L_5 R_1 R_4 + C_1 C_4 L_1 L_4 L_5 R_1 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (-C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_5 R_1 R_4 g_m - C_1 C_5 L_1 L_5 R_1 g_m - C_4 C_5 L_1 L_4 L_5 g_m) + s^3 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^2 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s (2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + 2R_1}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_1 L_4 L_5 g_m) + 2R_1}$$

9.909 X-INVALID-ORDER-909 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_1 L_4 R_5 g_m - C_4 C_5 L_1 L_4 + C_4 C_5 L_1 L_5 R_4 g_m) + s^3 (C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^2 (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_4 C_5 L_1 L_4 g_m + 2 C_4 C_5 L_1 L_5 g_m)}$$

9.910 X-INVALID-ORDER-910 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 - R_1 R_5 + s^5 (-C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_5 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + C_1 C_4 L_1 L_4 R_5 + 2C_1 C_1$$

9.911 X-INVALID-ORDER-911 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4 + C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_4 C_5 L_1 L_4 L_5 R_5 g_m - C_4 C_5 L_1 L_4 L_5) + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 R_1)}$$

9.912 X-INVALID-ORDER-912 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^5 (-C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + 2 C_4 C_5 L_1 L_4 L_5 g_m) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 R_5)}$$

9.913 X-INVALID-ORDER-913 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_1 R_4 R_5 g_m - C_1 L_1 L_4 R_1 R_4) + s^2 (L_1 L_4 R_4 R_5 g_m - L_1 L_4 R_4) + s (L_4 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 R_5) + s^3 (2C_1 L_1 L_4 R_1 R_4 g_m + 2C_1 L_1 L_4 R_1 R_5 g_m + 2C_1 L_1 L_4 R_1 + C_1 L_1 L_4 R_4 + 2C_1 L_1 L_4 R_5 + 2C_4 L_1 L_4 R_4 R_5 g_m + 2C_4 L_1 L_4 R_4) + s^2 (2C_1 L_1 R_1 R_4 R_5 g_m + 2C_1 L_1 R_1 R_4)}$$

9.914 X-INVALID-ORDER-914 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 R_4 s^4 + L_4 R_1 R_4 g_m s + s^3 (C_1 L_1 L_4 R_1 R_4 g_m - C_5 L_1 L_4 R_4) + s^2 (-C_5 L_4 R_4)}{2C_1 C_4 C_5 L_1 L_4 R_1 R_4 s^5 + 2R_1 R_4 g_m + 2R_4 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_4 + 2C_4 C_5 L_1 L_4 R_4) + s^3 (2C_1 C_5 L_1 R_1 R_4 + 2C_1 L_1 L_4 R_1 g_m + 2C_1 L_1 L_4 + 2C_4 C_5 L_4 R_1 R_4 + 2C_4 L_1 L_4 R_4 g_m + 2C_5 L_1 L_4 R_4)}$$

9.915 X-INVALID-ORDER-915 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4R_1R_4R_5s^5 + 2R_1R_4R_5g_m + 2R_1R_4 + 2R_4R_5 + s^4(2C_1C_4L_1L_4R_1R_4R_5g_m + 2C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_1L_4R_4R_5 + 2C_1C_5L_1L_4R_1R_4R_5g_m + 2C_1C_5L_1L_4R_1R_5 + C_1C_5L_1L_4R_4R_5 + 2C_4C_5L_1L_4R_4R_5) + s^3(2C_1C_5L_1R_1R_4R_5 + 2C_1L_1L_4R_1R_4g_m}$$

9.916 X-INVALID-ORDER-916 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{1}{2R_1R_4g_m + 2R_4 + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5) + s^4(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_1R_4g_m + 2C_1C_5L_1L_4R_1R_5g_m + 2C_1C_5L_1L_4R_1 + C_1C_5L_1L_4R_4 + 2C_1C_5L_1L_4R_5 + 2C_4C_5L_1L_4R_4R_5g_m)}$$

9.917 X-INVALID-ORDER-917 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_4 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_4) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 + 2C_1 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_5 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4$$

9.918 X-INVALID-ORDER-918 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1R_4s^6 + 2R_1R_4 + s^5(2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4 + 2C_4C_5L_1L_4L_5R_4) + s^4(2C_1C_4L_1L_4R_1R_4 + 2C_1C_5L_1L_5R_1R_4 + 2C_1L_1L_4L_5R_1g_m + 2C_1L_1L_4L_5 + 2C_4C_5L_4L_5R_1R_4)}{2C_1C_4C_5L_1L_4L_5R_1R_4s^6 + 2R_1R_4 + s^5(2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4 + 2C_4C_5L_1L_4L_5R_4) + s^4(2C_1C_4L_1L_4R_1R_4 + 2C_1C_5L_1L_5R_1R_4 + 2C_1L_1L_4L_5R_1g_m + 2C_1L_1L_4L_5 + 2C_4C_5L_4L_5R_1R_4)}$$

9.919 X-INVALID-ORDER-919 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5 + 2C_4C_5L_1L_4L_5R_4g_m) + s^4(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 +$$

9.920 X-INVALID-ORDER-920 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1R_4R_5s^6 + 2R_1R_4R_5 + s^5(2C_1C_4L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4L_1L_4L_5R_1R_4 + 2C_1C_4L_1L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_5L_1L_4L_5R_1R_5 + C_1C_5L_1L_4L_5R_4R_5 + 2C_4C_5L_1L_4L_5R_4R_5) + s^4(2C_1C_4L_1L_4R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_5)}{1}$$

9.921 X-INVALID-ORDER-921 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4R_5g_m + 2R_1R_4 + 2R_4R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^5(2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_5g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4 +$$

9.922 X-INVALID-ORDER-922 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4R_5g_m + 2R_1R_4 + 2R_4R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1R_5g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_5 +$$

9.933 X-INVALID-ORDER-933 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_1 R_4) + s^3 (C_4 L_1 L_4 R_4 R_5 g_m - C_4 L_1 L_4 R_4) +}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 R_1 R_4 + 2C_1 C_4 L_1 R_4 R_5 + 2C_4 L_1 L_4 R_4 g_m + 2C_4 L_1 L_4 R_5 g_m + 2C_4 L_1 L_4) + s^2 (2$$

9.934 X-INVALID-ORDER-934 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 R_4 s^5 + R_1 R_4 g_m + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 g_m - C_4 C_5 L_1 L_4 R_4) + s^3 (-C_1 C_5 L_1 R_1 R_4 - C_4 C_5 L_1 R_1 R_4)}{2 R_1 g_m + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_4 R_4) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_4 C_5 L_1 L_4 R_4 g_m + 2 C_4 C_5 L_1 L_4) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_4 + 2 C_1 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_4 C_5 L_1 R_4)}$$

9.935 X-INVALID-ORDER-935 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_5 + C_1C_4C_5L_1L_4R_4R_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + C_1C_4L_1L_4R_4 + 2C_1C_4L_1L_4R_5 + 2C_4C_5L_1L_4R_4}$$

9.936 X-INVALID-ORDER-936 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 R_4 g_m + s^5 (C_1 C_4 C_5 L_1 L_4)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 C_5 L_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_4 C_5 L_1 L_4 R_4 g_m + 2C_4 C_5 L_1 L_4)}$$

9.937 X-INVALID-ORDER-937 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m s^6 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) s^5 + (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) s^4 + (2C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 L_5 R_1 R_4 g_m) s^3 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4) s^2 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4) s + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m}{(2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) s^5 + (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_4 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_4 + 2C_4 C_5 L_1 L_4 L_5 g_m) s^4 + (2C_1 C_4 C_5 L_1 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 L_5 R_1 R_4 g_m) s^3 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4) s^2 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4) s + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m}.$$

9.938 X-INVALID-ORDER-938 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_1 + R_4 + s^6(2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 + C_1 C_4 C_5 L_1 L_4 L_5 R_4) + s^5(2C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_4 C_5 L_1 L_4 L_5 R_4 g_m + 2C_4 C_5 L_1 L_4 L_5) + s^4(2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L$$

9.939 X-INVALID-ORDER-939 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1g_m + s^6(2C_1C_4C_5L_1L_4L_5R_1g_m + 2C_1C_4C_5L_1L_4L_5) + s^5(2C_1C_4C_5L_1L_4R_1R_4g_m + 2C_1C_4C_5L_1L_4R_1R_5g_m + 2C_1C_4C_5L_1L_4R_1 + C_1C_4C_5L_1L_4R_4 + 2C_1C_4C_5L_1L_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4g_m + 2C_1C_4C_5L_1L_5R_4 + 2C_4C_5L_1L_4L_5g_m) + s^4(2C_1C_4C_5L_1R_1R_4g_m + 2C_1C_4C_5L_1R_1R_5g_m + 2C_1C_4C_5L_1R_1 + C_1C_4C_5L_1R_4 + 2C_1C_4C_5L_1R_5 + 2C_4C_5L_1L_4L_5g_m) + s^3(2C_1C_4C_5L_1R_4R_5g_m + 2C_1C_4C_5L_1R_4R_5 + 2C_1C_4C_5L_1R_5R_4 + 2C_4C_5L_1L_4L_5g_m) + s^2(2C_1C_4C_5L_1R_5R_4 + 2C_4C_5L_1L_4L_5g_m) + s(C_1C_4C_5L_1R_5R_4 + C_4C_5L_1L_4L_5g_m) + C_4C_5L_1L_4L_5g_m}{1}$$

9.940 X-INVALID-ORDER-940 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 R_1 + C_1 C_4 L_1 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_5)}{(s^7 + (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)s^6 + (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 R_1 + C_1 C_4 L_1 L_4 L_5 R_4 + 2C_1 C_4 L_1 L_4 L_5 R_5)s^5 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)s^4 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)s^3 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)s^2 + (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)s + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5}$$

9.941 X-INVALID-ORDER-941 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + C_1C_4C_5L_1L_4L_5R_4 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4R_5) + s^4(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s^3(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s^2(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + C_1C_4C_5L_1L_4L_5R_4 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4R_5) + s^4(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s^3(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s^2(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + s(2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5) + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5 + 2C_1C_4L_1L_5R_1R_4 + 2C_1C_4L_1L_5R_4R_5}$$

$$\mathbf{9.942 \quad X-INVALID-ORDER-942} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \frac{R_4 (C_4 L_4 s^2 + 1)}{C_4 L_4 s^2 + C_4 R_4 s + 1}, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 R_5)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 + C_1 C_4 C_5 L_1 L_4 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 R_4 R_5 + 2C_1 C_4 C_5 L_1 L_4 R_5)}$$

$$\mathbf{9.943 \quad X-INVALID-ORDER-943} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_4 s^3 + C_1 L_1 R_1 R_4 g_m s^2 - C_5 R_1 R_4 s + R_1 R_4 g_m}{2R_1 g_m + s^3 (2C_1 C_5 L_1 R_1 R_4 g_m + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4) + s^2 (C_1 C_5 R_1 R_4 + 2C_1 L_1 R_1 g_m + 2C_1 L_1) + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.944 \quad X-INVALID-ORDER-944} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_4 R_5 s^3 - C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^3 (2C_1 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_5 L_1 R_1 R_5 + C_1 C_5 L_1 R_4 R_5) + s^2 (C_1 C_5 R_1 R_4 R_5 + 2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 + C_1 L_1 R_4 + 2C_1 L_1 R_5) + s (C_1 R_1 R_4 + 2C_1 R_1 R_5 + 2C_5 R_1 R_4 R_5 g_m + 2C_5 R_1 R_5 + C_5 R_4 R_5)}$$

$$\mathbf{9.945 \quad X-INVALID-ORDER-945} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_4 g_m s^2 + R_1 R_4 g_m + s^3 (C_1 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_5 L_1 R_1 R_4) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^3 (2C_1 C_5 L_1 R_1 R_4 g_m + 2C_1 C_5 L_1 R_1 R_5 g_m + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5) + s^2 (C_1 C_5 R_1 R_4 + 2C_1 L_1 R_1 g_m + 2C_1 L_1) + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5 g_m + 2C_5 R_1 + C_5 R_4 + 2C_5 R_5) + 2}$$

$$\mathbf{9.946 \quad X-INVALID-ORDER-946} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 R_4 g_m s^4 - C_1 C_5 L_1 R_1 R_4 s^3 - C_5 R_1 R_4 s + R_1 R_4 g_m + s^2 (C_1 L_1 R_1 R_4 g_m + C_5 L_5 R_1 R_4 g_m)}{2R_1 g_m + s^4 (2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5) + s^3 (2C_1 C_5 L_1 R_1 R_4 g_m + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_5 R_1) + s^2 (C_1 C_5 R_1 R_4 + 2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_5 L_5 R_1 g_m + 2C_5 L_5) + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 + C_5 R_4) + 2}$$

$$\mathbf{9.947 \quad X-INVALID-ORDER-947} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_4 s^4 + C_1 L_1 L_5 R_1 R_4 g_m s^3 + L_5 R_1 R_4 g_m s - R_1 R_4 + s^2 (-C_1 L_1 R_1 R_4 - C_5 L_5 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 + R_4 + s^4 (2C_1 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_4) + s^3 (C_1 C_5 L_5 R_1 R_4 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 + C_1 L_1 R_4 + 2C_1 L_5 R_1 + 2C_5 L_5 R_1 R_4 g_m + 2C_5 L_5 R_1 + C_5 L_5 R_4) + s (C_1 R_1 R_4 + 2L_5 R_1 g_m + 2L_5 R_1)}$$

$$\mathbf{9.948 \quad X-INVALID-ORDER-948} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 R_4 g_m s^4 + R_1 R_4 g_m + s^3 (C_1 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_5 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 R_4 g_m + C_5 L_5 R_1 R_4 g_m) + s (C_5 R_1 R_4 R_5 g_m - C_5 R_1 R_4)}{2R_1 g_m + s^4 (2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5) + s^3 (2C_1 C_5 L_1 R_1 R_4 g_m + 2C_1 C_5 L_1 R_1 R_5 g_m + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_1 R_4 + 2C_1 C_5 L_1 R_5 + 2C_1 C_5 L_5 R_1) + s^2 (C_1 C_5 R_1 R_4 + 2C_1 C_5 R_1 R_5 + 2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_5 L_5 R_1 g_m + 2C_5 L_5) + s (2C_1 R_1 + 2C_5 R_1 R_4 g_m + 2C_5 R_1 R_5)}$$

$$\mathbf{9.949 \quad X-INVALID-ORDER-949} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_4 R_5 s^4 - R_1 R_4 R_5 + s^3 (C_1 L_1 L_5 R_1 R_4 R_5 g_m - C_1 L_1 L_5 R_1 R_4) + s^2 (-C_1 L_1 R_1 R_4 R_5 - C_5 L_5 R_1 R_4 R_5) + s (L_5 R_1 R_4 R_5)}{2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^4 (2C_1 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_5 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_4 R_5) + s^3 (C_1 C_5 L_5 R_1 R_4 R_5 + 2C_1 L_1 L_5 R_1 R_4 g_m + 2C_1 L_1 L_5 R_1 R_5 g_m + 2C_1 L_1 L_5 R_1 + C_1 L_1 L_5 R_4 + 2C_1 L_1 L_5 R_5) + s^2 (2C_1 L_1 R_1 R_4 R_5 g_m + 2C_1 L_1 R_1 R_5 + C_1 L_1 R_4 R_5)}$$

$$\mathbf{9.950 \quad X-INVALID-ORDER-950} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_5 R_1 R_4 g_m s^3 + L_5 R_1 R_4 g_m s + R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_1 R_4) + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^4 (2C_1 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5) + s^3 (C_1 C_5 L_5 R_1 R_4 + 2C_1 C_5 L_5 R_1 R_5 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_4 R_5)}$$

$$\mathbf{9.951 \quad X-INVALID-ORDER-951} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_4 R_5 s^3 - C_5 R_1 R_4 R_5 s + R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_1 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_5 L_1 L_5 R_1 R_4) + s^2 (C_1 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_1 R_4 R_5 + 2 C_1 C_5 L_1 R_1 R_4 R_5 s + 2 C_1 C_5 L_1 R_1 R_4 R_5 s^2 + 2 C_1 C_5 L_1 R_1 R_4 R_5 s^3 + 2 C_1 C_5 L_1 R_1 R_4 R_5 s^4)}{2 R_1 R_4 g_m + 2 R_1 R_5 g_m + 2 R_1 + R_4 + 2 R_5 + s^4 (2 C_1 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_4 + 2 C_1 C_5 L_1 L_5 R_5) + s^3 (2 C_1 C_5 L_1 R_1 R_4 R_5 g_m + 2 C_1 C_5 L_1 R_1 R_5 + C_1 C_5 L_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 + 2 C_1 C_5 L_5 R_1 R_5) + s^2 (C_1 C_5 R_1 R_4 R_5 + C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5) + s (C_1 R_1 R_4 R_5 + C_1 R_1 R_5 + C_1 R_4 R_5) + 1}$$

$$\mathbf{9.952 \quad X-INVALID-ORDER-952} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1)}{2 R_1 g_m + s^3 (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + 2 C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5) + 1}$$

$$\mathbf{9.953 \quad X-INVALID-ORDER-953} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 s^3 + C_1 L_1 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{2 C_1 C_4 C_5 L_1 R_1 s^4 + s^3 (2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.954 \quad X-INVALID-ORDER-954} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_5 s^3 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1)}{2 C_1 C_4 C_5 L_1 R_1 R_5 s^4 + 2 R_1 g_m + s^3 (2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_1 R_5 + 2 C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5) + s^2 (2 C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_5 + 2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 C_5 R_1 R_5) + s (C_1 R_1 + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + 2 C_4 R_5 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.955 \quad X-INVALID-ORDER-955} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^4 (2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + 2 C_1 C_4 C_5 L_1 R_5) + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 + 2 C_4 C_5 R_5) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.956 \quad X-INVALID-ORDER-956} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 - C_1 C_5 L_1 R_1 s^3 - C_5 R_1 s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_5 L_5 R_1 g_m)}{s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_1 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1) + s (2 C_4 R_1 g_m + 2 C_4 + 2 C_5 R_1 g_m + C_5) + 1}$$

$$\mathbf{9.957 \quad X-INVALID-ORDER-957} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 s^4 + C_1 L_1 L_5 R_1 g_m s^3 + L_5 R_1 g_m s - R_1 + s^2 (-C_1 L_1 R_1 - C_5 L_5 R_1)}{2 C_1 C_4 C_5 L_1 L_5 R_1 s^5 + 2 R_1 g_m + s^4 (2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_4 L_1 R_1 + 2 C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2 C_4 C_5 L_5 R_1) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 L_5 R_1 g_m + 2 C_4 L_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_4 R_1) + 1}$$

$$\mathbf{9.958 \quad X-INVALID-ORDER-958} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_5 L_5 R_1 g_m) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + 2 C_1 C_4 C_5 L_1 R_5 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_4 C_5 L_5 R_1 g_m + 2 C_4 C_5 L_5) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1 R_5) + s (C_1 R_1 + 2 C_4 R_1) + 1}$$

$$\mathbf{9.959 \quad X-INVALID-ORDER-959} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4 s}, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (C_1 L_1 L_5 R_1 R_5 g_m - C_1 L_1 L_5 R_1) + s^2 (-C_1 L_1 R_1 R_5 - C_5 L_5 R_1 R_5) + s (L_5 R_1 R_5 g_m - L_5 R_1 R_5)}{2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 s^5 + 2 R_1 R_5 g_m + R_5 + s^4 (2 C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 L_1 L_5 R_1 + 2 C_1 C_4 L_1 L_5 R_5 + 2 C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5) + s^3 (2 C_1 C_4 L_1 R_1 R_5 + 2 C_1 C_4 L_5 R_1 R_5 + C_1 C_5 L_5 R_1 R_5 + 2 C_1 L_1 L_5 R_1 g_m + C_1 L_1 L_5 + 2 C_4 C_5 L_5 R_1 R_5) + s^2 (2 C_1 L_1 R_1 R_5 + 2 C_1 L_1 R_5 + 2 C_4 L_5 R_1 R_5) + s (C_1 R_1 R_5 + C_1 R_5) + 1}$$

$$\mathbf{9.960 \quad X-INVALID-ORDER-960} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4s}, \quad \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1L_5R_1g_ms^3 + L_5R_1g_ms + R_1R_5g_m - R_1 + s^4(C_1C_5L_1L_5R_1R_5g_m - C_1C_5L_1L_5R_1) + s^2(C_1L_1R_1R_5g_m - C_1L_1R_1R_5)}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_5g_m + 2C_1C_4C_5L_1L_5R_1 + 2C_1C_4C_5L_1L_5R_5) + s^4(2C_1C_4C_5L_5R_1R_5 + 2C_1C_4L_1L_5R_1g_m + 2C_1C_4L_1L_5 + 2C_1C_5L_1L_5R_1g_m + C_1C_5L_1L_5) + s^3(2C_1C_4L_1R_1R_5g_m + 2C_1C_4L_1R_1 + 2C_1C_4L_1R_5 + 2C_1C_4L_5R_1 + C_1C_5L_5R_1 + 2C_4C_5L_5R_1) + s^2(2C_1C_4R_1R_4R_5 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1 + C_1L_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.961 \quad X-INVALID-ORDER-961} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{1}{C_4s}, \quad \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_5L_1R_1R_5s^3 - C_5R_1R_5s + R_1R_5g_m - R_1 + s^4(C_1C_5L_1L_5R_1R_5g_m - C_1C_5L_1L_5R_1) + s^2(C_1L_1R_1R_5g_m - C_1L_1R_1R_5)}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_5g_m + 2C_1C_4C_5L_1L_5R_1 + 2C_1C_4C_5L_1L_5R_5) + s^4(2C_1C_4C_5L_1R_1R_5 + 2C_1C_4C_5L_5R_1R_5 + 2C_1C_5L_1L_5R_1g_m + C_1C_5L_1L_5) + s^3(2C_1C_4L_1R_1R_5g_m + 2C_1C_4L_1R_1 + 2C_1C_4L_1R_5 + 2C_1C_4L_5R_1 + C_1C_5L_5R_1 + 2C_4C_5L_5R_1) + s^2(2C_1C_4R_1R_4R_5 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1 + C_1L_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.962 \quad X-INVALID-ORDER-962} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1R_4R_5g_m - R_1R_4 + s^2(C_1L_1R_1R_4R_5g_m - C_1L_1R_1R_4)}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^3(2C_1C_4L_1R_1R_4R_5g_m + 2C_1C_4L_1R_1R_4 + 2C_1C_4L_1R_4R_5) + s^2(2C_1C_4R_1R_4R_5 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1 + C_1L_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.963 \quad X-INVALID-ORDER-963} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_5L_1R_1R_4s^3 + C_1L_1R_1R_4g_ms^2 - C_5R_1R_4s + R_1R_4g_m}{2C_1C_4C_5L_1R_1R_4s^4 + 2R_1g_m + s^3(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_5L_1R_1R_4g_m + 2C_1C_5L_1R_1 + C_1C_5L_1R_4) + s^2(2C_1C_4R_1R_4 + C_1C_5R_1R_4 + 2C_1L_1R_1g_m + 2C_1L_1 + 2C_4C_5R_1R_4) + s(2C_1R_1 + 2C_4R_1R_4g_m + 2C_4R_4 + 2C_5R_1R_4g_m + 2C_5R_1 + C_5R_4) + C_1R_1R_4R_5}$$

$$\mathbf{9.964 \quad X-INVALID-ORDER-964} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad \frac{R_5}{C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_5L_1R_1R_4R_5s^3 - C_5R_1R_4R_5s + R_1R_4R_5g_m - R_1R_4 + s^2(C_1L_1R_1R_4R_5g_m - C_1L_1R_1R_4)}{2C_1C_4C_5L_1R_1R_4R_5s^4 + 2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^3(2C_1C_4L_1R_1R_4R_5g_m + 2C_1C_4L_1R_1R_4 + 2C_1C_4L_1R_4R_5 + 2C_1C_5L_1R_1R_4R_5g_m + 2C_1C_5L_1R_1R_5 + C_1C_5L_1R_4R_5) + s^2(2C_1C_4R_1R_4R_5 + C_1C_5R_1R_4R_5 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.965 \quad X-INVALID-ORDER-965} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1L_1R_1R_4g_ms^2 + R_1R_4g_m + s^3(C_1C_5L_1R_1R_4R_5g_m - C_1C_5L_1R_1R_4) + s(C_5R_1R_4R_5g_m - C_5R_1R_4R_5)}{2R_1g_m + s^4(2C_1C_4C_5L_1R_1R_4R_5g_m + 2C_1C_4C_5L_1R_1R_4 + 2C_1C_4C_5L_1R_4R_5) + s^3(2C_1C_4C_5R_1R_4R_5 + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_5L_1R_1R_4g_m + 2C_1C_5L_1R_1R_5g_m + 2C_1C_5L_1R_1 + C_1C_5L_1R_4 + 2C_1C_5L_1R_5) + s^2(2C_1C_4R_1R_4 + C_1C_5R_1R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.966 \quad X-INVALID-ORDER-966} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad L_5s + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1C_5L_1L_5R_1R_4g_ms^4 - C_1C_5L_1R_1R_4s^3 - C_5R_1R_4s + R_1R_4g_m + s^2(C_1L_1R_1R_4g_m + C_1L_1R_1R_5)}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_4g_m + 2C_1C_4C_5L_1L_5R_4) + s^4(2C_1C_4C_5L_1R_1R_4 + 2C_1C_4C_5L_5R_1R_4 + 2C_1C_5L_1L_5R_1g_m + 2C_1C_5L_1L_5) + s^3(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_5L_1R_1R_4g_m + 2C_1C_5L_1R_1 + C_1C_5L_1R_4 + 2C_1C_5L_5R_1 + 2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_1) + s^2(2C_1C_4R_1R_4 + 2C_1C_5R_1R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.967 \quad X-INVALID-ORDER-967} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad \frac{L_5s}{C_5L_5s^2+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_5L_1L_5R_1R_4s^4 + C_1L_1L_5R_1R_4g_ms^3 + L_5R_1R_4g_ms - R_1R_4 + s^2(-C_1L_1R_1R_4 - C_5L_5R_1R_4)}{2C_1C_4C_5L_1L_5R_1R_4s^5 + 2R_1R_4g_m + 2R_1 + R_4 + s^4(2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_4 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4) + s^3(2C_1C_4L_1R_1R_4 + 2C_1C_4L_5R_1R_4 + C_1C_5L_5R_1R_4 + 2C_1L_1L_5R_1g_m + 2C_1L_1L_5 + 2C_4C_5L_5R_1R_4) + s^2(2C_1C_4R_1R_4 + 2C_1C_5R_1R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

$$\mathbf{9.968 \quad X-INVALID-ORDER-968} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad \frac{R_4}{C_4R_4s+1}, \quad L_5s + R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1C_5L_1L_5R_1R_4g_ms^3 + L_5R_1R_4g_ms - R_1R_4 + s^2(-C_1L_1R_1R_4 - C_5L_5R_1R_4)}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_4g_m + 2C_1C_4C_5L_1L_5R_4) + s^4(2C_1C_4C_5L_1R_1R_4R_5g_m + 2C_1C_4C_5L_1R_1R_4 + 2C_1C_4C_5L_1R_4R_5 + 2C_1C_4C_5L_5R_1R_4 + 2C_1C_5L_1L_5R_1g_m + 2C_1C_5L_1L_5) + s^3(2C_1C_4C_5R_1R_4R_5 + 2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_4 + 2C_1C_5L_1R_1R_4g_m + 2C_1C_5L_1R_1R_5g_m + 2C_1C_5L_1R_1 + C_1C_5L_1R_4 + 2C_1C_5L_5R_1 + 2C_4C_5L_5R_1R_4g_m + 2C_4C_5L_5R_1) + s^2(2C_1C_4R_1R_4 + 2C_1C_5R_1R_4 + 2C_1L_1R_1R_4g_m + 2C_1L_1R_1R_5g_m + 2C_1L_1R_1R_4 + 2C_1L_1R_5) + s(C_1R_1R_4 + 2C_1R_1R_5 + 2C_4R_1R_4R_5g_m + 2C_4R_1R_4 + 2C_4R_1R_5) + C_1R_1R_4R_5}$$

9.969 X-INVALID-ORDER-969 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R}{2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 s^5 + 2R_1 R_4 R_5 g_m + 2R_1 R_5 + R_4 R_5 + s^4 (2C_1 C_4 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_4 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_5 L_1 L_5 R_1 R_5 + C_1 C_5 L_1 L_5 R_4 R_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 R_5 + 2C_1 C_4 L_5 R_1 R_4 R_5 + C_1 C_5 L_5 R_1 R_4 R_5 +$$

9.970 X-INVALID-ORDER-970 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{R_4}{C_4R_4s+1}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_4 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_4g_m + 2C_1C_5L_1L_5R_4R_5) + s^3(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^2(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + 2C_1C_4C_5L_1L_5R_1R_4R_5}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^4(2C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_4 + 2C_1C_5L_1L_5R_1R_4g_m + 2C_1C_5L_1L_5R_1R_5g_m + 2C_1C_5L_1L_5R_1 + C_1C_5L_1L_5R_4 + 2C_1C_5L_1L_5R_4g_m + 2C_1C_5L_1L_5R_4R_5) + s^3(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s^2(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + s(2C_1C_4C_5L_1L_5R_1R_4R_5 + 2C_1C_4C_5L_1L_5R_1R_4 + 2C_1C_4C_5L_1L_5R_4R_5) + 2C_1C_4C_5L_1L_5R_1R_4R_5}$$

9.971 X-INVALID-ORDER-971 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{R_4}{C_4 R_4 s + 1}, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_4 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_4 + 2C_1 C_5 L_1 L_5 R_5)}{(s^6 + (R_1 + R_4 + R_5)s^5 + (R_1 R_4 + R_1 R_5 + R_4 R_5)s^4 + (R_1 R_4 R_5)s^3 + (C_1 L_1 + C_1 L_4 + C_1 L_5)s^2 + C_1 L_1 L_4 L_5 s + C_1 L_1 L_4 L_5 R_1)$$

9.972 X-INVALID-ORDER-972 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_5 g_m + 2 C_4 R_1 + C_4 R_4 + 2 C_4 R_5) + 1}$$

9.973 X-INVALID-ORDER-973 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_1 R_4 s^4 + R_1 g_m + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4) + s (C_4 R_1 R_4 g_m - C_5 R_1)}{s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 R_1 R_4 g_m + 2C_4 C_5 R_1 + C_4 C_5 R_4) + s (2C_4 R_1 g_m + 2C_4 + 2C_5 R_1 g_m + C_5)}$$

9.974 X-INVALID-ORDER-974 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, R_4 + \frac{1}{C_4s}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 R_1 R_4 R_5 s^4 + R_1 R_5 g_m - R_1 + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4 - C_1 C_5 L_1 R_1 R_5) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 - C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_4 R_5)}{2R_1 g_m + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5) + s^2 (C_1 C_4 R_1 R_4 + 2C_1 C_4 R_1 R_5 + C_1 C_5 R_1 R_4 R_5)}$$

9.975 X-INVALID-ORDER-975 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_4 C_5 R_1 R_4 R_5 g_m - C_4 C_5 R_1 R_4) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_4 C_5 R_1)}{s^4 (2 C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2 C_1 C_4 C_5 L_1 R_5) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2 C_1 C_4 C_5 R_1 R_5 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s^2 (2 C_1 C_4 R_1 + C_1 C_5 R_1 + 2 C_4 C_5 R_1 R_4 g_m + 2 C_4 C_5 R_1 R_5 g_m + 2 C_4 C_5 R_1) + s (C_4 R_1 R_4 g_m + C_5 R_1 R_5 g_m - C_4 C_5 R_1)}$$

9.976 X-INVALID-ORDER-976 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m s^5 + R_1 g_m + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_5 L_1 L_5 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_5 R_1 R_4 g_m) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4 + C_5 L_5 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_5 L_5 R_1)}{s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_5 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_5 R_1 g_m + 2C_4 C_5 L_5) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 R_1 R_4 g_m - C_5 L_5 R_1) + s (C_4 R_1 R_4 g_m - C_5 L_5 R_1)}$$

9.977 X-INVALID-ORDER-977 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, R_4 + \frac{1}{C_4s}, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_5 R_1 R_4 s^5 - R_1 + s^4 (C_1 C_4 L_1 L_5 R_1 R_4 g_m - C_1 C_5 L_1 L_5 R_1) + s^3 (-C_1 C_4 L_1 R_1 R_4 + C_1 L_1 L_5 R_1 g_m - C_4 C_5 L_5 R_1 R_4) + s^2 (-C_1 L_1 R_1 R_4 + C_1 C_4 L_1 L_5 R_1 R_4 g_m + C_1 C_4 C_5 L_1 L_5 R_1 R_4) + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2C_4 C_5 L_5 R_1)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4) + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^3 (2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2C_1 C_4 L_5 R_1 + C_1 C_5 L_5 R_1 + 2C_4 C_5 L_5 R_1)}$$

$$\mathbf{9.978 \quad X-INVALID-ORDER-978} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4s}, \quad L_5s + R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4C_5L_1L_5R_1R_4g_ms^5 + R_1g_m + s^4(C_1C_4C_5L_1R_1R_4R_5g_m - C_1C_4C_5L_1R_1R_4 + C_1C_5L_1L_5R_1g_m) + s^3(C_1C_4L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}{s^5(2C_1C_4C_5L_1L_5R_1g_m + 2C_1C_4C_5L_1L_5) + s^4(2C_1C_4C_5L_1R_1R_4g_m + 2C_1C_4C_5L_1R_1R_5g_m + 2C_1C_4C_5L_1R_1 + C_1C_4C_5L_1R_4 + 2C_1C_4C_5L_1R_5 + 2C_1C_4C_5L_5R_1) + s^3(C_1C_4C_5R_1R_4 + 2C_1C_4C_5R_1R_5 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_1C_5L_1R_1g_m + C_1C_5L_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}$$

$$\mathbf{9.979 \quad X-INVALID-ORDER-979} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4s}, \quad \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_4C_5L_1L_5R_1R_4R_5s^5 - R_1R_5 + s^4(C_1C_4L_1L_5R_1R_4R_5g_m - C_1C_4C_5L_1L_5R_1R_4 + C_1C_5L_1L_5R_1g_m) + s^3(C_1C_4L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}{2R_1R_5g_m + R_5 + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_5 + C_1C_4C_5L_1L_5R_4R_5) + s^4(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_5R_1R_4g_m + 2C_1C_4L_1L_5R_1R_5g_m + 2C_1C_4L_1L_5R_1 + C_1C_4L_1L_5R_4 + 2C_1C_4L_1L_5R_5 + 2C_1C_5L_1L_5R_1R_5g_m + C_1C_5L_1L_5R_5) + s^3(C_1C_4C_5R_1R_4 + 2C_1C_4C_5R_1R_5 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_1C_5L_1R_1g_m + C_1C_5L_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}$$

$$\mathbf{9.980 \quad X-INVALID-ORDER-980} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4s}, \quad \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \quad \infty \right)$$

$$H(s) = \frac{R_1R_5g_m - R_1 + s^5(C_1C_4C_5L_1L_5R_1R_4R_5g_m - C_1C_4C_5L_1L_5R_1R_4) + s^4(C_1C_4L_1L_5R_1R_4g_m + C_1C_5L_1L_5R_1R_5g_m - C_1C_5L_1L_5R_1 + C_4C_5L_5R_1R_4g_m) + s^3(C_1C_4L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_4g_m + 2C_1C_4C_5L_1L_5R_1R_5g_m + 2C_1C_4C_5L_1L_5R_1 + C_1C_4C_5L_1L_5R_4 + 2C_1C_4C_5L_1L_5R_5) + s^4(C_1C_4C_5L_5R_1R_4 + 2C_1C_4C_5L_5R_1R_5 + 2C_1C_4L_1L_5R_1g_m + 2C_1C_4L_1L_5 + 2C_1C_5L_1L_5R_1g_m + C_1C_5L_1L_5) + s^3(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_1R_5g_m - 2C_1C_4L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}$$

$$\mathbf{9.981 \quad X-INVALID-ORDER-981} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad R_4 + \frac{1}{C_4s}, \quad \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{R_1R_5g_m - R_1 + s^5(C_1C_4C_5L_1L_5R_1R_4R_5g_m - C_1C_4C_5L_1L_5R_1R_4) + s^4(C_1C_4L_1L_5R_1R_4g_m + C_1C_5L_1L_5R_1R_5g_m - C_1C_5L_1L_5R_1 + C_4C_5L_5R_1R_4g_m) + s^3(C_1C_4L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}{2R_1g_m + s^5(2C_1C_4C_5L_1L_5R_1R_4g_m + 2C_1C_4C_5L_1L_5R_1R_5g_m + 2C_1C_4C_5L_1L_5R_1 + C_1C_4C_5L_1L_5R_4 + 2C_1C_4C_5L_1L_5R_5) + s^4(2C_1C_4C_5L_1R_1R_4R_5g_m + 2C_1C_4C_5L_1R_1R_5 + C_1C_4C_5L_1R_4R_5 + C_1C_4C_5L_5R_1R_4 + 2C_1C_4C_5L_5R_1R_5 + 2C_1C_5L_1L_5R_1g_m + C_1C_5L_1L_5) + s^3(2C_1C_4L_1R_1R_4g_m + 2C_1C_4L_1R_1R_5g_m - 2C_1C_4L_1R_1 + C_4C_5L_5R_1R_4g_m) + s^2(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + s(C_1L_1R_1R_4g_m + C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_5R_1R_4g_m) + R_1g_m}$$

$$\mathbf{9.982 \quad X-INVALID-ORDER-982} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad L_4s + \frac{1}{C_4s}, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1R_5g_m - R_1 + s^4(C_1C_4L_1L_4R_1R_5g_m - C_1C_4L_1L_4R_1) + s^2(C_1L_1R_1R_5g_m - C_1L_1R_1 + C_4L_4R_1R_5g_m - C_4L_4R_1) + s(C_1L_1R_1R_5g_m - C_1L_1R_1 + C_4L_4R_1R_5g_m - C_4L_4R_1) + R_1g_m}{2R_1g_m + s^4(2C_1C_4L_1L_4R_1g_m + C_1C_4L_1L_4) + s^3(2C_1C_4L_1R_1R_5g_m + 2C_1C_4L_1R_1 + 2C_1C_4L_1R_5 + C_1C_4L_4R_1) + s^2(2C_1C_4R_1R_5 + 2C_1L_1R_1g_m + C_1L_1 + 2C_4L_4R_1g_m + C_4L_4) + s(C_1R_1 + 2C_4R_1R_5g_m + 2C_4R_1 + 2C_4R_5) + 1}$$

$$\mathbf{9.983 \quad X-INVALID-ORDER-983} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad L_4s + \frac{1}{C_4s}, \quad \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_4C_5L_1L_4R_1s^5 + C_1C_4L_1L_4R_1g_ms^4 - C_5R_1s + R_1g_m + s^3(-C_1C_5L_1R_1 - C_4C_5L_4R_1) + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m) + s(C_1L_1R_1g_m + C_4L_4R_1g_m) + R_1g_m}{s^5(2C_1C_4C_5L_1L_4R_1g_m + C_1C_4C_5L_1L_4) + s^4(2C_1C_4C_5L_1R_1 + C_1C_4C_5L_4R_1) + s^3(2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_1C_5L_1R_1g_m + C_1C_5L_1 + 2C_4C_5L_4R_1g_m + C_4C_5L_4) + s^2(2C_1C_4R_1 + C_1C_5R_1 + 2C_4C_5R_1) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5) + R_1g_m}$$

$$\mathbf{9.984 \quad X-INVALID-ORDER-984} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad L_4s + \frac{1}{C_4s}, \quad \frac{R_5}{C_5R_5s+1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1C_4C_5L_1L_4R_1R_5s^5 - C_5R_1R_5s + R_1R_5g_m - R_1 + s^4(C_1C_4L_1L_4R_1R_5g_m - C_1C_4L_1L_4R_1) + s^3(-C_1C_5L_1R_1R_5 - C_4C_5L_4R_1R_5) + s^2(C_1L_1R_1R_5g_m + C_4L_4R_1R_5g_m) + s(C_1L_1R_1R_5g_m + C_4L_4R_1R_5g_m) + R_1R_5g_m}{2R_1g_m + s^5(2C_1C_4C_5L_1L_4R_1R_5g_m + C_1C_4C_5L_1L_4R_5) + s^4(2C_1C_4C_5L_1R_1R_5 + C_1C_4C_5L_4R_1R_5 + 2C_1C_4L_1L_4R_1g_m + C_1C_4L_1L_4) + s^3(2C_1C_4L_1R_1R_5g_m + 2C_1C_4L_1R_1 + 2C_1C_4L_1R_5 + C_1C_4L_4R_1 + 2C_1C_5L_1R_1R_5g_m + C_1C_5L_1R_5 + 2C_4C_5L_4R_1R_5g_m + C_4C_5L_4) + s^2(2C_1C_4R_1 + C_1C_5R_1 + 2C_4C_5R_1) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5) + R_1R_5g_m}$$

$$\mathbf{9.985 \quad X-INVALID-ORDER-985} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad L_4s + \frac{1}{C_4s}, \quad R_5 + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4L_1L_4R_1g_ms^4 + R_1g_m + s^5(C_1C_4C_5L_1L_4R_1R_5g_m - C_1C_4C_5L_1L_4R_1) + s^3(C_1C_5L_1R_1R_5g_m - C_1C_5L_1R_1 + C_4C_5L_4R_1R_5g_m - C_4C_5L_4R_1) + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m) + s(C_5R_1R_5g_m + C_4C_5L_4R_1R_5g_m) + R_1g_m}{s^5(2C_1C_4C_5L_1L_4R_1g_m + C_1C_4C_5L_1L_4) + s^4(2C_1C_4C_5L_1R_1R_5g_m + 2C_1C_4C_5L_1R_1 + 2C_1C_4C_5L_1R_5 + C_1C_4C_5L_4R_1) + s^3(2C_1C_4C_5R_1R_5 + 2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_1C_5L_1R_1g_m + C_1C_5L_1 + 2C_4C_5L_4R_1g_m + C_4C_5L_4) + s^2(2C_1C_4R_1 + C_1C_5R_1 + 2C_4C_5R_1) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5) + R_1g_m}$$

$$\mathbf{9.986 \quad X-INVALID-ORDER-986} \quad Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \quad \infty, \quad \infty, \quad L_4s + \frac{1}{C_4s}, \quad L_5s + \frac{1}{C_5s}, \quad \infty \right)$$

$$H(s) = \frac{C_1C_4C_5L_1L_4L_5R_1g_ms^6 - C_1C_4C_5L_1L_4R_1s^5 - C_5R_1s + R_1g_m + s^4(C_1C_4L_1L_4R_1g_m + C_1C_5L_1L_5R_1g_m + C_4C_5L_4L_5R_1g_m) + s^3(-C_1C_5L_1R_1 - C_4C_5L_4R_1) + s^2(C_1L_1R_1g_m + C_4L_4R_1g_m + C_5L_5R_1g_m) + s(C_1L_1R_1g_m + C_4L_4R_1g_m + C_5L_5R_1g_m) + R_1g_m}{s^5(2C_1C_4C_5L_1L_4R_1g_m + C_1C_4C_5L_1L_4 + 2C_1C_4C_5L_1L_5R_1g_m + 2C_1C_4C_5L_1L_5) + s^4(2C_1C_4C_5L_1R_1 + C_1C_4C_5L_4R_1 + 2C_1C_4C_5L_5R_1) + s^3(2C_1C_4L_1R_1g_m + 2C_1C_4L_1 + 2C_1C_5L_1R_1g_m + C_1C_5L_1 + 2C_4C_5L_4R_1g_m + C_4C_5L_4 + 2C_4C_5L_5R_1g_m + 2C_4C_5L_5) + s^2(2C_1C_4R_1 + C_1C_5R_1 + 2C_4C_5R_1) + s(2C_4R_1g_m + 2C_4 + 2C_5R_1g_m + C_5) + R_1g_m}$$

9.987 X-INVALID-ORDER-987 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + C_1 C_4 L_1 L_4 L_5 R_1 g_m s^5 + L_5 R_1 g_m s - R_1 + s^4 (-C_1 C_4 L_1 L_4 R_1 - C_1 C_5 L_1 L_5 R_1 - C_4 C_5 L_4 L_5 R_1) + s^3 (C_1 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 R_1 + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5) + s^2 (C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 L_1 L_4 + C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + R_1}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 R_1 + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5) + s^2 (C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_5 R_1 + C_1 C_5 L_1 L_5 R_1 + C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 L_1 L_4 + C_1 C_4 L_1 L_5 + C_1 C_5 L_1 L_5 + C_4 C_5 L_4 L_5) + R_1}$$

9.988 X-INVALID-ORDER-988 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1) + s^4 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_4)}{s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_5) + s^4 (2 C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 R_1 + 2 C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_1 + 2 C_1 C_4 C_5 L_5 R_1) + s^3 (2 C_1 C_4 C_5 R_1 R_5 + 2 C_1 C_4 L_1 R_1 g_m + 2 C_1 C_4 L_1 + 2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2 C_1 C_5 L_4)}$$

9.989 X-INVALID-ORDER-989 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 - R_1 R_5 + s^5 (C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_5 + C_1 C_4 C_5 L_4 L_5 R_1 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + C_1 C_4 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + C_1 C_4 L_1 L_4 R_5 + 2C_1 C_4 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_5 R_1 + 2C_1 C_4 L_1 L_5 R_5 +$$

9.990 X-INVALID-ORDER-990 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{C_1 C_4 L_1 L_4 L_5 R_1 g_m s^5 + L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 + C_1 C_4 L_1 L_4 R_5 R_1 + C_1 C_4 L_1 L_5 R_1 R_5 + C_1 C_4 L_1 L_5 R_5 R_1 + C_1 C_4 L_5 L_4 R_1 R_5 + C_1 C_4 L_5 L_4 R_5 R_1) + s^3 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_1 R_5 + C_1 C_4 C_5 L_1 L_5 R_5 R_1 + C_1 C_4 C_5 L_5 L_4 R_1 R_5 + C_1 C_4 C_5 L_5 L_4 R_5 R_1) + s^2 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s (2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^0 (2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 R_5 + 2 C_1 C_5 L_5 L_4 R_1 R_5 + 2 C_1 C_5 L_5 L_4 R_5 R_1)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2 C_1 C_4 C_5 L_5 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2 C_1 C_4 L_1 L_5 R_1 g_m + 2 C_1 C_4 L_1 L_5 + 2 C_1 C_5 L_1 L_5 R_1 g_m + 2 C_1 C_5 L_1 L_5 R_5 + 2 C_1 C_5 L_5 L_4 R_1 R_5 + 2 C_1 C_5 L_5 L_4 R_5 R_1)}$$

9.991 X-INVALID-ORDER-991 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + \frac{1}{C_4s}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + 2C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_5 + 2C_1 C_4 C_5 L_4 R_5 + C_1 C_4 C_5 L_5 R_5)}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + 2C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 C_5 L_5 R_1 R_5 + 2C_1 C_4 C_5 L_4 R_5 + C_1 C_4 C_5 L_5 R_5)}$$

9.992 X-INVALID-ORDER-992 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{s^3(C_1L_1L_4R_1R_5g_m - C_1L_1L_4R_1) + s(L_4R_1R_5g_m - L_4R_1)}{2R_1R_5g_m + 2R_1 + 2R_5 + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5) + s^3(2C_1C_4L_4R_1R_5 + 2C_1L_1L_4R_1g_m + C_1L_1L_4) + s^2(2C_1L_1R_1R_5g_m + 2C_1L_1R_1 + 2C_1L_1R_5 + C_1L_4R_1 + 2C_4L_4R_1R_5g_m + 2C_4L_4R_1 + 2C_4L_4R_5) + s(2C_1R_1R_5 + 2L_4R_1g_m)}$$

9.993 X-INVALID-ORDER-993 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 s^4 + C_1 L_1 L_4 R_1 g_m s^3 - C_5 L_4 R_1 s^2 + L_4 R_1 g_m s}{2C_1 C_4 C_5 L_1 L_4 R_1 s^5 + 2R_1 g_m + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_4 R_1 + 2C_1 C_5 L_1 R_1 + C_1 C_5 L_4 R_1 + 2C_4 C_5 L_4 R_1) + s^2 (2C_1 L_1 R_1 g_m + 2C_1 L_1 + 2C_4 L_4 R_1 g_m + 2C_4 L_4 + 2C_5 L_4 R_1 g_m + C_5 L_4) + s (2C_1 R_1 + 2C_5 R_1)}$$

9.994 X-INVALID-ORDER-994 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{L_4s}{C_4L_4s^2+1}, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 R_5 s^4 - C_5 L_4 R_1 R_5 s^2 + s^3 (C_1 L_1 L_4 R_1 R_5 g_m - C_1 L_1 L_4 R_1) + s (L_4 R_1 R_5 g_m - L_4 R_1)}{2C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + 2R_1 R_5 g_m + 2R_1 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_5 L_1 L_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_5 + 2C_1 C_5 L_1 R_1 R_5 + C_1 C_5 L_4 R_1 R_5 + 2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_4 C_5 L_4 R_1 R_5) + s^2 (2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5 + 2C_1 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_5 L_1 L_4 R_5) + s (L_4 R_1 R_5 g_m - L_4 R_1)}$$

9.995 X-INVALID-ORDER-995 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{L_4s}{C_4L_4s^2+1}, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_4 R_1 g_m s^3 + L_4 R_1 g_m s + s^4 (C_1 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_5 L_1 L_4 R_1) + s^2 (C_5 L_4 R_1 R_5 g)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4) + s^3 (2C_1 C_4 L_4 R_1 + 2C_1 C_5 L_1 R_1 R_5 g_m + 2C_1 C_5 L_1 R_1 + 2C_1 C_5 L_1 R_5 + C_1 C_5 L_4 R_1 + 2C_4 C}$$

9.996 X-INVALID-ORDER-996 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{L_4s}{C_4L_4s^2+1}, L_5s + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_1 g_m s^5 - C_1 C_5 L_1 L_4 R_1 s^4 - C_5 L_4 R_1 s^2 + L_4 R_1 g_m s + s^3 (C_1 L_1 L_4 R_1 g_m + 2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5))}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 + 2C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5) + s^3 (2C_1 C_4 L_1 L_4 R_1 g_m + 2C_1 C_4 L_1 L_4 + 2C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2C_1 C_5 L_1 L_5 R_1 g_m + 2C_1 C_5 L_1 L_5 + 2C_4 C_5 L_4 L_5 R_1 g_m + 2C_4 C_5 L_4 L_5)}$$

9.997 X-INVALID-ORDER-997 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_1 s^5 + C_1 L_1 L_4 L_5 R_1 g_m s^4 + L_4 L_5 R_1 g_m s^2 - L_4 R_1 s + s^3 (-C_1 L_1 L_4 R_1 - C_5 L_4 L_5 F)}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 + 2R_1 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_5 L_1 L_4 L_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_4 L_5 R_1 + 2C_1 C_5 L_1 L_5 R_1 + C_1 C_5 L_4 L_5 R_1 + 2C_4 C_5 L_4 L_5 R_1) + s^3 (2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4 + 2C_1 L_1 L_5 R_1 g_m + 2C_1 L_1 L_5 R_1)}$$

9.998 X-INVALID-ORDER-998 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_4 L_5 R_1 g_m s^5}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + 2 C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_4 R_1 + 2 C_1 C_4 C_5 L_1 L_4 R_5 + 2 C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2 C_1 C_4 C_5 L_4 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + 2 C_1 C_4 L_1 L_4 + 2 C_1 C_5 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_4 + 2 C_1 C_5 L_1 L_5 R_1 g_m)}$$

9.999 X-INVALID-ORDER-999 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 L_5 R_1}{2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 s^6 + 2R_1 R_5 + s^5 (2C_1 C_4 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 L_5 R_1 + 2C_1 C_4 L_1 L_4 L_5 R_5 + 2C_1 C_5 L_1 L_4 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_4 L_5 R_5) + s^4 (2C_1 C_4 L_1 L_4 R_1 R_5 + 2C_1 C_4 L_4 L_5 R_1 R_5 + 2C_1 C_5 L_1 L_5 R_1 R_5 + C_1 C_5 L_4 L_5 R_1 R_5 + 2C_1 L_1 L_4 L_5 R_1 g_m + C_1 L_1$$

9.1000 X-INVALID-ORDER-1000 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 s}{C_4 L_4 s^2 + 1}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2R_1 R_5 g_m + 2R_1 + 2R_5 + s^6(2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_5) + s^5(2C_1 C_4 C_5 L_4 L_5 R_1 R_5 + 2C_1 C_4 L_1 L_4 L_5 R_1 g_m + 2C_1 C_4 L_1 L_4 L_5 + 2C_1 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_5 L_1 L_4 L_5) + s^4(2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + 2C_1 C_4 L_1 L_4 R_5)}{(s^2 + 2s + 2)(s^2 + 2s + 2)}$$

9.1001 X-INVALID-ORDER-1001 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{L_4s}{C_4L_4s^2+1}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_4C_5L_4L_5R_1R_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s^3(2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s^2(2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s(C_1C_4L_1L_4R_1 + C_1C_4L_1L_4R_5 + C_1C_4L_1L_4L_5)}{2R_1R_5g_m + 2R_1 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_5 + 2C_1C_4C_5L_4L_5R_1R_5 + 2C_1C_5L_1L_4L_5R_1g_m + C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_5g_m + 2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s^3(2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s^2(2C_1C_4L_1L_4R_1 + 2C_1C_4L_1L_4R_5 + 2C_1C_4L_1L_4L_5) + s(C_1C_4L_1L_4R_1 + C_1C_4L_1L_4R_5 + C_1C_4L_1L_4L_5)}$$

9.1002 X-INVALID-ORDER-1002 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1) + s^3 (C_1 C_4 L_1 R_1 R_4 R_5 g_m - C_1 C_4 L_1 R_1 R_4) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_4 L_4 R_1 R_5 g_m - C_4 L_4 R_1) + s (C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4)}{2 R_1 g_m + s^4 (2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (2 C_1 C_4 L_1 R_1 R_4 g_m + 2 C_1 C_4 L_1 R_1 R_5 g_m + 2 C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 + 2 C_1 C_4 L_1 R_5 + C_1 C_4 L_4 R_1) + s^2 (C_1 C_4 R_1 R_4 + 2 C_1 C_4 R_1 R_5 + 2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_4 L_4 R_1 g_m + C_4 L_4) + s (C_1 R_1 + 2 C_4 R_1 R_4 g_m + 2 C_4 R_1 R_4)}$$

9.1003 X-INVALID-ORDER-1003 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 s^5 + R_1 g_m + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_4 R_1) + s^2 (C_1 L_1 R_1 g_m - C_4 C_5 R_1 R_4 + C_4 L_4 R_1 g_m) + s (C_4 R_1 R_4 g_m - C_4 C_5 L_4 R_1) + C_4 C_5 L_4 R_1}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_4 R_1 g_m + C_4 C_5 L_4) + s^2 (2C_1 C_4 R_1 + C_1 C_5 R_1 + 2C_4 C_5 L_4 R_1) + s (C_4 R_1 R_4 g_m - C_4 C_5 L_4 R_1) + C_4 C_5 L_4 R_1}.$$

9.1004 X-INVALID-ORDER-1004 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 R_1 R_5 s^5 + R_1 R_5 g_m - R_1 + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 L_1 L_4 R_1 R_5 g_m - C_1 C_4 L_1 L_4 R_1)}{2R_1 g_m + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s^3 (C_1 C_4 C_5 R_1 R_4 R_5 + 2C_1 C_4 L_1 R_1 R_4 g_m + 2C_1 C_4 L_1 R_1 R_5 g_m + 2C_1 C_4 L_1 R_1 + C_1 C_4 L_1 R_4 g_m + C_1 C_4 L_1 R_4 R_5) + s^2 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + s (C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4) + C_1 C_4 C_5 L_1 R_1 R_4 R_5 + C_1 C_4 C_5 L_1 R_1 R_5 + C_1 C_4 C_5 L_1 R_4 R_5 + C_1 C_4 C_5 L_4 R_1 R_5 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4}.$$

9.1005 X-INVALID-ORDER-1005 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m + C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_4 R_1 R_5 g_m - C_4 C_5 L_4)}$$

9.1006 X-INVALID-ORDER-1006 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (-C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m) + s^4 (-C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (C_1 C_4 L_1 R_1 R_4 g_m - C_1 C_5 L_1 R_1 - C_4 C_5 L_1)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_5 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 L_1 R_1 g_m + 2C_1 C_4 L_1 + 2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1 + 2C_4 C_5 L_1)}$$

9.1007 X-INVALID-ORDER-1007 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{-C_1 C_4 C_5 L_1 L_4 L_5 R_1 s^6 - R_1 + s^5 (-C_1 C_4 C_5 L_1 L_5 R_1 R_4 + C_1 C_4 L_1 L_4 L_5 R_1 g_m) + s^4 (-C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 L_5 R_1 g_m) + s^3 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^2 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 R_1) + C_1}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 R_1) + s^3 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^2 (2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + C_1 C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 C_5 L_5 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 L_1 L_4 + 2C_1 C_4 L_1 L_5 R_1 g_m + 2C_1 C_4 L_1 L_5 + 2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5 R_1) + C_1}.$$

9.1008 X-INVALID-ORDER-1008 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, L_5s + R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m s^6 + R_1 g_m + s^5 (C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 R_1 + C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m) + s^4 (C_1 C_4 C_5 L_1 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 R_1 R_4 + C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_5 L_1 L_5 R_1 g_m + C_4 C_5 L_4 L_5 R_1 g_m) + s^3 (C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_5 R_1 g_m + C_1 C_4 C_5 L_1 R_1 R_4 g_m + C_1 C_4 C_5 L_1 R_1 R_5 g_m + C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_1 + C_1 C_4 C_5 L_5 R_1) + s^2 (C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_5 R_1) + s^0 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_4 C_5 R_1)}{s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 + 2C_1 C_4 C_5 L_1 L_5 R_1 g_m + 2C_1 C_4 C_5 L_1 L_5) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 R_1 + C_1 C_4 C_5 L_1 R_4 + 2C_1 C_4 C_5 L_1 R_5 + C_1 C_4 C_5 L_4 R_1 + 2C_1 C_4 C_5 L_5 R_1) + s^3 (C_1 C_4 C_5 R_1 R_4 + 2C_1 C_4 C_5 R_1 R_5 + 2C_1 C_4 C_5 R_1)}$$

9.1009 X-INVALID-ORDER-1009 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2R_1R_5g_m + R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_5 + C_1C_4C_5L_1L_5R_4R_5 + C_1C_4C_5L_4L_5R_1R_5 + 2C_1C_4L_1L_4L_5R_1g_m + C_1C_4L_1L_4L_5) + s^4(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5g_m + C_1C_4L_1L_4R_1R_5) + s^3(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + s^2(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + s(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + C_1C_4C_5L_5R_1R_4R_5}{2R_1R_5g_m + R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_5g_m + C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_5R_1R_5 + C_1C_4C_5L_1L_5R_4R_5 + C_1C_4C_5L_4L_5R_1R_5 + 2C_1C_4L_1L_4L_5R_1g_m + C_1C_4L_1L_4L_5) + s^4(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5g_m + C_1C_4L_1L_4R_1R_5) + s^3(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + s^2(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + s(C_1C_4C_5L_5R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_5) + C_1C_4C_5L_5R_1R_4R_5}$$

9.1010 X-INVALID-ORDER-1010 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, L_4 s + R_4 + \frac{1}{C_4 s}, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^5 (C_1 C_4 C_5 L_1 L_5 R_1 R_4 R_5 g_m - C_1 C_4 C_5 L_1 L_5 R_1 R_4 + C_1 C_4 L_1 L_4 L_5 R_1 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^4 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^3 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^2 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^0 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1)}{2 R_1 g_m + s^6 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^3 (2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^2 (2 C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2 C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2 C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2 C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 + 2 C_1 C_4 L_1 L_4 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5 R_1) + s^0 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_5 g_m - C_1 C_4 C_5 L_1 L_4 L_5 R_1)}$$

9.1011 X-INVALID-ORDER-1011 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, L_4s + R_4 + \frac{1}{C_4s}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^6 (C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5)}{2R_1 g_m + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 g_m + C_1 C_4 C_5 L_1 L_4 L_5) + s^5 (2C_1 C_4 C_5 L_1 L_4 R_1 R_5 g_m + C_1 C_4 C_5 L_1 L_4 R_5 + 2C_1 C_4 C_5 L_1 L_5 R_1 R_4 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 R_5 g_m + 2C_1 C_4 C_5 L_1 L_5 R_1 + C_1 C_4 C_5 L_1 L_5 R_4 + 2C_1 C_4 C_5 L_1 L_5 R_5 + C_1 C_4 C_5 L_4 L_5 R_1) + s^4 (2C_1 C_4 C_5 L_1 R_1 R_4 R_5)}$$

9.1012 X-INVALID-ORDER-1012 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5, \infty \right)$

$$H(s) = \frac{s^3 (C_1 L_1 L_4 R_1 R_4 R_5 g_m - C_1 L_1 L_4 R_1 R_4) + s (L_4 R_1 R_4 R_5 g_m - L_4 R_1 R_4)}{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 R_4 + 2C_1 C_4 L_1 L_4 R_4 R_5) + s^3 (2C_1 C_4 L_4 R_1 R_4 R_5 + 2C_1 L_1 L_4 R_1 R_4 g_m + 2C_1 L_1 L_4 R_1 R_5 g_m + 2C_1 L_1 L_4 R_1 + C_1 L_1 L_4 R_4 + 2C_1 L_1 L_4 R_5) + s^2 (2C_1 L_1 R_1 R_4 R_5 g_m + 2C_1 L_1 R_1 R_4 + 2C_1 L_1 R_1 R_5)}$$

9.1013 X-INVALID-ORDER-1013 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_4 R_1 R_4 s^4 + C_1 L_1 L_4 R_1 R_4 g_m s^3 - C_5 L_4 R_1 R_4 s^2 + L_4 R_1 R_4 g_m s}{2C_1 C_4 C_5 L_1 L_4 R_1 R_4 s^5 + 2R_1 R_4 g_m + 2R_4 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_4 + 2C_1 C_5 L_1 L_4 R_1 R_4 g_m + 2C_1 C_5 L_1 L_4 R_1 + C_1 C_5 L_1 L_4 R_4) + s^3 (2C_1 C_4 L_4 R_1 R_4 + 2C_1 C_5 L_1 R_1 R_4 + C_1 C_5 L_4 R_1 R_4 + 2C_1 L_1 L_4 R_1 g_m + 2C_1 L_1 L_4 + 2C_4 C_5 L_4 R_1 R_4) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_4 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 R_4) + s (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_4 + 2C_4 C_5 L_4 R_1 R_4 g_m + 2C_4 C_5 L_4 R_1 R_4) + 2R_1 R_4 g_m + 2R_4}$$

$$\text{9.1014} \quad \text{X-INVALID-ORDER-1014} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{2C_1C_4C_5L_1L_4R_1R_4R_5s^5 + 2R_1R_4R_5g_m + 2R_1R_4 + 2R_4R_5 + s^4(2C_1C_4L_1L_4R_1R_4R_5g_m + 2C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_1L_4R_4R_5 + 2C_1C_5L_1L_4R_1R_4R_5g_m + 2C_1C_5L_1L_4R_1R_5 + C_1C_5L_1L_4R_4R_5) + s^3(2C_1C_4L_4R_1R_4R_5 + 2C_1C_5L_1R_1R_4R_5 + C_1C_5L_4R_1R_4R_5 -$$

9.1015 X-INVALID-ORDER-1015 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_4 + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5) + s^4(2C_1C_4C_5L_4R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_1R_4g_m + 2C_1C_5L_1L_4R_1R_5g_m + 2C_1C_5L_1L_4R_4 + C_1C_5L_1L_4R_5 + 2C_1C_5L_1L_4R_5)}{2R_1R_4g_m + 2R_4 + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5) + s^4(2C_1C_4C_5L_4R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_1R_4g_m + 2C_1C_5L_1L_4R_1R_5g_m + 2C_1C_5L_1L_4R_4 + C_1C_5L_1L_4R_5 + 2C_1C_5L_1L_4R_5)}$$

9.1016 X-INVALID-ORDER-1016 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_1R_4g_m + 2C_1C_5L_1L_4R_1 + C_1C_5L_1L_4R_4)}{2R_1R_4g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4L_1L_4R_1R_4g_m + 2C_1C_4L_1L_4R_4 + 2C_1C_5L_1L_4R_1R_4g_m + 2C_1C_5L_1L_4R_1 + C_1C_5L_1L_4R_4)}$$

9.1017 X-INVALID-ORDER-1017 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1R_4s^6 + 2R_1R_4 + s^5(2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4) + s^4(2C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_4L_5R_1R_4 + 2C_1C_5L_1L_5R_1R_4 + C_1C_5L_4L_5R_1R_4 + 2C_1L_1L_4L_5R_1g_m + 2C_1L_1L_4L_5R_4)}{2C_1C_4C_5L_1L_4L_5R_1R_4s^6 + 2R_1R_4 + s^5(2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1 + C_1C_5L_1L_4L_5R_4) + s^4(2C_1C_4L_1L_4R_1R_4 + 2C_1C_4L_4L_5R_1R_4 + 2C_1C_5L_1L_5R_1R_4 + C_1C_5L_4L_5R_1R_4 + 2C_1L_1L_4L_5R_1g_m + 2C_1L_1L_4L_5R_4)}$$

9.1018 X-INVALID-ORDER-1018 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5)}{2R_1R_4g_m + 2R_4 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_4) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_4L_5R_1R_4 + 2C_1C_5L_1L_4L_5R_1g_m + 2C_1C_5L_1L_4L_5) + s^4(2C_1C_4C_5L_4R_1R_4R_5 + 2C_1C_4L_1L_4R_1R_4}$$

9.1019 X-INVALID-ORDER-1019 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{2C_1C_4C_5L_1L_4L_5R_1R_4R_5s^6 + 2R_1R_4R_5 + s^5(2C_1C_4L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4L_1L_4L_5R_1R_4 + 2C_1C_4L_1L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_5L_1L_4L_5R_1R_5 + C_1C_5L_1L_4L_5R_4R_5) + s^4(2C_1C_4L_1L_4R_1R_4R_5 + 2C_1C_4L_4L_5R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_4R_5)}{2C_1C_4C_5L_1L_4L_5R_1R_4R_5s^6 + 2R_1R_4R_5 + s^5(2C_1C_4L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4L_1L_4L_5R_1R_4 + 2C_1C_4L_1L_4L_5R_4R_5 + 2C_1C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_5L_1L_4L_5R_1R_5 + C_1C_5L_1L_4L_5R_4R_5) + s^4(2C_1C_4L_1L_4R_1R_4R_5 + 2C_1C_4L_4L_5R_1R_4R_5 + 2C_1C_5L_1L_5R_1R_4R_5)}$$

9.1020 X-INVALID-ORDER-1020 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{L_4R_4s}{C_4L_4R_4s^2+L_4s+R_4}, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{2R_1R_4R_5g_m + 2R_1R_4 + 2R_4R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^5(2C_1C_4C_5L_4L_5R_1R_4R_5 + 2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1R_5g_m + 2C_1C_5L_1L_4L_5R_4R_5)}{s^6(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^5(2C_1C_4C_5L_4L_5R_1R_4R_5 + 2C_1C_4L_1L_4L_5R_1R_4g_m + 2C_1C_4L_1L_4L_5R_4 + 2C_1C_5L_1L_4L_5R_1R_4g_m + 2C_1C_5L_1L_4L_5R_1R_5g_m + 2C_1C_5L_1L_4L_5R_4R_5) + s^4(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^3(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s^2(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + s(2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5) + 2C_1C_4C_5L_1L_4L_5R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1R_4 + 2C_1C_4C_5L_1L_4L_5R_4R_5}$$

$$\text{9.1021} \quad \text{X-INVALID-ORDER-1021} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{L_4 R_4 s}{C_4 L_4 R_4 s^2 + L_4 s + R_4}, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{2R_1 R_4 R_5 g_m + 2R_1 R_4 + 2R_4 R_5 + s^6 (2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 R_5 g_m + 2C_1 C_4 C_5 L_1 L_4 L_5 R_1 R_4 + 2C_1 C_4 C_5 L_1 L_4 L_5 R_4 R_5)}{2C_1 C_4 C_5 L_1 L_4 R_1 R_4 R_5 + 2C_1 C_4 C_5 L_4 L_5 R_1 R_4 R_5 + 2C_1 C_5 L_1 L_4 L_5 R_1 R_4 g_m + 2C_1 C_5 L_1 L_4 L_5 R_1 R_5 g_m + 2C_1 C_5 L_1 L_4 L_5 R_1 + C_1 C_5 L_1 L_4 L_5}$$

9.1022 X-INVALID-ORDER-1022 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \frac{C_4 L_4 R_4 s^2 + L_4 s + R_4}{C_4 L_4 s^2 + 1}, R_5, \infty \right)$

$$H(s) = \frac{R_1 R_4 R_5 g_m - R_1 R_4 + s^4 (C_1 C_4 L_1 L_4 R_1 R_4 R_5 g_m - C_1 C_4 L_1 L_4 R_1 R_4) + s^3 (C_1 L_1 L_4 R_1 R_5 g_m - C_1 L_1 L_4 R_1) + s^2 (C_1 L_1 R_1 R_4 R_5 g_m - C_1 L_1 R_1 R_4 + C_4 L_4 R_1 R_4)}{2R_1 R_4 g_m + 2R_1 R_5 g_m + 2R_1 + R_4 + 2R_5 + s^4 (2C_1 C_4 L_1 L_4 R_1 R_4 g_m + 2C_1 C_4 L_1 L_4 R_1 R_5 g_m + 2C_1 C_4 L_1 L_4 R_1 + C_1 C_4 L_1 L_4 R_4 + 2C_1 C_4 L_1 L_4 R_5) + s^3 (C_1 C_4 L_4 R_1 R_4 + 2C_1 C_4 L_4 R_1 R_5 + 2C_1 L_1 L_4 R_1 g_m + C_1 L_1 L_4) + s^2 (2C_1 L_1 R_1 R_4 g_m + 2C_1 L_1 R_1 R_5 g_m + 2C_1 L_1 R_1 R_4)}$$

9.1041 X-INVALID-ORDER-1041 $Z(s) = \left(\frac{R_1(C_1L_1s^2+1)}{C_1L_1s^2+C_1R_1s+1}, \infty, \infty, \frac{R_4(C_4L_4s^2+1)}{C_4L_4s^2+C_4R_4s+1}, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + C_1C_4C_5L_1L_4L_5R_4 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_5 + C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_1L_4R_5R_4 + 2C_1C_4C_5L_1L_4R_5R_5) + s^4(2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_1R_5 + C_1C_4C_5L_1L_4R_4R_5 + C_1C_4C_5L_1L_4R_5R_4 + C_1C_4C_5L_1L_4R_5R_5) + s^3(2C_1C_4C_5L_1L_4R_1 + 2C_1C_4C_5L_1L_4R_4 + 2C_1C_4C_5L_1L_4R_5) + s^2(2C_1C_4C_5L_1L_4 + 2C_1C_4C_5L_1R_4 + 2C_1C_4C_5L_1R_5) + s(C_1C_4C_5L_1 + C_1C_4C_5R_4 + C_1C_4C_5R_5) + C_1C_4C_5}{2R_1R_4g_m + 2R_1R_5g_m + 2R_1 + R_4 + 2R_5 + s^6(2C_1C_4C_5L_1L_4L_5R_1R_4g_m + 2C_1C_4C_5L_1L_4L_5R_1R_5g_m + 2C_1C_4C_5L_1L_4L_5R_1 + C_1C_4C_5L_1L_4L_5R_4 + 2C_1C_4C_5L_1L_4L_5R_5) + s^5(2C_1C_4C_5L_1L_4R_1R_4R_5g_m + 2C_1C_4C_5L_1L_4R_1R_5 + C_1C_4C_5L_1L_4R_4R_5 + 2C_1C_4C_5L_1L_4R_5R_4 + 2C_1C_4C_5L_1L_4R_5R_5) + s^4(2C_1C_4C_5L_1L_4R_1R_4 + 2C_1C_4C_5L_1L_4R_1R_5 + C_1C_4C_5L_1L_4R_4R_5 + C_1C_4C_5L_1L_4R_5R_4 + C_1C_4C_5L_1L_4R_5R_5) + s^3(2C_1C_4C_5L_1L_4R_1 + 2C_1C_4C_5L_1L_4R_4 + 2C_1C_4C_5L_1L_4R_5) + s^2(2C_1C_4C_5L_1L_4 + 2C_1C_4C_5L_1R_4 + 2C_1C_4C_5L_1R_5) + s(C_1C_4C_5L_1 + C_1C_4C_5R_4 + C_1C_4C_5R_5) + C_1C_4C_5}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 R_1 R_4 R_5 s^2 + R_1 R_5 g_m - R_1 + s(C_4 R_1 R_4 R_5 g_m - C_4 R_1 R_4 - C_5 R_1 R_5)}{2R_1 g_m + s^2(2C_4 C_5 R_1 R_4 R_5 g_m + 2C_4 C_5 R_1 R_5 + C_4 C_5 R_4 R_5) + s(2C_4 R_1 R_4 g_m + 2C_4 R_1 R_5 g_m + 2C_4 R_1 + C_4 R_4 + 2C_4 R_5 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q:
$$\frac{2\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}}{2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5}$$

wo:
$$\sqrt{\frac{2R_1g_m+1}{2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_5+C_4C_5R_4R_5}}$$

bandwidth:
$$\frac{\sqrt{\frac{2R_1g_m+1}{2C_4C_5R_1R_4R_5g_m+2C_4C_5R_1R_5+C_4C_5R_4R_5}}(2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5)}{2\sqrt{C_4}\sqrt{C_5}R_1R_4\sqrt{R_5}g_m\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{C_4}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{C_4}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{2R_1g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}}$$

K-LP:
$$\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$$

K-HP:
$$-\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4}$$

K-BP:
$$\frac{C_4R_1R_4R_5g_m-C_4R_1R_4-C_5R_1R_5}{2C_4R_1R_4g_m+2C_4R_1R_5g_m+2C_4R_1+C_4R_4+2C_4R_5+2C_5R_1R_5g_m+C_5R_5}$$

Qz: None

Wz:
$$\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}\sqrt{R_5}}$$

10.2 X-INVALID-WZ-2 $Z(s) = \left(L_1 s, \infty, \infty, R_4 + \frac{1}{C_4 s}, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_4 C_5 L_1 R_4 s^2 + L_1 g_m + s(C_4 L_1 R_4 g_m - C_5 L_1)}{2C_4 + C_5 + s^2(2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1) + s(C_4 C_5 R_4 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_4g_m\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_4g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_4g_m+1}}}{C_4C_5R_4+2C_4L_1g_m+2C_5L_1g_m}$
wo: $\sqrt{2C_4+C_5}\sqrt{\frac{1}{2C_4C_5L_1R_4g_m+2C_4C_5L_1}}$
bandwidth: $\frac{\sqrt{2C_4+C_5}(C_4C_5R_4+2C_4L_1g_m+2C_5L_1g_m)\sqrt{\frac{1}{2C_4C_5L_1R_4g_m+2C_4C_5L_1}}}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_4g_m\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_4g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{\frac{1}{R_4g_m+1}}}$
K-LP: $\frac{L_1g_m}{2C_4+C_5}$
K-HP: $-\frac{R_4}{2R_4g_m+2}$
K-BP: $\frac{C_4L_1R_4g_m-C_5L_1}{C_4C_5R_4+2C_4L_1g_m+2C_5L_1g_m}$
Qz: None
Wz: $\frac{\sqrt{-g_m}}{\sqrt{C_4}\sqrt{C_5}\sqrt{R_4}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(L_1 s, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 g_m + s^2 (C_4 C_5 L_1 R_4 R_5 g_m - C_4 C_5 L_1 R_4) + s (C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1)}{2C_4 + C_5 + s^2 (2C_4 C_5 L_1 R_4 g_m + 2C_4 C_5 L_1 R_5 g_m + 2C_4 C_5 L_1) + s (C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_4g_m\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_5g_m\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}}{C_4C_5R_4+2C_4C_5R_5+2C_4L_1g_m+2C_5L_1g_m}$
 wo: $\sqrt{2C_4+C_5}\sqrt{\frac{1}{2C_4C_5L_1R_4g_m+2C_4C_5L_1R_5g_m+2C_4C_5L_1}}$
 bandwidth: $\frac{\sqrt{2C_4+C_5}(C_4C_5R_4+2C_4C_5R_5+2C_4L_1g_m+2C_5L_1g_m)\sqrt{2C_4C_5L_1R_4g_m+2C_4C_5L_1R_5g_m+2C_4C_5L_1}}{\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_4g_m\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}R_5g_m\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}+\sqrt{2}\sqrt{C_4}\sqrt{C_5}\sqrt{L_1}\sqrt{2C_4+C_5}\sqrt{R_4g_m+\frac{1}{R_5g_m+1}}}$

$$\begin{aligned}
\text{K-LP: } & \frac{L_1 g_m}{2C_4 + C_5} \\
\text{K-HP: } & \frac{R_4 R_5 g_m - R_4}{2R_4 g_m + 2R_5 g_m + 2} \\
\text{K-BP: } & \frac{C_4 L_1 R_4 g_m + C_5 L_1 R_5 g_m - C_5 L_1}{C_4 C_5 R_4 + 2C_4 C_5 R_5 + 2C_4 L_1 g_m + 2C_5 L_1 g_m} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \sqrt{\frac{g_m}{C_4 C_5 R_4 R_5 g_m - C_4 C_5 R_4}}
\end{aligned}$$

$$10.4 \quad \mathbf{X-INVALID-WZ-4} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 s^2 + R_4 g_m + s (C_1 R_1 R_4 g_m - C_5 R_4)}{2g_m + s^2 (2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5} \\
\text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1+C_1C_5R_4}} \\
\text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1+C_1C_5R_4}}(2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1+R_4}}} \\
\text{K-LP: } & \frac{R_4}{2} \\
\text{K-HP: } & -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\
\text{K-BP: } & \frac{C_1R_1R_4g_m-C_5R_4}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \frac{\sqrt{-g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}}
\end{aligned}$$

$$10.5 \quad \mathbf{X-INVALID-WZ-5} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_4 R_5 s^2 + R_4 R_5 g_m - R_4 + s (C_1 R_1 R_4 R_5 g_m - C_1 R_1 R_4 - C_5 R_4 R_5)}{2R_4 g_m + 2R_5 g_m + s^2 (2C_1 C_5 R_1 R_4 R_5 g_m + 2C_1 C_5 R_1 R_5 + C_1 C_5 R_4 R_5) + s (2C_1 R_1 R_4 g_m + 2C_1 R_1 R_5 g_m + 2C_1 R_1 + C_1 R_4 + 2C_1 R_5 + 2C_5 R_4 R_5 g_m + 2C_5 R_5) + 2}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{R_5g_m}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5} \\
\text{wo: } & \sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_1C_5R_1R_4R_5g_m+2C_1C_5R_1R_5+C_1C_5R_4R_5}} \\
\text{bandwidth: } & \frac{\sqrt{\frac{2R_4g_m+2R_5g_m+2}{2C_1C_5R_1R_4R_5g_m+2C_1C_5R_1R_5+C_1C_5R_4R_5}}(2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4\sqrt{R_5g_m}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{R_5}\sqrt{\frac{R_4g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{R_5g_m}{2R_1R_4g_m+2R_1+R_4}+\frac{1}{2R_1R_4g_m+2R_1+R_4}}} \\
\text{K-LP: } & \frac{R_4R_5g_m-R_4}{2R_4g_m+2R_5g_m+2} \\
\text{K-HP: } & -\frac{R_1R_4}{2R_1R_4g_m+2R_1+R_4} \\
\text{K-BP: } & \frac{C_1R_1R_4R_5g_m-C_1R_1R_4-C_5R_4R_5}{2C_1R_1R_4g_m+2C_1R_1R_5g_m+2C_1R_1+C_1R_4+2C_1R_5+2C_5R_4R_5g_m+2C_5R_5} \\
\text{Qz: } & \text{None} \\
\text{Wz: } & \frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}
\end{aligned}$$

$$10.6 \quad \mathbf{X-INVALID-WZ-6} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad R_4, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_4 g_m + s^2 (C_1 C_5 R_1 R_4 R_5 g_m - C_1 C_5 R_1 R_4) + s (C_1 R_1 R_4 g_m + C_5 R_4 R_5 g_m - C_5 R_4)}{2g_m + s^2 (2C_1 C_5 R_1 R_4 g_m + 2C_1 C_5 R_1 R_5 g_m + 2C_1 C_5 R_1 + C_1 C_5 R_4 + 2C_1 C_5 R_5) + s (2C_1 R_1 g_m + 2C_1 + 2C_5 R_4 g_m + 2C_5 R_5 g_m + 2C_5)}$$

Parameters:

$$\begin{aligned}
\text{Q: } & \frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5} \\
\text{wo: } & \sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1R_5g_m+2C_1C_5R_1+C_1C_5R_4+2C_1C_5R_5}} \\
\text{bandwidth: } & \frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_4g_m+2C_1C_5R_1R_5g_m+2C_1C_5R_1+C_1C_5R_4+2C_1C_5R_5}}(2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}} \\
\text{K-LP: } & \frac{R_4}{2} \\
\text{K-HP: } & \frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5} \\
\text{K-BP: } & \frac{C_1R_1R_4g_m+C_5R_4R_5g_m-C_5R_4}{2C_1R_1g_m+2C_1+2C_5R_4g_m+2C_5R_5g_m+2C_5}
\end{aligned}$$

Qz: None

Wz: $\sqrt{\frac{g_m}{C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1}}$

10.7 X-INVALID-WZ-7 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, R_4 + \frac{1}{C_4 s}, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 C_4 R_1 R_4 R_5 g_m - C_1 C_4 R_1 R_4) + s (C_1 R_1 R_5 g_m - C_1 R_1 + C_4 R_4 R_5 g_m - C_4 R_4) - 1}{2g_m + s^2 (2C_1 C_4 R_1 R_4 g_m + 2C_1 C_4 R_1 R_5 g_m + 2C_1 C_4 R_1 + C_1 C_4 R_4 + 2C_1 C_4 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_4 R_4 g_m + 2C_4 R_5 g_m + 2C_4)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}{2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_1R_5g_m+2C_1C_4R_1+C_1C_4R_4+2C_1C_4R_5}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_4R_1R_4g_m+2C_1C_4R_1R_5g_m+2C_1C_4R_1+C_1C_4R_4+2C_1C_4R_5}}(2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_4g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1R_5g_m\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_1\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_4\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}+2\sqrt{2}\sqrt{C_1}\sqrt{C_4}R_5\sqrt{\frac{g_m}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$

K-HP: $\frac{R_1R_4R_5g_m-R_1R_4}{2R_1R_4g_m+2R_1R_5g_m+2R_1+R_4+2R_5}$

K-BP: $\frac{C_1R_1R_5g_m-C_1R_1+C_4R_4R_5g_m-C_4R_4}{2C_1R_1g_m+C_1+2C_4R_4g_m+2C_4R_5g_m+2C_4}$

Qz: None

Wz: $\frac{1}{\sqrt{C_1}\sqrt{C_4}\sqrt{R_1}\sqrt{R_4}}$

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